

MATH 101: FINITE MATHEMATICS



University of St. Thomas

Math 101

Finite Mathematics

Compiled by Höft and Peterson

version 1.1



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Licensing

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CHAPTER OVERVIEW

1: Problem Solving

In previous math courses, you've no doubt run into the infamous "word problems." Unfortunately, these problems rarely resemble the type of problems we actually encounter in everyday life. In math books, you usually are told exactly which formula or procedure to use, and are given exactly the information you need to answer the question. In real life, problem solving requires identifying an appropriate formula or procedure, and determining what information you will need (and won't need) to answer the question.

In this chapter, we will review several basic but powerful algebraic ideas: percents, rates, and proportions. We will then focus on the problem solving process, and explore how to use these ideas to solve problems where we don't have perfect information.

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1.1: Rounding Decimals

✓ Learning Objectives

- be able to round a decimal number to a specified position

Rounding Decimal Numbers

We first considered the concept of rounding numbers in [\[link\]](#) where our concern with rounding was related to whole numbers only. With a few minor changes, we can apply the same rules of rounding to decimals.

To round a decimal to a particular position:

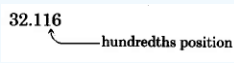
1. Mark the position of the round-off digit (with an arrow or check).
2. Note whether the digit to the immediate right of the marked digit is
 - a. *less than 5*. If so, leave the round-off digit unchanged.
 - b. *5 or greater*. If so, add 1 to the round-off digit.
3. If the round-off digit is
 - a. to the right of the decimal point, eliminate all the digits to its right.
 - b. to the left of the decimal point, replace all the digits between it and the decimal point with zeros and eliminate the decimal point and all the decimal digits.

✓ Sample Set A

Round each decimal to the specified position. (The numbers in parentheses indicate which step is being used.)

Round 32.116 to the nearest hundredth.

Solution

1: 

2b: The digit immediately to the right is 6, and $6 > 5$, so we add 1 to the round-off digit:

$$1 + 1 = 2$$

3a: The round-off digit is to the right of the decimal point, so we eliminate all digits to its right.

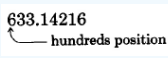
32.12

The number 32.116 rounded to the nearest hundredth is 32.12.

✓ Sample Set A

Round 633.14216 to the nearest hundred.

Solution

1: 

2a: The digit immediately to the right is 3, and $3 < 5$ so we leave the round-off digit unchanged.

3b: The round-off digit is to the left of 0, so we replace all the digits between it and the decimal point with zeros and eliminate the decimal point and all the decimal digits.

600

The number 633.14216 rounded to the nearest hundred is 600.

✓ Sample Set A

1,729.63 rounded to the nearest ten is 1,730.

✓ Sample Set A

1.0144 rounded to the nearest tenth is 1.0.

✓ Sample Set A

60.98 rounded to the nearest one is 61.

Sometimes we hear a phrase such as "round to three decimal places." This phrase means that the round-off digit is the third decimal digit (the digit in the thousandths position).

✓ Sample Set A

67.129 rounded to the second decimal place is 67.13.

✓ Sample Set A

67.129558 rounded to 3 decimal places is 67.130.

Practice Set A

Round each decimal to the specified position.

4.816 to the nearest hundredth.

Answer

4.82

Practice Set A

0.35928 to the nearest ten thousandths.

Answer

0.3593

Practice Set A

82.1 to the nearest one.

Answer

82

Practice Set A

753.98 to the nearest hundred.

Answer

800

Practice Set A

Round 43.99446 to three decimal places.

Answer

43.994

Practice Set A

Round 105.019997 to four decimal places.

Answer

105.0200

Practice Set A

Round 99.9999 to two decimal places.

Answer

100.00

Exercises

For the first 10 problems, complete the chart by rounding each decimal to the indicated positions.

Exercise 1.1.1

20.01071

Tenth	Hundredth	Thousandth	Ten Thousandth

Answer

Tenth	Hundredth	Thousandth	Ten Thousandth
20.0	20.01	20.011	20.0107

Exercise 1.1.2

3.52612

Tenth	Hundredth	Thousandth	Ten Thousandth
	3.53		

Exercise 1.1.3

531.21878

Tenth	Hundredth	Thousandth	Ten Thousandth

Answer

Tenth	Hundredth	Thousandth	Ten Thousandth
531.2	531.22	531.219	531.2188

Exercise 1.1.4

36.109053

Tenth	Hundredth	Thousandth	Ten Thousandth
36.1			

Exercise 1.1.5

1.999994

Tenth	Hundredth	Thousandth	Ten Thousandth

Answer

Tenth	Hundredth	Thousandth	Ten Thousandth
2.0	2.00	2.000	2.0000

Exercise 1.1.6

7.4141998

Tenth	Hundredth	Thousandth	Ten Thousandth
		7.414	

Exercise 1.1.7

0.000007

Tenth	Hundredth	Thousandth	Ten Thousandth

Answer

Tenth	Hundredth	Thousandth	Ten Thousandth
0.0	0.00	0.000	0.0000

Exercise 1.1.8

0.00008

Tenth	Hundredth	Thousandth	Ten Thousandth
			0.0001

Exercise 1.1.9

9.19191919

Tenth	Hundredth	Thousandth	Ten Thousandth

Answer

Tenth	Hundredth	Thousandth	Ten Thousandth
9.2	9.19	9.192	9.1919

Exercise 1.1.10

0.0876543

Tenth	Hundredth	Thousandth	Ten Thousandth

Calculator Problems

For the following 5 problems, round 18.4168095 to the indicated place.

Exercise 1.1.11

3 decimal places.

Answer

18.417

Exercise 1.1.12

1 decimal place.

Exercise 1.1.13

5 decimal places.

Answer

18.41681

Exercise 1.1.14

6 decimal places.

Exercise 1.1.15

2 decimal places.

Answer

18.42

Calculator Problems

For the following problems, perform each division using a calculator.

Exercise 1.1.16

$4 \div 3$ and round to 2 decimal places.

Exercise 1.1.17

$1 \div 8$ and round to 1 decimal place.

Answer

0.1

Exercise 1.1.18

$1 \div 27$ and round to 6 decimal places.

Exercise 1.1.19

$51 \div 61$ and round to 5 decimal places.

Answer

0.83607

Exercise 1.1.20

$3 \div 16$ and round to 3 decimal places.

Exercise 1.1.21

$16 \div 3$ and round to 3 decimal places.

Answer

5.333

Exercise 1.1.22

$26 \div 7$ and round to 5 decimal places.

Exercises for Review**Exercise 1.1.23**

What is the value of 2 in the number 421,916,017?

Answer

Ten million

Exercise 1.1.24

Perform the division: $378 \div 29$.

Exercise 1.1.25

Find the value of 4^4 .

Answer

256

Exercise 1.1.26

Convert $\frac{11}{3}$ to a mixed number.

Exercise 1.1.27

Convert 3.16 to a mixed number fraction.

Answer

$3\frac{4}{25}$

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1.2: Percents

In the 2004 vice-presidential debates, Edwards's claimed that US forces have suffered "90% of the coalition casualties" in Iraq. Cheney disputed this, saying that in fact Iraqi security forces and coalition allies "have taken almost 50 percent" of the casualties[1]. Who is correct? How can we make sense of these numbers?

Percent literally means "per 100," or "parts per hundred." When we write 40%, this is equivalent to the fraction $\frac{40}{100}$ or the decimal 0.40. Notice that 80 out of 200 and 10 out of 25 are also 40%, since $\frac{80}{200} = \frac{10}{25} = \frac{40}{100}$.

✓ Example 1

243 people out of 400 state that they like dogs. What percent is this?

Solution

$\frac{243}{400} = 0.6075 = \frac{60.75}{100}$. This is 60.75%.

Notice that the percent can be found from the equivalent decimal by moving the decimal point two places to the right.

✓ Example 2

Write each as a percent: a) $\frac{1}{4}$ b) 0.02 c) 2.35

Solution

a) $\frac{1}{4} = 0.25 = 25\%$ b) $0.02 = 2\%$ c) $2.35 = 235\%$

📌 Percents

If we have a *part* that is some *percent* of a *whole*, then

$\text{percent} = \frac{\text{part}}{\text{whole}}$, or equivalently, $\text{part} = \text{percent} \cdot \text{whole}$

To do the calculations, we write the percent as a decimal.

✓ Example 3

The sales tax in a town is 9.4%. How much tax will you pay on a \$140 purchase?

Solution

Here, \$140 is the whole, and we want to find 9.4% of \$140. We start by writing the percent as a decimal by moving the decimal point two places to the left (which is equivalent to dividing by 100). We can then compute:

$\text{tax} = 0.094(140) = \13.16 in tax.

✓ Example 4

In the news, you hear "tuition is expected to increase by 7% next year." If tuition this year was \$1200 per quarter, what will it be next year?

Solution

The tuition next year will be the current tuition plus an additional 7%, so it will be 107% of this year's tuition:

$\$1200(1.07) = \1284

Alternatively, we could have first calculated 7% of \$1200: $\$1200(0.07) = \84

Notice this is *not* the expected tuition for next year (we could only wish). Instead, this is the expected *increase*, so to calculate the expected tuition, we'll need to add this change to the previous year's tuition:

$\$1200 + \$84 = \$1284$

? Try it Now 1

A TV originally priced at \$799 is on sale for 30% off. There is then a 9.2% sales tax. Find the price after including the discount and sales tax.

Answer

The sale price is $\$799(0.70) = \559.30 . After tax, the price is $\$559.30(1.092) = \610.76

✓ Example 5

The value of a car dropped from \$7400 to \$6800 over the last year. What percent decrease is this?

Solution

To compute the percent change, we first need to find the dollar value change: $\$6800 - \$7400 = -\$600$. Often we will take the absolute value of this amount, which is called the **absolute change**: $|-600| = 600$.

Since we are computing the decrease relative to the starting value, we compute this percent out of \$7400:

$\frac{600}{7400} = 0.081 = 8.1\%$ decrease. This is called a **relative change**.

📌 Absolute and Relative Change

Given two quantities,

Absolute change = |ending quantity – starting quantity|

Relative change: $\frac{\text{absolute change}}{\text{starting quantity}}$

Absolute change has the same units as the original quantity.

Relative change gives a percent change.

The starting quantity is called the **base** of the percent change.

The base of a percent is very important. For example, while Nixon was president, it was argued that marijuana was a “gateway” drug, claiming that 80% of marijuana smokers went on to use harder drugs like cocaine. The problem is, this isn’t true. The true claim is that 80% of harder drug users first smoked marijuana. The difference is one of base: 80% of marijuana smokers using hard drugs, vs. 80% of hard drug users having smoked marijuana. These numbers are not equivalent. As it turns out, only one in 2,400 marijuana users actually go on to use harder drugs[2].

✓ Example 6

There are about 75 QFC supermarkets in the U.S. Albertsons has about 215 stores. Compare the size of the two companies.

Solution

When we make comparisons, we must ask first whether an absolute or relative comparison. The absolute difference is $215 - 75 = 140$. From this, we could say “Albertsons has 140 more stores than QFC.” However, if you wrote this in an article or paper, that number does not mean much. The relative difference may be more meaningful. There are two different relative changes we could calculate, depending on which store we use as the base:

Using QFC as the base, $\frac{140}{75} = 1.867$.

This tells us Albertsons is 186.7% larger than QFC.

Using Albertsons as the base, $\frac{140}{215} = 0.651$.

This tells us QFC is 65.1% smaller than Albertsons.

Notice both of these are showing percent *differences*. We could also calculate the size of Albertsons relative to QFC: , which tells us Albertsons is 2.867 times the size of QFC. Likewise, we could calculate the size of QFC relative to Albertsons: , which tells us that QFC is 34.9% of the size of Albertsons.

✓ Example 7

Suppose a stock drops in value by 60% one week, then increases in value the next week by 75%. Is the value higher or lower than where it started?

Solution

To answer this question, suppose the value started at \$100. After one week, the value dropped by 60%:

$$\$100 - \$100(0.60) = \$100 - \$60 = \$40$$

In the next week, notice that base of the percent has changed to the new value, \$40. Computing the 75% increase:

$$\$40 + \$40(0.75) = \$40 + \$30 = \$70.$$

In the end, the stock is still \$30 lower, or $\frac{\$30}{\$100} = 30\%$ lower, valued than it started.

? Try it Now 2

The U.S. federal debt at the end of 2001 was \$5.77 trillion, and grew to \$6.20 trillion by the end of 2002. At the end of 2005 it was \$7.91 trillion, and grew to \$8.45 trillion by the end of 2006[3]. Calculate the absolute and relative increase for 2001-2002 and 2005-2006. Which year saw a larger increase in federal debt?

Answer

2001-2002: Absolute change: \$0.43 trillion. Relative change: 7.45%

2005-2006: Absolute change: \$0.54 trillion. Relative change: 6.83%

2005-2006 saw a larger absolute increase, but a smaller relative increase.

✓ Example 8

A Seattle Times article on high school graduation rates reported “The number of schools graduating 60 percent or fewer students in four years – sometimes referred to as “dropout factories” – decreased by 17 during that time period. The number of kids attending schools with such low graduation rates was cut in half.”

a) Is the “decrease by 17” number a useful comparison?

b) Considering the last sentence, can we conclude that the number of “dropout factories” was originally 34?

Solution

a) This number is hard to evaluate, since we have no basis for judging whether this is a larger or small change. If the number of “dropout factories” dropped from 20 to 3, that’d be a very significant change, but if the number dropped from 217 to 200, that’d be less of an improvement.

b) The last sentence provides relative change which helps put the first sentence in perspective. We can estimate that the number of “dropout factories” was probably previously around 34. However, it’s possible that students simply moved schools rather than the school improving, so that estimate might not be fully accurate.

✓ Example 9

In the 2004 vice-presidential debates, Edwards's claimed that US forces have suffered "90% of the coalition casualties" in Iraq. Cheney disputed this, saying that in fact Iraqi security forces and coalition allies "have taken almost 50 percent" of the casualties. Who is correct?

Solution

Without more information, it is hard for us to judge who is correct, but we can easily conclude that these two percents are talking about different things, so one does not necessarily contradict the other. Edward's claim was a percent with coalition forces as the base of the percent, while Cheney's claim was a percent with both coalition and Iraqi security forces as the base of the percent. It turns out both statistics are in fact fairly accurate.

? Try it Now 3

In the 2012 presidential elections, one candidate argued that “the president’s plan will cut \$716 billion from Medicare, leading to fewer services for seniors,” while the other candidate rebuts that “our plan does not cut current spending and actually expands benefits for seniors, while implementing cost saving measures.” Are these claims in conflict, in agreement, or not comparable because they’re talking about different things?

Answer

Without more information, it is hard to judge these arguments. This is compounded by the complexity of Medicare. As it turns out, the \$716 billion is not a cut in current spending, but a cut in future increases in spending, largely reducing future growth in health care payments. In this case, at least the numerical claims in both statements could be considered at least partially true. Here is one source of more information if you’re interested: <http://factcheck.org/2012/08/a-campaign-full-of-medicare/>

We’ll wrap up our review of percents with a couple cautions. First, when talking about a change of quantities that are already measured in percents, we have to be careful in how we describe the change.

✓ Example 10

A politician’s support increases from 40% of voters to 50% of voters. Describe the change.

Solution

We could describe this using an absolute change: $|50\% - 40\%| = 10\%$. Notice that since the original quantities were percents, this change also has the units of percent. In this case, it is best to describe this as an increase of 10 **percentage points**.

In contrast, we could compute the percent change: $\frac{10\%}{40\%} = 0.25 = 25\%$ increase. This is the relative change, and we’d say the politician’s support has increased by 25%.

Lastly, a caution against averaging percents.

✓ Example 11

A basketball player scores on 40% of 2-point field goal attempts, and on 30% of 3-point of field goal attempts. Find the player’s overall field goal percentage.

Solution

It is very tempting to average these values, and claim the overall average is 35%, but this is likely not correct, since most players make many more 2-point attempts than 3-point attempts. We don’t actually have enough information to answer the question. Suppose the player attempted 200 2-point field goals and 100 3-point field goals. Then they made $200(0.40) = 80$ 2-point shots and $100(0.30) = 30$ 3-point shots. Overall, they made 110 shots out of 300, for a $\frac{110}{300} = 0.367 = 36.7\%$ overall field goal percentage.

[1] www.factcheck.org/cheney_edwards_mangle_facts.html

[2] <http://tvtropes.org/pmwiki/pmwiki.php/Main/LiesDamnedLiesAndStatistics>

[3] www.whitehouse.gov/sites/default/files/hist07z1.xls

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1.3: Proportions and Rates

If you wanted to power the city of Seattle using wind power, how many windmills would you need to install? Questions like these can be answered using rates and proportions.

✚ Rates

A rate is the ratio (fraction) of two quantities.

A **unit rate** is a rate with a denominator of one.

✓ Example 12

Your car can drive 300 miles on a tank of 15 gallons. Express this as a rate.

Solution

Expressed as a rate, $\frac{300 \text{ miles}}{15 \text{ gallons}}$. We can divide to find a unit rate: $\frac{20 \text{ miles}}{1 \text{ gallon}}$, which we could also write as $20 \frac{\text{miles}}{\text{gallon}}$, or just 20 miles per gallon.

✚ Proportion Equation

A proportion equation is an equation showing the equivalence of two rates or ratios.

✓ Example 13

Solve the proportion $\frac{5}{3} = \frac{x}{6}$ for the unknown value x .

Solution

This proportion is asking us to find a fraction with denominator 6 that is equivalent to the fraction $\frac{5}{3}$. We can solve this by multiplying both sides of the equation by 6, giving $x = \frac{5}{3} \cdot 6 = 10$.

✓ Example 14

A map scale indicates that $\frac{1}{2}$ inch on the map corresponds with 3 real miles. How many miles apart are two cities that are $2\frac{1}{4}$ inches apart on the map?

Solution

We can set up a proportion by setting equal two $\frac{\text{map inches}}{\text{real miles}}$ rates, and introducing a variable, x , to represent the unknown quantity – the mile distance between the cities.

$$\frac{\frac{1}{2} \text{ map inch}}{3 \text{ miles}} = \frac{2\frac{1}{4} \text{ map inches}}{x \text{ miles}} \quad \text{Multiply both sides by } x \text{ and rewriting the mixed number}$$

$$\frac{1}{2} \cdot x = \frac{9}{4} \quad \text{Multiply both sides by 3}$$

$$\frac{1}{2}x = \frac{27}{4} \quad \text{Multiply both sides by 2 (or divide by } \frac{1}{2})$$

$$x = \frac{27}{2} = 13\frac{1}{2} \text{ miles}$$

Many proportion problems can also be solved using **dimensional analysis**, the process of multiplying a quantity by rates to change the units.

✓ Example 15

Your car can drive 300 miles on a tank of 15 gallons. How far can it drive on 40 gallons?

Solution

We could certainly answer this question using a proportion: $\$ \frac{300 \text{ miles}}{15 \text{ gallons}} = \frac{x \text{ miles}}{40 \text{ gallons}} \$$.

However, we earlier found that 300 miles on 15 gallons gives a rate of 20 miles per gallon. If we multiply the given 40 gallon quantity by this rate, the *gallons* unit “cancels” and we’re left with a number of miles:

$$40 \text{ gallons} \cdot \frac{20 \text{ miles}}{1 \text{ gallon}} = \frac{40 \text{ gallons}}{1} \cdot \frac{20 \text{ miles}}{\text{gallon}} = 800 \text{ miles}$$

Notice if instead we were asked “how many gallons are needed to drive 50 miles?” we could answer this question by inverting the 20 mile per gallon rate so that the *miles* unit cancels and we’re left with gallons:

$$50 \text{ miles} \cdot \frac{1 \text{ gallon}}{20 \text{ miles}} = \frac{50 \text{ miles}}{1} \cdot \frac{1 \text{ gallon}}{20 \text{ miles}} = \frac{50 \text{ gallons}}{20} = 2.5 \text{ gallons}$$

Dimensional analysis can also be used to do unit conversions. Here are some unit conversions for reference.

Unit Conversions

Length

1 foot (ft) = 12 inches (in) 1 yard (yd) = 3 feet (ft)
 1 mile = 5,280 feet
 1000 millimeters (mm) = 1 meter (m) 100 centimeters (cm) = 1 meter
 1000 meters (m) = 1 kilometer (km) 2.54 centimeters (cm) = 1 inch

Weight and Mass

1 pound (lb) = 16 ounces (oz) 1 ton = 2000 pounds
 1000 milligrams (mg) = 1 gram (g) 1000 grams = 1 kilogram (kg)
 1 kilogram = 2.2 pounds (on earth)

Capacity

1 cup = 8 fluid ounces (fl oz)* 1 pint = 2 cups
 1 quart = 2 pints = 4 cups 1 gallon = 4 quarts = 16 cups
 1000 milliliters (ml) = 1 liter (L)

*Fluid ounces are a capacity measurement for liquids. 1 fluid ounce $\approx$ 1 ounce (weight) for water only.

Example 16

A bicycle is traveling at 15 miles per hour. How many feet will it cover in 20 seconds?

Solution

To answer this question, we need to convert 20 seconds into feet. If we know the speed of the bicycle in feet per second, this question would be simpler. Since we don’t, we will need to do additional unit conversions. We will need to know that 5280 ft = 1 mile. We might start by converting the 20 seconds into hours:

$$20 \text{ seconds} \cdot \frac{1 \text{ minute}}{60 \text{ seconds}} \cdot \frac{1 \text{ hour}}{60 \text{ minutes}} = \frac{1}{180} \text{ hour} \quad \text{Now we can multiply by the 15 miles/hr}$$

$$\frac{1}{180} \text{ hour} \cdot \frac{15 \text{ miles}}{1 \text{ hour}} = \frac{1}{12} \text{ mile} \quad \text{Now we can convert to feet}$$

$$\frac{1}{12} \text{ mile} \cdot \frac{5280 \text{ feet}}{1 \text{ mile}} = 440 \text{ feet}$$

We could have also done this entire calculation in one long set of products:

$$20 \text{ seconds} \cdot \frac{1 \text{ minute}}{60 \text{ seconds}} \cdot \frac{1 \text{ hour}}{60 \text{ minutes}} \cdot \frac{15 \text{ miles}}{1 \text{ hour}} \cdot \frac{5280 \text{ feet}}{1 \text{ mile}} = 440 \text{ feet}$$

? Try it Now 4

A 1000 foot spool of bare 12-gauge copper wire weighs 19.8 pounds. How much will 18 inches of the wire weigh, in ounces?

Answer

$$18 \text{ inches} \cdot \frac{1 \text{ foot}}{12 \text{ inches}} \cdot \frac{19.8 \text{ pounds}}{1000 \text{ feet}} \cdot \frac{16 \text{ ounces}}{1 \text{ pound}} \approx 0.475 \text{ ounces}$$

Notice that with the miles per gallon example, if we double the miles driven, we double the gas used. Likewise, with the map distance example, if the map distance doubles, the real-life distance doubles. This is a key feature of proportional relationships, and one we must confirm before assuming two things are related proportionally.

✓ Example 17

Suppose you're tiling the floor of a 10 ft by 10 ft room, and find that 100 tiles will be needed. How many tiles will be needed to tile the floor of a 20 ft by 20 ft room?

Solution

In this case, while the width the room has doubled, the area has quadrupled. Since the number of tiles needed corresponds with the area of the floor, not the width, 400 tiles will be needed. We could find this using a proportion based on the areas of the rooms:

$$\frac{100 \text{ tiles}}{100\text{ft}^2} = \frac{n \text{ tiles}}{400\text{ft}^2}$$

Other quantities just don't scale proportionally at all.

✓ Example 18

Suppose a small company spends \$1000 on an advertising campaign, and gains 100 new customers from it. How many new customers should they expect if they spend \$10,000?

Solution

While it is tempting to say that they will gain 1000 new customers, it is likely that additional advertising will be less effective than the initial advertising. For example, if the company is a hot tub store, there are likely only a fixed number of people interested in buying a hot tub, so there might not even be 1000 people in the town who would be potential customers.

Sometimes when working with rates, proportions, and percents, the process can be made more challenging by the magnitude of the numbers involved. Sometimes, large numbers are just difficult to comprehend.

✓ Example 19

Compare the 2010 U.S. military budget of \$683.7 billion to other quantities.

Here we have a very large number, about \$683,700,000,000 written out. Of course, imagining a billion dollars is very difficult, so it can help to compare it to other quantities.

If that amount of money was used to pay the salaries of the 1.4 million Walmart employees in the U.S., each would earn over \$488,000.

There are about 300 million people in the U.S. The military budget is about \$2,200 per person.

If you were to put \$683.7 billion in \$100 bills, and count out 1 per second, it would take 216 years to finish counting it.

✓ Example 20

Compare the electricity consumption per capita in China to the rate in Japan.

To address this question, we will first need data. From the CIA[1] website we can find the electricity consumption in 2011 for China was 4,693,000,000,000 KWH (kilowatt-hours), or 4.693 trillion KWH, while the consumption for Japan was 859,700,000,000, or 859.7 billion KWH. To find the rate per capita (per person), we will also need the population of the two countries. From the World Bank[2], we can find the population of China is 1,344,130,000, or 1.344 billion, and the population of Japan is 127,817,277, or 127.8 million.

Solution

Computing the consumption per capita for each country:

$$\text{China: } \frac{4,693,000,000,000 \text{ KWH}}{1,344,130,000 \text{ people}} \approx 3491.5 \text{ KWH per person}$$

$$\text{Japan: } \frac{859,700,000,000 \text{ KWH}}{127,817,277 \text{ people}} \approx 6726 \text{ KWH per person}$$

While China uses more than 5 times the electricity of Japan overall, because the population of Japan is so much smaller, it turns out Japan uses almost twice the electricity per person compared to China.

[1] www.cia.gov/library/publicat.../2042rank.html

[2] <http://data.worldbank.org/indicator/SP.POP.TOT>

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1.4: Problem Solving and Estimating

Finally, we will bring together the mathematical tools we've reviewed, and use them to approach more complex problems. In many problems, it is tempting to take the given information, plug it into whatever formulas you have handy, and hope that the result is what you were supposed to find. Chances are, this approach has served you well in other math classes.

This approach does not work well with real life problems. Instead, problem solving is best approached by first starting at the end: identifying exactly what you are looking for. From there, you then work backwards, asking "what information and procedures will I need to find this?" Very few interesting questions can be answered in one mathematical step; often times you will need to chain together a solution pathway, a series of steps that will allow you to answer the question.

Problem Solving Process

1. Identify the question you're trying to answer.
2. Work backwards, identifying the information you will need and the relationships you will use to answer that question.
3. Continue working backwards, creating a solution pathway.
4. If you are missing necessary information, look it up or estimate it. If you have unnecessary information, ignore it.
5. Solve the problem, following your solution pathway.

In most problems we work, we will be approximating a solution, because we will not have perfect information. We will begin with a few examples where we will be able to approximate the solution using basic knowledge from our lives.

✓ Example 24

How many times does your heart beat in a year?

Solution

This question is asking for the rate of heart beats per year. Since a year is a long time to measure heart beats for, if we knew the rate of heart beats per minute, we could scale that quantity up to a year. So the information we need to answer this question is heart beats per minute. This is something you can easily measure by counting your pulse while watching a clock for a minute.

Suppose you count 80 beats in a minute. To convert this beats per year:

$$\frac{80 \text{ beats}}{1 \text{ minute}} \cdot \frac{60 \text{ minutes}}{1 \text{ hour}} \cdot \frac{24 \text{ hours}}{1 \text{ day}} \cdot \frac{365 \text{ days}}{1 \text{ year}} = 42,048,000 \text{ beats per year}$$

✓ Example 25

How thick is a single sheet of paper? How much does it weigh?

Solution

While you might have a sheet of paper handy, trying to measure it would be tricky. Instead we might imagine a stack of paper, and then scale the thickness and weight to a single sheet. If you've ever bought paper for a printer or copier, you probably bought a ream, which contains 500 sheets. We could estimate that a ream of paper is about 2 inches thick and weighs about 5 pounds. Scaling these down,

$$\frac{2 \text{ inches}}{\text{ream}} \cdot \frac{1 \text{ ream}}{500 \text{ pages}} = 0.004 \text{ inches per sheet}$$

$$\frac{5 \text{ pounds}}{\text{ream}} \cdot \frac{1 \text{ ream}}{500 \text{ pages}} = 0.01 \text{ pounds per sheet, or } 0.16 \text{ ounces per sheet.}$$

✓ Example 26

A recipe for zucchini muffins states that it yields 12 muffins, with 250 calories per muffin. You instead decide to make mini-muffins, and the recipe yields 20 muffins. If you eat 4, how many calories will you consume?

Solution

There are several possible solution pathways to answer this question. We will explore one.

To answer the question of how many calories 4 mini-muffins will contain, we would want to know the number of calories in each mini-muffin. To find the calories in each mini-muffin, we could first find the total calories for the entire recipe, then divide it by the number of mini-muffins produced. To find the total calories for the recipe, we could multiply the calories per standard muffin by the number per muffin. Notice that this produces a multi-step solution pathway. It is often easier to solve a problem in small steps, rather than trying to find a way to jump directly from the given information to the solution.

We can now execute our plan:

$$12 \text{ muffins} \cdot \frac{250 \text{ calories}}{\text{muffin}} = 3000 \text{ calories for the whole recipe}$$

$$\frac{3000 \text{ calories}}{20 \text{ mini - muffins}} \text{ gives } 150 \text{ calories per mini-muffin}$$

$$4 \text{ mini muffins} \cdot \frac{150 \text{ calories}}{\text{mini - muffin}} \text{ totals } 600 \text{ calories consumed.}$$

✓ Example 27

You need to replace the boards on your deck. About how much will the materials cost?

Solution

There are two approaches we could take to this problem: 1) estimate the number of boards we will need and find the cost per board, or 2) estimate the area of the deck and find the approximate cost per square foot for deck boards. We will take the latter approach.

For this solution pathway, we will be able to answer the question if we know the cost per square foot for decking boards and the square footage of the deck. To find the cost per square foot for decking boards, we could compute the area of a single board, and divide it into the cost for that board. We can compute the square footage of the deck using geometric formulas. So first we need information: the dimensions of the deck, and the cost and dimensions of a single deck board.

Suppose that measuring the deck, it is rectangular, measuring 16 ft by 24 ft, for a total area of 384ft^2 .

From a visit to the local home store, you find that an 8 foot by 4 inch cedar deck board costs about \$7.50. The area of this board, doing the necessary conversion from inches to feet, is:

$$8 \text{ feet} \cdot 4 \text{ inches} \cdot \frac{1 \text{ foot}}{12 \text{ inches}} = 2.667\text{ft}^2. \text{ The cost per square foot is then}$$

$$\frac{\$7.50}{2.667\text{ft}^2} = \$2.8125 \text{ per ft}^2.$$

This will allow us to estimate the material cost for the whole 384ft^2 deck

$$\$384\text{ft}^2 \cdot \frac{\$2.8125}{\text{ft}^2} = \$1080 \text{ total cost.}$$

Of course, this cost estimate assumes that there is no waste, which is rarely the case. It is common to add at least 10% to the cost estimate to account for waste.

✓ Example 28

Is it worth buying a Hyundai Sonata hybrid instead the regular Hyundai Sonata?

Solution

To make this decision, we must first decide what our basis for comparison will be. For the purposes of this example, we'll focus on fuel and purchase costs, but environmental impacts and maintenance costs are other factors a buyer might consider.

It might be interesting to compare the cost of gas to run both cars for a year. To determine this, we will need to know the miles per gallon both cars get, as well as the number of miles we expect to drive in a year. From that information, we can find the number of gallons required from a year. Using the price of gas per gallon, we can find the running cost.

From Hyundai's website, the 2013 Sonata will get 24 miles per gallon (mpg) in the city, and 35 mpg on the highway. The hybrid will get 35 mpg in the city, and 40 mpg on the highway.

An average driver drives about 12,000 miles a year. Suppose that you expect to drive about 75% of that in the city, so 9,000 city miles a year, and 3,000 highway miles a year.

We can then find the number of gallons each car would require for the year.

Sonata:

$$9000 \text{ city miles} \cdot \frac{1 \text{ gallon}}{24 \text{ city miles}} + 3000 \text{ highway miles} \cdot \frac{1 \text{ gallon}}{35 \text{ highway miles}} = 460.7 \text{ gallons}$$

Hybrid:

$$9000 \text{ city miles} \cdot \frac{1 \text{ gallon}}{35 \text{ city miles}} + 3000 \text{ highway miles} \cdot \frac{1 \text{ gallon}}{40 \text{ highway miles}} = 332.1 \text{ gallons}$$

If gas in your area averages about \$3.50 per gallon, we can use that to find the running cost:

$$\text{Sonata: } 460.7 \text{ gallons} \cdot \frac{\$3.50}{\text{gallon}} = \$1612.45$$

$$\text{Hybrid: } 332.1 \text{ gallons} \cdot \frac{\$3.50}{\text{gallon}} = \$1162.35$$

The hybrid will save \$450.10 a year. The gas costs for the hybrid are about $\frac{\$450.10}{\$1612.45} = 0.279 = 27.9\%$ lower than the costs for the standard Sonata.

While both the absolute and relative comparisons are useful here, they still make it hard to answer the original question, since “is it worth it” implies there is some tradeoff for the gas savings. Indeed, the hybrid Sonata costs about \$25,850, compared to the base model for the regular Sonata, at \$20,895.

To better answer the “is it worth it” question, we might explore how long it will take the gas savings to make up for the additional initial cost. The hybrid costs \$4965 more. With gas savings of \$451.10 a year, it will take about 11 years for the gas savings to make up for the higher initial costs.

We can conclude that if you expect to own the car 11 years, the hybrid is indeed worth it. If you plan to own the car for less than 11 years, it may still be worth it, since the resale value of the hybrid may be higher, or for other non-monetary reasons. This is a case where math can help guide your decision, but it can’t make it for you.

? Try it Now 6

If traveling from Seattle, WA to Spokane WA for a three-day conference, does it make more sense to drive or fly?

Answer

There is not enough information provided to answer the question, so we will have to make some assumptions, and look up some values.

Assumptions:

- a) We own a car. Suppose it gets 24 miles to the gallon. We will only consider gas cost.
- b) We will not need to rent a car in Spokane, but will need to get a taxi from the airport to the conference hotel downtown and back.
- c) We can get someone to drop us off at the airport, so we don’t need to consider airport parking.
- d) We will not consider whether we will lose money by having to take time off work to drive.

Values looked up (your values may be different)

- a) Flight cost: \$184
- b) Taxi cost: \$25 each way (estimate, according to hotel website)
- c) Driving distance: 280 miles each way
- d) Gas cost: \$3.79 a gallon

Cost for flying: \$184 flight cost + \$50 in taxi fares = \$234.

Cost for driving: 560 miles round trip will require 23.3 gallons of gas, costing \$88.31

Based on these assumptions, driving is cheaper. However, our assumption that we only include gas cost may not be a good one. Tax law allows you deduct \$0.55 (in 2012) for each mile driven, a value that accounts for gas as well as a portion of

the car cost, insurance, maintenance, etc. Based on this number, the cost of driving would be \$319

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1.5: Extension - Taxes

Governments collect taxes to pay for the services they provide. In the United States, federal income taxes help fund the military, the environmental protection agency, and thousands of other programs. Property taxes help fund schools. Gasoline taxes help pay for road improvements. While very few people enjoy paying taxes, they are necessary to pay for the services we all depend upon.

Taxes can be computed in a variety of ways, but are typically computed as a percentage of a sale, of one's income, or of one's assets.

✓ Example 1

The sales tax rate in a city is 9.3%. How much sales tax will you pay on a \$140 purchase?

Solution

The sales tax will be 9.3% of \$140. To compute this, we multiply \$140 by the percent written as a decimal:
 $\$140(0.093) = \13.02

When taxes are not given as a fixed percentage rate, sometimes it is necessary to calculate the **effective rate**.

📌 Effective rate

The effective tax rate is the equivalent percent rate of the tax paid out of the dollar amount the tax is based on.

✓ Example 2

Joan paid \$3,200 in property taxes on her house valued at \$215,000 last year. What is the effective tax rate?

Solution

We can compute the equivalent percentage: $3200/215000 = 0.01488$ or about 1.49% effective rate.

Taxes are often referred to as progressive, regressive, or flat.

📌 Tax categories

A **flat tax**, or proportional tax, charges a constant percentage rate.

A **progressive tax** increases the percent rate as the base amount increases.

A **regressive tax** decreases the percent rate as the base amount increases.

✓ Example 3

The United States federal income tax on earned wages is an example of a progressive tax. People with a higher wage income pay a higher percent tax on their income.

Solution

For a single person in 2011, adjusted gross income (income after deductions) under \$8,500 was taxed at 10%. Income over \$8,500 but under \$34,500 was taxed at 15%.

A person earning \$10,000 would pay 10% on the portion of their income under \$8,500, and 15% on the income over \$8,500, so they'd pay:

$$8500(0.10) = 850 \quad 10\% \text{ of } 8500$$

$$1500(0.15) = 225 \quad 15\% \text{ of the remaining } \$1500 \text{ of income}$$

$$\text{Total tax: } = \$1075$$

$$\text{The effective tax rate paid is } 1075/10000 = 10.75\%$$

A person earning \$30,000 would also pay 10% on the portion of their income under \$8,500, and 15% on the income over \$8,500, so they'd pay:

$$8500(0.10) = 850 \quad 10\% \text{ of } 8500$$

$$21500(0.15) = 3225 \quad 15\% \text{ of the remaining } \$21500 \text{ of income}$$

$$\text{Total tax:} = \$4075$$

The effective tax rate paid is $4075/30000 = 13.58\%$

Notice that the effective rate has increased with income, showing this is a progressive tax.

✓ Example 4

A gasoline tax is a flat tax when considered in terms of consumption, a tax of, say, \$0.30 per gallon is proportional to the amount of gasoline purchased. Someone buying 10 gallons of gas at \$4 a gallon would pay \$3 in tax, which is $\$3/\$40 = 7.5\%$. Someone buying 30 gallons of gas at \$4 a gallon would pay \$9 in tax, which is $\$9/\$120 = 7.5\%$, the same effective rate.

Solution

However, in terms of income, a gasoline tax is often considered a regressive tax. It is likely that someone earning \$30,000 a year and someone earning \$60,000 a year will drive about the same amount. If both pay \$60 in gasoline taxes over a year, the person earning \$30,000 has paid 0.2% of their income, while the person earning \$60,000 has paid 0.1% of their income in gas taxes.

? Try it Now 1

A sales tax is a fixed percentage tax on a person's purchases. Is this a flat, progressive, or regressive tax?

Answer

While sales tax is a flat percentage rate, it is often considered a regressive tax for the same reasons as the gasoline tax.

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1.6: Exercises

1. Out of 230 racers who started the marathon, 212 completed the race, 14 gave up, and 4 were disqualified. What percentage did not complete the marathon?
2. Patrick left an \$8 tip on a \$50 restaurant bill. What percent tip is that?
3. Ireland has a 23% VAT (value-added tax, similar to a sales tax). How much will the VAT be on a purchase of a €250 item?
4. Employees in 2012 paid 4.2% of their gross wages towards social security (FICA tax), while employers paid another 6.2%. How much will someone earning \$45,000 a year pay towards social security out of their gross wages?
5. A project on Kickstarter.com was aiming to raise \$15,000 for a precision coffee press. They ended up with 714 supporters, raising 557% of their goal. How much did they raise?
6. Another project on Kickstarter for an iPad stylus raised 1,253% of their goal, raising a total of \$313,490 from 7,511 supporters. What was their original goal?
7. The population of a town increased from 3,250 in 2008 to 4,300 in 2010. Find the absolute and relative (percent) increase.
8. The number of CDs sold in 2010 was 114 million, down from 147 million the previous year[1]. Find the absolute and relative (percent) decrease.
9. A company wants to decrease their energy use by 15%.
 - a. If their electric bill is currently \$2,200 a month, what will their bill be if they're successful?
 - b. If their next bill is \$1,700 a month, were they successful? Why or why not?
10. A store is hoping an advertising campaign will increase their number of customers by 30%. They currently have about 80 customers a day.
 - a. How many customers will they have if their campaign is successful?
 - b. If they increase to 120 customers a day, were they successful? Why or why not?
11. An article reports "attendance dropped 6% this year, to 300." What was the attendance before the drop?
12. An article reports "sales have grown by 30% this year, to \$200 million." What were sales before the growth?
13. The Walden University had 47,456 students in 2010, while Kaplan University had 77,966 students. Complete the following statements:
 - a. Kaplan's enrollment was ____% larger than Walden's.
 - b. Walden's enrollment was ____% smaller than Kaplan's.
 - c. Walden's enrollment was ____% of Kaplan's.
14. In the 2012 Olympics, Usain Bolt ran the 100m dash in 9.63 seconds. Jim Hines won the 1968 Olympic gold with a time of 9.95 seconds.
 - a. Bolt's time was ____% faster than Hines'.
 - b. Hine' time was ____% slower than Bolt's.
 - c. Hine' time was ____% of Bolt's.
15. A store has clearance items that have been marked down by 60%. They are having a sale, advertising an additional 30% off clearance items. What percent of the original price do you end up paying?
16. Which is better: having a stock that goes up 30% on Monday than drops 30% on Tuesday, or a stock that drops 30% on Monday and goes up 30% on Tuesday? In each case, what is the net percent gain or loss?
17. Are these two claims equivalent, in conflict, or not comparable because they're talking about different things?
 - a. "16.3% of Americans are without health insurance"[2]
 - b. "only 55.9% of adults receive employer provided health insurance"[3]

18. Are these two claims equivalent, in conflict, or not comparable because they're talking about different things?
- "We mark up the wholesale price by 33% to come up with the retail price"
 - "The store has a 25% profit margin"
19. Are these two claims equivalent, in conflict, or not comparable because they're talking about different things?
- "Every year since 1950, the number of American children gunned down has doubled."
 - "The number of child gunshot deaths has doubled from 1950 to 1994."
20. Are these two claims equivalent, in conflict, or not comparable because they're talking about different things?[4]
- "75 percent of the federal health care law's taxes would be paid by those earning less than \$120,000 a year"
 - "76 percent of those who would pay the penalty [health care law's taxes] for not having insurance in 2016 would earn under \$120,000"
21. Are these two claims equivalent, in conflict, or not comparable because they're talking about different things?
- "The school levy is only a 0.1% increase of the property tax rate."
 - "This new levy is a 12% tax hike, raising our total rate to \$9.33 per \$1000 of value."
22. Are the values compared in this statement comparable or not comparable? "Guns have murdered more Americans here at home in recent years than have died on the battlefields of Iraq and Afghanistan. In support of the two wars, more than 6,500 American soldiers have lost their lives. During the same period, however, guns have been used to murder about 100,000 people on American soil"[5]
23. A high school currently has a 30% dropout rate. They've been tasked to decrease that rate by 20%. Find the equivalent percentage point drop.
24. A politician's support grew from 42% by 3 percentage points to 45%. What percent (relative) change is this?
25. Marcy has a 70% average in her class going into the final exam. She says "I need to get a 100% on this final so I can raise my score to 85%." Is she correct?
26. Suppose you have one quart of water/juice mix that is 50% juice, and you add 2 quarts of juice. What percent juice is the final mix?
27. Find a unit rate: You bought 10 pounds of potatoes for \$4.
28. Find a unit rate: Joel ran 1500 meters in 4 minutes, 45 seconds.
29. Solve: $\frac{2}{5} = \frac{6}{x}$.
30. Solve: $\frac{n}{5} = \frac{16}{20}$.
31. A crepe recipe calls for 2 eggs, 1 cup of flour, and 1 cup of milk. How much flour would you need if you use 5 eggs?
32. An 8ft length of 4 inch wide crown molding costs \$14. How much will it cost to buy 40ft of crown molding?
33. Four 3-megawatt wind turbines can supply enough electricity to power 3000 homes. How many turbines would be required to power 55,000 homes?
34. A highway had a landslide, where 3,000 cubic yards of material fell on the road, requiring 200 dump truck loads to clear. On another highway, a slide left 40,000 cubic yards on the road. How many dump truck loads would be needed to clear this slide?
35. Convert 8 feet to inches.
36. Convert 6 kilograms to grams.
37. A wire costs \$2 per meter. How much will 3 kilometers of wire cost?
38. Sugar contains 15 calories per teaspoon. How many calories are in 1 cup of sugar?
39. A car is driving at 100 kilometers per hour. How far does it travel in 2 seconds?
40. A chain weighs 10 pounds per foot. How many ounces will 4 inches weigh?

41. The table below gives data on three movies. Gross earnings is the amount of money the movie brings in. Compare the net earnings (money made after expenses) for the three movies.[6]

Movie Earnings

Movie	Release Date	Budget	Gross Earnings
Saw	10/29/2004	\$1,200,000	\$103,096,345
Titanic	10/29/2004	\$200,000,000	\$1,842,879,955
Jurassic Park	6/11/1993	\$63,000,000	\$923,863,984

42. For the movies in the previous problem, which provided the best return on investment?

43. The population of the U.S. is about 309,975,000, covering a land area of 3,717,000 square miles. The population of India is about 1,184,639,000, covering a land area of 1,269,000 square miles. Compare the population densities of the two countries.

44. The GDP (Gross Domestic Product) of China was \$5,739 billion in 2010, and the GDP of Sweden was \$435 billion. The population of China is about 1,347 million, while the population of Sweden is about 9.5 million. Compare the GDP per capita of the two countries.

45. In June 2012, Twitter was reporting 400 million tweets per day. Each tweet can consist of up to 140 characters (letter, numbers, etc.). Create a comparison to help understand the amount of tweets in a year by imagining each character was a drop of water and comparing to filling something up.

46. The photo sharing site Flickr had 2.7 billion photos in June 2012. Create a comparison to understand this number by assuming each picture is about 2 megabytes in size, and comparing to the data stored on other media like DVDs, iPods, or flash drives.

47. Your chocolate milk mix says to use 4 scoops of mix for 2 cups of milk. After pouring in the milk, you start adding the mix, but get distracted and accidentally put in 5 scoops of mix. How can you adjust the mix if:

- There is still room in the cup?
- The cup is already full?

48. A recipe for sabayon calls for 2 egg yolks, 3 tablespoons of sugar, and $\frac{1}{4}$ cup of white wine. After cracking the eggs, you start measuring the sugar, but accidentally put in 4 tablespoons of sugar. How can you compensate?

49. The Deepwater Horizon oil spill resulted in 4.9 million barrels of oil spilling into the Gulf of Mexico. Each barrel of oil can be processed into about 19 gallons of gasoline. How many cars could this have fueled for a year? Assume an average car gets 20 miles to the gallon, and drives about 12,000 miles in a year.

50. The store is selling lemons at 2 for \$1. Each yields about 2 tablespoons of juice. How much will it cost to buy enough lemons to make a 9-inch lemon pie requiring $\frac{1}{2}$ cup of lemon juice?

51. A piece of paper can be made into a cylinder in two ways: by joining the short sides together, or by joining the long sides together[7]. Which cylinder would hold more? How much more?

52. Which of these glasses contains more liquid? How much more?

In the next 4 questions, estimate the values by making reasonable approximations for unknown values, or by doing some research to find reasonable values.

53. Estimate how many gallons of water you drink in a year.

54. Estimate how many times you blink in a day.

55. How much does the water in a 6-person hot tub weigh?

56. How many gallons of paint would be needed to paint a two-story house 40 ft long and 30 ft wide?

57. During the landing of the Mars Science Laboratory *Curiosity*, it was reported that the signal from the rover would take 14 minutes to reach earth. Radio signals travel at the speed of light, about 186,000 miles per second. How far was Mars from Earth when *Curiosity* landed?



58. It is estimated that a driver takes, on average, 1.5 seconds from seeing an obstacle to reacting by applying the brake or swerving. How far will a car traveling at 60 miles per hour travel (in feet) before the driver reacts to an obstacle?
59. The flash of lightning travels at the speed of light, which is about 186,000 miles per second. The sound of lightning (thunder) travels at the speed of sound, which is about 750 miles per hour.
- If you see a flash of lightning, then hear the thunder 4 seconds later, how far away is the lightning?
 - Now let's generalize that result. Suppose it takes n seconds to hear the thunder after a flash of lightning. How far away is the lightning, in terms of n ?
60. Sound travels about 750 miles per hour. If you stand in a parking lot near a building and sound a horn, you will hear an echo.
- Suppose it takes about $\frac{1}{2}$ a second to hear the echo. How far away is the building?[8]?
 - Now let's generalize that result. Suppose it takes n seconds to hear the echo. How far away is the building, in terms of n ?
61. It takes an air pump 5 minutes to fill a twin sized air mattress (39 by 8.75 by 75 inches). How long will it take to fill a queen sized mattress (60 by 8.75 by 80 inches)?
62. It takes your garden hose 20 seconds to fill your 2-gallon watering can. How long will it take to fill
- An inflatable pool measuring 3 feet wide, 8 feet long, and 1 foot deep.[9]
 - A circular inflatable pool 13 feet in diameter and 3 feet deep.[10]
63. You want to put a 2" thick layer of topsoil for a new 20'x30' garden. The dirt store sells by the cubic yards. How many cubic yards will you need to order?
64. A box of Jell-O costs \$0.50, and makes 2 cups. How much would it cost to fill a swimming pool 4 feet deep, 8 feet wide, and 12 feet long with Jell-O? (1 cubic foot is about 7.5 gallons)
65. You read online that a 15 ft by 20 ft brick patio would cost about \$2,275 to have professionally installed. Estimate the cost of having a 18 by 22 ft brick patio installed.
66. I was at the store, and saw two sizes of avocados being sold. The regular size sold for \$0.88 each, while the jumbo ones sold for \$1.68 each. Which is the better deal?



67. The grocery store has bulk pecans on sale, which is great since you're planning on making 10 pecan pies for a wedding. Your recipe calls for $1\frac{3}{4}$ cups pecans per pie. However, in the bulk section there's only a scale available, not a measuring cup. You run over to the baking aisle and find a bag of pecans, and look at the nutrition label to gather some info. How many pounds of pecans should you buy?

Nutrition Facts	
Serving Size: 1 cup, halves (99 g)	
Servings per Container: about 2	
Amount Per Serving	
Calories 684	Calories from Fat 596
% Daily Value*	
Total Fat 71g	110%
Saturated Fat 6g	31%
Trans Fat	
Cholesterol 0mg	0%

68. Soda is often sold in 20 ounce bottles. The nutrition label for one of these bottles is shown to the right. A packet of sugar (the kind they have at restaurants for your coffee or tea) typically contain 4 grams of sugar in the U.S. Drinking a 20 oz soda is equivalent to eating how many packets of sugar?[11]

Nutrition Facts	
Serving Size: 8 fl oz (240 mL)	
Servings Per Container: about 2.5	
Amount Per Serving	
Calories 110	
	% Daily Value*
Total Fat 0g	0%
Sodium 70mg	3%
Total Carbohydrate 31g	10%
Sugars 30g	
Protein 0g	

For the next set of questions, *first* identify the information you need to answer the question, and *then* turn to the end of the section to find that information. The details may be imprecise; answer the question the best you can with the provided information. Be sure to justify your decision.

69. You're planning on making 6 meatloafs for a party. You go to the store to buy breadcrumbs, and see they are sold by the canister. How many canisters do you need to buy?

70. Your friend wants to cover their car in bottle caps, like in this picture.[12] How many bottle caps are you going to need?



71. You need to buy some chicken for dinner tonight. You found an ad showing that the store across town has it on sale for \$2.99 a pound, which is cheaper than your usual neighborhood store, which sells it for \$3.79 a pound. Is it worth the extra drive?

72. I have an old gas furnace, and am considering replacing it with a new, high efficiency model. Is upgrading worth it?

73. Janine is considering buying a water filter and a reusable water bottle rather than buying bottled water. Will doing so save her money?

74. Marcus is considering going car-free to save money and be more environmentally friendly. Is this financially a good decision?

For the next set of problems, research or make educated estimates for any unknown quantities needed to answer the question.

75. You want to travel from Tacoma, WA to Chico, CA for a wedding. Compare the costs and time involved with driving, flying, and taking a train. Assume that if you fly or take the train you'll need to rent a car while you're there. Which option is best?

76. You want to paint the walls of a 6ft by 9ft storage room that has one door and one window. You want to put on two coats of paint. How many gallons and/or quarts of paint should you buy to paint the room as cheaply as possible?

77. A restaurant in New York tiled their floor with pennies[13]. Just for the materials, is this more expensive than using a more traditional material like ceramic tiles? If each penny has to be laid by hand, estimate how long it would take to lay the pennies for a 12ft by 10ft room. Considering material and labor costs, are pennies a cost-effective replacement for ceramic tiles?

78. You are considering taking up part of your back yard and turning it into a vegetable garden, to grow broccoli, tomatoes, and zucchini. Will doing so save you money, or cost you more than buying vegetables from the store?

79. Barry is trying to decide whether to keep his 1993 Honda Civic with 140,000 miles, or trade it in for a used 2008 Honda Civic. Consider gas, maintenance, and insurance costs in helping him make a decision.

80. Some people claim it costs more to eat vegetarian, while some claim it costs less. Examine your own grocery habits, and compare your current costs to the costs of switching your diet (from omnivore to vegetarian or vice versa as appropriate). Which

diet is more cost effective based on your eating habits?

Info for the breadcrumbs question

How much breadcrumbs does the recipe call for?

It calls for $1\frac{1}{2}$ cups of breadcrumbs.

How many meatloafs does the recipe make?

It makes 1 meatloaf.

How many servings does that recipe make?

It says it serves 8.

How big is the canister?

It is cylindrical, 3.5 inches across and 7 inches tall.

What is the net weight of the contents of 1 canister?

15 ounces.

How much does a cup of breadcrumbs weigh?

I'm not sure, but maybe something from the nutritional label will help.

Nutrition Facts	
Serving Size: 1/3 cup (30g)	
Servings per Container: about 14	
Amount Per Serving	
Calories 110	Calories from Fat 15
% Daily Value*	
Total Fat 1.5g	2%

How much does a canister cost?

\$2.39

Info for bottle cap car

What kind of car is that?

A 1993 Honda Accord.

How big is that car / what are the dimensions? Here is some details from MSN autos:

Weight: 2800lb Length: 185.2 in Width: 67.1 in Height: 55.2 in

How much of the car was covered with caps?

Everything but the windows and the underside.

How big is a bottle cap?

Caps are 1 inch in diameter.

Info for chicken problem

How much chicken will you be buying?

Four pounds

How far are the two stores?

My neighborhood store is 2.2 miles away, and takes about 7 minutes. The store across town is 8.9 miles away, and takes about 25 minutes.

What kind of mileage does your car get?

It averages about 24 miles per gallon in the city.

How many gallons does your car hold?

About 14 gallons

How much is gas?

About \$3.69/gallon right now.

Info for furnace problem

How efficient is the current furnace?

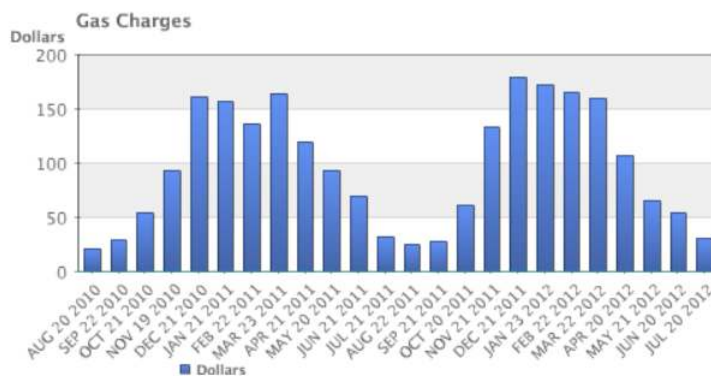
It is a 60% efficient furnace.

How efficient is the new furnace?

It is 94% efficient.

What is your gas bill?

Here is the history for 2 years:

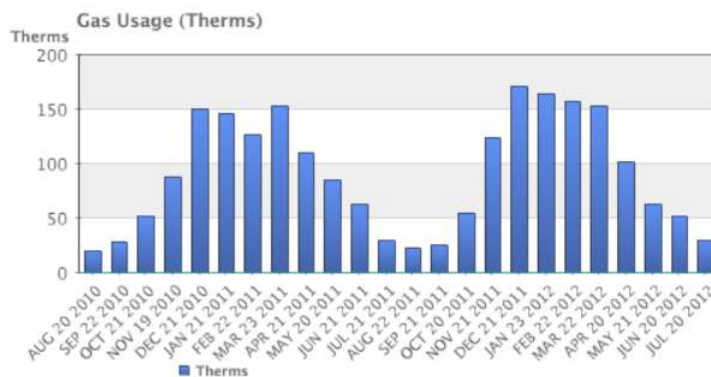


How much do you pay for gas?

There is \$10.34 base charge, plus \$0.39097 per Therm for a delivery charge, and \$0.65195 per Therm for cost of gas.

How much gas do you use?

Here is the history for 2 years:



How much does the new furnace cost?

It will cost \$7,450.

How long do you plan to live in the house?

Probably at least 15 years.

Info for water filter problem

How much water does Janine drink in a day?

She normally drinks 3 bottles a day, each 16.9 ounces.

How much does a bottle of water cost?

She buys 24-packs of 16.9 ounce bottles for \$3.99.

How much does a reusable water bottle cost?

About \$10.

How long does a reusable water bottle last?

Basically forever (or until you lose it).

How much does a water filter cost? How much water will they filter?

- A faucet-mounted filter costs about \$28. Refill filters cost about \$33 for a 3-pack. The box says each filter will filter up to 100 gallons (378 liters)
- A water filter pitcher costs about \$22. Refill filters cost about \$20 for a 4-pack. The box says each filter lasts for 40 gallons or 2 months
- An under-sink filter costs \$130. Refill filters cost about \$60 each. The filter lasts for 500 gallons.

Info for car-free problem

Where does Marcus currently drive? He:

- Drives to work 5 days a week, located 4 miles from his house.
- Drives to the store twice a week, located 7 miles from his house.
- Drives to other locations on average 5 days a week, with locations ranging from 1 mile to 20 miles.
- Drives to his parent's house 80 miles away once a month.

How will he get to these locations without a car?

- For work, he can walk when it's sunny and he gets up early enough. Otherwise he can take a bus, which takes about 20 minutes
- For the store, he can take a bus, which takes about 35 minutes.
- Some of the other locations he can bus to. Sometimes he'll be able to get a friend to pick him up. A few locations he is able to walk to. A couple locations are hard to get to by bus, but there is a ZipCar (short term car rental) location within a few blocks.
- He'll need to get a ZipCar to visit his parents.

How much does gas cost?

About \$3.69/gallon.

How much does he pay for insurance and maintenance?

- He pays \$95/month for insurance.
- He pays \$30 every 3 months for an oil change, and has averaged about \$300/year for other maintenance costs.

How much is he paying for the car?

- He's paying \$220/month on his car loan right now, and has 3 years left on the loan.
- If he sold the car, he'd be able to make enough to pay off the loan.
- If he keeps the car, he's planning on trading the car in for a newer model in a couple years.

What mileage does his car get?

About 26 miles per gallon on average.

How much does a bus ride cost?

\$2.50 per trip, or \$90 for an unlimited monthly pass.

How much does a ZipCar rental cost?

- The "occasional driving plan": \$25 application fee and \$60 annual fee, with no monthly commitment. Monday-Thursday the cost is \$8/hour, or \$72 per day. Friday-Sunday the cost is \$8/hour or \$78/day. Gas, insurance, and 180 miles are included in the cost. Additional miles are \$0.45/mile.
- The "extra value plan": Same as above, but with a \$50 monthly commitment, getting you a 10% discount on the usage costs.

- [1] <http://www.cnn.com/2010/SHOWBIZ/Musi...les/index.html>
- [2] <http://www.cnn.com/2012/06/27/politi...are/index.html>
- [3] <http://www.politico.com/news/stories/0712/78134.html>
- [4] <http://factcheck.org/2012/07/twistin...th-care-taxes/>
- [5] www.northjersey.com/news/opin...tml?c=y&page=2
- [6] <http://www.the-numbers.com/movies/records/budgets.php>
- [7] vimeo.com/42501010
- [8] vimeo.com/40377128
- [9] <http://www.youtube.com/watch?v=DlkwefReHZc>
- [10] <http://www.youtube.com/watch?v=p9SABH7Yg9M>
- [11] <http://www.youtube.com/watch?v=62JMfv0tf3Q>
- [12] Photo credit: <http://www.flickr.com/photos/swayze/>, CC-BY
- [13] <http://www.notcot.com/archives/2009/...-of-pennie.php>

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CHAPTER OVERVIEW

2: Sets and Counting

Learning Objectives

In this chapter, you will learn to:

- Use set theory and Venn diagrams to solve counting problems.
- Use the Multiplication Axiom to solve counting problems.
- Use Permutations to solve counting problems.
- Use Combinations to solve counting problems.
- Use the Binomial Theorem to expand $(x + y)^n$

2.1: Sets, Set Operations, and Venn Diagrams

2.1.1: Basics of Sets

2.1.2: Union, Intersection, and Complement

2.1.3: Venn Diagrams

2.2: Cardinality

2.3: Tree Diagrams and the Multiplication Axiom

2.4: Permutations and Combinations

2.4.1: Permutations

2.4.2: Combinations

2.4.3: Combinations- Involving Several Sets

2.4.4: Circular Permutations and Permutations with Similar Elements

2.5: Chapter Review Exercises

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2.1: Sets, Set Operations, and Venn Diagrams

In this section we introduce sets, set operations, and Venn diagrams:

- [1.1.1: Basics of Sets](#)
- [1.1.2: Union, Intersection, and Complement](#)
- [1.1.3: Venn Diagrams](#)

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2.1.1: Basics of Sets

An art collector might own a collection of paintings, while a music lover might keep a collection of CDs. Any collection of items can form a **set**.

📌 Set

A **set** is a collection of distinct objects, called **elements** of the set

A set can be defined by describing the contents, or by listing the elements of the set, enclosed in curly brackets.

✓ Example 1

Some examples of sets defined by describing the contents:

- The set of all even numbers
- The set of all books written about travel to Chile

Some examples of sets defined by listing the elements of the set:

- $\{1, 3, 9, 12\}$
- $\{\text{red, orange, yellow, green, blue, indigo, purple}\}$

A set simply specifies the contents; order is not important. The set represented by $\{1, 2, 3\}$ is equivalent to the set $\{3, 1, 2\}$.

📌 Notation

Commonly, we will use a variable to represent a set, to make it easier to refer to that set later.

The symbol $\in$ means “is an element of”.

A set that contains no elements, $\{\}$, is called the **empty set** and is notated $\emptyset$

✓ Example 2

Let $A = \{1, 2, 3, 4\}$

To notate that 2 is element of the set, we'd write $2 \in A$

Sometimes a collection might not contain all the elements of a set. For example, Chris owns three Madonna albums. While Chris's collection is a set, we can also say it is a **subset** of the larger set of all Madonna albums.

📌 Subset

A **subset** of a set A is another set that contains only elements from the set A , but may not contain all the elements of A .

If B is a subset of A , we write $B \subseteq A$

A **proper subset** is a subset that is not identical to the original set – it contains fewer elements.

If B is a proper subset of A , we write $B \subset A$

✓ Example 3

Consider these three sets

$A = \text{the set of all even numbers}$ $B = \{2, 4, 6\}$ $C = \{2, 3, 4, 6\}$

Here $B \subset A$ since every element of B is also an even number, so is an element of A .

More formally, we could say $B \subset A$ since if $x \in B$, then $x \in A$

It is also true that $B \subset C$.

C is not a subset of A , since C contains an element, 3, that is not contained in A .

✓ Example 4

Suppose a set contains the plays “Much Ado About Nothing”, “MacBeth”, and “A Midsummer’s Night Dream”. What is a larger set this might be a subset of?

Solution

There are many possible answers here. One would be the set of plays by Shakespeare. This is also a subset of the set of all plays ever written. It is also a subset of all British literature.

? Try it Now 1

The set $A = \{1, 3, 5\}$. What is a larger set this might be a subset of?

Answer

There are several answers: The set of all odd numbers less than 10. The set of all odd numbers. The set of all integers. The set of all real numbers.

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2.1.2: Union, Intersection, and Complement

Commonly sets interact. For example, you and a new roommate decide to have a house party, and you both invite your circle of friends. At this party, two sets are being combined, though it might turn out that there are some friends that were in both sets.

Union, Intersection, and Complement

The **union** of two sets contains all the elements contained in either set (or both sets).

The union is notated $A \cup B$

More formally, $x \in A \cup B$ if $x \in A$ or $x \in B$ (or both)

The **intersection** of two sets contains only the elements that are in both sets.

The intersection is notated $A \cap B$

More formally, $x \in A \cap B$ if $x \in A$ and $x \in B$

The **complement** of a set A contains everything that is *not* in the set A .

The complement is notated A , or A^c , or sometimes $\sim A$.

Example 5

Consider the sets:

$$A = \{ \text{red, green, blue} \} \quad B = \{ \text{red, yellow, orange} \} \quad C = \{ \text{red, orange, yellow, green, blue, purple} \}$$

- Find $A \cup B$
- Find $A \cap B$
- Find $A^c \cap C$

Solution

a) The union contains all the elements in either set: $A \cup B = \{ \text{red, green, blue, yellow, orange} \}$

Notice we only list red once.

b) The intersection contains all the elements in both sets: $A \cap B = \{ \text{red} \}$

c) Here we're looking for all the elements that are not in set A and are also in C .

$$A^c \cap C = \{ \text{orange, yellow, purple} \}$$

Try it Now 2

Using the sets from the previous example, find $A \cup C$ and $B^c \cap A$

Answer

$$A \cup C = \{ \text{red, orange, yellow, green, blue, purple} \}$$

$$B^c \cap A = \{ \text{green, blue} \}$$

Notice that in the example above, it would be hard to just ask for A^c , since everything from the color fuchsia to puppies and peanut butter are included in the complement of the set. For this reason, complements are usually only used with intersections, or when we have a universal set in place.

Universal Set

A **universal set** is a set that contains all the elements we are interested in. This would have to be defined by the context.

A complement is relative to the universal set, so A^c contains all the elements in the universal set that are not in A .

✓ Example 6

- If we were discussing searching for books, the universal set might be all the books in the library.
- If we were grouping your Facebook friends, the universal set would be all your Facebook friends.
- If you were working with sets of numbers, the universal set might be all whole numbers, all integers, or all real numbers

✓ Example 7

Suppose the universal set is $U =$ all whole numbers from 1 to 9. If $A = \{1, 2, 4\}$, then

$$A^c = \{3, 5, 6, 7, 8, 9\}$$

As we saw earlier with the expression $A^c \cap C$, set operations can be grouped together. Grouping symbols can be used like they are with arithmetic - to force an order of operations.

✓ Example 8

Suppose

$$H = \{ \text{cat, dog, rabbit, mouse} \}, F = \{ \text{dog, cow, duck, pig, rabbit} \} \quad W = \{ \text{duck, rabbit, deer, frog, mouse} \}$$

- Find $(H \cap P) \cup W$
- Find $H \cap (F \cup W)$
- Find $(H \cap P) \cap W$

Solution

a) We start with the intersection: $H \cap F = \{ \text{dog, rabbit} \}$

Now we union that result with W : $(H \cap F) \cup W = \{ \text{dog, duck, rabbit, deer, frog, mouse} \}$

b) We start with the union: $F \cup W = \{ \text{dog, cow, rabbit, duck, pig, deer, frog, mouse} \}$

Now we intersect that result with H : $H \cap (F \cup W) = \{ \text{dog, rabbit, mouse} \}$

c) We start with the intersection: $H \cap F = \{ \text{dog, rabbit} \}$

Now we want to find the elements of W that are not in $H \cap F$

$$(H \cap F)^c \cap W = \{ \text{duck, deer, frog, mouse} \}$$

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2.1.3: Venn Diagrams

To visualize the interaction of sets, John Venn in 1880 thought to use overlapping circles, building on a similar idea used by Leonhard Euler in the 18th century. These illustrations now called **Venn Diagrams**.

📌 Venn Diagram

A Venn diagram represents each set by a circle, usually drawn inside of a containing box representing the universal set. Overlapping areas indicate elements common to both sets.

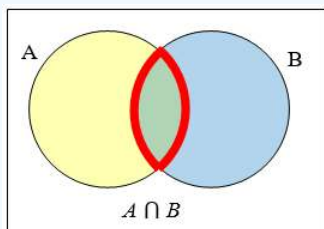
Basic Venn diagrams can illustrate the interaction of two or three sets.

✓ Example 9

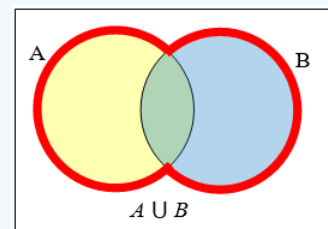
Create Venn diagrams to illustrate $A \cup B$, $A \cap B$, and $A^c \cap B$

$A \cup B$ contains all elements in either set.

$A \cup B$ contains all elements in either set.



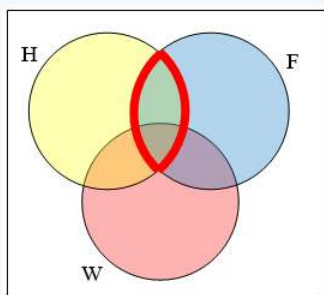
$A \cap B$ contains only those elements in both sets - in the overlap of the circles.



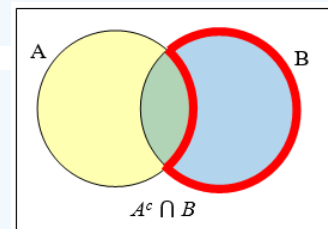
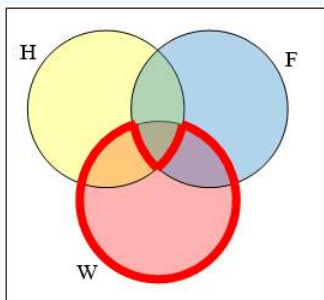
✓ Example 10

Use a Venn diagram to illustrate $(H \cap P)^c \cap W$

We'll start by identifying everything in the set $H \cap P$



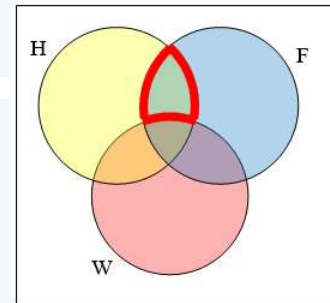
Now, $(H \cap P)^c \cap W$ will contain everything not in the set identified above that is also in set W



✓ Example 11

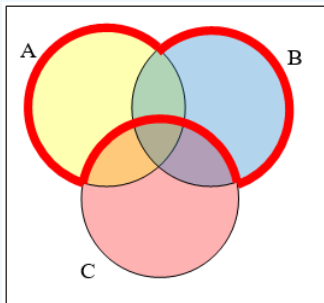
Create an expression to represent the outlined part of the Venn diagram shown.

The elements in the outlined set are in sets H and F , but are not in set W . So we could represent this set as $H \cap F \cap W^c$



? Try it Now 3

Create an expression to represent the outlined portion of the Venn diagram shown



Answer

$$A \cup B \cap C^c$$

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2.2: Cardinality

Often times we are interested in the number of items in a set or subset. This is called the cardinality of the set.

Cardinality

The number of elements in a set is the cardinality of that set.

The cardinality of the set A is often notated as $|A|$ or $n(A)$

✓ Example 12

Let $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{2, 4, 6, 8\}$

What is the cardinality of B ? $A \cup B$, $A \cap B$?

Solution

The cardinality of B is 4, since there are 4 elements in the set.

The cardinality of $A \cup B$ is 7, since $A \cup B = \{1, 2, 3, 4, 5, 6, 8\}$, which contains 7 elements.

The cardinality of $A \cap B$ is 3, since $A \cap B = \{2, 4, 6\}$, which contains 3 elements.

✓ Example 13

What is the cardinality of P = the set of English names for the months of the year?

Solution

The cardinality of this set is 12, since there are 12 months in the year.

Sometimes we may be interested in the cardinality of the union or intersection of sets, but not know the actual elements of each set. This is common in surveying.

✓ Example 14

A survey asks 200 people “What beverage do you drink in the morning”, and offers choices:

- Tea only
- Coffee only
- Both coffee and tea

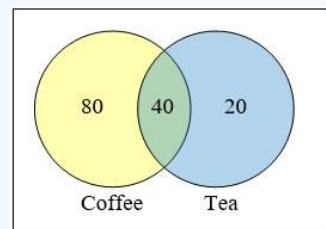
Suppose 20 report tea only, 80 report coffee only, 40 report both. How many people drink tea in the morning? How many people drink neither tea or coffee?

Solution

This question can most easily be answered by creating a Venn diagram. We can see that we can find the people who drink tea by adding those who drink only tea to those who drink both: 60 people.

We can also see that those who drink neither are those not contained in the any of the three other groupings, so we can count those by subtracting from the cardinality of the universal set, 200.

$200 - 20 - 80 - 40 = 60$ people who drink neither.



✓ Example 15

A survey asks: Which online services have you used in the last month:

- Twitter
- Facebook
- Have used both

The results show 40% of those surveyed have used Twitter, 70% have used Facebook, and 20% have used both. How many people have used neither Twitter or Facebook?

Solution

Let T be the set of all people who have used Twitter, and F be the set of all people who have used Facebook. Notice that while the cardinality of F is 70% and the cardinality of T is 40%, the cardinality of $F \cup T$ is not simply $70\% + 40\%$, since that would count those who use both services twice. To find the cardinality of $F \cup T$, we can add the cardinality of F and the cardinality of T , then subtract those in intersection that we've counted twice. In symbols,

$$n(F \cup T) = n(F) + n(T) - n(F \cap T)$$

$$n(F \cup T) = 70\% + 40\% - 20\% = 90\%$$

Now, to find how many people have not used either service, we're looking for the cardinality of $(F \cup T)^c$. since the universal set contains 100% of people and the cardinality of $F \cup T = 90\%$, the cardinality of $(F \cup T)^c$ must be the other 10%

The previous example illustrated two important properties

📌 Cardinality properties

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A^c) = n(U) - n(A)$$

Notice that the first property can also be written in an equivalent form by solving for the cardinality of the intersection:

$$n(A \cap B) = n(A) + n(B) - n(A \cup B)$$

✓ Example 16

Add text here. Fifty students were surveyed, and asked if they were taking a social science (SS), humanities (HM) or a natural science (NS) course the next quarter.

21 were taking a SS course	26 were taking a HM course
19 were taking a NS course	9 were taking SS and HM
7 were taking SS and NS	10 were taking HM and NS
3 were taking all three	7 were taking none

How many students are only taking a SS course?

Solution

It might help to look at a Venn diagram.

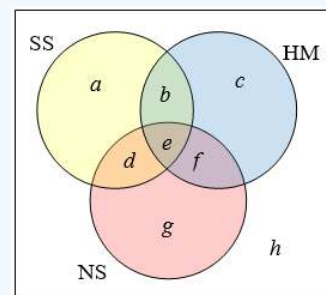
From the given data, we know that there are 3 students in region e and 7 students in region h

since 7 students were taking a SS and NS course, we know that $n(d) + n(e) = 7$. since we know there are 3 students in region 3, there must be

$$7 - 3 = 4 \text{ students in region } d$$

Similarly, since there are 10 students taking HM and NS, which includes regions e and f , there must be

$$10 - 3 = 7 \text{ students in region } f$$



Since 9 students were taking SS and HM, there must be $9 - 3 = 6$ students in region b

Now, we know that 21 students were taking a SS course. This includes students from regions a, b, d , and e . since we know the number of students in all but region a , we can determine that $21 - 6 - 4 - 3 = 8$ students are in region a

8 students are taking only a SS course.

? Try it Now 4

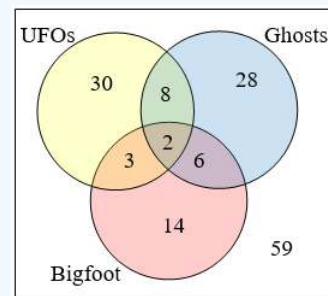
One hundred fifty people were surveyed and asked if they believed in UFOs, ghosts, and Bigfoot.

43 believed in UFOs 44 believed in ghosts
25 believed in Bigfoot 10 believed in UFOs and ghosts
8 believed in ghosts and Bigfoot 5 believed in UFOs and Bigfoot
2 believed in all three

How many people surveyed believed in at least one of these things?

Answer

Starting with the intersection of all three circles, we work our way out. since 10 people believe in UFOs and Ghosts, and 2 believe in all three, that leaves 8 that believe in only UFOs and Ghosts. We work our way out, filling in all the regions. Once we have, we can add up all those regions, getting 91 people in the union of all three sets. This leaves $150 - 91 = 59$ who believe in none.



Exercises

1. List out the elements of the set "The letters of the word Mississippi"
2. List out the elements of the set "Months of the year"
3. Write a verbal description of the set $\{3, 6, 9\}$
4. Write a verbal description of the set $\{a, i, e, o, u\}$
5. Is $\{1, 3, 5\}$ a subset of the set of odd integers?
6. Is $\{A, B, C\}$ a subset of the set of letters of the alphabet?

For problems 7-12, consider the sets below, and indicate if each statement is true or false.

$A = \{1, 2, 3, 4, 5\}$ $B = \{1, 3, 5\}$ $C = \{4, 6\}$ $U = \{\text{numbers from 0 to 10}\}$

7. $3 \in B$
8. $5 \in C$
9. $B \subset A$
10. $C \subset A$
11. $C \subset B$
12. $C \subset U$

Using the sets from above, and treating U as the Universal set, find each of the following:

13. $A \cup B$
14. $A \cup C$
15. $A \cap C$

16. $B \cap C$

17. A^c

18. B^c

Let $D = \{b, a, c, k\}$, $E = \{t, a, s, k\}$, $F = \{b, a, t, h\}$. Using these sets, find the following:

19. $D^c \cap E$

20. $F^c \cap D$

21. $(D \cap E) \cup F$

22. $D \cap (E \cup P)$

23. $(F \cap E)^c \cap D$

24. $(D \cup E)^c \cap F$

Create a Venn diagram to illustrate each of the following:

25. $(F \cap E) \cup D$

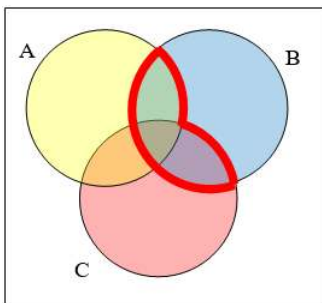
26. $(D \cup E)^c \cap F$

27. $(F^c \cap E') \cap D$

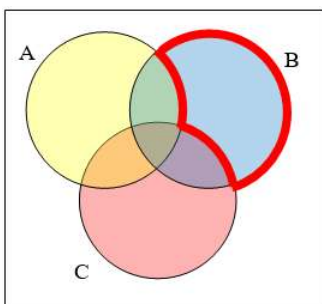
28. $(D \cup E) \cup F$

Write an expression for the shaded region.

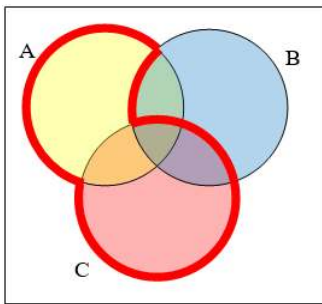
29.



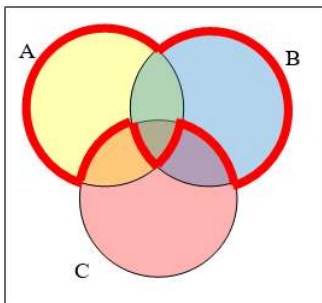
30.



31.



32.



Let $A = \{1, 2, 3, 4, 5\}$ $B = \{1, 3, 5\}$ $C = \{4, 6\}$ Find the cardinality of the given set.

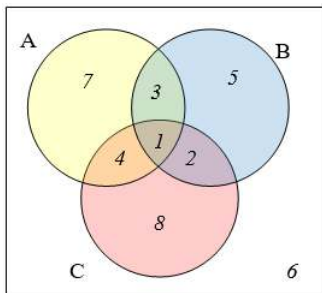
33. $n(A)$

34. $n(B)$

35. $n(A \cup C)$

36. $n(A \cap C)$

The Venn diagram here shows the cardinality of each set. Use this in 37-40 to find the cardinality of given set.



37. $n(A \cap C)$

38. $n(B \cup C)$

39. $n(A \cap B \cap C)$

40. $n(A \cap B^c \cap C)$

41. If $n(G) = 20$, $n(H) = 30$, $n(G \cap H) = 5$, find $n(G \cup H)$

42. If $n(G) = 5$, $n(H) = 8$, $n(G \cap H) = 4$, find $n(G \cup H)$

43. A survey was given asking whether they watch movies at home from Netflix, Redbox, or a video store. Use the results to determine how many people use Redbox.

52 only use Netflix	62 only use Redbox
24 only use a video store	16 use only a video store and Redbox
48 use only Netflix and Redbox	30 use only a video store and Netflix
10 use all three	25 use none of these

44. A survey asked buyers whether color, size, or brand influenced their choice of cell phone. The results are below. How many people were influenced by brand?

5 only said color	8 only said size
16 only said brand	20 said only color and size
42 said only color and brand	53 said only size and brand
102 said all three	20 said none of these

45. Use the given information to complete a Venn diagram, then determine: a) how many students have seen exactly one of these movies, and b) how many had seen only *Star Wars*.

18 had seen The Matrix (M)	24 had seen Star Wars (SW)
20 had seen Lord of the Rings (LotR)	10 had seen M and SW
14 had seen LotR and SW	12 had seen M and LotR
6 had seen all three	

46. A survey asked people what alternative transportation modes they use. Using the data to complete a Venn diagram, then determine: a) what percent of people only ride the bus, and b) how many people don't use any alternate transportation.

30% use the bus	20% ride a bicycle
25% walk	5% use the bus and ride a bicycle
10% ride a bicycle and walk	12% use the bus and walk
2% use all three	

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2.3: Tree Diagrams and the Multiplication Axiom

Learning Objectives

In this section you will learn to

1. Use trees to count possible outcomes in a multi-step process
2. Use the multiplication axiom to count possible outcomes in a multi-step process.

In this chapter, we are trying to develop counting techniques that will be used in the next chapter to study probability. One of the most fundamental of such techniques is called the Multiplication Axiom. Before we introduce the multiplication axiom, we first look at some examples.

✓ Example 2.3.1

If a woman has two blouses and three skirts, how many different outfits consisting of a blouse and a skirt can she wear?

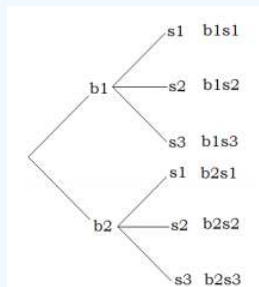
Solution

Suppose we call the blouses b_1 and b_2 , and skirts s_1 , s_2 , and s_3 .

We can have the following six outfits.

$$b_1 s_1, b_1 s_2, b_1 s_3, b_2 s_1, b_2 s_2, b_2 s_3$$

Alternatively, we can draw a tree diagram:



The tree diagram gives us all six possibilities. The method involves two steps. First the woman chooses a blouse. She has two choices: blouse one or blouse two. If she chooses blouse one, she has three skirts to match it with; skirt one, skirt two, or skirt three. Similarly if she chooses blouse two, she can match it with each of the three skirts, again. The tree diagram helps us visualize these possibilities.

The reader should note that the process involves two steps. For the first step of choosing a blouse, there are two choices, and for each choice of a blouse, there are three choices of choosing a skirt. So altogether there are $2 \cdot 3 = 6$ possibilities.

If, in the previous example, we add the shoes to the outfit, we have the following problem.

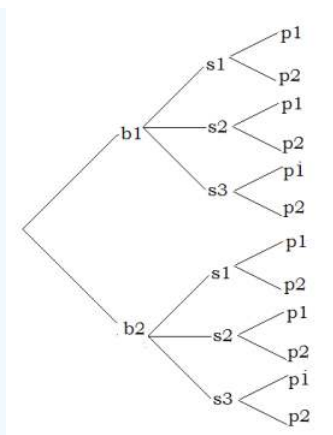
✓ Example 2.3.2

If a woman has two blouses, three skirts, and two pumps, how many different outfits consisting of a blouse, a skirt, and a pair of pumps can she wear?

Solution

Suppose we call the blouses b_1 and b_2 , the skirts s_1 , s_2 , and s_3 , and the pumps p_1 , and p_2 .

The following tree diagram results.



We count the number of branches in the tree, and see that there are 12 different possibilities.

This time the method involves three steps. First, the woman chooses a blouse. She has two choices: blouse one or blouse two. Now suppose she chooses blouse one. This takes us to step two of the process which consists of choosing a skirt. She has three choices for a skirt, and let us suppose she chooses skirt two. Now that she has chosen a blouse and a skirt, we have moved to the third step of choosing a pair of pumps. Since she has two pairs of pumps, she has two choices for the last step. Let us suppose she chooses pumps two. She has chosen the outfit consisting of blouse one, skirt two, and pumps two, or $b_1s_2p_2$. By looking at the different branches on the tree, one can easily see the other possibilities.

The important thing to observe here, again, is that this is a three step process. There are two choices for the first step of choosing a blouse. For each choice of a blouse, there are three choices of choosing a skirt, and for each combination of a blouse and a skirt, there are two choices of selecting a pair of pumps.

All in all, we have $2 \cdot 3 \cdot 2 = 12$ different possibilities.

Tree diagrams help us visualize the different possibilities, but they are not practical when the possibilities are numerous. Besides, we are mostly interested in finding the number of elements in the set and not the actual list of all possibilities; once the problem is envisioned, we can solve it without a tree diagram. The two examples we just solved may have given us a clue to do just that.

Let us now try to solve Example 2.3.2 without a tree diagram. The problem involves three steps: choosing a blouse, choosing a skirt, and choosing a pair of pumps. The number of ways of choosing each are listed below. By multiplying these three numbers we get 12, which is what we got when we did the problem using a tree diagram.

The number of ways of choosing a blouse	The number of ways of choosing a skirt	The number of ways of choosing pumps
2	3	2

The procedure we just employed is called the multiplication axiom.

The Multiplication Axiom

If a task can be done in m ways, and a second task can be done in n ways, then the operation involving the first task followed by the second can be performed in $m \cdot n$ ways.

The general multiplication axiom is not limited to just two tasks and can be used for any number of tasks.

✓ Example 2.3.3

A truck license plate consists of a letter followed by four digits. How many such license plates are possible?

Solution

Since there are 26 letters and 10 digits, we have the following choices for each.

Letter	Digit	Digit	Digit	Digit
26	10	10	10	10

Therefore, the number of possible license plates is $26 \cdot 10 \cdot 10 \cdot 10 \cdot 10 = 260,000$.

✓ Example 2.3.4

In how many different ways can a 3-question true-false test be answered?

Solution

Since there are two choices for each question, we have

Question 1	Question 2	Question 3
2	2	2

Applying the multiplication axiom, we get $2 \cdot 2 \cdot 2 = 8$ different ways.

We list all eight possibilities: TTT, TTE, TFT, TFE, FTT, FTE, FFT, FFF

The reader should note that the first letter in each possibility is the answer corresponding to the first question, the second letter corresponds to the answer to the second question, and so on. For example, TFE, says that the answer to the first question is given as true, and the answers to the second and third questions false.

✓ Example 2.3.5

In how many different ways can four people be seated in a row?

Solution

Suppose we put four chairs in a row, and proceed to put four people in these seats.

There are four choices for the first chair we choose. Once a person sits down in that chair, there are only three choices for the second chair, and so on. We list as shown below.

4	3	2	1
---	---	---	---

So there are altogether $4 \cdot 3 \cdot 2 \cdot 1 = 24$ different ways.

✓ Example 2.3.6

How many three-letter word sequences can be formed using the letters { A, B, C } if no letter is to be repeated?

Solution

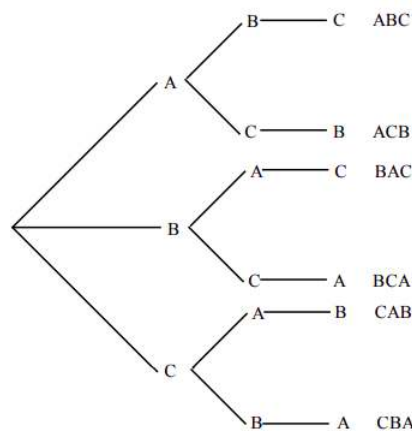
The problem is very similar to the previous example.

Imagine a child having three building blocks labeled A, B, and C. Suppose he puts these blocks on top of each other to make word sequences. For the first letter he has three choices, namely A, B, or C. Let us suppose he chooses the first letter to be a B, then for the second block which must go on top of the first, he has only two choices: A or C. And for the last letter he has only one choice. We list the choices below.

3	2	1
---	---	---

Therefore, 6 different word sequences can be formed.

Finally, we'd like to illustrate this with a tree diagram showing all six possibilities.



Exercises

Do the following problems using a tree diagram or the multiplication axiom.

1. A man has 3 shirts, and 2 pairs of pants. Use a tree diagram to determine the number of possible outfits.	2. In a city election, there are 2 candidates for mayor, and 3 for supervisor. Use a tree diagram to find the number of ways to fill the two offices.
3. There are 4 roads from Town A to Town B, 2 roads from Town B to Town C. Use a tree diagram to find the number of ways one can travel from Town A to Town C.	4. Brown Home Construction offers a selection of 3 floor plans, 2 roof types, and 2 exterior wall types. Use a tree diagram to determine the number of possible homes available.
5. For lunch, a small restaurant offers 2 types of soups, three kinds of sandwiches, and two types of soft drinks. Use a tree diagram to determine the number of possible meals consisting of a soup, sandwich, and a soft drink.	6. A California license plate consists of a number from 1 to 5, then three letters followed by three digits. How many such plates are possible?

Do the following problems using the Multiplication Axiom

7. A license plate consists of three letters followed by three digits. How many license plates are possible if no letter may be repeated?	8. How many different 4-letter radio station call letters can be made if the first letter must be K or W and no letters can be repeated?
9. How many seven-digit telephone numbers are possible if the first two digits cannot be ones or zeros?	10. How many 3-letter word sequences can be formed using the letters {a, b, c, d} if no letter is to be repeated?

Use a tree diagram for questions 11 and 12:

11. A family has two children, use a tree diagram to determine all four possibilities of outcomes by gender.	12. A coin is tossed three times and the sequence of heads and tails is recorded. Use a tree diagram to list all the possible outcomes.
--	---

Do the following problems using the Multiplication Axiom

13. In how many ways can a 4-question true-false test be answered?	14. In how many ways can three people be arranged to stand in a straight line?
15. A combination lock is opened by first turning to the left, then to the right, and then to the left again. If there are 30 digits on the dial, how many possible combinations are there?	16. How many different answers are possible for a multiple-choice test with 10 questions and five possible answers for each question?

17. In the past, a college required students to use a 4 digit PIN (Personal Identification Number) as their password for its registration system. How many different PINs are possible if each must have 4 digits with no restrictions on selection or arrangement of the digits used?

18. The college decided that a more secure password system is needed. New passwords must have 3 numerical digits followed by 6 letters. There are no restrictions on the selection of the numerical digits. However, the letters I and O are not permitted. How many different passwords are possible?

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2.4: Permutations and Combinations

In this section we introduce and explore the related concepts of permutations and combinations of elements from a set:

- [1.4.1: Permutations](#)
 - [1.4.2: Combinations](#)
 - [1.4.3: Combinations- Involving Several Sets](#)
 - [1.4.4: Circular Permutations and Permutations with Similar Elements](#)
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2.4.1: Permutations

Learning Objectives

In this section you will learn to

1. Count the number of possible permutations (ordered arrangement) of n items taken r at a time
2. Count the number of possible permutations when there are conditions imposed on the arrangements
3. Perform calculations using factorials

In Example 7.2.6 of section 7.2, we were asked to find the word sequences formed by using the letters $\{ A, B, C \}$ if no letter is to be repeated. The tree diagram gave us the following six arrangements.

ABC, ACB, BAC, BCA, CAB, and CBA.

Arrangements like these, where order is important and no element is repeated, are called permutations.

Definition: Permutations

A permutation of a set of elements is an ordered arrangement where each element is used once.

✓ Example 2.4.1.1

How many three-letter word sequences can be formed using the letters $\{ A, B, C, D \}$?

Solution

There are four choices for the first letter of our word, three choices for the second letter, and two choices for the third.

4

3

2

Applying the multiplication axiom, we get $4 \cdot 3 \cdot 2 = 24$ different arrangements.

✓ Example 2.4.1.2

How many permutations of the letters of the word ARTICLE have consonants in the first and last positions?

Solution

In the word ARTICLE, there are 4 consonants.

Since the first letter must be a consonant, we have four choices for the first position, and once we use up a consonant, there are only three consonants left for the last spot. We show as follows:

4

3

Since there are no more restrictions, we can go ahead and make the choices for the rest of the positions.

So far we have used up 2 letters, therefore, five remain. So for the next position there are five choices, for the position after that there are four choices, and so on. We get

4

5

4

3

2

1

3

So the total permutations are $4 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 3 = 1440$.

✓ Example 2.4.1.3

Given five letters { A, B, C, D, E }. Find the following:

- The number of four-letter word sequences.
- The number of three-letter word sequences.
- The number of two-letter word sequences.

Solution

The problem is easily solved by the multiplication axiom, and answers are as follows:

- The number of four-letter word sequences is $5 \cdot 4 \cdot 3 \cdot 2 = 120$.
- The number of three-letter word sequences is $5 \cdot 4 \cdot 3 = 60$.
- The number of two-letter word sequences is $5 \cdot 4 = 20$.

We often encounter situations where we have a set of n objects and we are selecting r objects to form permutations. We refer to this as **permutations of n objects taken r at a time**, and we write it as **nPr** .

Therefore, the above example can also be answered as listed below.

- The number of four-letter word sequences is $5P4 = 120$.
- The number of three-letter word sequences is $5P3 = 60$.
- The number of two-letter word sequences is $5P2 = 20$.

Before we give a formula for nPr , we'd like to introduce a symbol that we will use a great deal in this as well as in the next chapter.

Definition: Factorial

$$n! = n(n-1)(n-2)(n-3) \cdots 3 \cdot 2 \cdot 1$$

where n is a natural number.

$$0! = 1$$

Now we define nPr .

Definition: nPr

The Number of Permutations of n Objects Taken r at a Time

$$nPr = n(n-1)(n-2)(n-3) \cdots (n-r+1)$$

or

$$nPr = \frac{n!}{(n-r)!}$$

where n and r are natural numbers.

The reader should become familiar with both formulas and should feel comfortable in applying either.

✓ Example 2.4.1.4

Compute the following using both formulas.

- $6P3$
- $7P2$

Solution

We will identify n and r in each case and solve using the formulas provided.

a. $6P3 = 6 \cdot 5 \cdot 4 = 120$, alternately

$$6P3 = \frac{6!}{(6-3)!} = \frac{6!}{3!} = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1} = 120$$

b. $7P2 = 7 \cdot 6 = 42$, or

$$7P2 = \frac{7!}{5!} = \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 42$$

Next we consider some more permutation problems to get further insight into these concepts.

✓ Example 2.4.1.5

In how many different ways can 4 people be seated in a straight line if two of them insist on sitting next to each other?

Solution

Let us suppose we have four people A, B, C, and D. Further suppose that A and B want to sit together. For the sake of argument, we tie A and B together and treat them as one person.

The four people are $\boxed{AB}$ CD. Since $\boxed{AB}$ is treated as one person, we have the following possible arrangements.

$$\boxed{AB}CD, \boxed{AB}DC, C\boxed{AB}D, D\boxed{AB}C, CD\boxed{AB}, DC\boxed{AB}$$

Note that there are six more such permutations because A and B could also be tied in the order BA. And they are

$$\boxed{BA}CD, \boxed{BA}DC, C\boxed{BA}D, D\boxed{BA}C, CD\boxed{BA}, DC\boxed{BA}$$

So altogether there are 12 different permutations.

Let us now do the problem using the multiplication axiom.

After we tie two of the people together and treat them as one person, we can say we have only three people. The multiplication axiom tells us that three people can be seated in $3!$ ways. Since two people can be tied together $2!$ ways, there are $3! \cdot 2! = 12$ different arrangements

✓ Example 2.4.1.6

You have 4 math books and 5 history books to put on a shelf that has 5 slots. In how many ways can the books be shelved if the first three slots are filled with math books and the next two slots are filled with history books?

Solution

We first do the problem using the multiplication axiom.

Since the math books go in the first three slots, there are 4 choices for the first slot, 3 choices for the second and 2 choices for the third.

The fourth slot requires a history book, and has five choices. Once that choice is made, there are 4 history books left, and therefore, 4 choices for the last slot. The choices are shown below.

4	3	2	5	4
---	---	---	---	---

Therefore, the number of permutations are $4 \cdot 3 \cdot 2 \cdot 5 \cdot 4 = 480$.

Alternately, we can see that $4 \cdot 3 \cdot 2$ is really same as $4P3$, and $5 \cdot 4$ is $5P2$.

So the answer can be written as $(4P3) (5P2) = 480$.

Clearly, this makes sense. For every permutation of three math books placed in the first three slots, there are $5P2$ permutations of history books that can be placed in the last two slots. Hence the multiplication axiom applies, and we have the answer $(4P3) (5P2)$.

We summarize the concepts of this section:

Note

1. Permutations

A permutation of a set of elements is an ordered arrangement where each element is used once.

2. Factorial

$$n! = n(n-1)(n-2)(n-3) \cdots 3 \cdot 2 \cdot 1$$

Where n is a natural number.

$$0! = 1$$

3. Permutations of n Objects Taken r at a Time

$${}_n P_r = n(n-1)(n-2)(n-3) \cdots (n-r+1)$$

or

$${}_n P_r = \frac{n!}{(n-r)!}$$

where n and r are natural numbers.

Exercises

Do the following problems using permutations.

1. How many three-letter words can be made using the letters { a, b, c, d, e } if no repetitions are allowed?	2. A grocery store has five checkout counters, and seven clerks. How many different ways can the 7 clerks be assigned to the 5 counters?
3. A group of fifteen people who are members of an investment club wish to choose a president, and a secretary. How many different ways can this be done?	4. Compute the following. a. ${}_9 P_2$ b. ${}_6 P_4$ c. ${}_8 P_3$ d. ${}_7 P_4$
5. In how many ways can the letters of the word CUPERTINO be arranged if each letter is used only once in each arrangement?	6. How many permutations of the letters of the word PROBLEM end in a vowel?
7. How many permutations of the letters of the word SECURITY end in a consonant?	8. How many permutations of the letters PRODUCT have consonants in the second and third positions?
9. How many three-digit numbers are there?	10. How many three-digit odd numbers are there?
11. In how many different ways can five people be seated in a row if two of them insist on sitting next to each other?	12. In how many different ways can five people be seated in a row if two of them insist on not sitting next to each other?
13. In how many ways can 3 English, 3 history, and 2 math books be set on a shelf, if the English books are set on the left, history books in the middle, and math books on the right?	14. In how many ways can 3 English, 3 history, and 2 math books be set on a shelf, if they are grouped by subject?
15. You have 5 math books and 6 history books to put on a shelf with five slots. In how many ways can you put the books on the shelf if the first two slots are to be filled with math books and the next three with history books?	16. You have 5 math books and 6 history books to put on a shelf with five slots. In how many ways can you put the books on the shelf if the first two slots are to be filled with the books of one subject and the next three slots are to be filled with the books of the other subject?

17. A bakery has 9 different fancy cakes. In how many ways can 5 of the 9 fancy cakes be lined up in a row in the bakery display case?	18. A landscaper has 6 different flowering plants. She needs to plant 4 of them in a row in a garden. How many different ways can 4 of the 6 plants be arranged in a row?
19. At an auction of used construction vehicles, there are 7 different vehicles for sale. In how many orders could these 7 vehicles be listed in the auction program?	20. A landscaper has 6 different flowering plants and 4 different non-flowering bushes. She needs to plant a row of 6 plants in a garden. There must be a bush at each end, and four flowering plants in a row in between the bushes. How many different arrangements in a row are possible?
21. In how many ways can all 7 letters of the word QUIETLY be arranged if the letters Q and U must be next to each other in the order QU?	22. a. In how many ways can the letters ABCDEXY be arranged if the X and Y must be next to each other in either order XY or YX? b. In how many ways can the letters ABCDEXY be arranged if the X and Y can not be next to each other?

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2.4.2: Combinations

Learning Objectives

In this section you will learn to

1. Count the number of combinations of r out of n items (selections without regard to arrangement)
2. Use factorials to perform calculations involving combinations

Suppose we have a set of three letters $\{ A, B, C \}$, and we are asked to make two-letter word sequences. We have the following six permutations.

AB BA BC CB AC CA

Now suppose we have a group of three people $\{ A, B, C \}$ as Al, Bob, and Chris, respectively, and we are asked to form committees of two people each. This time we have only three committees, namely,

AB BC AC

When forming committees, the order is not important, because the committee that has Al and Bob is no different than the committee that has Bob and Al. As a result, we have only three committees and not six.

Forming word sequences is an example of permutations, while forming committees is an example of **combinations** - the topic of this section.

Permutations are those arrangements where order is important, while combinations are those arrangements where order is not significant. From now on, this is how we will tell permutations and combinations apart.

In the above example, there were six permutations, but only three combinations.

Just as the symbol nPr represents the number of permutations of n objects taken r at a time, nCr represents the number of combinations of n objects taken r at a time.

So in the above example, $3P2 = 6$, and $3C2 = 3$.

Our next goal is to determine the relationship between the number of combinations and the number of permutations in a given situation.

In the above example, if we knew that there were three combinations, we could have found the number of permutations by multiplying this number by $2!$. That is because each combination consists of two letters, and that makes $2!$ permutations.

✓ Example 2.4.2.1

Given the set of letters $\{ A, B, C, D \}$. Write the number of combinations of three letters, and then from these combinations determine the number of permutations.

Solution

We have the following four combinations.

ABC BCD CDA BDA

Since every combination has three letters, there are $3!$ permutations for every combination. We list them below.

ABC	BCD	CDA	BDA
ACB	BDC	CAD	BAD
BAC	CDB	DAC	DAB
BCA	CBD	DCA	DBA
CAB	DCB	ACD	ADB
CBA	DBC	ADC	ABD

The number of permutations are $3!$ times the number of combinations; that is

$$4P3 = 3! \cdot 4C3$$

or

$$4C3 = \frac{4P3}{3!}$$

In general,

$$nC_r = \frac{nPr}{r!}$$

Since

$$nPr = \frac{n!}{(n-r)!}$$

We have,

$$nC_r = \frac{n!}{(n-r)!r!}$$

Summarizing,

Note

1. Combinations

A combination of a set of elements is an arrangement where each element is used once, and order is not important.

2. The Number of Combinations of n Objects Taken r at a Time

$$nC_r = \frac{n!}{(n-r)!r!}$$

where n and r are natural numbers.

✓ Example 2.4.2.2

Compute:

a. $5C3$

b. $7C3$

Solution

We use the above formula.

$$5C3 = \frac{5!}{(5-3)!3!} = \frac{5!}{2!3!} = 10$$

$$7C3 = \frac{7!}{(7-3)!3!} = \frac{7!}{4!3!} = 35$$

✓ Example 2.4.2.3

In how many different ways can a student select to answer five questions from a test that has seven questions, if the order of the selection is not important?

Solution

Since the order is not important, it is a combination problem, and the answer is

$$7C5 = 21$$

✓ Example 2.4.2.4

How many line segments can be drawn by connecting any two of the six points that lie on the circumference of a circle?

Solution

Since the line that goes from point A to point B is same as the one that goes from B to A, this is a combination problem.

It is a combination of 6 objects taken 2 at a time. Therefore, the answer is

$${}^6C_2 = \frac{6!}{4!2!} = 15$$

✓ Example 2.4.2.5

There are ten people at a party. If they all shake hands, how many hand-shakes are possible?

Solution

Note that between any two people there is only one hand shake. Therefore, we have

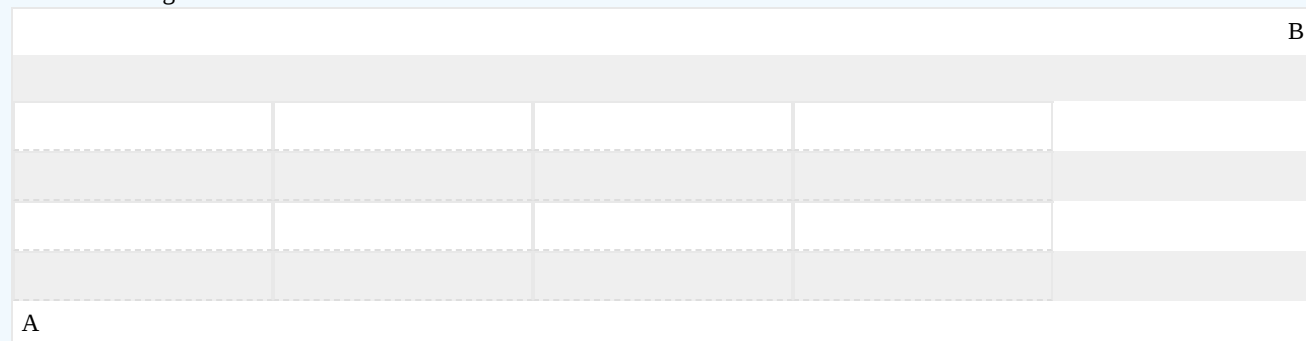
$${}^{10}C_2 = 45 \text{ hand-shakes.}$$

✓ Example 2.4.2.6

The shopping area of a town is in the shape of square that is 5 blocks by 5 blocks. How many different routes can a taxi driver take to go from one corner of the shopping area to the opposite cater-corner?

Solution

Let us suppose the taxi driver drives from the point A, the lower left hand corner, to the point B, the upper right hand corner as shown in the figure below.



To reach his destination, he has to travel ten blocks; five horizontal, and five vertical. So if out of the ten blocks he chooses any five horizontal, the other five will have to be the vertical blocks, and vice versa.

Therefore, all he has to do is to choose 5 out of ten to be the horizontal blocks

The answer is ${}^{10}C_5$, or 252.

Alternately, the problem can be solved by permutations with similar elements.

The taxi driver's route consists of five horizontal and five vertical blocks. If we call a horizontal block H, and a vertical block a V, then one possible route may be as follows.

HHHHHVVVVV

Clearly there are $\frac{10!}{5!5!} = 252$ permutations.

Further note that by definition ${}^{10}C_5 = \frac{10!}{5!5!}$.

✓ Example 2.4.2.7

If a coin is tossed six times, in how many ways can it fall four heads and two tails?

Solution

First we solve this problem using section 6.5 technique-permutations with similar elements.

We need 4 heads and 2 tails, that is

HHHHTT

There are $\frac{6!}{4!2!} = 15$ permutations.

Now we solve this problem using combinations.

Suppose we have six spots to put the coins on. If we choose any four spots for heads, the other two will automatically be tails. So the problem is simply

$${}^6C_4 = 15.$$

Incidentally, we could have easily chosen the two tails, instead. In that case, we would have gotten

$${}^6C_2 = 15.$$

Further observe that by definition

$${}^6C_4 = \frac{6!}{2!4!}$$

and

$${}^6C_2 = \frac{6!}{4!2!}$$

Which implies ${}^6C_4 = {}^6C_2$.

Exercises

Do the following problems using combinations.

1. How many different 3-people committees can be chosen from ten people?	2. How many different 5-player teams can be chosen from eight players?
3. In how many ways can a person chose to vote for three out of five candidates on a ballot for a school board election?	4. Compute the following: a. 9C_2 b. 6C_4 c. 8C_3 d. 7C_4
5. How many 5-card hands can be chosen from a deck of cards?	6. How many 13-card bridge hands can be chosen from a deck of cards?
7. There are twelve people at a party. If they all shake hands, how many different hand-shakes are there?	8. In how many ways can a student choose to do four questions out of five on a test?
9. Five points lie on a circle. How many chords can be drawn through them?	10. How many diagonals does a hexagon have?
11. There are five team in a league. How many games are played if every team plays each other twice?	12. A team plays 15 games a season. In how many ways can it have 8 wins and 7 losses?
13. In how many different ways can a 4-child family have 2 boys and 2 girls?	14. A coin is tossed five times. In how many ways can it fall three heads and two tails?

15. The shopping area of a town is a square that is six blocks by six blocks. How many different routes can a taxi driver take to go from one corner of the shopping area to the opposite cater-corner?	16. If the shopping area in the previous problem has a rectangular form of 5 blocks by 3 blocks, then how many different routes can a taxi driver take to drive from one end of the shopping area to the opposite kitty corner end?
17. A team of 7 workers is assigned to a project. In how many ways can 3 of the 7 workers be selected to make a presentation to the management about their progress on the project?	18. A real estate company has 12 houses listed for sale by their clients. In how many ways can 5 of the 12 houses be selected to be featured in an advertising brochures?
19. A frozen yogurt store has 9 toppings to choose from. In how many ways can 3 of the 9 toppings be selected ?	20. A kindergarten teacher has 14 books about a holiday. In how many ways can she select 4 of the books to read to her class in the week before the holiday?

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2.4.3: Combinations- Involving Several Sets

Learning Objectives

In this section you will learn to

1. count the number of items selected from more than one set
2. count the number of items selected when there are restrictions on the selections

So far we have solved the basic combination problem of r objects chosen from n different objects. Now we will consider certain variations of this problem.

✓ Example 2.4.3.1

How many five-people committees consisting of 2 men and 3 women can be chosen from a total of 4 men and 4 women?

Solution

We list 4 men and 4 women as follows:

$$M_1 M_2 M_3 M_4 W_1 W_2 W_3 W_4$$

Since we want 5-people committees consisting of 2 men and 3 women, we'll first form all possible two-man committees and all possible three-woman committees. Clearly there are $4C2 = 6$ two-man committees, and $4C3 = 4$ three-woman committees, we list them as follows:

2-Man Committees	3-Woman Committees
$M_1 M_2$	$W_1 W_2 W_3$
$M_1 M_3$	$W_1 W_2 W_4$
$M_1 M_4$	$W_1 W_3 W_4$
$M_2 M_3$	$W_2 W_3 W_4$
$M_2 M_4$	
$M_3 M_4$	

For every 2-man committee there are four 3-woman committees that can be chosen to make a 5-person committee. If we choose $M_1 M_2$ as our 2-man committee, then we can choose any of $W_1 W_2 W_3$, $W_1 W_2 W_4$, $W_1 W_3 W_4$, or $W_2 W_3 W_4$ as our 3-woman committees. As a result, we get

$$\boxed{M_1 M_2} W_1 W_2 W_3, \boxed{M_1 M_2} W_1 W_2 W_4, \boxed{M_1 M_2} W_1 W_3 W_4, \boxed{M_1 M_2} W_2 W_3 W_4$$

Similarly, if we choose $M_1 M_3$ as our 2-man committee, then, again, we can choose any of $W_1 W_2 W_3$, $W_1 W_2 W_4$, $W_1 W_3 W_4$, or $W_2 W_3 W_4$ as our 3-woman committees.

$$\boxed{M_1 M_3} W_1 W_2 W_3, \boxed{M_1 M_3} W_1 W_2 W_4, \boxed{M_1 M_3} W_1 W_3 W_4, \boxed{M_1 M_3} W_2 W_3 W_4$$

And so on.

Since there are six 2-man committees, and for every 2-man committee there are four 3-woman committees, there are altogether $6 \cdot 4 = 24$ five-people committees.

In essence, we are applying the multiplication axiom to the different combinations.

✓ Example 2.4.3.2

A high school club consists of 4 freshmen, 5 sophomores, 5 juniors, and 6 seniors. How many ways can a committee of 4 people be chosen that includes

- a. One student from each class?
- b. All juniors?
- c. Two freshmen and 2 seniors?
- d. No freshmen?
- e. At least three seniors?

Solution

a. Applying the multiplication axiom to the combinations involved, we get

$$(4C1)(5C1)(5C1)(6C1) = 600$$

b. We are choosing all 4 members from the 5 juniors, and none from the others.

$$5C4 = 5$$

c. $4C2 \cdot 6C2 = 90$

d. Since we don't want any freshmen on the committee, we need to choose all members from the remaining 16. That is

$$16C4 = 1820$$

e. Of the 4 people on the committee, we want at least three seniors. This can be done in two ways. We could have three seniors, and one non-senior, or all four seniors.

$$(6C3)(14C1) + 6C4 = 295$$

✓ Example 2.4.3.3

How many five-letter word sequences consisting of 2 vowels and 3 consonants can be formed from the letters of the word INTRODUCE?

Solution

First we select a group of five letters consisting of 2 vowels and 3 consonants.

Since there are 4 vowels and 5 consonants, we have

$$(4C2)(5C3)$$

Since our next task is to make word sequences out of these letters, we multiply these by 5!.

$$(4C2)(5C3)(5!) = 7200.$$

✓ Example 2.4.3.4

A standard deck of playing cards has 52 cards consisting of 4 suits each with 13 cards. In how many different ways can a 5-card hand consisting of four cards of one suit and one of another be drawn?

Solution

We will do the problem using the following steps.

Step 1. Select a suit.

Step 2. Select four cards from this suit.

Step 3. Select another suit.

Step 4. Select a card from that suit.

Applying the multiplication axiom, we have

Ways of selecting the first suit	Ways of selecting 4 cards from this suit	Ways of selecting the next suit	Ways of selecting a card from that suit
4C1	13C4	3C1	13C1

$$(4C1)(13C4)(3C1)(13C1) = 111,540.$$

A STANDARD DECK OF 52 PLAYING CARDS

As in the previous example, many examples and homework problems in this book refer to a standard deck of 52 playing cards. Before we end this section, we take a minute to describe a standard deck of playing cards, as some readers may not be familiar with this.

A standard deck of 52 playing cards has 4 suits with 13 cards in each suit.

♦ diamonds ♥ hearts ♠ spades ♣ clubs

Each suit is associated with a color, either black (spades, clubs) or red (diamonds, hearts)

Each suit contains 13 denominations (or values) for cards:

nine numbers 2, 3, 4, ..., 10 and Jack(J), Queen (Q), King (K), Ace (A).

The Jack, Queen and King are called “face cards” because they have pictures on them. Therefore a standard deck has 12 face cards: (3 values JQK) x (4 suits ♦♥♠♣)

We can visualize the 52 cards by the following display

Suit	Color	Values (Denominations)
♦ Diamonds	Red	2 3 4 5 6 7 8 9 10 J Q K A
♥ Hearts	Red	2 3 4 5 6 7 8 9 10 J Q K A
♠ Spades	Black	2 3 4 5 6 7 8 9 10 J Q K A
♣ Clubs	Black	2 3 4 5 6 7 8 9 10 J Q K A

Exercises

Following problems involve combinations from several different sets.

1. How many 5-people committees consisting of three boys and two girls can be chosen from a group of four boys and four girls?	2. A club has 4 men, 5 women, 8 boys and 10 girls as members. In how many ways can a group of 2 men, 3 women, 4 boys and 4 girls be chosen?
3. How many 4-people committees chosen from 4 men and 6 women will have at least 3 men?	4. A batch contains 10 transistors of which three are defective. If three are chosen, in how many ways can they be selected with two defective?
5. In how many ways can five counters labeled A, B, C, D and E at a store be staffed by two men and three women chosen from a group of four men and six women?	6. How many 4-letter word sequences consisting of two vowels and two consonants can be made from the letters of the word PHOENIX if no letter is repeated?

Three marbles are chosen from an urn that contains 5 red, 4 white, and 3 blue marbles. How many samples of the following type are possible?

7. All three white.	8. Two blue and one white
9. One of each color.	10. All three of the same color.
11. At least two red.	12. None red.

The following problems involve combinations from several different sets.

Five coins are chosen from a bag that contains 4 dimes, 5 nickels, and 6 pennies. How many samples of five coins of the following types are possible?

13. At least four nickels.	14. No pennies.
----------------------------	-----------------

15. Five of a kind.	16. Four of a kind.
17. Two of one kind and two of another kind.	18. Three of one kind and two of another kind.

Find the number of different ways draw a 5-card hand from a deck to have the following combinations.

19. Three face cards.	20. A heart flush (all hearts).
21. Two hearts and three diamonds	22. Two cards of one suit, and three of another suit.
23. Two kings and three queens.	24. 2 cards of one value and 3 of another value

The party affiliation of the 100 United States Senators in the 114th Congress, January 2015, was:

44 Democrats, 54 Republicans, and 2 Independents.

25. In how many ways could a 10 person committee be selected if it is to contain 4 Democrats, 5 Republicans, and 1 Independent?	26. In how many different ways could a 10 person committee be selected with 6 or 7 Republicans and the Democrats (with no Independents)?
---	--

The 100 United States Senators in the 114th Congress, January 2015, included 80 men and 20 women. Suppose a committee senators is working on legislation about wage discrimination by gender.

27. In how many ways could a 12 person committee be selected to contain equal numbers of men and women.	28. In how many ways could a 6 person committee be selected to contain fewer women than men?
---	--

Jorge has 6 rock songs, 7 rap songs and 4 country songs that he likes to listen to while he exercises. He randomly selects six (6) of these songs to create a playlist to listen to today while he exercises.

How many different playlists of 6 songs can be selected that satisfy each of the following: (We care which songs are selected to be on the playlist, but not what order they are selected or listed in.)

29. Playlist has 2 songs of each type	30. Playlist has no country songs
31. Playlist has 3 rocks, 2 raps, and 1 country song	32. Playlist has 3 or 4 rock songs and all the rest are rap songs

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2.4.4: Circular Permutations and Permutations with Similar Elements

Learning Objectives

In this section you will learn to

1. Count the number of possible permutations of items arranged in a circle
2. Count the number of possible permutations when there are repeated items

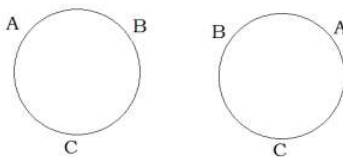
In this section we will address the following two problems.

1. In how many different ways can five people be seated in a circle?
2. In how many different ways can the letters of the word MISSISSIPPI be arranged?

The first problem comes under the category of Circular Permutations, and the second under Permutations with Similar Elements.

Circular Permutations

Suppose we have three people named A, B, and C. We have already determined that they can be seated in a straight line in $3!$ or 6 ways. Our next problem is to see how many ways these people can be seated in a circle. We draw a diagram.



It happens that there are only two ways we can seat three people in a circle, relative to each other's positions. This kind of permutation is called a circular permutation. In such cases, no matter where the first person sits, the permutation is not affected. Each person can shift as many places as they like, and the permutation will not be changed. We are interested in the position of each person in relation to the others. Imagine the people on a merry-go-round; the rotation of the permutation does not generate a new permutation. So in circular permutations, the first person is considered a place holder, and where he sits does not matter.

Definition: Circular Permutations

The number of permutations of n elements in a circle is $(n - 1)!$

✓ Example 2.4.4.1

In how many different ways can five people be seated at a circular table?

Solution

We have already determined that the first person is just a place holder. Therefore, there is only one choice for the first spot. We have

1	4	3	2	1
---	---	---	---	---

So the answer is 24.

✓ Example 2.4.4.2

In how many ways can four couples be seated at a round table if the men and women want to sit alternately?

Solution

We again emphasize that the first person can sit anywhere without affecting the permutation.

So there is only one choice for the first spot. Suppose a man sat down first. The chair next to it must belong to a woman, and there are 4 choices. The next chair belongs to a man, so there are three choices and so on. We list the choices below.

1	4	3	3	2	2	1	1
---	---	---	---	---	---	---	---

So the answer is 144.

PERMUTATIONS WITH SIMILAR ELEMENTS

Let us determine the number of distinguishable permutations of the letters ELEMENT.

Suppose we make all the letters different by labeling the letters as follows.

$$E_1 L E_2 M E_3 N T$$

Since all the letters are now different, there are $7!$ different permutations.

Let us now look at one such permutation, say

$$L E_1 M E_2 N E_3 T$$

Suppose we form new permutations from this arrangement by only moving the E's. Clearly, there are $3!$ or 6 such arrangements. We list them below.

$$\begin{aligned} &L E_1 M E_2 N E_3 \\ &L E_1 M E_3 N E_2 \\ &L E_2 M E_1 N E_3 \\ &L E_2 M E_3 N E_1 \\ &L E_3 M E_2 N E_1 \\ &L E_3 M E_1 N E_2 \end{aligned}$$

Because the E's are not different, there is only one arrangement LEMENET and not six. This is true for every permutation.

Let us suppose there are n different permutations of the letters ELEMENT.

Then there are $n \cdot 3!$ permutations of the letters $E_1 L E_2 M E_3 N T$.

But we know there are $7!$ permutations of the letters $E_1 L E_2 M E_3 N T$.

Therefore, $n \cdot 3! = 7!$

Or $n = \frac{7!}{3!}$.

This gives us the method we are looking for.

Definition: Permutations with Similar Elements

The number of permutations of n elements taken n at a time, with r_1 elements of one kind, r_2 elements of another kind, and so on, is

$$\frac{n!}{r_1! r_2! \dots r_k!}$$

✓ Example 2.4.4.3

Find the number of different permutations of the letters of the word MISSISSIPPI.

Solution

The word MISSISSIPPI has 11 letters. If the letters were all different there would have been $11!$ different permutations. But MISSISSIPPI has 4 S's, 4 I's, and 2 P's that are alike.

So the answer is $\frac{11!}{4!4!2!} = 34,650$.

✓ Example 2.4.4.4

If a coin is tossed six times, how many different outcomes consisting of 4 heads and 2 tails are there?

Solution

Again, we have permutations with similar elements.

We are looking for permutations for the letters HHHHTT.

The answer is $\frac{6!}{4!2!} = 15$.

✓ Example 2.4.4.5

In how many different ways can 4 nickels, 3 dimes, and 2 quarters be arranged in a row?

Solution

Assuming that all nickels are similar, all dimes are similar, and all quarters are similar, we have permutations with similar elements. Therefore, the answer is

$$\frac{9!}{4!3!2!} = 1260$$

✓ Example 2.4.4.6

A stock broker wants to assign 20 new clients equally to 4 of its salespeople. In how many different ways can this be done?

Solution

This means that each sales person gets 5 clients. The problem can be thought of as an ordered partitions problem. In that case, using the formula we get

$$\frac{20!}{5!5!5!5!} = 11,732,745,024$$

✓ Example 2.4.4.7

A shopping mall has a straight row of 5 flagpoles at its main entrance plaza. It has 3 identical green flags and 2 identical yellow flags. How many distinct arrangements of flags on the flagpoles are possible?

Solution

The problem can be thought of as distinct permutations of the letters GGGYY; that is arrangements of 5 letters, where 3 letters are similar, and the remaining 2 letters are similar:

$$\frac{5!}{3!2!} = 10$$

Just to provide a little more insight into the solution, we list all 10 distinct permutations:

GGGY, GGYG, GGYG, GYGG, GYGY, GYGY, YGGG, YGGY, YGYG, YGGG

We summarize.

📌 Summary

1. Circular Permutations

The number of permutations of n elements in a circle is

$$(n-1)!$$

2. Permutations with Similar Elements

The number of permutations of n elements taken n at a time, with r_1 elements of one kind, r_2 elements of another kind, and so on, such that $n = r_1 + r_2 + \dots + r_k$ is

$$\frac{n!}{r_1! r_2! \dots r_k!}$$

This is also referred to as **ordered partitions**.

Exercises

Do the following problems using the techniques learned in this section.

1. In how many different ways can five children hold hands to play "Ring Around the Rosy"?	2. In how many ways can three people be made to sit at a round table?
3. In how many different ways can six children ride a "Merry Go Around" with six horses?	4. In how many ways can three couples be seated at a round table, so that men and women sit alternately?
5. In how many ways can six trinkets be arranged on a chain?	6. In how many ways can five keys be put on a key ring?
7. Find the number of different permutations of the letters of the word MASSACHUSETTS.	8. Find the number of different permutations of the letters of the word MATHEMATICS.
9. Seven flags are to be flown on seven poles: 3 flags are red, 2 are white, and 2 are blue,. How many different arrangements are possible?	10. How many different ways can 3 pennies, 2 nickels and 5 dimes be arranged in a row?
11. How many four-digit numbers can be made using two 2's and two 3's?	12. How many five-digit numbers can be made using two 6's and three 7's?
13. If a coin is tossed 5 times, how many different outcomes of 3 heads and 2 tails are possible?	14. If a coin is tossed 10 times, how many different outcomes of 7 heads and 3 tails are possible?
15. If a team plays ten games, how many different outcomes of 6 wins and 4 losses are possible?	16. If a team plays ten games, how many different ways can the team have a winning season?

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2.5: Chapter Review Exercises

1. Suppose of the 4,000 freshmen at a college everyone is enrolled in a mathematics or an English class during a given quarter. If 2,000 are enrolled in a mathematics class, and 3,000 in an English class, how many are enrolled in both a mathematics class and an English class?
2. In a survey of 250 people, it was found that 125 had read Time magazine, 175 had read Newsweek, 100 had read U. S. News, 75 had read Time and Newsweek, 60 had read Newsweek and U. S. News, 55 had read Time and U. S. News, and 25 had read all three.
 - a. How many had read Time but not the other two?
 - b. How many had read Time or Newsweek but not the U. S. News And World Report?
 - c. How many had read none of these three magazines?
3. At a manufacturing plant, a product goes through assembly, testing, and packing. If a plant has three assembly stations, two testing stations, and two packing stations, in how many different ways can a product achieve its completion?
4. Six people are to line up for a photograph. How many different lineups are possible if three of them insist on standing next to each other?
5. How many four-letter word sequences can be made from the letters of the word CUPERTINO?
6. In how many different ways can a 20-question multiple choice test be designed so that its answers contain 2 A's, 4 B's, 9 C's, 3 D's, and 2 E's?
7. The U. S. Supreme Court has nine judges. In how many different ways can the judges cast a six-to-three decision in favor of a ruling?
8. In how many different ways can a coach choose a linebacker, a guard, and a tackle from five players on the bench, if all five can play any of the three positions?
9. How many three digit even numbers can be formed from the digits 1, 2, 3, 4, 5 if no repetitions are allowed?
10. Compute:
 - a. 9C_4
 - b. 8P_3
 - c. $\frac{10!}{4!(10-4)!}$
11. In how many ways can 3 English, 3 Math, and 4 Spanish books be set on a shelf if the books are grouped by subject?
12. In how many ways can a 10-question multiple choice test with four possible answers for each question be answered?
13. On a soccer team three fullbacks can play any of the three fullback positions, left, center, and right. The three halfbacks can play any of the three halfback positions, the four forwards can play any of the four positions, and the goalkeeper plays only his position. How many different arrangements of the 11 players are possible?
14. From a group of 6 people, 3 are assigned to cleaning, 2 to hauling and one to garbage collecting. How many different ways can this be done?
15. How many three-letter word sequences can be made from the letters of the word OXYGEN?
16. In how many ways can 3 books be selected from 4 English and 2 History books if at least one English book must be chosen?
17. Five points lie on the rim of a circle. Choosing the points as vertices, how many different triangles can be drawn?
18. A club consists of six men and nine women. In how many ways can a president, a vice president and a treasurer be chosen if the two of the officers must be women?
19. Of its 12 sales people, a company wants to assign 4 to its Western territory, 5 to its Northern territory, and 3 to its Southern territory. How many ways can this be done?
20. How many permutations of the letters of the word OUTSIDE have consonants in the first and last place?
21. How many distinguishable permutations are there in the word COMMUNICATION?
22. How many five-card poker hands consisting of the following distribution are there?
 - a. A flush(all five cards of a single suit)
 - b. Three of a kind(e.g. three aces and two other cards)
 - c. Two pairs(e.g. two aces, two kings and one other card)
 - d. A straight(all five cards in a sequence)
23. Company stocks on an exchange are given symbols consisting of three letters. How many different three-letter symbols are possible?
24. How many four-digit odd numbers are there?

25. In how many ways can 7 people be made to stand in a straight line? In a circle?
26. A United Nations delegation consists of 6 Americans, 5 Russians, and 4 Chinese. Answer the following questions.
 - a. How many committees of five people are there?
 - b. How many committees of three people consisting of at least one American are there?
 - c. How many committees of four people having no Russians are there?
 - d. How many committees of three people have more Americans than Russians?
 - e. How many committees of three people do not have all three Americans?
27. If a coin is flipped five times, in how many different ways can it show up three heads?
28. To reach his destination, a man is to walk three blocks north and four blocks west. How many different routes are possible?
29. All three players of the women's beach volleyball team, and all three players of the men's beach volleyball team are to line up for a picture with all members of the women's team lined together and all members of men's team lined up together. How many ways can this be done?
30. From a group of 6 Americans, 5 Japanese and 4 German delegates, two Americans, two Japanese and a German are chosen to line up for a photograph. In how many different ways can this be done?
31. Find the fourth term of the expansion $(2x - 3y)^8$.
32. Find the coefficient of the a^5b^4 term in the expansion of $(a - 2b)^9$.

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CHAPTER OVERVIEW

3: Probability

Learning Objectives

In this chapter, you will learn to:

1. Write sample spaces.
2. Determine whether two events are mutually exclusive.
3. Use the Addition Rule.
4. Calculate probabilities using both tree diagrams and combinations.
5. Do problems involving conditional probability.
6. Determine whether two events are independent.

[3.1: Sample Spaces and Probability](#)

[3.2: Mutually Exclusive Events and the Addition Rule](#)

[3.3: Probability Using Tree Diagrams and Combinations](#)

[3.4: Conditional Probability and Independent Events](#)

[3.4.1: Conditional Probability](#)

[3.4.2: Independent Events](#)

[3.5: Bayes' Formula](#)

[3.6: Chapter Review Exercises](#)

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3.1: Sample Spaces and Probability

Learning Objectives

In this section, you will learn to:

1. Write sample spaces.
2. Calculate probabilities by examining simple events in sample spaces.

If two coins are tossed, what is the probability that both coins will fall heads? The problem seems simple enough, but it is not uncommon to hear the incorrect answer $1/3$. A student may incorrectly reason that if two coins are tossed there are three possibilities, one head, two heads, or no heads. Therefore, the probability of two heads is one out of three. The answer is wrong because if we toss two coins there are four possibilities and not three. For clarity, assume that one coin is a penny and the other a nickel. Then we have the following four possibilities.

HH HT TH TT

The possibility HT, for example, indicates a head on the penny and a tail on the nickel, while TH represents a tail on the penny and a head on the nickel. It is for this reason, we emphasize the need for understanding sample spaces.

Sample Spaces

An act of flipping coins, rolling dice, drawing cards, or surveying people are referred to as a probability **experiment**. A **sample space** of an experiment is the set of all possible outcomes.

✓ Example 3.1.1

If a die is rolled, write a sample space.

Solution

A die has six faces each having an equally likely chance of appearing. Therefore, the set of all possible outcomes S is

$\{ 1, 2, 3, 4, 5, 6 \}$.

✓ Example 3.1.2

A family has three children. Write a sample space.

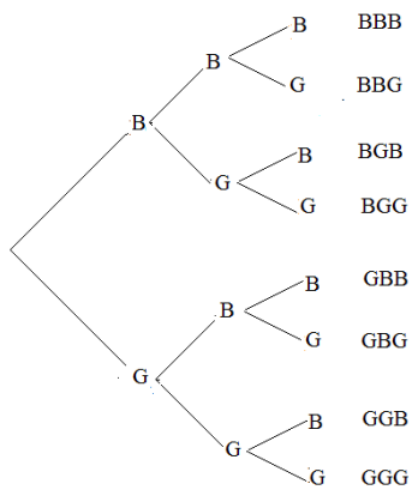
Solution

The sample space consists of eight possibilities.

$\{ BBB, BBG, BGB, BGG, GBB, GBG, GGB, GGG \}$

The possibility BGB, for example, indicates that the first born is a boy, the second born a girl, and the third a boy.

We illustrate these possibilities with a tree diagram.



✓ Example 3.1.3

Two dice are rolled. Write the sample space.

Solution

We assume one of the dice is red, and the other green. We have the following 36 possibilities.

Green						
Red	1	2	3	4	5	6
1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

The entry (2, 5), for example, indicates that the red die shows a 2, and the green a 5.

Probability

Now that we understand the concept of a sample space, we will define probability.

Definition: Probability

For a sample space S , and an outcome A of S , the following two properties are satisfied.

1. If A is an outcome of a sample space, then the probability of A , denoted by $P(A)$, is between 0 and 1, inclusive.

$$0 \leq P(A) \leq 1$$

2. The sum of the probabilities of all the outcomes in S equals 1.

The probability $P(A)$ of an event A describes the chance or likelihood of that event occurring.

- If $P(A) = 0$, event A is certain not to occur. If $P(A) = 1$, event A is certain to occur.
- If $P(A) = 0.5$, then event A is equally likely to occur or not occur.
- If we toss a fair coin that is equally likely to land on heads or tails, then $P(\text{Head}) = 0.50$.

- If the weather forecast says there is a 70% chance of rain today, then $P(\text{Rain}) = 0.70$, indicating it is more likely to rain than to not rain.

✓ Example 3.1.4

If two dice, one red and one green, are rolled, find the probability that the red die shows a 3 and the green shows a six.

Solution

Since two dice are rolled, there are 36 possibilities. The probability of each outcome, listed in Example 3.1.3, is equally likely. Since (3, 6) is one such outcome, the probability of obtaining (3, 6) is $1/36$.

The example we just considered consisted of only one outcome of the sample space. We are often interested in finding probabilities of several outcomes represented by an event.

An **event** is a subset of a sample space. If an event consists of only one outcome, it is called a **simple event**.

✓ Example 3.1.5

If two dice are rolled, find the probability that the sum of the faces of the dice is 7.

Solution

Let E represent the event that the sum of the faces of two dice is 7.

The possible cases for the sum to be equal to 7 are: (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), and (6, 1), so event E is

$$E = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$$

The probability of the event E is

$$P(E) = 6/36 \text{ or } 1/6.$$

✓ Example 3.1.6

A jar contains 3 red, 4 white, and 3 blue marbles. If a marble is chosen at random, what is the probability that the marble is a red marble or a blue marble?

Solution

We assume the marbles are $r_1, r_2, r_3, w_1, w_2, w_3, w_4, b_1, b_2, b_3$. Let the event C represent that the marble is red or blue.

The sample space $S = \{r_1, r_2, r_3, w_1, w_2, w_3, w_4, b_1, b_2, b_3\}$.

And the event $C = \{r_1, r_2, r_3, b_1, b_2, b_3\}$

Therefore, the probability of C ,

$$P(C) = 6/10 \text{ or } 3/5$$

✓ Example 3.1.7

A jar contains three marbles numbered 1, 2, and 3. If two marbles are drawn **without replacement**, what is the probability that the sum of the numbers is 5?

Note: The two marbles in this example are drawn consecutively **without replacement**. That means that after a marble is drawn it is not replaced in the jar, and therefore is no longer available to select on the second draw.

Solution

Since two marbles are drawn without replacement, the sample space consists of the following six possibilities.

$$S = \{(1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2)\}$$

Note that (1,1), (2,2) and (3,3) are not listed in the sample space. These outcomes are not possible when drawing without replacement, because once the first marble is drawn but not replaced into the jar, that marble is not available in the jar to be selected again on the second draw.

Let the event E represent that the sum of the numbers is five. Then

$$E = \{(2, 3), (3, 2)\}$$

Therefore, the probability of E is

$$P(E) = 2/6 \text{ or } 1/3.$$

✓ Example 3.1.8

A jar contains three marbles numbered 1, 2, and 3. If two marbles are drawn **without replacement**, what is the probability that the sum of the numbers is *at least* 4?

Solution

The sample space, as in Example 3.1.7, consists of the following six possibilities.

$$S = \{(1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2)\}$$

Let the event F represent that the sum of the numbers is at least four. Then

$$F = \{(1, 3), (3, 1), (2, 3), (3, 2)\}$$

Therefore, the probability of F is

$$P(F) = 4/6 \text{ or } 2/3$$

✓ Example 3.1.9

A jar contains three marbles numbered 1, 2, and 3. If two marbles are drawn **with replacement**, what is the probability that the sum of the numbers is 5?

Note: The two marbles in this example are drawn consecutively **with replacement**. That means that after a marble is drawn it is replaced in the jar, and therefore is available to select again on the second draw.

Solution

When two marbles are drawn with replacement, the sample space consists of the following nine possibilities.

$$S = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$$

Note that (1,1), (2,2) and (3,3) are listed in the sample space. These outcomes are possible when drawing with replacement, because once the first marble is drawn and replaced, that marble is not available in the jar to be drawn again.

Let the event E represent that the sum of the numbers is four. Then

$$E = (2, 3), (3, 2)$$

Therefore, the probability of F is $P(E) = 2/9$

Note that in Example 3.1.9 when we selected marbles with replacement, the probability has changed from Example 3.1.7 where we selected marbles without replacement.

✓ Example 3.1.10

A jar contains three marbles numbered 1, 2, and 3. If two marbles are drawn **with replacement**, what is the probability that the sum of the numbers is *at least* 4?

Solution

The sample space when drawing with replacement consists of the following nine possibilities.

$$S = (1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)$$

Let the event F represent that the sum of the numbers is at least four. Then

$$F = (1, 3), (3, 1), (2, 3), (3, 2), (2, 2), (3, 3)$$

Therefore, the probability of F is

$$P(F) = 6/9 \text{ or } 2/3.$$

Note that in Example 3.1.10 when we selected marbles with replacement, the probability is the same as in Example 3.1.8 where we selected marbles without replacement.

Thus sampling with or without replacement MAY change the probabilities, but may not, depending on the situation in the particular problem under consideration. We'll re-examine the concepts of sampling with and without replacement in Section 8.3.

✓ Example 3.1.11

One 6 sided die is rolled once. Find the probability that the result is greater than 4.

Solution

The sample space consists of the following six possibilities in set S : $S = 1, 2, 3, 4, 5, 6$

Let E be the event that the number rolled is greater than four: $E = 5, 6$

Therefore, the probability of E is: $P(E) = 2/6$ or $1/3$.

Exercises

SECTION 8.1 PROBLEM SET: SAMPLE SPACES AND PROBABILITY

In problems 1 - 6, write a sample space for the given experiment.

1) A die is rolled.	2) A penny and a nickel are tossed.
3) A die is rolled, and a coin is tossed.	4) Three coins are tossed.
5) Two dice are rolled.	6) A jar contains four marbles numbered 1, 2, 3, and 4. Two marbles are drawn.

In problems 7 - 12, one card is randomly selected from a deck. Find the following probabilities.

7) $P(\text{an ace})$	8) $P(\text{a red card})$
9) $P(\text{a club})$	10) $P(\text{a face card})$
11) $P(\text{a jack or a spade})$	12) $P(\text{a jack and a spade})$

For problems 13 - 16: A jar contains 6 red, 7 white, and 7 blue marbles. If one marble is chosen at random, find the following probabilities.

13) $P(\text{red})$	14) $P(\text{white})$
15) $P(\text{red or blue})$	16) $P(\text{red and blue})$

For problems 17 - 22: Consider a family of three children. Find the following probabilities.

17) $P(\text{two boys and a girl})$	18) $P(\text{at least one boy})$
-------------------------------------	----------------------------------

19) P(children of both sexes)	20) P(at most one girl)
21) P(first and third children are male)	22) P(all children are of the same gender)

For problems 23 - 27: Two dice are rolled. Find the following probabilities.

23) P(the sum of the dice is 5)	24) P(the sum of the dice is 8)
25) P(the sum is 3 or 6)	26) P(the sum is more than 10)
27) P(the result is a double) (Hint: a double means that both dice show the same value)	

For problems 28-31: A jar contains four marbles numbered 1, 2, 3, and 4. Two marbles are drawn randomly WITHOUT REPLACEMENT. That means that after a marble is drawn it is NOT replaced in the jar before the second marble is selected. Find the following probabilities.

28) P(the sum of the numbers is 5)	29) P(the sum of the numbers is odd)
30) P(the sum of the numbers is 9)	31) P(one of the numbers is 3)

For problems 32-33: A jar contains four marbles numbered 1, 2, 3, and 4. Two marbles are drawn randomly WITH REPLACEMENT. That means that after a marble is drawn it is replaced in the jar before the second marble is selected. Find the following probabilities.

32) P(the sum of the numbers is 5)	33) P(the sum of the numbers is 2)
------------------------------------	------------------------------------

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3.2: Mutually Exclusive Events and the Addition Rule

Learning Objectives

In this section, you will learn to:

1. Define compound events using union, intersection, and complement.
2. Identify mutually exclusive events
3. Use the Addition Rule to calculate probability for unions of events.

In the last chapter, we learned to find the union, intersection, and complement of a set. We will now use these set operations to describe events.

- The **union** of two events E and F , $E \cup F$, is the set of outcomes that are in E or in F or in both.
- The **intersection** of two events E and F , $E \cap F$, is the set of outcomes that are in both E and F .
- The **complement** of an event E , denoted by E^c , is the set of outcomes in the sample space S that are not in E .

It is worth noting that $P(E^c) = 1 - P(E)$. This follows from the fact that if the sample space has n elements and E has k elements, then E^c has $n - k$ elements. Therefore,

$$P(E^c) = \frac{n - k}{n} = 1 - \frac{k}{n} = 1 - P(E)$$

Of particular interest to us are the events whose outcomes do not overlap. We call these events mutually exclusive.

Two events E and F are said to be **mutually exclusive** if they do not intersect: $E \cap F = \emptyset$.

Next we'll determine whether a given pair of events are mutually exclusive.

✓ Example 3.2.1

A card is drawn from a standard deck. Determine whether the pair of events given below is mutually exclusive.

$E = \{\text{The card drawn is an Ace}\}$

$F = \{\text{The card drawn is a heart}\}$

Solution

Clearly the ace of hearts belongs to both sets. That is

$$E \cap F = \{\text{Ace of hearts}\} \neq \emptyset$$

Therefore, the events E and F are not mutually exclusive.

✓ Example 3.2.2

Two dice are rolled. Determine whether the pair of events given below is mutually exclusive.

$G = \{\text{The sum of the faces is six}\}$

$H = \{\text{One die shows a four}\}$

Solution

For clarity, we list the elements of both sets.

$G = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$ and $H = \{(2, 4), (4, 2)\}$

Clearly, $G \cap H = \{(2, 4), (4, 2)\} \neq \emptyset$.

Therefore, the two sets are not mutually exclusive.

✓ Example 3.2.3

A family has three children. Determine whether the following pair of events are mutually exclusive.

$M = \{\text{The family has at least one boy}\}$

$N = \{\text{The family has all girls}\}$

Solution

Although the answer may be clear, we list both the sets.

$M = \{ \text{BBB, BBG, BGB, BGG, GBB, GBG, GGB} \}$ and $N = \{ \text{GGG} \}$

Clearly, $M \cap N = \emptyset$

Therefore, events M and N are mutually exclusive.

We will now consider problems that involve the union of two events.

Given two events, E , F , then finding the probability of $E \cup F$, is the same as finding the probability that E will happen, or F will happen, or both will happen.

✓ Example 3.2.4

If a die is rolled, what is the probability of obtaining an even number or a number greater than four?

Solution

Let E be the event that the number shown on the die is an even number, and let F be the event that the number shown is greater than four.

The sample space $S = \{ 1, 2, 3, 4, 5, 6 \}$. The event $E = \{ 2, 4, 6 \}$, and event $F = \{ 5, 6 \}$

We need to find $P(E \cup F)$.

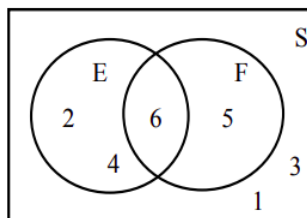
Since $P(E) = 3/6$, and $P(F) = 2/6$, a student may say $P(E \cup F) = 3/6 + 2/6$. This will be incorrect because the element 6, which is in both E and F has been counted twice, once as an element of E and once as an element of F . In other words, the set $E \cup F$ has only four elements and not five: set $E \cup F = \{ 2, 4, 5, 6 \}$

Therefore, $P(E \cup F) = 4/6$ and not $5/6$.

This can be illustrated by a Venn diagram. We'll use the Venn Diagram to re-examine Example 3.2.4 and derive a probability rule that we can use to calculate probabilities for unions of events.

The sample space S , the events E and F , and $E \cap F$ are listed below.

$S = \{1, 2, 3, 4, 5, 6\}$, $E = \{2, 4, 6\}$, $F = \{5, 6\}$, and $E \cap F = \{6\}$.



The above figure shows S , E , F , and $E \cap F$.

Finding the probability of $E \cup F$, is the same as finding the probability that E will happen, or F will happen, or both will happen.

If we count the number of elements $n(E)$ in E , and add to it the number of elements $n(F)$ in F , the points in both E and F are counted twice, once as elements of E and once as elements of F . Now if we subtract from the sum, $n(E) + n(F)$, the number $n(E \cap F)$, we remove the duplicity and get the correct answer. So as a rule,

$$n(E \cup F) = n(E) + n(F) - n(E \cap F)$$

By dividing the entire equation by $n(S)$, we get

$$\frac{n(E \cup F)}{n(S)} = \frac{n(E)}{n(S)} + \frac{n(F)}{n(S)} - \frac{n(E \cap F)}{n(S)}$$

Since the probability of an event is the number of elements in that event divided by the number of all possible outcomes, we have

$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

Applying the above for Example 3.2.4, we get

$$P(E \cup F) = 3/6 + 2/6 - 1/6 = 4/6$$

This is because, when we add $P(E)$ and $P(F)$, we have added $P(E \cap F)$ twice. Therefore, we must subtract $P(E \cap F)$, once.

This gives us the general formula, called **the Addition Rule**, for finding the probability of the union of two events. Because event $E \cup F$ is the event that E will happen, OR F will happen, OR both will happen, we sometimes call this the **Addition Rule for OR Events**. It states

Addition Rule

$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

If, and only if, two events E and F are mutually exclusive, then $E \cap F = \emptyset$ and $P(E \cap F) = 0$, and we get $P(E \cup F) = P(E) + P(F)$

✓ Example 3.2.5

If a card is drawn from a deck, use the addition rule to find the probability of obtaining an ace or a heart.

Solution

Let A be the event that the card is an ace, and H the event that it is a heart.

Since there are four aces, and thirteen hearts in the deck,

$$P(A) = 4/52 \text{ and } P(H) = 13/52.$$

Furthermore, since the intersection of two events consists of only one card, the ace of hearts, we now have:

$$P(A \cap H) = 1/52$$

We need to find $P(A \cup H)$:

$$\begin{aligned} P(A \cup H) &= P(A) + P(H) - P(A \cap H) \\ &= 4/52 + 13/52 - 1/52 = 16/52 \end{aligned}$$

✓ Example 3.2.6

Two dice are rolled, and the events F and T are as follows:

$F = \{\text{The sum of the dice is four}\}$ and $T = \{\text{At least one die shows a three}\}$

Find $P(F \cup T)$.

Solution

We list F and T , and $F \cap T$ as follows:

$F = \{(1, 3), (2, 2), (3, 1)\}$

$T = \{(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (1, 3), (2, 3), (4, 3), (5, 3), (6, 3)\}$

$F \cap T = \{(1, 3), (3, 1)\}$

Since $P(F \cup T) = P(F) + P(T) - P(F \cap T)$

We have $P(F \cup T) = 3/36 + 11/36 - 2/36 = 12/36$.

✓ Example 3.2.7

Mr. Washington is seeking a mathematics instructor's position at his favorite community college in Cupertino. His employment depends on two conditions: whether the board approves the position, and whether the hiring committee selects him. There is a 80% chance that the board will approve the position, and there is a 70% chance that the hiring committee will select him. If there is a 90% chance that at least one of the two conditions, the board approval or his selection, will be met, what is the probability that Mr. Washington will be hired?

Solution

Let A be the event that the board approves the position, and S be the event that Mr. Washington gets selected. We have,

$P(A) = .80$, $P(S) = .70$, and $P(A \cup S) = .90$.

We need to find, $P(A \cap S)$.

The addition formula states that,

$$P(A \cup S) = P(A) + P(S) - P(A \cap S)$$

Substituting the known values, we get

$$.90 = .80 + .70 - P(A \cap S)$$

Therefore, $P(A \cap S) = .60$.

✓ Example 3.2.8

The probability that this weekend will be cold is .6, the probability that it will be rainy is .7, and probability that it will be both cold and rainy is .5. What is the probability that it will be neither cold nor rainy?

Solution

Let C be the event that the weekend will be cold, and R be event that it will be rainy. We are given that

$$P(C) = .6, \quad P(R) = .7, \quad P(C \cap R) = .5$$

First we find $P(C \cup R)$ using the Addition Rule.

$$P(C \cup R) = P(C) + P(R) - P(C \cap R) = .6 + .7 - .5 = .8$$

Then we find $P((C \cup R)^c)$ using the Complement Rule.

$$P((C \cup R)^c) = 1 - P(C \cup R) = 1 - .8 = .2$$

We summarize this section by listing the important rules.

📌 Summary

The Addition Rule

For Two Events E and F, $P(E \cup F) = P(E) + P(F) - P(E \cap F)$

The Addition Rule for Mutually Exclusive Events

If Two Events E and F are Mutually Exclusive, then $P(E \cup F) = P(E) + P(F)$

The Complement Rule

If E^c is the Complement of Event E, then $P(E^c) = 1 - P(E)$

Exercises

SECTION 8.2 PROBLEM SET: MUTUALLY EXCLUSIVE EVENTS AND THE ADDITION RULE

Determine whether the following pair of events are mutually exclusive.

1) $A = \{\text{A person earns more than \$25,000}\}$ $B = \{\text{A person earns less than \$20,000}\}$	2) A card is drawn from a deck. $C = \{\text{It is a King}\}$ $D = \{\text{It is a heart}\}$.
3) A die is rolled. $E = \{\text{An even number shows}\}$ $F = \{\text{A number greater than 3 shows}\}$	4) Two dice are rolled. $G = \{\text{The sum of dice is 8}\}$ $H = \{\text{One die shows a 6}\}$
5) Three coins are tossed. $I = \{\text{Two heads come up}\}$ $J = \{\text{At least one tail comes up}\}$	6) A family has three children. $K = \{\text{First born is a boy}\}$ $L = \{\text{The family has children of both sexes}\}$

Use the Addition Rule to find the following probabilities.

7) A card is drawn from a deck. Events C and D are: $C = \{\text{It is a king}\}$ $D = \{\text{It is a heart}\}$ Find $P(C \text{ or } D)$.	8) A die is rolled. The events E and F are: $E = \{\text{An even number shows}\}$ $F = \{\text{A number greater than 3 shows}\}$ Find $P(E \text{ or } F)$.
9) Two dice are rolled. Events G and H are: $G = \{\text{The sum of dice is 8}\}$ $H = \{\text{Exactly one die shows a 6}\}$ Find $P(G \text{ or } H)$.	10) Three coins are tossed. Events I and J are: $I = \{\text{Two heads come up}\}$ $J = \{\text{At least one tail comes up}\}$ Find $P(I \text{ or } J)$.
11) At a college, 20% of the students take Finite Mathematics, 30% take Statistics and 10% take both. What percent of students take Finite Mathematics or Statistics?	12) This quarter, there is a 50% chance that Jason will pass Accounting, a 60% chance that he will pass English, and 80% chance that he will pass at least one of these two courses. What is the probability that he will pass both Accounting and English?

Questions 13 - 20 refer to the following: The table shows the distribution of Democratic and Republican U.S by gender in the 114th Congress as of January 2015.

	MALE(M)	FEMALE(F)	TOTAL
DEMOCRATS (D)	30	14	44
REPUBLICANS(R)	48	6	54
OTHER (T)	2	0	2
TOTALS	80	20	100

Use this table to determine the following probabilities.

13) $P(M \text{ and } D)$	14) $P(F \text{ and } R)$
15) $P(M \text{ or } D)$	16) $P(F \text{ or } R)$
17) $P(Mc \text{ or } R)$	18) $P(M \text{ or } F)$
19) Are the events F, R mutually exclusive? Use probabilities to support your conclusions.	20) Are the events F, T mutually exclusive? Use probabilities to support your conclusion.

SECTION 8.2 PROBLEM SET: MUTUALLY EXCLUSIVE EVENTS AND THE ADDITION RULE

Use the Addition Rule to find the following probabilities.

21) If $P(E) = .5$, $P(F) = .4$, E and F are mutually exclusive, find $P(E \text{ and } F)$.	22) If $P(E) = .4$, $P(F) = .2$, E and F are mutually exclusive, find $P(E \text{ or } F)$.
23) If $P(E) = .3$, $P(E \text{ or } F) = .6$, $P(E \text{ and } F) = .2$, find $P(F)$.	24) If $P(E) = .4$, $P(F) = .5$, $P(E \text{ or } F) = .7$, find $P(E \text{ and } F)$.
25) In a box of assorted cookies, 36% of cookies contain chocolate and 12% of cookies contain nuts. 8% of cookies have both chocolates and nuts. Sean is allergic to chocolate and nuts. Find the probability that a cookie has chocolate chips or nuts (he can't eat it).	26) At a college, 72% of courses have final exams and 46% of courses require research papers. 32% of courses have both a research paper and a final exam. Let F be the event that a course has a final exam and R be the event that a course requires a research paper. Find the probability that a course requires a final exam or a research paper.

Questions 25 and 26 are adapted from *Introductory Statistics* from OpenStax under a *creative Commons Attribution 3.0 Unported License*, available for download free at cnx.org/content/col11562/latest

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3.3: Probability Using Tree Diagrams and Combinations

Learning Objectives

In this section, you will learn to:

1. Use probability tree diagrams to calculate probabilities
2. Use combinations to calculate probabilities

In this section, we will apply previously learnt counting techniques in calculating probabilities, and use tree diagrams to help us gain a better understanding of what is involved.

USING TREE DIAGRAMS TO CALCULATE PROBABILITIES

We already used tree diagrams to list events in a sample space. Tree diagrams can be helpful in organizing information in probability problems; they help provide a structure for understanding probability. In this section we expand our previous use of tree diagrams to situations in which the events in the sample space are not all equally likely.

We assign the appropriate probabilities to the events shown on the branches of the tree.

By multiplying probabilities along a path through the tree, we can find probabilities for “and” events, which are intersections of events.

We begin with an example.

✓ Example 3.3.1

Suppose a jar contains 3 red and 4 white marbles. If two marbles are drawn with replacement, what is the probability that both marbles are red?

Solution

Let E be the event that the first marble drawn is red, and let F be the event that the second marble drawn is red.

We need to find $P(E \cap F)$.

By the statement, “two marbles are drawn with replacement,” we mean that the first marble is replaced before the second marble is drawn.

There are 7 choices for the first draw. And since the first marble is replaced before the second is drawn, there are, again, seven choices for the second draw. Using the multiplication axiom, we conclude that the sample space S consists of 49 ordered pairs. Of the 49 ordered pairs, there are $3 \times 3 = 9$ ordered pairs that show red on the first draw and, also, red on the second draw. Therefore,

$$P(E \cap F) = \frac{9}{49}$$

Further note that in this particular case

$$P(E \cap F) = \frac{9}{49} = \frac{3}{7} \cdot \frac{3}{7}$$

giving us the result that in this example: $P(E \cap F) = P(E) \cdot P(F)$

✓ Example 3.3.2

If in Example 3.3.1, the two marbles are drawn without replacement, then what is the probability that both marbles are red?

Solution

By the statement, “two marbles are drawn without replacement,” we mean that the first marble is not replaced before the second marble is drawn.

Again, we need to find $P(E \cap F)$.

There are, again, 7 choices for the first draw. And since the first marble is not replaced before the second is drawn, there are only six choices for the second draw. Using the multiplication axiom, we conclude that the sample space S consists of 42 ordered pairs. Of the 42 ordered pairs, there are $3 \times 2 = 6$ ordered pairs that show red on the first draw and red on the second draw. Therefore,

$$P(E \cap F) = \frac{6}{42}$$

Note that we can break this calculation down as

$$P(E \cap F) = \frac{6}{42} = \frac{3}{7} \cdot \frac{2}{6}$$

Here $3/7$ represents $P(E)$, and $2/6$ represents the probability of drawing a red on the second draw, given that the first draw resulted in a red.

We write the latter as $P(\text{red on the second} \mid \text{red on first})$ or $P(F|E)$. The "|" represents the word "given" or "if". This leads to the result that:

$$P(E \cap F) = P(E) \cdot P(F|E)$$

This is an important result, called the **Multiplication Rule**, which will appear again in later sections.

We now demonstrate the above results with a tree diagram.

✓ Example 3.3.3

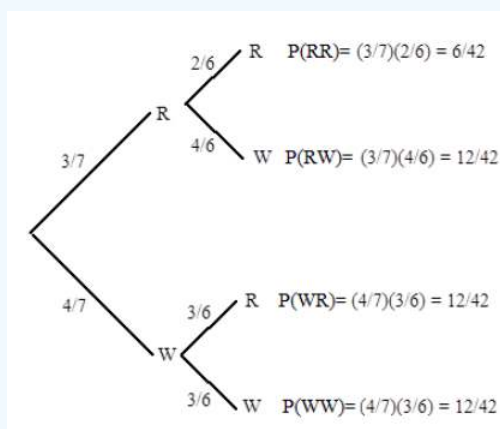
Suppose a jar contains 3 red and 4 white marbles. If two marbles are drawn without replacement, find the following probabilities using a tree diagram.

- The probability that both marbles are red.
- The probability that the first marble is red and the second white.
- The probability that one marble is red and the other white.

Solution

Let R be the event that the marble drawn is red, and let W be the event that the marble drawn is white.

We draw the following tree diagram.



- The probability that both marbles are red is $P(RR) = 6/42$
- The probability that the first marble is red and the second is white is $P(RW) = 12/42$
- For the probability that one marble is red and the other is white, we observe that this can be satisfied if the first is red and the second is white, **or** if the first is white and the second is red. The "or" tells us we'll be using the Addition Rule from Section 7.2.

Furthermore events RW and WR are mutually exclusive events, so we use the form of the Addition Rule that applies to mutually exclusive events.

Therefore

$P(\text{one marble is red and the other marble is white})$

$$\begin{aligned} &= P(RW \text{ or } WR) \\ &= P(RW) + P(WR) \\ &= 12/42 + 12/42 = 24/42 \end{aligned}$$

USING COMBINATIONS TO FIND PROBABILITIES

Although the tree diagrams give us better insight into a problem, they are not practical for problems where more than two or three things are chosen. In such cases, we use the concept of combinations that we learned in the last chapter. This method is best suited for problems where the order in which the objects are chosen is not important, and the objects are chosen without replacement.

✓ Example 3.3.4

Suppose a jar contains 3 red, 2 white, and 3 blue marbles. If three marbles are drawn without replacement, find the following probabilities.

- $P(\text{Two red and one white})$
- $P(\text{One of each color})$
- $P(\text{None blue})$
- $P(\text{At least one blue})$

Solution

Let us suppose the marbles are labeled as $R_1, R_2, R_3, W_1, W_2, B_1, B_2, B_3$.

- $P(\text{Two red and one white})$

Since we are choosing 3 marbles from a total of 8, there are $8C3 = 56$ possible combinations. Of these 56 combinations, there are $3C2 \times 2C1 = 6$ combinations consisting of 2 red and one white. Therefore,

$$P(\text{Two red and one white}) = \frac{3C2 \times 2C1}{8C3} = \frac{6}{56}.$$

- $P(\text{One of each color})$

Again, there are $8C3 = 56$ possible combinations. Of these 56 combinations, there are $3C1 \times 2C1 \times 3C1 = 18$ combinations consisting of one red, one white, and one blue. Therefore,

$$P(\text{One of each color}) = \frac{3C1 \times 2C1 \times 3C1}{8C3} = \frac{18}{56}$$

- $P(\text{None blue})$

There are 5 non-blue marbles, therefore

$$P(\text{None blue}) = \frac{5C3}{8C3} = \frac{10}{56} = \frac{5}{28}$$

- $P(\text{At least one blue})$

By "at least one blue marble," we mean the following: one blue marble and two non-blue marbles, OR two blue marbles and one non-blue marble, OR all three blue marbles. So we have to find the sum of the probabilities of all three cases.

$$P(\text{At least one blue}) = P(1 \text{ blue, 2 non-blue}) + P(2 \text{ blue, 1 non-blue}) + P(3 \text{ blue})$$

$$P(\text{At least one blue}) = \frac{3C1 \times 5C2}{8C3} + \frac{3C2 \times 5C1}{8C3} + \frac{3C3}{8C3}$$

$$P(\text{At least one blue}) = 30/56 + 15/56 + 1/56 = 46/56 = 23/28$$

Alternately, we can use the fact that $P(E) = 1 - P(E^c)$. If the event E = At least one blue, then E^c = None blue. But from part c of this example, we have $P(E^c) = 5/28$, so $P(E) = 1 - 5/28 = 23/28$.

✓ Example 3.3.5

Five cards are drawn from a deck. Find the probability of obtaining two pairs, that is, two cards of one value, two of another value, and one other card.

Solution

Let us first do an easier problem-the probability of obtaining a pair of kings and queens.

Since there are four kings, and four queens in the deck, the probability of obtaining two kings, two queens and one other card is

$$P(\text{A pair of kings and queens}) = \frac{4C2 \times 4C2 \times 44C1}{52C5}$$

To find the probability of obtaining two pairs, we have to consider all possible pairs.

Since there are altogether 13 values, that is, aces, deuces, and so on, there are $13C2$ different combinations of pairs.

$$P(\text{Two pairs}) = 13C2 \cdot \frac{4C2 \times 4C2 \times 44C1}{52C5} = .04754$$

✓ Example 3.3.6

A cell phone store receives a shipment of 15 cell phones that contains 8 iPhones and 7 Android phones. Suppose that 6 cell phones are randomly selected from this shipment. Find the probability that a randomly selected set of 6 cell phones consists of 2 iPhones and 4 Android phones.

Solution

There are $8C2$ ways of selecting 2 out of the 8 iPhones.

and $7C4$ ways of selecting 4 out of the 7 Android phones

But altogether there are $15C6$ ways of selecting 6 out of 15 cell phones.

Therefore we have

$$P(2 \text{ iPhones and } 4 \text{ Android phones}) = \frac{8C2 \times 7C4}{15C6} = \frac{(28)(35)}{5005} = \frac{980}{5005} = 0.1958$$

✓ Example 3.3.7

One afternoon, a bagel store still has 53 bagels remaining: 20 plain, 15 poppyseed, and 18 sesame seed bagels. Suppose that the store owner packages up a bag of 9 bagels to bring home for tomorrow's breakfast, and selects the bagels randomly. Find the probability that the bag contains 4 plain, 3 poppyseed, and 2 sesame seed.

Solution

There are $20C4$ ways of selecting 4 out of the 20 plain bagels,

and $15C3$ ways of selecting 3 out of the 15 poppyseed bagels,

and $18C2$ ways of selecting 2 out of the 18 sesame seed bagels.

But altogether there are $53C9$ ways of selecting 9 out of the 53 bagels.

$$\begin{aligned} P(4 \text{ plain, } 3 \text{ poppyseed, and } 2 \text{ sesame seed}) &= \frac{20C4 \times 15C3 \times 18C2}{53C9} \\ &= \frac{(4845)(455)(153)}{4431613550} \\ &= 0.761 \end{aligned} \tag{3.3.1}$$

We end the section by solving a famous problem called the **Birthday Problem**.

✓ Example 3.3.8: Birthday Problem

If there are 25 people in a room, what is the probability that at least two people have the same birthday?

Solution

Let event E represent that at least two people have the same birthday.

We first find the probability that no two people have the same birthday.

We analyze as follows.

Suppose there are 365 days to every year. According to the multiplication axiom, there are 365^{25} possible birthdays for 25 people. Therefore, the sample space has 365^{25} elements. We are interested in the probability that no two people have the same birthday. There are 365 possible choices for the first person and since the second person must have a different birthday, there are 364 choices for the second, 363 for the third, and so on. Therefore,

$$P(\text{No two have the same birthday}) = \frac{365 \cdot 364 \cdot 363 \cdots 341}{365^{25}} = \frac{365P_{25}}{365^{25}}$$

Since $P(\text{at least two people have the same birthday}) = 1 - P(\text{No two have the same birthday})$,

$$P \text{ at least two people have the same birthday} = 1 - \frac{365P_{25}}{365^{25}} = .5687$$

Exercises

SECTION 8.3 PROBLEM SET: PROBABILITIES USING TREE DIAGRAMS AND COMBINATIONS

Two apples are chosen from a basket containing five red and three yellow apples.

Draw a tree diagram below, and find the following probabilities.

1) P(both red)	2) P(one red, one yellow)
3) P(both yellow)	4) P(First red and second yellow)

A basket contains six red and four blue marbles. Three marbles are drawn at random.

Find the following probabilities using the method shown in Example 8.3.2. Do not use combinations.

5) P(All three red)	6) P(two red, one blue)
7) P(one red, two blue)	8) P(first red, second blue, third red)

Three marbles are drawn from a jar containing five red, four white, and three blue marbles.

Find the following probabilities using combinations.

9) P(all three red)	10) P(two white and 1 blue)
11) P(none white)	12) P(at least one red)

A committee of four is selected from a total of 4 freshmen, 5 sophomores, and 6 juniors. Find the probabilities for the following events.

13) At least three freshmen.	14) No sophomores.
15) All four of the same class.	16) Not all four from the same class.
17) Exactly three of the same class.	18) More juniors than freshmen and sophomores combined.

Five cards are drawn from a deck. Find the probabilities for the following events.

19) Two hearts, two spades, and one club.	20) A flush of any suit (<i>all cards of a single suit</i>).
21) A full house of nines and tens (<i>3 nines and 2 tens</i>).	22) Any full house.
23) A pair of nines and a pair of tens (<i>and the fifth card is not a nine or ten</i>).	24) Any two pairs (<i>two cards of one value, two more cards of another value, and the fifth card does not have the same value as either pair</i>).

Jorge has 6 rock songs, 7 rap songs and 4 country songs that he likes to listen to while he exercises. He randomly selects six (6) of these songs to create a playlist to listen to today while he exercises.

Find the following probabilities:

25) P(playlist has 2 songs of each type)	26) P(playlist has no country songs)
27) P(playlist has 3 rock, 2 rap, and 1 country song)	28) P(playlist has 3 or 4 rock songs and the rest are rap songs)

A project is staffed 12 people: 5 engineers, 4 salespeople, and 3 customer service representatives. A committee of 5 people is selected to make a presentation to senior management.

Find the probabilities of the following events.

29) The committee has 2 engineers, 2 salespeople, and 1 customer service representative.	30) The committee contains 3 engineer and 2 salespeople.
31) The committee has no engineers.	32) The committee has all salespeople.

Do the following birthday problems.

33) If there are 5 people in a room, what is the probability that no two have the same birthday?	34) If there are 5 people in a room, find the probability that at least 2 have the same birthday.
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3.4: Conditional Probability and Independent Events

In this section we introduce conditional probability and the related idea of independent events:

- [2.4.1: Conditional Probability](#)
- [2.4.2: Independent Events](#)

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3.4.1: Conditional Probability

Learning Objectives

In this section, you will learn to:

1. recognize situations involving conditional probability
2. calculate conditional probabilities

Suppose a friend asks you the probability that it will snow today.

If you are in Boston, Massachusetts in the winter, the probability of snow today might be quite substantial. If you are in Cupertino, California in summer, the probability of snow today is very tiny, this probability is pretty much 0.

Let:

- A = the event that it will snow today
- B = the event that today you are in Boston in wintertime
- C = the event that today you are in Cupertino in summertime

Because the probability of snow is affected by the location and time of year, we can't just write $P(A)$ for the probability of snow. We need to indicate the other information we know -location and time of year. We need to use **conditional probability**.

The event we are interested in is event A for snow. The other event is called the condition, representing location and time of year in this case.

We represent conditional probability using a vertical line $|$ that means “if”, or “given that”, or “if we know that”. The event of interest appears on the left of the $|$. The condition appears on the right side of the $|$.

The probability it will snow given that (if) you are in Boston in the winter is represented by $P(A|B)$. In this case, the condition is B .

The probability that it will snow given that (if) you are in Cupertino in the summer is represented by $P(A|C)$. In this case, the condition is C .

Now, let's examine a situation where we can calculate some probabilities.

Suppose you and a friend play a game that involves choosing a single card from a well-shuffled deck. Your friend deals you one card, face down, from the deck and offers you the following deal: If the card is a king, he will pay you \$5, otherwise, you pay him \$1. Should you play the game?

You reason in the following manner. Since there are four kings in the deck, the probability of obtaining a king is $4/52$ or $1/13$. So, probability of not obtaining a king is $12/13$. This implies that the ratio of your winning to losing is 1 to 12, while the payoff ratio is only \$1 to \$5. Therefore, you determine that you should not play.

But consider the following scenario. While your friend was dealing the card, you happened to get a glance of it and noticed that the card was a face card. Should you, now, play the game?

Since there are 12 face cards in the deck, the total elements in the sample space are no longer 52, but just 12. This means the chance of obtaining a king is $4/12$ or $1/3$. So your chance of winning is $1/3$ and of losing $2/3$. This makes your winning to losing ratio 1 to 2 which fares much better with the payoff ratio of \$1 to \$5. This time, you determine that you should play.

In the second part of the above example, we were finding the probability of obtaining a king knowing that a face card had shown. This is an example of **conditional probability**. Whenever we are finding the probability of an event E under the condition that another event F has happened, we are finding conditional probability.

The symbol $P(E|F)$ denotes the problem of finding the probability of E given that F has occurred. We read $P(E|F)$ as "the probability of E , given F ."

✓ Example 3.4.1.1

A family has three children. Find the conditional probability of having two boys and a girl given that the first born is a boy.

Solution

Let event E be that the family has two boys and a girl, and F that the first born is a boy.

First, we the sample space for a family of three children as follows.

$$S = \{BBB, BBG, BGB, BGG, GBB, GBG, GGB, GGG\}$$

Since we know the first born is a boy, our possibilities narrow down to four outcomes: BBB, BBG, BGB, and BGG.

Among the four, BBG and BGB represent two boys and a girl.

Therefore, $P(E|F) = 2/4$ or $1/2$.

✓ Example 3.4.1.2

One six sided die is rolled once.

- Find the probability that the result is even.
- Find the probability that the result is even given that the result is greater than three.

Solution

The sample space is $S = 1, 2, 3, 4, 5, 6$

Let event E be that the result is even and T be that the result is greater than 3.

- $P(E) = 3/6$ because $E = 2, 4, 6$
- Because $T = 4, 5, 6$, we know that 1, 2, 3 cannot occur; only outcomes 4, 5, 6 are possible. Therefore of the values in E , only 4, 6 are possible.

Therefore, $P(E|T) = 2/3$

✓ Example 3.4.1.3

A fair coin is tossed twice.

- Find the probability that the result is is two heads.
- Find the probability that the result is two heads given that at least one head is obtained.

Solution

The sample space is $S = HH, HT, TH, TT$

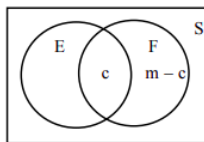
Let event E be that the two heads are obtained and F be at least one head is obtained

- $P(E) = 1/4$ because $E = HH$ and the sample space S has 4 outcomes.
- $F = HH, HT, TH$. Since at least one head was obtained, TT did not occur. We are interested in the probability event $E = HH$ out of the 3 outcomes in the reduced sample space F .

Therefore, $P(E|F) = 1/3$

Let us now develop a formula for the conditional probability $P(E|F)$.

Suppose an experiment consists of n equally likely events. Further suppose that there are m elements in F , and c elements in $E \cap F$, as shown in the following Venn diagram.



If the event F has occurred, the set of all possible outcomes is no longer the entire sample space, but instead, the subset F . Therefore, we only look at the set F and at nothing outside of F . Since F has m elements, the denominator in the calculation of $P(E|F)$ is m . We may think that the numerator for our conditional probability is the number of elements in E . But clearly we cannot consider the elements of E that are not in F . We can only count the elements of E that are in F , that is, the elements in $E \cap F$. Therefore,

$$P(E|F) = \frac{c}{m}$$

Dividing both the numerator and the denominator by n , we get

$$P(E|F) = \frac{c/n}{m/n}$$

But $c/n = P(E \cap F)$, and $m/n = P(F)$.

Substituting, we derive the following formula for $P(E|F)$.

Conditional Probability Rule

For two events E and F , the probability of "E Given F" is

$$P(E|F) = \frac{P(E \cap F)}{P(F)}$$

✓ Example 3.4.1.4

A single die is rolled. Use the above formula to find the conditional probability of obtaining an even number given that a number greater than three has shown.

Solution

Let E be the event that an even number shows, and F be the event that a number greater than three shows. We want $P(E|F)$.

$E = 2, 4, 6$ and $F = 4, 5, 6$. Which implies, $E \cap F = 4, 6$

Therefore, $P(F) = 3/6$, and $P(E \cap F) = 2/6$

$$P(E|F) = \frac{P(E \cap F)}{P(F)} = \frac{2/6}{3/6} = \frac{2}{3}.$$

✓ Example 3.4.1.5

The following table shows the distribution by gender of students at a community college who take public transportation and the ones who drive to school.

	Male(M)	Female(F)	Total
Public Transportation(T)	8	13	21
Drive(D)	39	40	79
Total	47	53	100

The events M , F , T , and D are self explanatory. Find the following probabilities.

- a. $P(D|M)$
- b. $P(F|D)$
- c. $P(M|T)$

Solution 1

Conditional probabilities can often be found directly from a contingency table. If the condition corresponds to only one row or only one column in the table, then you can ignore the rest of the table and read the conditional probability directly from the row or column indicated by the condition.

- a. The condition is event M; we can look at only the “Male” column of the table and ignore the rest of the table:

$$P(D|M) = \frac{39}{47}.$$

- b. The condition is event D; we can look at only the “Drive” row of the table and ignore the rest of the table: $P(F|D) = \frac{40}{79}.$

- c. The condition is event T; we can look at only the “Public Transportation” row of the table and ignore the rest of the table:

$$P(M|T) = \frac{8}{21}.$$

Solution 2

We use the conditional probability formula $P(E|F) = \frac{P(E \cap F)}{P(F)}.$

- a.
$$P(D|M) = \frac{P(D \cap M)}{P(M)} = \frac{39/100}{47/100} = \frac{39}{47}.$$
- b.
$$P(F|D) = \frac{P(F \cap D)}{P(D)} = \frac{40/100}{79/100} = \frac{40}{79}.$$
- c.
$$P(M|T) = \frac{P(M \cap T)}{P(T)} = \frac{8/100}{21/100} = \frac{8}{21}.$$

✓ Example 3.4.1.6

Given $P(E) = .5$, $P(F) = .7$, and $P(E \cap F) = .3$. Find the following:

- a. $P(E|F)$
- b. $P(F|E)$

Solution

We use the conditional probability formula.

- a. $P(E|F) = \frac{P(E \cap F)}{P(F)} = \frac{.3}{.7} = \frac{3}{7}$
- b. $P(F|E) = \frac{P(E \cap F)}{P(E)} = .3/.5 = 3/5$

✓ Example 3.4.1.7

E and F are mutually exclusive events such that $P(E) = .4$, $P(F) = .9$. Find $P(E|F)$.

Solution

E and F are mutually exclusive, so $P(E \cap F) = 0$.

Therefore $P(E|F) = \frac{P(E \cap F)}{P(F)} = \frac{0}{.9} = 0.$

✓ Example 3.4.1.8

Given $P(F|E) = .5$, and $P(E \cap F) = .3$. Find $P(E)$.

Solution

Using the conditional probability formula $P(E|F) = \frac{P(E \cap F)}{P(F)}$, we get

$$P(F|E) = \frac{P(E \cap F)}{P(E)}$$

Substituting and solving:

$$.5 = \frac{.3}{P(E)} \quad \text{or} \quad P(E) = 3/5$$

✓ Example 3.4.1.9

In a family of three children, find the conditional probability of having two boys and a girl, given that the family has at least two boys.

Solution

Let event E be that the family has two boys and a girl, and let F be the probability that the family has at least two boys. We want $P(E|F)$.

We list the sample space along with the events E and F.

$$\begin{aligned} S &= \{BBB, BBG, BGB, BGG, GBB, GBG, GGB, GGG\} \\ E &= \{BBG, BGB, GBB\} \text{ and } F = \{BBB, BBG, BGB, GBB\} \\ E \cap F &= \{BBG, BGB, GBB\} \end{aligned}$$

Therefore, $P(F) = 4/8$, and $P(E \cap F) = 3/8$, and

$$P(E|F) = \frac{P(E \cap F)}{P(F)} = \frac{3/8}{4/8} = \frac{3}{4}.$$

✓ Example 3.4.1.10

At a community college 65% of the students subscribe to Amazon Prime, 50% subscribe to Netflix, and 20% subscribe to both. If a student is chosen at random, find the following probabilities:

- the student subscribes to Amazon Prime given that he subscribes to Netflix
- the student subscribes to Netflix given that he subscribes to Amazon Prime

Solution

Let A be the event that the student subscribes to Amazon Prime, and N be the event that the student subscribes to Netflix.

First identify the probabilities and events given in the problem.

$$P(\text{student subscribes to Amazon Prime}) = P(A) = 0.65$$

$$P(\text{student subscribes to Netflix}) = P(N) = 0.50$$

$$P(\text{student subscribes to both Amazon Prime and Netflix}) = P(A \cap N) = 0.20$$

Then use the conditional probability rule:

$$\text{a. } P(A|N) = \frac{P(A \cap N)}{P(N)} = \frac{.20}{.50} = \frac{2}{5}$$

$$\text{b. } P(N|A) = \frac{P(A \cap N)}{P(A)} = \frac{.20}{.65} = \frac{4}{13}$$

Exercises

SECTION 8.4 PROBLEM SET: CONDITIONAL PROBABILITY

Questions 1 - 4: Do these problems using the conditional probability formula: $P(A|B) = \frac{P(A \cap B)}{P(B)}$.

1. A card is drawn from a deck. Find the conditional probability of $P(\text{a queen} \text{a face card})$.	2. A card is drawn from a deck. Find the conditional probability of $P(\text{a queen} \text{a club})$.
3. A die is rolled. Find the conditional probability that it shows a three if it is known that an odd number has shown.	4. If $P(A) = .3$, $P(B) = .4$, $P(A \text{ and } B) = .12$, find: a. $P(A B)$ b. $P(B A)$

Questions 5 - 8 refer to the following: The table shows the distribution of Democratic and Republican U.S. Senators by gender in the 114th Congress as of January 2015.

	MALE(M)	FEMALE(F)	TOTAL
DEMOCRATS (D)	30	14	44
REPUBLICANS(R)	48	6	54
OTHER (T)	2	0	2
TOTALS	80	20	100

Use this table to determine the following probabilities:

5. $P(M D)$	6. $P(D M)$
7. $P(F R)$	8. $P(R F)$

Do the following conditional probability problems.

9. At a college, 20% of the students take Finite Math, 30% take History, and 5% take both Finite Math and History. If a student is chosen at random, find the following conditional probabilities. a. He is taking Finite Math given that he is taking History. b. He is taking History assuming that he is taking Finite Math.	10. At a college, 60% of the students pass Accounting, 70% pass English, and 30% pass both of these courses. If a student is selected at random, find the following conditional probabilities. a. He passes Accounting given that he passed English. b. He passes English assuming that he passed Accounting.
11. If $P(F) = .4$, $P(E F) = .3$, find $P(E \text{ and } F)$.	12. $P(E) = .3$, $P(F) = .3$; E and F are mutually exclusive. Find $P(E F)$.
13. If $P(E) = .6$, $P(E \text{ and } F) = .24$, find $P(F E)$.	14. If $P(E \text{ and } F) = .04$, $P(E F) = .1$, find $P(F)$.

At a college, 72% of courses have final exams and 46% of courses require research papers. 32% of courses have both a research paper and a final exam. Let F be the event that a course has a final exam and R be the event that a course requires a research paper.

15. Find the probability that a course has a final exam given that it has a research paper.	16. Find the probability that a course has a research paper if it has a final exam.
---	---

SECTION 8.4 PROBLEM SET: CONDITIONAL PROBABILITY

Consider a family of three children. Find the following probabilities.

17. $P(\text{two boys} \text{first born is a boy})$	18. $P(\text{all girls} \text{at least one girl is born})$
19. $P(\text{children of both sexes} \text{first born is a boy})$	20. $P(\text{all boys} \text{there are children of both sexes})$

Questions 21 - 26 refer to the following:

The table shows highest attained educational status for a sample of US residents age 25 or over:

--

	(D) Did not Complete High School	(H) High School Graduate	(C) Some College	(A) Associate Degree	(B) Bachelor Degree	(G) Graduate Degree	TOTAL
25-44 (R)	95	228	143	81	188	61	796
45-64 (S)	83	256	136	80	150	67	772
65+ (T)	96	191	84	36	80	41	528
Total	274	675	363	197	418	169	2096

Use this table to determine the following probabilities:

21. $P(C T)$	22. $P(S A)$	23. $P(C \text{ and } T)$
24. $P(R B)$	25. $P(B R)$	26. $P(G S)$

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3.4.2: Independent Events

Learning Objectives

In this section, you will:

1. Define independent events
2. Identify whether two events are independent or dependent

In the last section, we considered conditional probabilities. In some examples, the probability of an event changed when additional information was provided. This is not always the case. The additional information may or may not alter the probability of the event.

In Example 3.4.2.1 we revisit the discussion at the beginning of the previous section and then contrast that with Example 3.4.2.2

✓ Example 3.4.2.1

A card is drawn from a deck. Find the following probabilities.

- a. The card is a king.
- b. The card is a king given that the card is a face card.

Solution

a. Clearly, $P(\text{The card is a king}) = 4/52 = 1/13$.

b. To find $P(\text{The card is a king} \mid \text{The card is a face card})$, we reason as follows:

There are 12 face cards in a deck of cards. There are 4 kings in a deck of cards.

$$P(\text{The card is a king} \mid \text{The card is a face card}) = 4/12 = 1/3.$$

The reader should observe that in the above example,

$$P(\text{The card is a king} \mid \text{The card is a face card}) \neq P(\text{The card is a king})$$

In other words, the additional information, knowing that the card selected is a face card changed the probability of obtaining a king.

✓ Example 3.4.2.2

A card is drawn from a deck. Find the following probabilities.

- a. The card is a king.
- b. The card is a king given that a red card has shown.

Solution

a. Clearly, $P(\text{The card is a king}) = 4/52 = 1/13$.

b. To find $P(\text{The card is a king} \mid \text{A red card has shown})$, we reason as follows:

Since a red card has shown, there are only twenty six possibilities. Of the 26 red cards, there are two kings. Therefore,

$$P(\text{The card is a king} \mid \text{A red card has shown}) = 2/26 = 1/13.$$

The reader should observe that in the above example,

$$P(\text{The card is a king} \mid \text{A red card has shown}) = P(\text{The card is a king})$$

In other words, the additional information, a red card has shown, did not affect the probability of obtaining a king.

Whenever the probability of an event E is not affected by the occurrence of another event F , and vice versa, we say that the two events E and F are **independent**. This leads to the following definition.

Definition: Independent

Two Events E and F are **independent** if and only if at least one of the following two conditions is true.

1. $P(E|F) = P(E)$ or
2. $P(F|E) = P(F)$

If the events are not independent, then they are dependent.

If one of these conditions is true, then both are true.

We can use the definition of independence to determine if two events are independent.

We can use that definition to develop another way to test whether two events are independent.

Recall the conditional probability formula:

$$P(E|F) = \frac{P(E \cap F)}{P(F)}$$

Multiplying both sides by $P(F)$, we get

$$P(E \cap F) = P(E|F)P(F)$$

Now if the two events are independent, then by definition

$$P(E|F) = P(E)$$

Substituting, $P(E \cap F) = P(E)P(F)$

We state it formally as follows.

Test For Independence

Two events E and F are independent if and only if

$$P(E \cap F) = P(E)P(F)$$

In the Examples 3.4.2.3 and 3.4.2.4 we'll examine how to check for independence using both methods:

- Examine the probability of intersection of events to check whether $P(E \cap F) = P(E)P(F)$
- Examine conditional probabilities to check whether $P(E|F) = P(E)$ or $P(F|E) = P(F)$

We need to use only **one** of these methods. Both methods, if used properly, will always give results that are consistent with each other.

Use the method that seems easier based on the information given in the problem.

✓ Example 3.4.2.3

The table below shows the distribution of color-blind people by gender.

	Male(M)	Female(F)	Total
Color-Blind(C)	6	1	7
Not Color-Blind(N)	46	47	93
Total	52	48	100

where M represents male, F represents female, C represents color-blind, and N represents not color-blind. Are the events color-blind and male independent?

Solution 1: According to the test for independence, C and M are independent if and only if $P(C \cap M) = P(C)P(M)$.

From the table: $P(C) = 7/100$, $P(M) = 52/100$ and $P(C \cap M) = 6/100$

So $P(C)P(M) = (7/100)(52/100) = .0364$

which is **not** equal to $P(C \cap M) = 6/100 = .06$

Therefore, the two events are not independent. We may say they are dependent.

Solution 2: C and M are independent if and only if $P(C|M) = P(C)$.

From the total column $P(C) = 7/100 = 0.07$

From the male column $P(C|M) = 6/52 = 0.1154$

Therefore $P(C|M) \neq P(C)$, indicating that the two events are not independent.

✓ Example 3.4.2.4

In a city with two airports, 100 flights were surveyed. 20 of those flights departed late.

- 45 flights in the survey departed from airport A; 9 of those flights departed late.
- 55 flights in the survey departed from airport B; 11 flights departed late.

Are the events "depart from airport A" and "departed late" independent?

Solution 1

Let A be the event that a flight departs from airport A, and L the event that a flight departs late. We have

$$P(A \cap L) = 9/100, P(A) = 45/100 \text{ and } P(L) = 20/100$$

In order for two events to be independent, we must have $P(A \cap L) = P(A)P(L)$

Since $P(A \cap L) = 9/100 = 0.09$

and $P(A)P(L) = (45/100)(20/100) = 900/10000 = 0.09$

the two events "departing from airport A" and "departing late" are independent.

Solution 2

The definition of independent events states that two events are independent if $P(E|F) = P(E)$.

In this problem we are given that

$$P(L|A) = 9/45 = 0.2 \text{ and } P(L) = 20/100 = 0.2$$

$P(L|A) = P(L)$, so events "departing from airport A" and "departing late" are independent.

✓ Example 3.4.2.5

A coin is tossed three times, and the events E , F and G are defined as follows:

E : The coin shows a head on the first toss.

F : At least two heads appear.

G : Heads appear in two successive tosses.

Determine whether the following events are independent.

- E and F
- F and G
- E and G

Solution

We list the sample space, the events, their intersections and the probabilities.

$$\begin{aligned}
 S &= \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\} \\
 E &= \{HHH, HHT, HTH, HTT\}, & P(E) &= 4/8 \text{ or } 1/2 \\
 F &= \{HHH, HHT, HTH, THH\}, & P(F) &= 4/8 \text{ or } 1/2 \\
 G &= \{HHT, THH\}, & P(G) &= 2/8 \text{ or } 1/4 \\
 E \cap F &= \{HHH, HHT, HTH\}, & P(E \cap F) &= 3/8 \\
 F \cap G &= \{HHT, THH\}, & P(F \cap G) &= 2/8 \text{ or } 1/4 \\
 E \cap G &= \{HHT\} & P(E \cap G) &= 1/8
 \end{aligned}$$

a. E and F will be independent if and only if $P(E \cap F) = P(E)P(F)$

$$P(E \cap F) = 3/8 \text{ and } P(E)P(F) = 1/2 \cdot 1/2 = 1/4.$$

Since $3/8 \neq 1/4$, we have $P(E \cap F) \neq P(E)P(F)$.

Events E and F are not independent.

b. F and G will be independent if and only if $P(F \cap G) = P(F)P(G)$.

$$P(F \cap G) = 1/4 \text{ and } P(F)P(G) = 1/2 \cdot 1/4 = 1/8.$$

Since $1/4 \neq 1/8$, we have $P(F \cap G) \neq P(F)P(G)$.

Events F and G are not independent.

c. E and G will be independent if $P(E \cap G) = P(E)P(G)$

$$P(E \cap G) = 1/8 \text{ and } P(E)P(G) = 1/2 \cdot 1/4 = 1/8$$

Events E and G are independent events because $P(E \cap G) = P(E)P(G)$

✓ Example 3.4.2.6

The probability that Jaime will visit his aunt in Baltimore this year is .30, and the probability that he will go river rafting on the Colorado river is .50. If the two events are independent, what is the probability that Jaime will do both?

Solution

Let A be the event that Jaime will visit his aunt this year, and R be the event that he will go river rafting.

We are given $P(A) = .30$ and $P(R) = .50$, and we want to find $P(A \cap R)$.

Since we are told that the events A and R are independent,

$$P(A \cap R) = P(A)P(R) = (.30)(.50) = .15$$

✓ Example 3.4.2.7

Given $P(B|A) = .4$. If A and B are independent, find $P(B)$.

Solution

If A and B are independent, then by definition $P(B|A) = P(B)$

Therefore, $P(B) = .4$

✓ Example 3.4.2.8

Given $P(A) = .7$, $P(B|A) = .5$. Find $P(A \cap B)$.

Solution 1

By definition $P(B|A) = \frac{P(A \cap B)}{P(A)}$

Substituting, we have

$$.5 = \frac{P(A \cap B)}{.7}$$

Therefore, $P(A \cap B) = .35$

Solution 2

Again, start with $P(B|A) = \frac{P(A \cap B)}{P(A)}$

Multiplying both sides by $P(A)$ gives

$$P(A \cap B) = P(B|A)P(A) = (.5)(.7) = .35$$

Both solutions to Example 3.4.2.8 are actually the same, except that in Solution 2 we delayed substituting the values into the equation until after we solved the equation for $P(A \cap B)$. That gives the following result:

Multiplication Rule for events that are NOT independent

If events E and F are not independent

$$P(E \cap F) = P(E|F)P(F) \quad \text{and} \quad P(E \cap F) = P(F|E)P(E)$$

✓ Example 3.4.2.9

Given $P(A) = .5$, $P(A \cup B) = .7$, if A and B are independent, find $P(B)$.

Solution

The addition rule states that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Since A and B are independent, $P(A \cap B) = P(A)P(B)$

We substitute for $P(A \cap B)$ in the addition formula and get

$$P(A \cup B) = P(A) + P(B) - P(A)P(B)$$

By letting $P(B) = x$, and substituting values, we get

$$.7 = .5 + x - .5x$$

$$.7 = .5 + .5x$$

$$.2 = .5x$$

$$.4 = x$$

Therefore, $P(B) = .4$

Exercises

SECTION 8.5 PROBLEM SET: INDEPENDENT EVENTS

The distribution of the number of fiction and non-fiction books checked out at a city's main library and at a smaller branch on a given day is as follows.

	MAIN (M)	BRANCH (B)	TOTAL
FICTION (F)	300	100	400
NON-FICTION (N)	150	50	200
TOTALS	450	150	600

Use this table to determine the following probabilities:

1. $P(F)$	2. $P(M F)$
3. $P(N B)$	4. Is the fact that a person checks out a fiction book independent of the main library? Use probabilities to justify your conclusion.

For a two-child family, let the events E , F , and G be as follows.

E : The family has at least one boy

F : The family has children of both sexes

G : The family's first born is a boy

5. Find the following. a. $P(E)$ b. $P(F)$ c. $P(E \cap F)$ d. Are E and F independent? Use probabilities to justify your conclusion.	6. Find the following. a. $P(F)$ b. $P(G)$ c. $P(F \cap G)$ d. Are F and G independent? Use probabilities to justify your conclusion.
---	---

Do the following problems involving independence.

7. If $P(E) = .6$, $P(F) = .2$, and E and F are independent, find $P(E \text{ and } F)$.	8. If $P(E) = .6$, $P(F) = .2$, and E and F are independent, find $P(E \text{ or } F)$.
9. If $P(E) = .9$, $P(F E) = .36$, and E and F are independent, find $P(F)$.	10. If $P(E) = .6$, $P(E \text{ or } F) = .8$, and E and F are independent, find $P(F)$.
11. In a survey of 100 people, 40 were casual drinkers, and 60 did not drink. Of the ones who drank, 6 had minor headaches. Of the non-drinkers, 9 had minor headaches. Are the events "drinkers" and "had headaches" independent?	12. It is known that 80% of the people wear seat belts, and 5% of the people quit smoking last year. If 4% of the people who wear seat belts quit smoking, are the events, wearing a seat belt and quitting smoking, independent?
13. John's probability of passing statistics is 40%, and Linda's probability of passing the same course is 70%. If the two events are independent, find the following probabilities. a. $P(\text{both of them will pass statistics})$ b. $P(\text{at least one of them will pass statistics})$	14. Jane is flying home for the Christmas holidays. She has to change planes twice. There is an 80% chance that she will make the first connection, and a 90% chance that she will make the second connection. If the two events are independent, find the probabilities: a. $P(\text{Jane will make both connections})$ b. $P(\text{Jane will make at least one connection})$

For a three-child family, let the events E , F , and G be as follows.

E : The family has at least one boy

F : The family has children of both sexes

G : The family's first born is a boy

15. Find the following. a. $P(E)$ b. $P(F)$ c. $P(E \cap F)$ d. Are E and F independent?	16. Find the following. a. $P(F)$ b. $P(G)$ c. $P(F \cap G)$ d. Are F and G independent?
--	--

SECTION 8.5 PROBLEM SET: INDEPENDENT EVENTS

17. $P(K|D) = 0.7$, $P(D) = 0.25$ and $P(K) = 0.7$
- Are events K and D independent? Use probabilities to justify your conclusion.
 - Find $P(K \cap D)$

19. At a college:
- 54% of students are female
 - 25% of students are majoring in engineering.
 - 15% of female students are majoring in engineering.
- Event E = student is majoring in engineering
Event F = student is female
- Are events E and F independent? Use probabilities to justify your conclusion.
 - Find $P(E \cap F)$

18. $P(R|S) = 0.4$, $P(S) = 0.2$ and $P(R) = 0.3$
- Are events R and S independent? Use probabilities to justify your conclusion.
 - Find $P(R \cap S)$

20. At a college:
- 54% of all students are female
 - 60% of all students receive financial aid.
 - 60% of female students receive financial aid.
- Event A = student receives financial aid
Event F = student is female
- Are events A and F independent? Use probabilities to justify your conclusion.
 - Find $P(A \cap F)$

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3.5: Bayes' Formula

Learning Objectives

In this section, you will learn to:

1. Find probabilities using Bayes' formula
2. Use a probability tree to find and represent values needed when using Bayes' formula.

In this section, we will develop and use Bayes' Formula to solve an important type of probability problem. Bayes' formula is a method of calculating the conditional probability $P(F|E)$ from $P(E|F)$. The ideas involved here are not new, and most of these problems can be solved using a tree diagram. However, Bayes' formula does provide us with a tool with which we can solve these problems without a tree diagram.

We begin with an example.

✓ Example 3.5.1

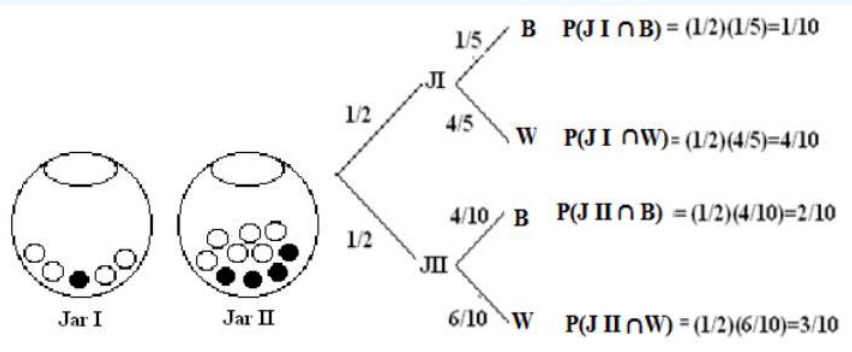
Suppose you are given two jars. Jar I contains one black and 4 white marbles, and Jar II contains 4 black and 6 white marbles. If a jar is selected at random and a marble is chosen,

- a. What is the probability that the marble chosen is a black marble?
- b. If the chosen marble is black, what is the probability that it came from Jar I?
- c. If the chosen marble is black, what is the probability that it came from Jar II?

Solution

Let J_I be the event that Jar I is chosen, J_{II} be the event that Jar II is chosen, B be the event that a black marble is chosen and W the event that a white marble is chosen.

We illustrate using a tree diagram.



- a. The probability that a black marble is chosen is $P(B) = 1/10 + 2/10 = 3/10$.
- b. To find $P(J_I|B)$, we use the definition of conditional probability, and we get

$$P(J_I|B) = \frac{P(J_I \cap B)}{P(B)} = \frac{1/10}{3/10} = \frac{1}{3}$$

- c. Similarly, $P(J_{II}|B) = \frac{P(J_{II} \cap B)}{P(B)} = \frac{2/10}{3/10} = \frac{2}{3}$

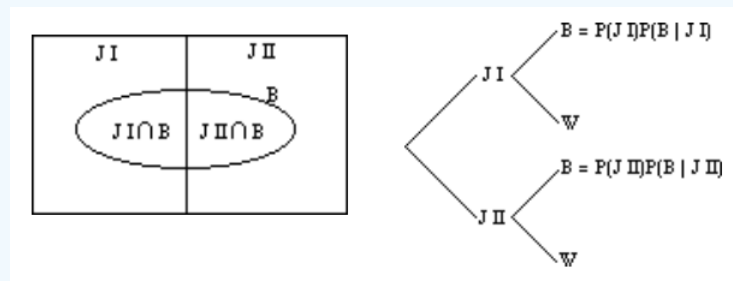
In parts b and c, the reader should note that the denominator is the sum of all probabilities of all branches of the tree that produce a black marble, while the numerator is the branch that is associated with the particular jar in question.

We will soon discover that this is a statement of Bayes' formula .

Let us first visualize the problem.

We are given a sample space S and two mutually exclusive events J_I and J_{II} . That is, the two events, J_I and J_{II} , divide the sample space into two parts such that $J_I \cup J_{II} = S$. Furthermore, we are given an event B that has elements in both J_I and

JII , as shown in the Venn diagram below.



From the Venn diagram, we can see that $B = (B \cap JI) \cup (B \cap JII)$ Therefore:

$$P(B) = P(B \cap JI) + P(B \cap JII) \quad (3.5.1)$$

But the product rule in chapter 7 gives us

$$P(B \cap JI) = P(JI) \cdot P(B|JI) \quad \text{and} \quad P(B \cap JII) = P(JII) \cdot P(B|JII)$$

Substituting in 3.5.1, we get

$$P(B) = P(JI) \cdot P(B|JI) + P(JII) \cdot P(B|JII)$$

The conditional probability formula gives us

$$P(JI|B) = \frac{P(JI \cap B)}{P(B)}$$

$$\text{Therefore, } P(JI|B) = \frac{P(JI) \cdot P(B|JI)}{P(B)}$$

or

$$P(JI|B) = \frac{P(JI) \cdot P(B|JI)}{P(JI) \cdot P(B|JI) + P(JII) \cdot P(B|JII)}$$

The last statement is Bayes' Formula for the case where the sample space is divided into two partitions.

The following is the generalization of Bayes' formula for n partitions.

Bayes' Formula for n partitions

Let S be a sample space that is divided into n partitions, $A_1, A_2, \dots, A_n$. If E is any event in S , then

$$P(A_i|E) = \frac{P(A_i)P(E|A_i)}{P(A_1)P(E|A_1) + P(A_2)P(E|A_2) + \dots + P(A_n)P(E|A_n)}$$

We begin with the following example.

✓ Example 3.5.2

A department store buys 50% of its appliances from Manufacturer A, 30% from Manufacturer B, and 20% from Manufacturer C. It is estimated that 6% of Manufacturer A's appliances, 5% of Manufacturer B's appliances, and 4% of Manufacturer C's appliances need repair before the warranty expires. An appliance is chosen at random. If the appliance chosen needed repair before the warranty expired, what is the probability that the appliance was manufactured by Manufacturer A? Manufacturer B? Manufacturer C?

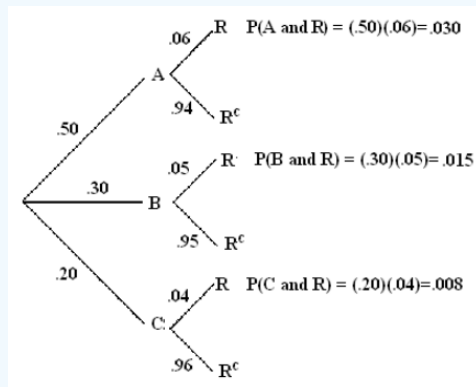
Solution

Let A, B and C be the events that the appliance is manufactured by Manufacturer A, Manufacturer B, and Manufacturer C, respectively. Further, suppose that the event R denotes that the appliance needs repair before the warranty expires.

We need to find $P(A | R)$, $P(B | R)$ and $P(C | R)$.

We will do this problem both by using a tree diagram and by using Bayes' formula.

We draw a tree diagram.



The probability $P(A | R)$, for example, is a fraction whose denominator is the sum of all probabilities of all branches of the tree that result in an appliance that needs repair before the warranty expires, and the numerator is the branch that is associated with Manufacturer A. $P(B | R)$ and $P(C | R)$ are found in the same way.

$$P(A|R) = \frac{.030}{(.030)+(.015)+(.008)} = \frac{.030}{.053} = .566$$

$$P(B|R) = \frac{.015}{.053} = .283 \text{ and } P(C|R) = \frac{.008}{.053} = .151$$

Alternatively, using Bayes' formula,

$$\begin{aligned} P(A|R) &= \frac{P(A)P(R|A)}{P(A)P(R|A) + P(B)P(R|B) + P(C)P(R|C)} \\ &= \frac{.030}{(.030) + (.015) + (.008)} = \frac{.030}{.053} = .566 \end{aligned}$$

$P(B | R)$ and $P(C | R)$ can be determined in the same manner.

✓ Example 3.5.3

There are five Jacy's department stores in San Jose. The distribution of number of employees by gender is given in the table below.

Store Number	Number of Employees	Percent of Women Employees
1	300	.40
2	150	.65
3	200	.60
4	250	.50
5	100	.70
	Total = 1000	

If an employee chosen at random is a woman, what is the probability that the employee works at store III?

Solution

Let $k = 1, 2, \dots, 5$ be the event that the employee worked at store k , and W be the event that the employee is a woman. Since there are a total of 1000 employees at the five stores,

$$P(1) = .30 \quad P(2) = .15 \quad P(3) = .20 \quad P(4) = .25 \quad P(5) = .10$$

Using Bayes' formula,

$$\begin{aligned} P(3|W) &= \frac{P(3)P(W|3)}{P(1)P(W|1)+P(2)P(W|2)+P(3)P(W|3)+P(4)P(W|4)+P(5)P(W|5)} \\ &= \frac{(.20)(.60)}{(.30)(.40)+(.15)(.65)+(.20)(.60)+(.25)(.50)+(.10)(.70)} \\ &= .2254 \end{aligned}$$

Exercises

SECTION 9.2 PROBLEM SET: BAYES' FORMULA

1. Jar I contains five red and three white marbles, and Jar II contains four red and two white marbles. A jar is picked at random and a marble is drawn. Draw a tree diagram below, and find the following probabilities.

- $P(\text{marble is red})$
- $P(\text{It came from Jar II} \mid \text{marble is white})$
- $P(\text{Red} \mid \text{Jar I})$

3. A city has 60% Democrats, and 40% Republicans. In the last mayoral election, 60% of the Democrats voted for their Democratic candidate while 95% of the Republicans voted for their candidate. Which party's mayor runs city hall?

5. A test for a certain disease gives a positive result 95% of the time if the person actually carries the disease. However, the test also gives a positive result 3% of the time when the individual is not carrying the disease. It is known that 10% of the population carries the disease. If a person tests positive, what is the probability that he or she has the disease?

7. A computer company buys its chips from three different manufacturers. Manufacturer I provides 60% of the chips and is known to produce 5% defective; Manufacturer II supplies 30% of the chips and makes 4% defective; while the rest are supplied by Manufacturer III with 3% defective chips. If a chip is chosen at random, find the following probabilities:

- $P(\text{the chip is defective})$
- $P(\text{chip is from Manufacturer II} \mid \text{defective})$
- $P(\text{defective} \mid \text{chip is from manufacturer III})$

2. In Mr. Symons' class, if a student does homework most days, the chance of passing the course is 90%. On the other hand, if a student does not do homework most days, the chance of passing the course is only 20%.

H = event that the student did homework

C = event that the student passed the course

Mr. Symons claims that 80% of his students do homework on a regular basis. If a student is chosen at random from Mr. Symons' class, find the following probabilities.

- $P(C)$
- $P(H|C)$
- $P(C|H)$

4. In a certain population of 48% men and 52% women, 56% of the men and 8% of the women are color-blind.

- What percent of the people are color-blind?
- If a person is found to be color-blind, what is the probability that the person is a male?

6. A person has two coins: a fair coin and a two-headed coin. A coin is selected at random, and tossed. If the coin shows a head, what is the probability that the coin is fair?

8. Lincoln Union High School District is made up of three high schools: Monterey, Fremont, and Kennedy, with an enrollment of 500, 300, and 200, respectively. On a given day, the percentage of students absent at Monterey High School is 6%, at Fremont 4%, and at Kennedy 5%. If a student is chosen at random, find the probabilities below: *Hint: Convert the enrollments into percentages.*

- $P(\text{the student is absent})$
- $P(\text{student is from Kennedy} \mid \text{student is absent})$
- $P(\text{student is absent} \mid \text{student is from Fremont})$

9. At a retail store, 20% of customers use the store's online app to assist them when shopping in the store ; 80% of store shoppers don't use the app.

Of those customers that use the online app while in the store, 50% are very satisfied with their purchases, 40% are moderately satisfied, and 10% are dissatisfied.

Of those customers that do not use the online app while in the store, 30% are very satisfied with their purchases, 50% are moderately satisfied and 20% are dissatisfied.

Indicate the events by the following:

A = shopper uses the app in the store

N = shopper does not use the app in the store

V = very satisfied with purchase

M = moderately satisfied

D = dissatisfied

a. Find $P(A \text{ and } D)$, the probability that a store customer uses the app and is dissatisfied

b. Find $P(A|D)$, the probability that a store customer uses the app if the customer is dissatisfied.

10. A medical clinic uses a pregnancy test to confirm pregnancy in patients who suspect they are pregnant. Historically data has shown that overall, 70% of the women at this clinic who are given the pregnancy test are pregnant, but 30% are not.

The test's manufacturer indicates that if a woman is pregnant, the test will be positive 92% of the time.

But if a woman is not pregnant, the test will be positive only 2% of the time and will be negative 98% of the time.

a. Find the probability that a woman at this clinic is pregnant and tests positive.

b. Find the probability that a woman at this clinic is actually pregnant given that she tests positive.

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3.6: Chapter Review Exercises

SECTION 8.6 PROBLEM SET: CHAPTER REVIEW

- Two dice are rolled. Find the probability that the sum of the dice is
 - four
 - five
- A jar contains 3 red, 4 white, and 5 blue marbles. If a marble is chosen at random, find the following probabilities:
 - $P(\text{red or blue})$
 - $P(\text{not blue})$
- A card is drawn from a standard deck. Find the following probabilities:
 - $P(\text{a jack or a king})$
 - $P(\text{a jack or a spade})$
- A basket contains 3 red and 2 yellow apples. Two apples are chosen at random. Find the following probabilities:
 - $P(\text{one red, one yellow})$
 - $P(\text{at least one red})$
- A basket contains 4 red, 3 white, and 3 blue marbles. Three marbles are chosen at random. Find the following probabilities:
 - $P(\text{two red, one white})$
 - $P(\text{first red, second white, third blue})$
 - $P(\text{at least one red})$
 - $P(\text{none red})$
- Given a family of four children. Find the following probabilities:
 - $P(\text{All boys})$
 - $P(1 \text{ boy and } 3 \text{ girls})$
- Consider a family of three children. Find the following:
 - $P(\text{children of both sexes} \mid \text{first born is a boy})$
 - $P(\text{all girls} \mid \text{children of both sexes})$
- Mrs. Rossetti is flying from San Francisco to New York. On her way to the San Francisco Airport she encounters heavy traffic and determines that there is a 20% chance that she will be late to the airport and will miss her flight. Even if she makes her flight, there is a 10% chance that she will miss her connecting flight at Chicago. What is the probability that she will make it to New York as scheduled?
- At a college, twenty percent of the students take history, thirty percent take math, and ten percent take both. What percent of the students take at least one of these two courses?
- In a T-maze, a mouse may run to the right (R) or may run to the left (L). A mouse goes up the maze three times, and events E and F are described as follows:

E: Runs to the right on the first trial F: Runs to the left two consecutive times

Determine whether the events E and F are independent.

- A college has found that 20% of its students take advanced math courses, 40% take advanced English courses and 15% take both advanced math and advanced English courses. If a student is selected at random, what is the probability that
 - he is taking English given that he is taking math?
 - he is taking math or English?
- If there are 35 students in a class, what is the probability that at least two have the same birthday?
- A student feels that her probability of passing accounting is .62, of passing mathematics is .45, and her passing accounting or mathematics is .85. Find the probability that the student passes both accounting and math.
- There are nine judges on the U. S. Supreme Court. Suppose that five are conservative and four are liberal. This year the court will act on six major cases. What is the probability that out of six cases the court will favor the conservatives in at least four?
- Five cards are drawn from a deck. Find the probability of obtaining
 - four cards of a single suit

- b. two cards of one suit, two of another suit, and one from the remaining
- c. a pair(e.g. two aces and three other cards)
- d. a straight flush(five in a row of a single suit but not a royal flush)

16. The following table shows a distribution of drink preferences by gender.

	Coke(C)	Pepsi(P)	Seven Up(S)	TOTALS
Male(M)	60	50	22	132
Female(F)	50	40	18	108
TOTALS	110	90	40	240

The events M, F, C, P and S are defined as Male, Female, Coca Cola, Pepsi, and Seven Up, respectively. Find the following:

- a. $P(F | S)$
 - b. $P(P | F)$
 - c. $P(C | M)$
 - d. $P(M | P \cup C)$
 - e. Are the events F and S mutually exclusive?
 - f. Are the events F and S independent?
17. At a clothing outlet 20% of the clothes are irregular, 10% have at least a button missing and 4% are both irregular and have a button missing. If Martha found a dress that has a button missing, what is the probability that it is irregular?
18. A trade delegation consists of four Americans, three Japanese and two Germans. Three people are chosen at random. Find the following probabilities:
- a. $P(\text{two Americans and one Japanese})$
 - b. $P(\text{at least one American})$
 - c. $P(\text{One of each nationality})$
 - d. $P(\text{no German})$
19. A coin is tossed three times, and the events E and F are as follows.

E: It shows a head on the first toss F: Never turns up a tail

Are the events E and F independent?

- 20. If $P(E) = .6$ and $P(F) = .4$ and E and F are mutually exclusive, find $P(E \text{ and } F)$.
- 21. If $P(E) = .5$ and $P(F) = .3$ and E and F are independent, find $P(E \cup F)$.
- 22. If $P(F) = .9$ and $P(E|F) = .36$ and E and F are independent, find $P(E)$.
- 23. If $P(E) = .4$ and $P(E \text{ or } F) = .9$ and E and F are independent, find $P(F)$.
- 24. If $P(E) = .4$ and $P(F|E) = .5$, find $P(E \text{ and } F)$.
- 25. If $P(E) = .6$ and $P(E \text{ and } F) = .3$, find $P(F|E)$.
- 26. If $P(E) = .3$ and $P(F) = .4$ and E and F are independent, find $P(E|F)$.

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CHAPTER OVERVIEW

4: Probability Distributions and Statistics

4.1: Random Variables

4.2: Probability Distributions for Discrete Random Variables

4.3: The Binomial Distribution

4.4: Continuous Random Variables

4.5: The Standard Normal Distribution

4.6: Probability Computations for General Normal Random Variables

4.7: Exercises

4.7.1: Discrete Random Variables

4.7.2: Continuous Random Variables

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4.1: Random Variables

Learning Objectives

- To learn the concept of a random variable.
- To learn the distinction between discrete and continuous random variables.

Definition: random variable

A random variable is a numerical quantity that is generated by a random experiment.

We will denote random variables by capital letters, such as X or Z , and the actual values that they can take by lowercase letters, such as x and z .

Table 4.1.1 gives four examples of random variables. In the second example, the three dots indicates that every counting number is a possible value for X . Although it is highly unlikely, for example, that it would take 50 tosses of the coin to observe heads for the first time, nevertheless it is conceivable, hence the number 50 is a possible value. The set of possible values is infinite, but is still at least countable, in the sense that all possible values can be listed one after another. In the last two examples, by way of contrast, the possible values cannot be individually listed, but take up a whole interval of numbers. In the fourth example, since the light bulb could conceivably continue to shine indefinitely, there is no natural greatest value for its lifetime, so we simply place the symbol ∞ for infinity as the right endpoint of the interval of possible values.

Table 4.1.1: Four Random Variables

Experiment	Number X	Possible Values of X
Roll two fair dice	Sum of the number of dots on the top faces	2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
Flip a fair coin repeatedly	Number of tosses until the coin lands heads	1, 2, 3, 4, ...
Measure the voltage at an electrical outlet	Voltage measured	$118 \leq x \leq 122$
Operate a light bulb until it burns out	Time until the bulb burns out	$0 \leq x < \infty$

Definition: discrete random variable

A random variable is called discrete if it has either a finite or a countable number of possible values. A random variable is called continuous if its possible values contain a whole interval of numbers.

The examples in the table are typical in that discrete random variables typically arise from a counting process, whereas continuous random variables typically arise from a measurement.

Key Takeaway

- A random variable is a number generated by a random experiment.
- A random variable is called discrete if its possible values form a finite or countable set.
- A random variable is called continuous if its possible values contain a whole interval of numbers.

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4.2: Probability Distributions for Discrete Random Variables

Learning Objectives

- To learn the concept of the probability distribution of a discrete random variable.
- To learn the concepts of the mean, variance, and standard deviation of a discrete random variable, and how to compute them.

Associated to each possible value x of a discrete random variable X is the probability $P(x)$ that X will take the value x in one trial of the experiment.

Definition: probability distribution

The probability distribution of a discrete random variable X is a list of each possible value of X together with the probability that X takes that value in one trial of the experiment.

The probabilities in the probability distribution of a random variable X must satisfy the following two conditions:

- Each probability $P(x)$ must be between 0 and 1:

$$0 \leq P(x) \leq 1.$$

- The sum of all the possible probabilities is 1:

$$\sum P(x) = 1.$$

✓ Example 4.2.1: two Fair Coins

A fair coin is tossed twice. Let X be the number of heads that are observed.

- Construct the probability distribution of X .
- Find the probability that at least one head is observed.

Solution

- The possible values that X can take are 0, 1, and 2. Each of these numbers corresponds to an event in the sample space $S = \{hh, ht, th, tt\}$ of equally likely outcomes for this experiment:

$X = 0$ to $\{tt\}$, $X = 1$ to $\{ht, th\}$, and $X = 2$ to hh .

The probability of each of these events, hence of the corresponding value of X , can be found simply by counting, to give

x	0	1	2
$P(x)$	0.25	0.50	0.25

This table is the probability distribution of X .

- “At least one head” is the event $X \geq 1$, which is the union of the mutually exclusive events $X = 1$ and $X = 2$. Thus

$$\begin{aligned} P(X \geq 1) &= P(1) + P(2) = 0.50 + 0.25 \\ &= 0.75 \end{aligned}$$

A histogram that graphically illustrates the probability distribution is given in Figure 4.2.1.

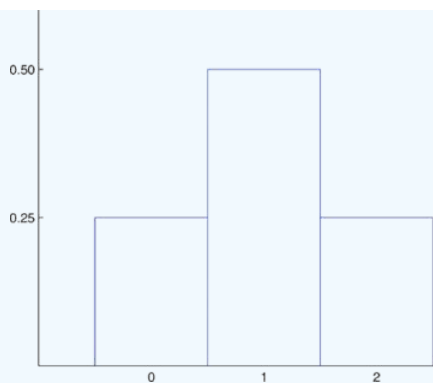


Figure 4.2.1: Probability Distribution for Tossing a Fair Coin Twice

✓ Example 4.2.2: Two Fair Dice

A pair of fair dice is rolled. Let X denote the sum of the number of dots on the top faces.

- Construct the probability distribution of X for a pair of fair dice.
- Find $P(X \geq 9)$.
- Find the probability that X takes an even value.

Solution

The sample space of equally likely outcomes is

11	12	13	14	15	16
21	22	23	24	25	26
31	32	33	34	35	36
41	42	43	44	45	46
51	52	53	54	55	56
61	62	63	64	65	66

where the first digit is die 1 and the second number is die 2.

- The possible values for X are the numbers 2 through 12. $X = 2$ is the event $\{11\}$, so $P(2) = 1/36$. $X = 3$ is the event $\{12, 21\}$, so $P(3) = 2/36$. Continuing this way we obtain the following table

x	2	3	4	5	6	7	8	9	10	11	12
$P(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

This table is the probability distribution of X .

- The event $X \geq 9$ is the union of the mutually exclusive events $X = 9$, $X = 10$, $X = 11$, and $X = 12$. Thus

$$\begin{aligned}
 P(X \geq 9) &= P(9) + P(10) + P(11) + P(12) \\
 &= \frac{4}{36} + \frac{3}{36} + \frac{2}{36} + \frac{1}{36} \\
 &= \frac{10}{36} \\
 &= 0.2\bar{7}
 \end{aligned}$$

- Before we immediately jump to the conclusion that the probability that X takes an even value must be 0.5, note that X takes six different even values but only five different odd values. We compute

$$\begin{aligned}
 P(X \text{ is even}) &= P(2) + P(4) + P(6) + P(8) + P(10) + P(12) \\
 &= \frac{1}{36} + \frac{3}{36} + \frac{5}{36} + \frac{5}{36} + \frac{3}{36} + \frac{1}{36} \\
 &= \frac{18}{36} \\
 &= 0.5
 \end{aligned}$$

A histogram that graphically illustrates the probability distribution is given in Figure 4.2.2

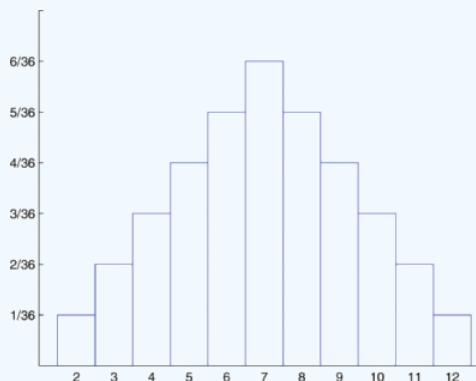


Figure 4.2.2 Probability Distribution for Tossing Two Fair Dice

The Mean and Standard Deviation of a Discrete Random Variable

Definition: mean

The *mean* (also called the "expectation value" or "expected value") of a discrete random variable X is the number

$$\mu = E(X) = \sum xP(x) \quad (4.2.1)$$

The mean of a random variable may be interpreted as the average of the values assumed by the random variable in repeated trials of the experiment.

✓ Example 4.2.3

Find the mean of the discrete random variable X whose probability distribution is

x	-2	1	2	3.5
$P(x)$	0.21	0.34	0.24	0.21

Solution

Using the definition of mean (Equation 4.2.1) gives

$$\begin{aligned}
 \mu &= \sum xP(x) \\
 &= (-2)(0.21) + (1)(0.34) + (2)(0.24) + (3.5)(0.21) \\
 &= 1.135
 \end{aligned}$$

✓ Example 4.2.4

A service organization in a large town organizes a raffle each month. One thousand raffle tickets are sold for \$1 each. Each has an equal chance of winning. First prize is \$300, second prize is \$200, and third prize is \$100. Let X denote the net gain from the purchase of one ticket.

- Construct the probability distribution of X .
- Find the probability of winning any money in the purchase of one ticket.

c. Find the expected value of X , and interpret its meaning.

Solution

a. If a ticket is selected as the first prize winner, the net gain to the purchaser is the \$300 prize less the \$1 that was paid for the ticket, hence $X = 300 - 1 = 299$. There is one such ticket, so $P(299) = 0.001$. Applying the same “income minus outgo” principle to the second and third prize winners and to the 997 losing tickets yields the probability distribution:

x	299	199	99	-1
$P(x)$	0.001	0.001	0.001	0.997

b. Let W denote the event that a ticket is selected to win one of the prizes. Using the table

$$\begin{aligned} P(W) &= P(299) + P(199) + P(99) = 0.001 + 0.001 + 0.001 \\ &= 0.003 \end{aligned}$$

c. Using the definition of expected value (Equation 4.2.1),

$$\begin{aligned} E(X) &= (299) \cdot (0.001) + (199) \cdot (0.001) + (99) \cdot (0.001) + (-1) \cdot (0.997) \\ &= -0.4 \end{aligned}$$

The negative value means that one loses money on the average. In particular, if someone were to buy tickets repeatedly, then although he would win now and then, on average he would lose 40 cents per ticket purchased.

The concept of expected value is also basic to the insurance industry, as the following simplified example illustrates.

✓ Example 4.2.5

A life insurance company will sell a \$200,000 one-year term life insurance policy to an individual in a particular risk group for a premium of \$195. Find the expected value to the company of a single policy if a person in this risk group has a 99.97% chance of surviving one year.

Solution

Let X denote the net gain to the company from the sale of one such policy. There are two possibilities: the insured person lives the whole year or the insured person dies before the year is up. Applying the “income minus outgo” principle, in the former case the value of X is $195 - 0$; in the latter case it is $195 - 200,000 = -199,805$. Since the probability in the first case is 0.9997 and in the second case is $1 - 0.9997 = 0.0003$, the probability distribution for X is:

x	195	-199,805
$P(x)$	0.9997	0.0003

Therefore

$$\begin{aligned} E(X) &= \sum xP(x) \\ &= (195) \cdot (0.9997) + (-199,805) \cdot (0.0003) \\ &= 135 \end{aligned}$$

Occasionally (in fact, 3 times in 10,000) the company loses a large amount of money on a policy, but typically it gains \$195, which by our computation of $E(X)$ works out to a net gain of \$135 per policy sold, on average.

Definition: variance

The *variance* (σ^2) of a discrete random variable X is the number

$$\sigma^2 = \sum (x - \mu)^2 P(x) \quad (4.2.2)$$

which by algebra is equivalent to the formula

$$\sigma^2 = \left[\sum x^2 P(x) \right] - \mu^2 \quad (4.2.3)$$

 Definition: standard deviation

The standard deviation, σ , of a discrete random variable X is the square root of its variance, hence is given by the formulas

$$\sigma = \sqrt{\sum (x - \mu)^2 P(x)} = \sqrt{\left[\sum x^2 P(x) \right] - \mu^2} \quad (4.2.4)$$

The variance and standard deviation of a discrete random variable X may be interpreted as measures of the variability of the values assumed by the random variable in repeated trials of the experiment. The units on the standard deviation match those of X .

✓ Example 4.2.6

A discrete random variable X has the following probability distribution:

x	-1	0	1	4
$P(x)$	0.2	0.5	a	0.1

(4.2.5)

A histogram that graphically illustrates the probability distribution is given in Figure 4.2.3.

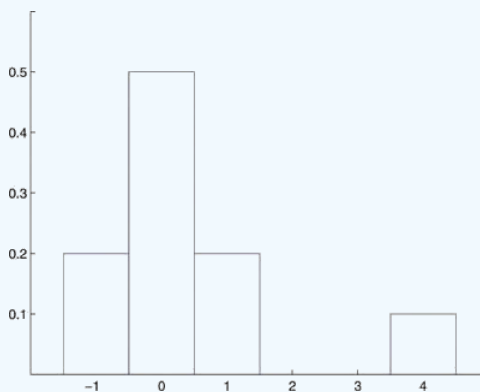


Figure 4.2.3 Probability Distribution of a Discrete Random Variable

Compute each of the following quantities.

- a. a .
- b. $P(0)$.
- c. $P(X > 0)$.
- d. $P(X \geq 0)$.
- e. $P(X \leq -2)$.
- f. The mean μ of X .
- g. The variance σ^2 of X .
- h. The standard deviation σ of X .

Solution

- a. Since all probabilities must add up to 1,

$$a = 1 - (0.2 + 0.5 + 0.1) = 0.2$$

- b. Directly from the table, $P(0) = 0.5$

$$P(0) = 0.5$$

- c. From Table 4.2.5,

$$P(X > 0) = P(1) + P(4) = 0.2 + 0.1 = 0.3$$

- d. From Table 4.2.5,

$$P(X \geq 0) = P(0) + P(1) + P(4) = 0.5 + 0.2 + 0.1 = 0.8$$

e. Since none of the numbers listed as possible values for X is less than or equal to -2 , the event $X \leq -2$ is impossible, so

$$P(X \leq -2) = 0$$

f. Using the formula in the definition of μ (Equation 4.2.1)

$$\begin{aligned}\mu &= \sum xP(x) \\ &= (-1) \cdot (0.2) + (0) \cdot (0.5) + (1) \cdot (0.2) + (4) \cdot (0.1) \\ &= 0.4\end{aligned}$$

g. Using the formula in the definition of σ^2 (Equation 4.2.2) and the value of μ that was just computed,

$$\begin{aligned}\sigma^2 &= \sum (x - \mu)^2 P(x) \\ &= (-1 - 0.4)^2 \cdot (0.2) + (0 - 0.4)^2 \cdot (0.5) + (1 - 0.4)^2 \cdot (0.2) + (4 - 0.4)^2 \cdot (0.1) \\ &= 1.84\end{aligned}$$

h. Using the result of part (g), $\sigma = \sqrt{1.84} = 1.3565$

Summary

- The probability distribution of a discrete random variable X is a listing of each possible value x taken by X along with the probability $P(x)$ that X takes that value in one trial of the experiment.
- The mean μ of a discrete random variable X is a number that indicates the average value of X over numerous trials of the experiment. It is computed using the formula $\mu = \sum xP(x)$.
- The variance σ^2 and standard deviation σ of a discrete random variable X are numbers that indicate the variability of X over numerous trials of the experiment. They may be computed using the formula $\sigma^2 = [\sum x^2 P(x)] - \mu^2$.

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4.3: The Binomial Distribution

Learning Objectives

- To learn the concept of a binomial random variable.
- To learn how to recognize a random variable as being a binomial random variable.

The experiment of tossing a fair coin three times and the experiment of observing the genders according to birth order of the children in a randomly selected three-child family are completely different, but the random variables that count the number of heads in the coin toss and the number of boys in the family (assuming the two genders are equally likely) are the same random variable, the one with probability distribution

x	0	1	2	3
$P(x)$	0.125	0.375	0.375	0.125

A histogram that graphically illustrates this probability distribution is given in Figure 4.3.1. What is common to the two experiments is that we perform three identical and independent trials of the same action, each trial has only two outcomes (heads or tails, boy or girl), and the probability of success is the same number, 0.5, on every trial. The random variable that is generated is called the binomial random variable with parameters $n = 3$ and $p = 0.5$. This is just one case of a general situation.

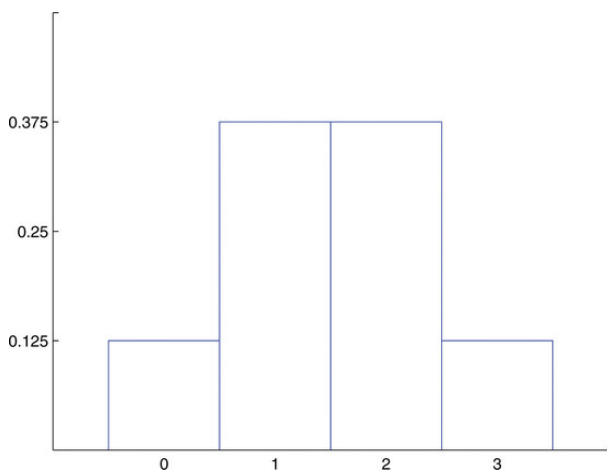


Figure 4.3.1: Probability Distribution for Three Coins and Three Children

Definition: binomial distribution

Suppose a random experiment has the following characteristics.

- There are n identical and independent trials of a common procedure.
- There are exactly two possible outcomes for each trial, one termed “success” and the other “failure.”
- The probability of success on any one trial is the same number p .

Then the discrete random variable X that counts the number of successes in the n trials is the binomial random variable with parameters n and p . We also say that X has a binomial distribution with parameters n and p .

The following four examples illustrate the definition. Note how in every case “success” is the outcome that is counted, not the outcome that we prefer or think is better in some sense.

1. A random sample of 125 students is selected from a large college in which the proportion of students who are females is 57%. Suppose X denotes the number of female students in the sample. In this situation there are $n = 125$ identical and independent trials of a common procedure, selecting a student at random; there are exactly two possible outcomes for each trial, “success” (what we are counting, that the student be female) and “failure,” and finally the probability of success on any one trial is the same number $p = 0.57$. X is a binomial random variable with parameters $n = 125$ and $p = 0.57$.

2. A multiple-choice test has 15 questions, each of which has five choices. An unprepared student taking the test answers each of the questions completely randomly by choosing an arbitrary answer from the five provided. Suppose X denotes the number of answers that the student gets right. X is a binomial random variable with parameters $n = 15$ and $p = 1/5 = 0.20$.
3. In a survey of 1,000 registered voters each voter is asked if he intends to vote for a candidate Titania Queen in the upcoming election. Suppose X denotes the number of voters in the survey who intend to vote for Titania Queen. X is a binomial random variable with $n = 1000$ and p equal to the true proportion of voters (surveyed or not) who intend to vote for Titania Queen.
4. An experimental medication was given to 30 patients with a certain medical condition. Suppose X denotes the number of patients who develop severe side effects. X is a binomial random variable with $n = 30$ and p equal to the true probability that a patient with the underlying condition will experience severe side effects if given that medication.

Probability Formula for a Binomial Random Variable

Often the most difficult aspect of working a problem that involves the binomial random variable is recognizing that the random variable in question has a binomial distribution. Once that is known, probabilities can be computed using the following formula.

If X is a binomial random variable with parameters n and p , then

$$P(x) = \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

where $q = 1 - p$ and where for any counting number m , $m!$ (read “m factorial”) is defined by

$$0! = 1, 1! = 1, 2! = 1 \cdot 2, 3! = 1 \cdot 2 \cdot 3$$

and in general

$$m! = 1 \cdot 2 \cdots (m-1) \cdot m$$

✓ Example 4.3.1

Seventeen percent of victims of financial fraud know the perpetrator of the fraud personally.

- a. Use the formula to construct the probability distribution for the number X of people in a random sample of five victims of financial fraud who knew the perpetrator personally.
- b. A investigator examines five cases of financial fraud every day. Find the most frequent number of cases each day in which the victim knew the perpetrator.
- c. A investigator examines five cases of financial fraud every day. Find the average number of cases per day in which the victim knew the perpetrator.

Solution

The random variable X is binomial with parameters $n = 5$ and $p = 0.17$; $q = 1 - p = 0.83$. The possible values of X are 0, 1, 2, 3, 4, and 5

$$\begin{aligned} P(0) &= \frac{5!}{0!5!} (0.17)^0 (0.83)^5 \\ &= \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5}{(1)(1 \cdot 2 \cdot 3 \cdot 4 \cdot 5)} 1 \cdot (0.3939040643) \\ &= 0.3939040643 \approx 0.3939 \end{aligned}$$

$$\begin{aligned} P(1) &= \frac{5!}{1!4!} (0.17)^1 (0.83)^4 \\ &= \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5}{(1)(1 \cdot 2 \cdot 3 \cdot 4)} (0.17) \cdot (0.47458321) \\ &= 5 \cdot (0.17) \cdot (0.47458321) \\ &= 0.4033957285 \approx 0.4034 \end{aligned}$$

$$\begin{aligned}
 P(2) &= \frac{5!}{2!3!} (0.17)^2 (0.83)^3 \\
 &= \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5}{(1 \cdot 2)(1 \cdot 2 \cdot 3)} (0.0289) \cdot (0.571787) \\
 &= 10 \cdot (0.0289) \cdot (0.571787) \\
 &= 0.165246443 \approx 0.1652
 \end{aligned}$$

The remaining three probabilities are computed similarly, to give the probability distribution

x	0	1	2	3	4	5
$P(x)$	0.3939	0.4034	0.1652	0.0338	0.0035	0.0001

The probabilities do not add up to exactly 1 because of rounding.

This probability distribution is represented by the histogram in Figure 4.3.2, which graphically illustrates just how improbable the events $X = 4$ and $X = 5$ are. The corresponding bar in the histogram above the number 4 is barely visible, if visible at all, and the bar above 5 is far too short to be visible.

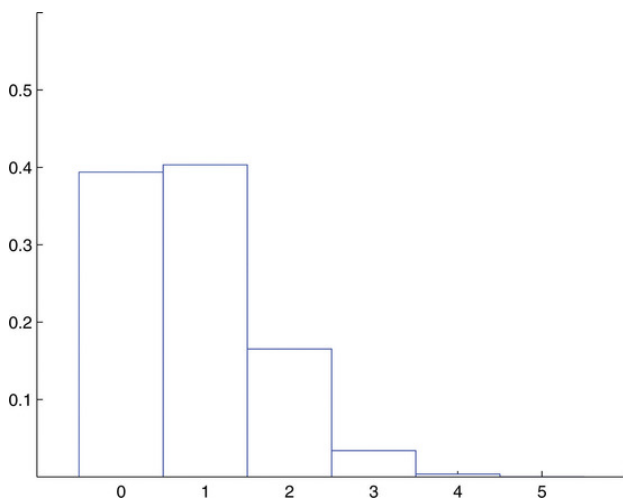


Figure 4.3.2 Probability Distribution of the Binomial Random

Variable in Example 4.3.1

The value of X that is most likely is $X = 1$, so the most frequent number of cases seen each day in which the victim knew the perpetrator is one.

The average number of cases per day in which the victim knew the perpetrator is the mean of X , which is

$$\mu = \sum xP(x) \quad (4.3.1)$$

$$= 0 \cdot 0.3939 + 1 \cdot 0.4034 + 2 \cdot 0.1652 + 3 \cdot 0.0338 + 4 \cdot 0.0035 + 5 \cdot 0.0001 \quad (4.3.2)$$

$$= 0.8497 \quad (4.3.3)$$

Special Formulas for the Mean and Standard Deviation of a Binomial Random Variable

Since a binomial random variable is a discrete random variable, the formulas for its mean, variance, and standard deviation given in the previous section apply to it, as we just saw in Example 4.3.2 in the case of the mean. However, for the binomial random variable there are much simpler formulas.

If X is a binomial random variable with parameters n and p , then

$$\mu = np$$

$$\sigma^2 = npq$$

$$\sigma = \sqrt{npq}$$

where $q = 1 - p$.

✓ Example 4.3.2

Find the mean and standard deviation of the random variable X of Example 4.3.1.

Solution

The random variable X is binomial with parameters $n = 5$ and $p = 0.17$, and $q = 1 - p = 0.83$. Thus its mean and standard deviation are

$$\mu = np = (5) \cdot (0.17) = 0.85 \text{ (exactly)}$$

and

$$\sigma = \sqrt{npq} = \sqrt{(5) \cdot (0.17) \cdot (0.83)} = \sqrt{0.7055} \approx 0.8399$$

The Cumulative Probability Distribution of a Binomial Random Variable

In order to allow a broader range of more realistic problems contains probability tables for binomial random variables for various choices of the parameters n and p . These tables are not the probability distributions that we have seen so far, but are **cumulative probability distributions**. In the place of the probability $P(x)$ the table contains the probability

$$P(X \leq x) = P(0) + P(1) + \dots + P(x)$$

This is illustrated in Figure 4.3.3. The probability entered in the table corresponds to the area of the shaded region. The reason for providing a cumulative table is that in practical problems that involve a binomial random variable typically the probability that is sought is of the form $P(X \leq x)$ or $P(X \geq x)$. The cumulative table is much easier to use for computing $P(X \leq x)$ since all the individual probabilities have already been computed and added. The one table suffices for both $P(X \leq x)$ or $P(X \geq x)$ and can be used to readily obtain probabilities of the form $P(x)$, too, because of the following formulas. The first is just the Probability Rule for Complements.

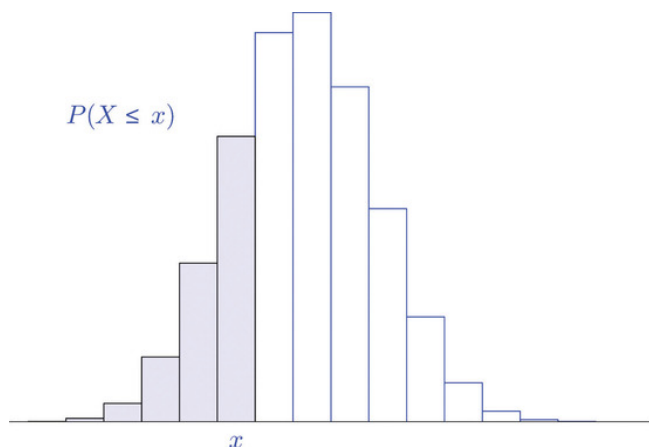


Figure 4.3.3 Cumulative Probabilities

If X is a discrete random variable, then

$$P(X \geq x) = 1 - P(X \leq x - 1)$$

and

$$P(x) = P(X \leq x) - P(X \leq x - 1)$$

✓ Example 4.3.3

A student takes a ten-question true/false exam.

- Find the probability that the student gets exactly six of the questions right simply by guessing the answer on every question.
- Find the probability that the student will obtain a passing grade of 60% or greater simply by guessing.

Solution

Let X denote the number of questions that the student guesses correctly. Then X is a binomial random variable with parameters $n = 10$ and $p = 0.50$.

- a. The probability sought is $P(6)$. The formula gives

$$P(6) = 10!(6!)(4!)(.5)^6 = 0.205078125$$

Using the table,

$$P(6) = P(X \leq 6) - P(X \leq 5) = 0.8281 - 0.6230 = 0.2051$$

- b. The student must guess correctly on at least 60% of the questions, which is $(0.60) \cdot (10) = 6$ questions. The probability sought is not $P(6)$ (an easy mistake to make), but

$$P(X \geq 6) = P(6) + P(7) + P(8) + P(9) + P(10)$$

Instead of computing each of these five numbers using the formula and adding them we can use the table to obtain

$$P(X \geq 6) = 1 - P(X \leq 5) = 1 - 0.6230 = 0.3770$$

which is much less work and of sufficient accuracy for the situation at hand.

✓ Example 4.3.4

An appliance repairman services five washing machines on site each day. One-third of the service calls require installation of a particular part.

- The repairman has only one such part on his truck today. Find the probability that the one part will be enough today, that is, that at most one washing machine he services will require installation of this particular part.
- Find the minimum number of such parts he should take with him each day in order that the probability that he have enough for the day's service calls is at least 95%.

Solution

Let X denote the number of service calls today on which the part is required. Then X is a binomial random variable with parameters $n = 5$ and $p = 1/3 = 0.\bar{3}$

- a. Note that the probability in question is not $P(1)$, but rather $P(X \leq 1)$. Using the cumulative distribution table,

$$P(X \leq 1) = 0.4609$$

- b. The answer is the smallest number x such that the table entry $P(X \leq x)$ is at least 0.9500. Since $P(X \leq 2) = 0.7901$ is less than 0.95, two parts are not enough. Since $P(X \leq 3) = 0.9547$ is as large as 0.95, three parts will suffice at least 95% of the time. Thus the minimum needed is three.

Summary

- The discrete random variable X that counts the number of successes in n identical, independent trials of a procedure that always results in either of two outcomes, "success" or "failure," and in which the probability of success on each trial is the same number p , is called the binomial random variable with parameters n and p .
- There is a formula for the probability that the binomial random variable with parameters n and p will take a particular value x .
- There are special formulas for the mean, variance, and standard deviation of the binomial random variable with parameters n and p that are much simpler than the general formulas that apply to all discrete random variables.
- Cumulative probability distribution tables, when available, facilitate computation of probabilities encountered in typical practical situations.

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4.4: Continuous Random Variables

Learning Objectives

- To learn the concept of the probability distribution of a continuous random variable, and how it is used to compute probabilities.
- To learn basic facts about the family of normally distributed random variables.

The Probability Distribution of a Continuous Random Variable

For a discrete random variable X the probability that X assumes one of its possible values on a single trial of the experiment makes good sense. This is not the case for a continuous random variable. For example, suppose X denotes the length of time a commuter just arriving at a bus stop has to wait for the next bus. If buses run every 30 minutes without fail, then the set of possible values of X is the interval denoted $[0, 30]$, the set of all decimal numbers between 0 and 30. But although the number 7.211916 is a possible value of X , there is little or no meaning to the concept of the probability that the commuter will wait precisely 7.211916 minutes for the next bus. If anything the probability should be zero, since if we could meaningfully measure the waiting time to the nearest millionth of a minute it is practically inconceivable that we would ever get exactly 7.211916 minutes. More meaningful questions are those of the form: What is the probability that the commuter's waiting time is less than 10 minutes, or is between 5 and 10 minutes? In other words, with continuous random variables one is concerned not with the event that the variable assumes a single particular value, but with the event that the random variable assumes a value in a particular interval.

Definition: density function

The probability distribution of a continuous random variable X is an assignment of probabilities to intervals of decimal numbers using a function $f(x)$, called a density function, in the following way: the probability that X assumes a value in the interval $[a, b]$ is equal to the area of the region that is bounded above by the graph of the equation $y = f(x)$, bounded below by the x -axis, and bounded on the left and right by the vertical lines through a and b , as illustrated in Figure 4.4.1.

$$P(a < X < b) = \text{area of shaded region}$$

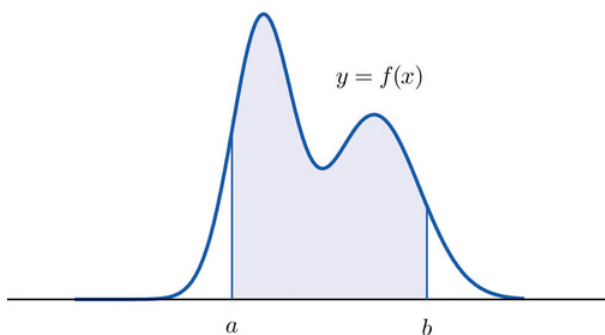


Figure 4.4.1: Probability Given as Area of a Region under a Curve

This definition can be understood as a natural outgrowth of the discussion in Section 2.1.3. There we saw that if we have in view a population (or a very large sample) and make measurements with greater and greater precision, then as the bars in the relative frequency histogram become exceedingly fine their vertical sides merge and disappear, and what is left is just the curve formed by their tops, as shown in Figure 2.1.5. Moreover the total area under the curve is 1, and the proportion of the population with measurements between two numbers a and b is the area under the curve and between a and b , as shown in Figure 2.1.6. If we think of X as a measurement to infinite precision arising from the selection of any one member of the population at random, then $P(a < X < b)$ is simply the proportion of the population with measurements between a and b , the curve in the relative frequency histogram is the density function for X , and we arrive at the definition just above.

- Every density function $f(x)$ must satisfy the following two conditions:
- For all numbers x , $f(x) \geq 0$, so that the graph of $y = f(x)$ never drops below the x -axis.
- The area of the region under the graph of $y = f(x)$ and above the x -axis is 1.

Because the area of a line segment is 0, the definition of the probability distribution of a continuous random variable implies that for any particular decimal number, say a , the probability that X assumes the exact value a is 0. This property implies that whether or not the endpoints of an interval are included makes no difference concerning the probability of the interval.

For any continuous random variable X :

$$P(a \leq X \leq b) = P(a < X \leq b) = P(a \leq X < b) = P(a < X < b)$$

✓ Example 4.4.1

A random variable X has the uniform distribution on the interval $[0, 1]$: the density function is $f(x) = 1$ if x is between 0 and 1 and $f(x) = 0$ for all other values of x , as shown in Figure 4.4.2.

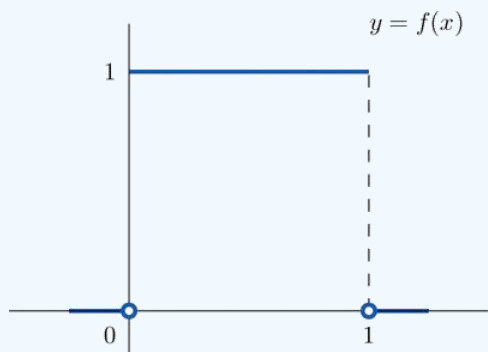
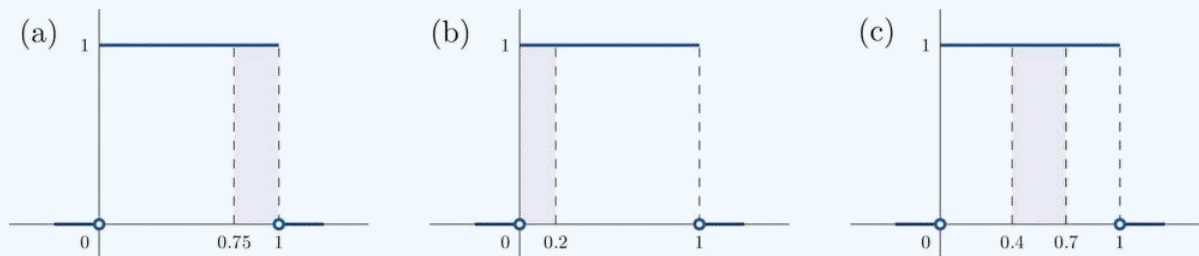


Figure 4.4.2 Uniform Distribution on $[0,1]$.

1. Find $P(X > 0.75)$, the probability that X assumes a value greater than 0.75.
2. Find $P(X \leq 0.2)$, the probability that X assumes a value less than or equal to 0.2.
3. Find $P(0.4 < X < 0.7)$, the probability that X assumes a value between 0.4 and 0.7.

Solution

1. $P(X > 0.75)$ is the area of the rectangle of height 1 and base length $1 - 0.75 = 0.25$, hence is $\text{base} \times \text{height} = (0.25) \cdot (1) = 0.25$. See Figure 4.4.3a.
2. $P(X \leq 0.2)$ is the area of the rectangle of height 1 and base length $0.2 - 0 = 0.2$, hence is $\text{base} \times \text{height} = (0.2) \cdot (1) = 0.2$. See Figure 4.4.3b.
3. $P(0.4 < X < 0.7)$ is the area of the rectangle of height 1 and length $0.7 - 0.4 = 0.3$, hence is $\text{base} \times \text{height} = (0.3) \cdot (1) = 0.3$. See Figure 4.4.3c.



Figure

4.4.3 Probabilities from the Uniform Distribution on $[0,1]$

✓ Example 4.4.2

A man arrives at a bus stop at a random time (that is, with no regard for the scheduled service) to catch the next bus. Buses run every 30 minutes without fail, hence the next bus will come any time during the next 30 minutes with evenly distributed probability (a uniform distribution). Find the probability that a bus will come within the next 10 minutes.

Solution

The graph of the density function is a horizontal line above the interval from 0 to 30 and is the x -axis everywhere else. Since the total area under the curve must be 1, the height of the horizontal line is $1/30$ (Figure 4.4.4). The probability sought is $P(0 \leq X \leq 10)$. By definition, this probability is the area of the rectangular region bounded above by the horizontal line $f(x) = 1/30$, bounded below by the x -axis, bounded on the left by the vertical line at 0 (the y -axis), and bounded on the right

by the vertical line at 10. This is the shaded region in Figure 4.4.4. Its area is the base of the rectangle times its height, $(10) \cdot (1/30) = 1/3$. Thus $P(0 \leq X \leq 10) = 1/3$.

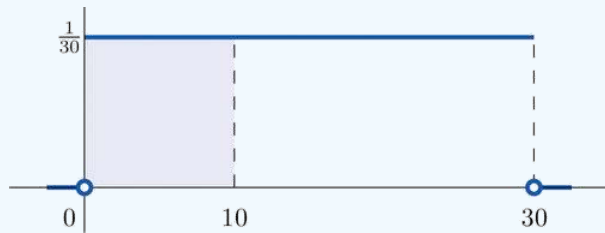


Figure 4.4.4 Probability of Waiting At Most 10 Minutes for a Bus

Normal Distributions

Most people have heard of the “bell curve.” It is the graph of a specific density function $f(x)$ that describes the behavior of continuous random variables as different as the heights of human beings, the amount of a product in a container that was filled by a high-speed packing machine, or the velocities of molecules in a gas. The formula for $f(x)$ contains two parameters μ and σ that can be assigned any specific numerical values, so long as σ is positive. We will not need to know the formula for $f(x)$, but for those who are interested it is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(\mu-x)^2/\sigma^2}$$

where $\pi \approx 3.14159$ and $e \approx 2.71828$ is the base of the natural logarithms.

Each different choice of specific numerical values for the pair μ and σ gives a different bell curve. The value of μ determines the location of the curve, as shown in Figure 4.4.5. In each case the curve is symmetric about μ .

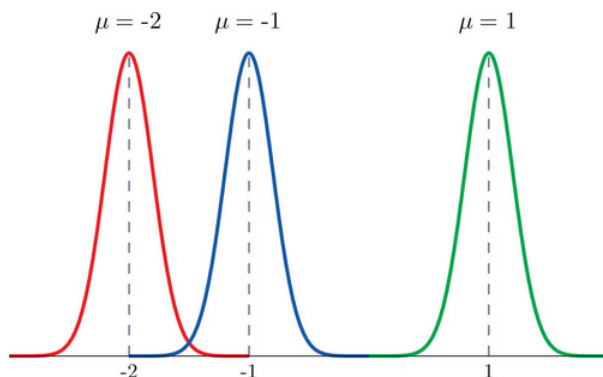


Figure 4.4.5 Bell Curves with $\sigma = 0.25$ and Different Values of μ

The value of σ determines whether the bell curve is tall and thin or short and squat, subject always to the condition that the total area under the curve be equal to 1. This is shown in Figure 4.4.6, where we have arbitrarily chosen to center the curves at $\mu = 6$.

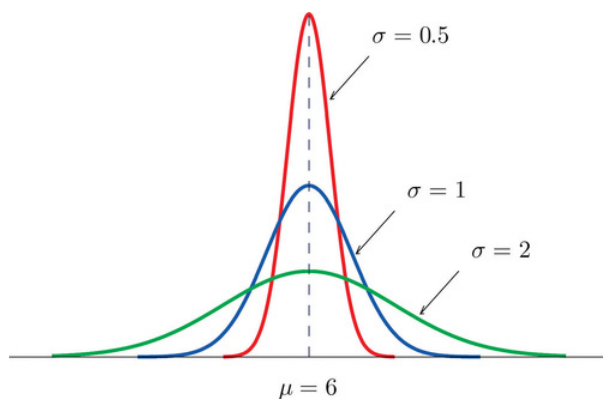


Figure 4.4.6 Bell Curves with $\mu = 6$ and Different Values of σ .

Definition: normal distribution

The probability distribution corresponding to the density function for the bell curve with parameters μ and σ is called the normal distribution with mean μ and standard deviation σ .

Definition: normally distributed random variable

A continuous random variable whose probabilities are described by the normal distribution with mean μ and standard deviation σ is called a normally distributed random variable, or a normal random variable for short, with mean μ and standard deviation σ .

Figure 4.4.7 shows the density function that determines the normal distribution with mean μ and standard deviation σ . We repeat an important fact about this curve: **The density curve for the normal distribution is symmetric about the mean.**

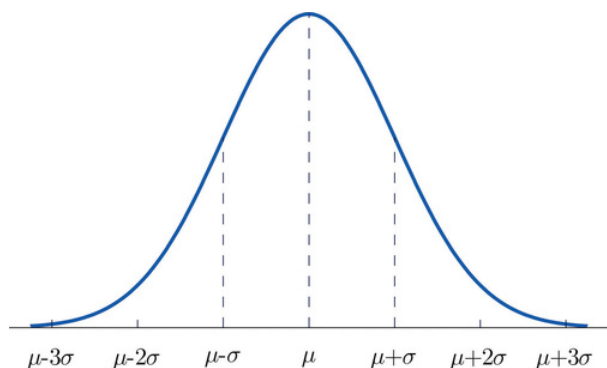


Figure 4.4.7 Density Function for a Normally Distributed Random Variable with Mean μ and Standard Deviation σ

✓ Example 4.4.3

Heights of 25-year-old men in a certain region have mean 69.75 inches and standard deviation 2.59 inches. These heights are approximately normally distributed. Thus the height X of a randomly selected 25-year-old man is a normal random variable with mean $\mu = 69.75$ and standard deviation $\sigma = 2.59$. Sketch a qualitatively accurate graph of the density function for X . Find the probability that a randomly selected 25-year-old man is more than 69.75 inches tall.

Solution

The distribution of heights looks like the bell curve in Figure 4.4.8. The important point is that it is centered at its mean, 69.75, and is symmetric about the mean.

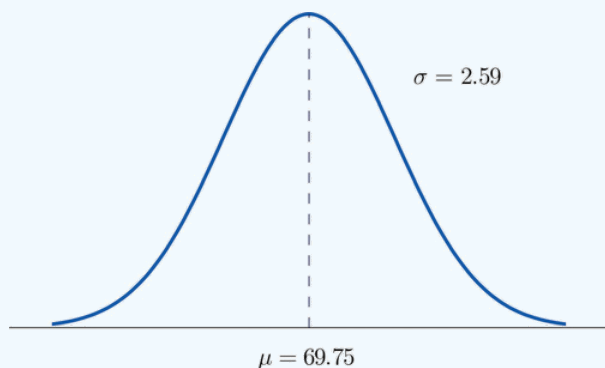


Figure 4.4.8 Density Function for Heights of 25-Year-Old Men

Since the total area under the curve is 1, by symmetry the area to the right of 69.75 is half the total, or 0.5. But this area is precisely the probability $P(X > 69.75)$, the probability that a randomly selected 25-year-old man is more than 69.75 inches tall. We will learn how to compute other probabilities in the next two sections.

Key Takeaway

- For a continuous random variable X the only probabilities that are computed are those of X taking a value in a specified interval.
- The probability that X take a value in a particular interval is the same whether or not the endpoints of the interval are included.
- The probability $P(a < X < b)$, that X take a value in the interval from a to b , is the area of the region between the vertical lines through a and b , above the x -axis, and below the graph of a function $f(x)$ called the density function.
- A normally distributed random variable is one whose density function is a bell curve.
- Every bell curve is symmetric about its mean and lies everywhere above the x -axis, which it approaches asymptotically (arbitrarily closely without touching).

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4.5: The Standard Normal Distribution

Learning Objectives

- To learn what a standard normal random variable is.
- To learn how to compute probabilities related to a standard normal random variable.

Definition: standard normal random variable

A *standard normal random variable* is a normally distributed random variable with mean $\mu = 0$ and standard deviation $\sigma = 1$. It will always be denoted by the letter Z .

The density function for a standard normal random variable is shown in Figure 4.5.1.

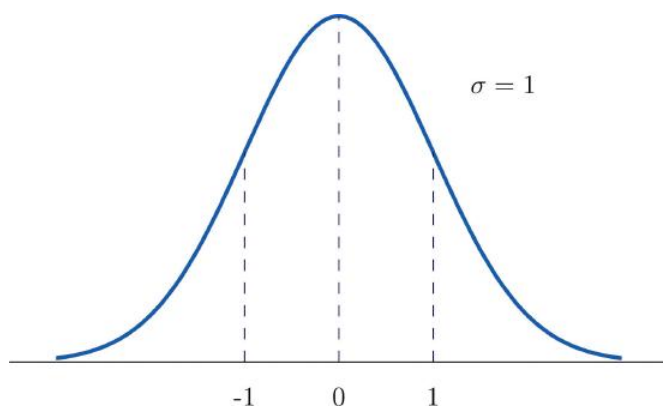


Figure 4.5.1: Density Curve for a Standard Normal Random

Variable

To compute probabilities for Z we will not work with its density function directly but instead read probabilities out of Figure 4.5.2. The tables are tables of *cumulative* probabilities; their entries are probabilities of the form $P(Z < z)$. The use of the tables will be explained by the following series of examples.

✓ Example 4.5.1

Find the probabilities indicated, where as always Z denotes a standard normal random variable.

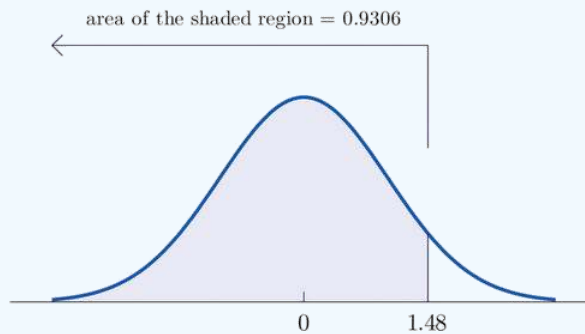
1. $P(Z < 1.48)$.
2. $P(Z < -0.25)$.

Solution

1. Figure 4.5.3 shows how this probability is read directly from the table without any computation required. The digits in the ones and tenths places of 1.48, namely 1.4, are used to select the appropriate row of the table; the hundredths part of 1.48, namely 0.08, is used to select the appropriate column of the table. The four decimal place number in the interior of the table that lies in the intersection of the row and column selected, 0.9306 is the probability sought:

$$P(Z < 1.48) = 0.9306$$

...	0.08	...
...	...	...
1.4	...	0.9306
...	...	...



Figure

4.5.3 Computing Probabilities Using the Cumulative Table

- The minus sign in -0.25 makes no difference in the procedure; the table is used in exactly the same way as in part (a): the probability sought is the number that is in the intersection of the row with heading -0.2 and the column with heading 0.05 , the number 0.4013 . Thus $P(Z < -0.25) = 0.4013$.

✓ Example 4.5.2

Find the probabilities indicated.

- $P(Z > 1.60)$.
- $P(Z > -1.02)$.

Solution

- Because the events $Z > 1.60$ and $Z \leq 1.60$ are complements, the *Probability Rule for Complements* implies that

$$P(Z > 1.60) = 1 - P(Z \leq 1.60)$$

Since inclusion of the endpoint makes no difference for the continuous random variable Z , $P(Z \leq 1.60) = P(Z < 1.60)$, which we know how to find from the table in Figure 4.5.2. The number in the row with heading 1.6 and in the column with heading 0.00 is 0.9452 . Thus $P(Z < 1.60) = 0.9452$ so

$$P(Z > 1.60) = 1 - P(Z \leq 1.60) = 1 - 0.9452 = 0.0548$$

Figure 4.5.4 illustrates the ideas geometrically. Since the total area under the curve is 1 and the area of the region to the left of 1.60 is (from the table) 0.9452 , the area of the region to the right of 1.60 must be $1 - 0.9452 = 0.0548$.

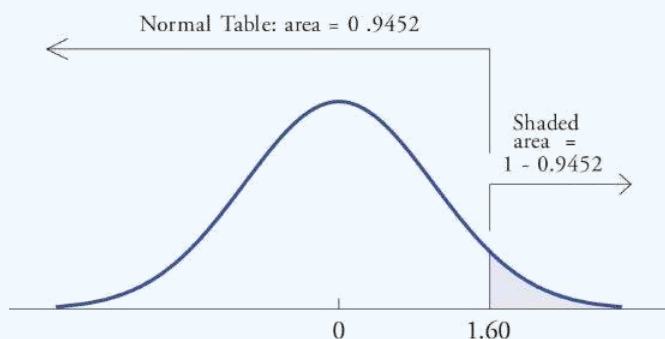


Figure 4.5.4 Computing a Probability for a Right Half-Line

- The minus sign in -1.02 makes no difference in the procedure; the table is used in exactly the same way as in part (a). The number in the intersection of the row with heading -1.0 and the column with heading 0.02 is 0.1539 . This means that $P(Z < -1.02) = P(Z \leq -1.02) = 0.1539$. Hence

$$P(Z > -1.02) = P(Z \leq -1.02) = 1 - 0.1539 = 0.8461$$

✓ Example 4.5.3

Find the probabilities indicated.

1. $P(0.5 < Z < 1.57)$.
2. $P(-2.55 < Z < 0.09)$.

Solution

1. Figure 4.5.5 illustrates the ideas involved for intervals of this type. First look up the areas in the table that correspond to the numbers 0.5 (which we think of as 0.50 to use the table) and 1.57. We obtain 0.6915 and 0.9418 respectively. From the figure it is apparent that we must take the difference of these two numbers to obtain the probability desired. In symbols,

$$P(0.5 < Z < 1.57) = P(Z < 1.57) - P(Z < 0.50) = 0.9418 - 0.6915 = 0.2503$$

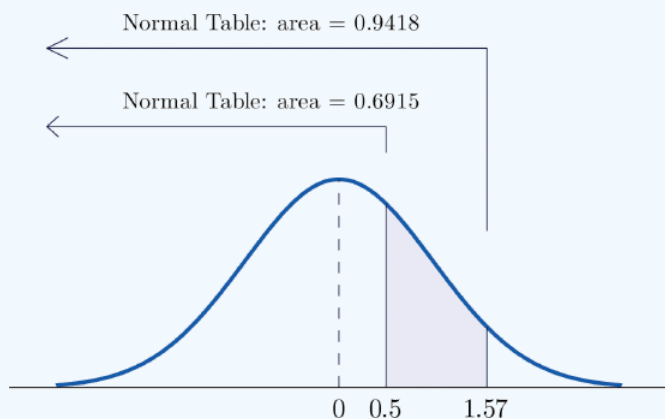


Figure 4.5.5 Computing a Probability for an Interval of Finite Length

2. The procedure for finding the probability that Z takes a value in a finite interval whose endpoints have opposite signs is exactly the same procedure used in part (a), and is illustrated in Figure 4.5.6 "Computing a Probability for an Interval of Finite Length". In symbols the computation is

$$P(-2.55 < Z < 0.09) = P(Z < 0.09) - P(Z < -2.55) = 0.5359 - 0.0054 = 0.5305$$

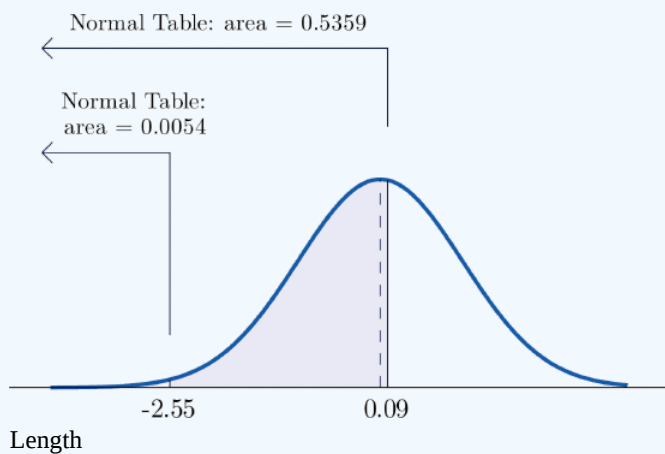


Figure 4.5.6 Computing a Probability for an Interval of Finite Length

The next example shows what to do if the value of Z that we want to look up in the table is not present there.

✓ Example 4.5.4

Find the probabilities indicated.

1. $P(1.13 < Z < 4.16)$.
2. $P(-5.22 < Z < 2.15)$.

Solution

1. We attempt to compute the probability exactly as in Example 4.5.3 by looking up the numbers 1.13 and 4.16 in the table. We obtain the value 0.8708 for the area of the region under the density curve to left of 1.13 without any problem, but when we go to look up the number 4.16 in the table, it is not there. We can see from the last row of numbers in the table that the area to the left of 4.16 must be so close to 1 that to four decimal places it rounds to 1.0000. Therefore

$$P(1.13 < Z < 4.16) = 1.0000 - 0.8708 = 0.1292$$

2. Similarly, here we can read directly from the table that the area under the density curve and to the left of 2.15 is 0.9842 but -5.22 is too far to the left on the number line to be in the table. We can see from the first line of the table that the area to the left of -5.22 must be so close to 0 that to four decimal places it rounds to 0.0000. Therefore

$$P(-5.22 < Z < 2.15) = 0.9842 - 0.0000 = 0.9842$$

The final example of this section explains the origin of the proportions given in the Empirical Rule.

✓ Example 4.5.5

Find the probabilities indicated.

1. $P(-1 < Z < 1)$.
2. $P(-2 < Z < 2)$.
3. $P(-3 < Z < 3)$.

Solution

1. Using the table as was done in Example 4.5.3 we obtain

$$P(-1 < Z < 1) = 0.8413 - 0.1587 = 0.6826$$

Since Z has mean 0 and standard deviation 1, for Z to take a value between -1 and 1 means that Z takes a value that is within one standard deviation of the mean. Our computation shows that the probability that this happens is about 0.68, the proportion given by the Empirical Rule for histograms that are mound shaped and symmetrical, like the bell curve.

2. Using the table in the same way,

$$P(-2 < Z < 2) = 0.9772 - 0.0228 = 0.9544$$

This corresponds to the proportion 0.95 for data within two standard deviations of the mean.

3. Similarly,

$$P(-3 < Z < 3) = 0.9987 - 0.0013 = 0.9974$$

which corresponds to the proportion 0.997 for data within three standard deviations of the mean.

Key takeaway

- A standard normal random variable Z is a normally distributed random variable with mean $\mu = 0$ and standard deviation $\sigma = 1$.
- Probabilities for a standard normal random variable are computed using Figure 4.5.2.

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4.6: Probability Computations for General Normal Random Variables

Learning Objectives

- To learn how to compute probabilities related to any normal random variable.

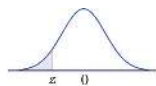
If X is any normally distributed normal random variable then Figure 4.6.1 can also be used to compute a probability of the form $P(a < X < b)$ by means of the following equality.

equality

If X is a normally distributed random variable with mean μ and standard deviation σ , then

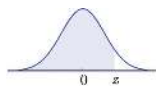
$$P(a < X < b) = P\left(\frac{a - \mu}{\sigma} < Z < \frac{b - \mu}{\sigma}\right)$$

where Z denotes a standard normal random variable. a can be any decimal number or $-\infty$; b can be any decimal number or ∞ .



Cumulative Probability $P(Z \leq z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
-3.8	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
-3.7	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
-3.6	0.0002	0.0002	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
-3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641



Cumulative Probability $P(Z \leq z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981

2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Figure 4.6.1 Cumulative Normal Probability

The new endpoints $\frac{a-\mu}{\sigma}$ and $\frac{b-\mu}{\sigma}$ are the z -scores of a and b as defined in Chapter 2.

Figure 4.6.2 illustrates the meaning of the equality geometrically: the two shaded regions, one under the density curve for X and the other under the density curve for Z , have the same area. Instead of drawing both bell curves, though, we will always draw a single generic bell-shaped curve with both an x -axis and a z -axis below it.

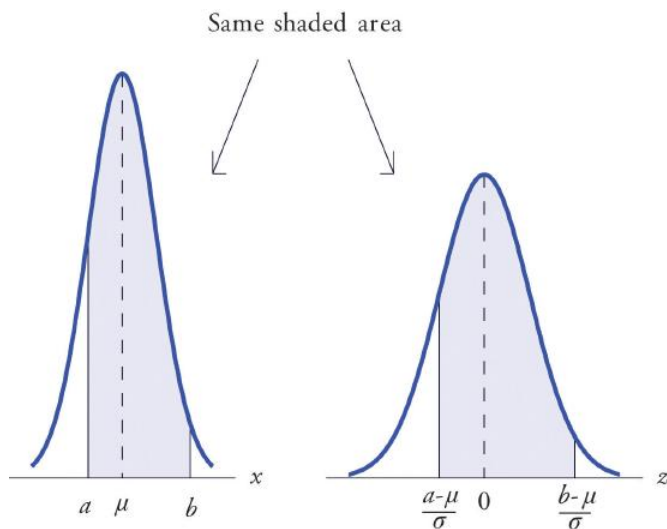


Figure 4.6.2 Probability for an Interval of Finite Length

✓ Example 4.6.1

Let X be a normal random variable with mean $\mu = 10$ and standard deviation $\sigma = 2.5$. Compute the following probabilities.

1. $P(X < 14)$.
2. $P(8 < X < 14)$.

Solution

1. See Figure 4.6.3 "Probability Computation for a General Normal Random Variable".

$$\begin{aligned}
 P(X < 14) &= P\left(Z < \frac{14 - \mu}{\sigma}\right) \\
 &= P\left(Z < \frac{14 - 10}{2.5}\right) \\
 &= P(Z < 1.60) \\
 &= 0.9452
 \end{aligned}$$

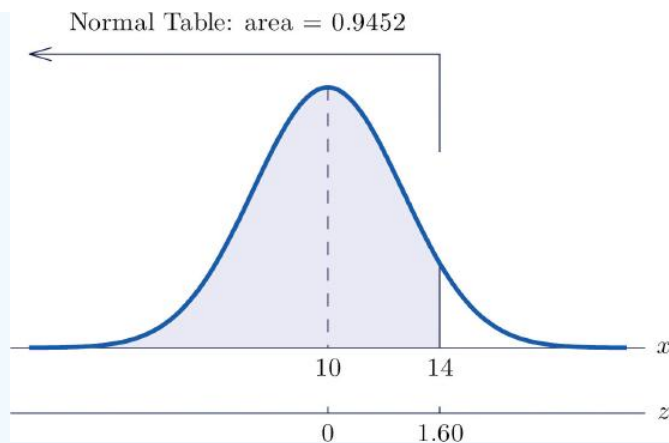


Figure 4.6.3 Probability Computation for a General Normal

Random Variable

2. See Figure 4.6.4 "Probability Computation for a General Normal Random Variable".

$$\begin{aligned} P(8 < X < 14) &= P\left(\frac{8-10}{2.5} < Z < \frac{14-10}{2.5}\right) \\ &= P(-0.80 < Z < 1.60) \\ &= 0.9452 - 0.2119 \\ &= 0.7333 \end{aligned}$$

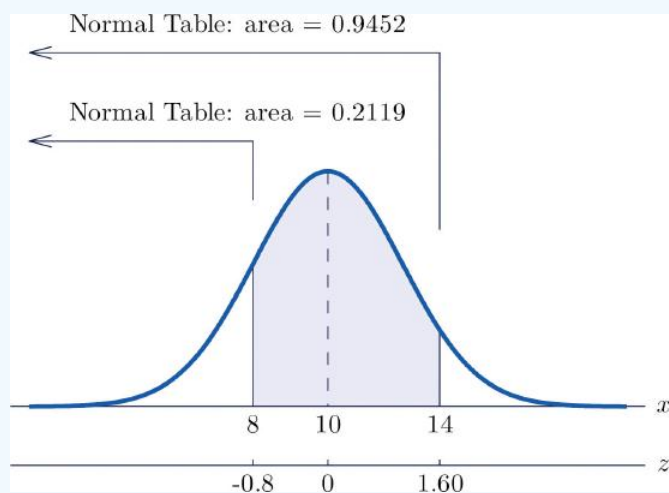


Figure 4.6.4 Probability Computation for a General Normal

Random Variable

✓ Example 4.6.2

The lifetimes of the tread of a certain automobile tire are normally distributed with mean 37,500 miles and standard deviation 4,500 miles. Find the probability that the tread life of a randomly selected tire will be between 30,000 and 40,000 miles.

Solution

Let X denote the tread life of a randomly selected tire. To make the numbers easier to work with we will choose thousands of miles as the units. Thus $\mu = 37.5$, $\sigma = 4.5$, and the problem is to compute $P(30 < X < 40)$. Figure 4.6.5 "Probability Computation for Tire Tread Wear" illustrates the following computation:

$$\begin{aligned}
 P(30 < X < 40) &= P\left(\frac{30 - \mu}{\sigma} < Z < \frac{40 - \mu}{\sigma}\right) \\
 &= P\left(\frac{30 - 37.5}{4.5} < Z < \frac{40 - 37.5}{4.5}\right) \\
 &= P(-1.67 < Z < 0.56) \\
 &= 0.7123 - 0.0475 \\
 &= 0.6648
 \end{aligned}$$

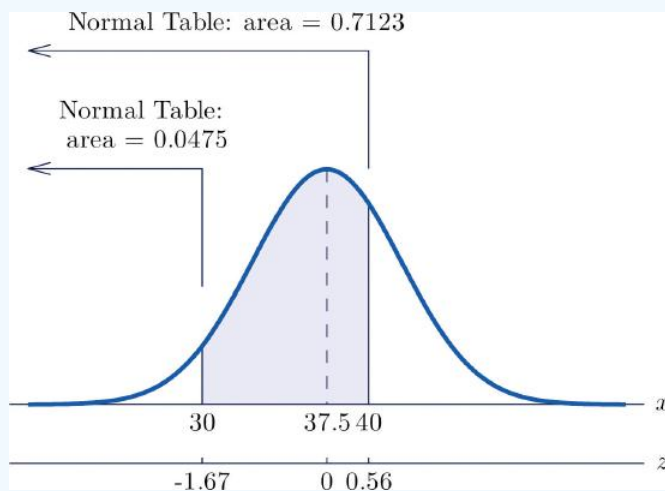


Figure 4.6.5 Probability Computation for Tire Tread Wear

Note that the two z -scores were rounded to two decimal places in order to use Figure 4.6.1 "Cumulative Normal Probability".

✓ Example 4.6.3

Scores on a standardized college entrance examination (*CEE*) are normally distributed with mean 510 and standard deviation 60. A selective university considers for admission only applicants with *CEE* scores over 650. Find percentage of all individuals who took the *CEE* who meet the university's *CEE* requirement for consideration for admission.

Solution

Let X denote the score made on the *CEE* by a randomly selected individual. Then X is normally distributed with mean 510 and standard deviation 60. The probability that X lie in a particular interval is the same as the proportion of all exam scores that lie in that interval. Thus the solution to the problem is $P(X > 650)$, expressed as a percentage. Figure 4.6.6 "Probability Computation for Exam Scores" illustrates the following computation:

$$\begin{aligned}
 P(X > 650) &= P\left(Z > \frac{650 - \mu}{\sigma}\right) \\
 &= P\left(Z > \frac{650 - 510}{60}\right) \\
 &= P(Z > 2.33) \\
 &= 1 - 0.9901 \\
 &= 0.0099
 \end{aligned}$$

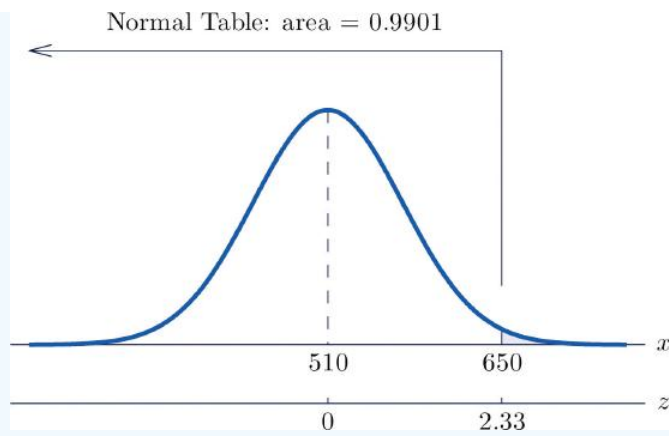


Figure 4.6.6 Probability Computation for Exam Scores

The proportion of all CEE scores that exceed 650 is 0.0099, hence 0.99% or about 1% do.

- Probabilities for a general normal random variable are computed using Figure 4.6.1 after converting x -values to z -scores.

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4.7: Exercises

Here we organize exercises for this chapter by the type of random variable:

- [3.7.1: Discrete Random Variables](#)
 - [3.7.2: Continuous Random Variables](#)
-

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4.7.1: Discrete Random Variables

These are homework exercises to accompany the Textmap created for "Introductory Statistics" by Shafer and Zhang.

4.1: Random Variables

Basic

1. Classify each random variable as either discrete or continuous.
 - a. The number of arrivals at an emergency room between midnight and 6 : 00 *a. m.*
 - b. The weight of a box of cereal labeled "18 ounces."
 - c. The duration of the next outgoing telephone call from a business office.
 - d. The number of kernels of popcorn in a 1-pound container.
 - e. The number of applicants for a job.
2. Classify each random variable as either discrete or continuous.
 - a. The time between customers entering a checkout lane at a retail store.
 - b. The weight of refuse on a truck arriving at a landfill.
 - c. The number of passengers in a passenger vehicle on a highway at rush hour.
 - d. The number of clerical errors on a medical chart.
 - e. The number of accident-free days in one month at a factory.
3. Classify each random variable as either discrete or continuous.
 - a. The number of boys in a randomly selected three-child family.
 - b. The temperature of a cup of coffee served at a restaurant.
 - c. The number of no-shows for every 100 reservations made with a commercial airline.
 - d. The number of vehicles owned by a randomly selected household.
 - e. The average amount spent on electricity each July by a randomly selected household in a certain state.
4. Classify each random variable as either discrete or continuous.
 - a. The number of patrons arriving at a restaurant between 5 : 00 *p. m.* and 6 : 00 *p. m.*
 - b. The number of new cases of influenza in a particular county in a coming month.
 - c. The air pressure of a tire on an automobile.
 - d. The amount of rain recorded at an airport one day.
 - e. The number of students who actually register for classes at a university next semester.
5. Identify the set of possible values for each random variable. (Make a reasonable estimate based on experience, where necessary.)
 - a. The number of heads in two tosses of a coin.
 - b. The average weight of newborn babies born in a particular county one month.
 - c. The amount of liquid in a 12-ounce can of soft drink.
 - d. The number of games in the next World Series (best of up to seven games).
 - e. The number of coins that match when three coins are tossed at once.
6. Identify the set of possible values for each random variable. (Make a reasonable estimate based on experience, where necessary.)
 - a. The number of hearts in a five-card hand drawn from a deck of 52 cards that contains 13 hearts in all.
 - b. The number of pitches made by a starting pitcher in a major league baseball game.
 - c. The number of breakdowns of city buses in a large city in one week.
 - d. The distance a rental car rented on a daily rate is driven each day.
 - e. The amount of rainfall at an airport next month.

Answers

1.
 - a. discrete
 - b. continuous
 - c. continuous
 - d. discrete

- e. discrete
2. a. discrete
b. continuous
c. discrete
d. discrete
e. continuous
3. a. $\{0.1.2\}$
b. an interval (a, b) (answers vary)
c. an interval (a, b) (answers vary)
d. $\{4, 5, 6, 7\}$
e. $\{2, 3\}$

4.2: Probability Distributions for Discrete Random Variables

Basic

1. Determine whether or not the table is a valid probability distribution of a discrete random variable. Explain fully.

a.

x	-2	0	2	4
$P(x)$	0.3	0.5	0.2	0.1

(4.7.1.1)

b.

x	0.5	0.25	0.25
$P(x)$	-0.4	0.6	0.8

(4.7.1.2)

c.

x	1.1	2.5	4.1	4.6	5.3
$P(x)$	0.16	0.14	0.11	0.27	0.22

(4.7.1.3)

2. Determine whether or not the table is a valid probability distribution of a discrete random variable. Explain fully.

a.

x	0	1	2	3	4
$P(x)$	-0.25	0.50	0.35	0.10	0.30

(4.7.1.4)

b.

x	1	2	3
$P(x)$	0.325	0.406	0.164

(4.7.1.5)

c.

x	25	26	27	28	29
$P(x)$	0.13	0.27	0.28	0.18	0.14

(4.7.1.6)

3. A discrete random variable X has the following probability distribution:

x	77	78	79	80	81
$P(x)$	0.15	0.15	0.20	0.40	0.10

(4.7.1.7)

Compute each of the following quantities.

- a. $P(80)$.
 - b. $P(X > 80)$.
 - c. $P(X \leq 80)$.
 - d. The mean μ of X .
 - e. The variance σ^2 of X .
 - f. The standard deviation σ of X .
4. A discrete random variable X has the following probability distribution:

x	13	18	20	24	27
$P(x)$	0.22	0.25	0.20	0.17	0.16

(4.7.1.8)

Compute each of the following quantities.

- $P(18)$.
 - $P(X > 18)$.
 - $P(X \leq 18)$.
 - The mean μ of X .
 - The variance σ^2 of X .
 - The standard deviation σ of X .
5. If each die in a pair is “loaded” so that one comes up half as often as it should, six comes up half again as often as it should, and the probabilities of the other faces are unaltered, then the probability distribution for the sum X of the number of dots on the top faces when the two are rolled is

x	2	3	4	5	6	7
$P(x)$	$\frac{1}{144}$	$\frac{4}{144}$	$\frac{8}{144}$	$\frac{12}{144}$	$\frac{16}{144}$	$\frac{22}{144}$

(4.7.1.9)

x	8	9	10	11	12
$P(x)$	$\frac{24}{144}$	$\frac{20}{144}$	$\frac{16}{144}$	$\frac{12}{144}$	$\frac{9}{144}$

(4.7.1.10)

Compute each of the following.

- $P(5 \leq X \leq 9)$.
- $P(X \geq 7)$.
- The mean μ of X . (For fair dice this number is 7).
- The standard deviation σ of X . (For fair dice this number is about 2.415).

Applications

6. Borachio works in an automotive tire factory. The number X of sound but blemished tires that he produces on a random day has the probability distribution

x	2	3	4	5
$P(x)$	0.48	0.36	0.12	0.04

(4.7.1.11)

- Find the probability that Borachio will produce more than three blemished tires tomorrow.
 - Find the probability that Borachio will produce at most two blemished tires tomorrow.
 - Compute the mean and standard deviation of X . Interpret the mean in the context of the problem.
7. In a hamster breeder's experience the number X of live pups in a litter of a female not over twelve months in age who has not borne a litter in the past six weeks has the probability distribution

x	3	4	5	6	7	8	9
$P(x)$	0.04	0.10	0.26	0.31	0.22	0.05	0.02

(4.7.1.12)

- Find the probability that the next litter will produce five to seven live pups.
 - Find the probability that the next litter will produce at least six live pups.
 - Compute the mean and standard deviation of X . Interpret the mean in the context of the problem.
8. The number X of days in the summer months that a construction crew cannot work because of the weather has the probability distribution

x	6	7	8	9	10
$P(x)$	0.03	0.08	0.15	0.20	0.19

(4.7.1.13)

x	11	12	13	14
$P(x)$	0.16	0.10	0.07	0.02

(4.7.1.14)

- Find the probability that no more than ten days will be lost next summer.
- Find the probability that from 8 to 12 days will be lost next summer.
- Find the probability that no days at all will be lost next summer.
- Compute the mean and standard deviation of X . Interpret the mean in the context of the problem.

9. Let X denote the number of boys in a randomly selected three-child family. Assuming that boys and girls are equally likely, construct the probability distribution of X .
10. Let X denote the number of times a fair coin lands heads in three tosses. Construct the probability distribution of X .
11. Five thousand lottery tickets are sold for \$1 each. One ticket will win \$1,000, two tickets will win \$500 each, and ten tickets will win \$100 each. Let X denote the net gain from the purchase of a randomly selected ticket.
 - a. Construct the probability distribution of X .
 - b. Compute the expected value $E(X)$ of X . Interpret its meaning.
 - c. Compute the standard deviation σ of X .
12. Seven thousand lottery tickets are sold for \$5 each. One ticket will win \$2,000, two tickets will win \$750 each, and five tickets will win \$100 each. Let X denote the net gain from the purchase of a randomly selected ticket.
 - a. Construct the probability distribution of X .
 - b. Compute the expected value $E(X)$ of X . Interpret its meaning.
 - c. Compute the standard deviation σ of X .
13. An insurance company will sell a \$90,000 one-year term life insurance policy to an individual in a particular risk group for a premium of \$478. Find the expected value to the company of a single policy if a person in this risk group has a 99.62% chance of surviving one year.
14. An insurance company will sell a \$10,000 one-year term life insurance policy to an individual in a particular risk group for a premium of \$368. Find the expected value to the company of a single policy if a person in this risk group has a 97.25% chance of surviving one year.
15. An insurance company estimates that the probability that an individual in a particular risk group will survive one year is 0.9825. Such a person wishes to buy a \$150,000 one-year term life insurance policy. Let C denote how much the insurance company charges such a person for such a policy.
 - a. Construct the probability distribution of X . (Two entries in the table will contain C).
 - b. Compute the expected value $E(X)$ of X .
 - c. Determine the value C must have in order for the company to break even on all such policies (that is, to average a net gain of zero per policy on such policies).
 - d. Determine the value C must have in order for the company to average a net gain of \$250 per policy on all such policies.
16. An insurance company estimates that the probability that an individual in a particular risk group will survive one year is 0.99. Such a person wishes to buy a \$75,000 one-year term life insurance policy. Let C denote how much the insurance company charges such a person for such a policy.
 - a. Construct the probability distribution of X . (Two entries in the table will contain C).
 - b. Compute the expected value $E(X)$ of X .
 - c. Determine the value C must have in order for the company to break even on all such policies (that is, to average a net gain of zero per policy on such policies).
 - d. Determine the value C must have in order for the company to average a net gain of \$150 per policy on all such policies.
17. A roulette wheel has 38 slots. Thirty-six slots are numbered from 1 to 36; half of them are red and half are black. The remaining two slots are numbered 0 and 00 and are green. In a \$1 bet on red, the bettor pays \$1 to play. If the ball lands in a red slot, he receives back the dollar he bet plus an additional dollar. If the ball does not land on red he loses his dollar. Let X denote the net gain to the bettor on one play of the game.
 - a. Construct the probability distribution of X .
 - b. Compute the expected value $E(X)$ of X , and interpret its meaning in the context of the problem.
 - c. Compute the standard deviation of X .
18. A roulette wheel has 38 slots. Thirty-six slots are numbered from 1 to 36; the remaining two slots are numbered 0 and 00. Suppose the "number" 00 is considered not to be even, but the number 0 is still even. In a \$1 bet on even, the bettor pays \$1 to play. If the ball lands in an even numbered slot, he receives back the dollar he bet plus an additional dollar. If the ball does not land on an even numbered slot, he loses his dollar. Let X denote the net gain to the bettor on one play of the game.
 - a. Construct the probability distribution of X .
 - b. Compute the expected value $E(X)$ of X , and explain why this game is not offered in a casino (where 0 is not considered even).
 - c. Compute the standard deviation of X .

19. The time, to the nearest whole minute, that a city bus takes to go from one end of its route to the other has the probability distribution shown. As sometimes happens with probabilities computed as empirical relative frequencies, probabilities in the table add up only to a value other than 1.00 because of round-off error.

x	42	43	44	45	46	47
$P(x)$	0.10	0.23	0.34	0.25	0.05	0.02

(4.7.1.15)

- Find the average time the bus takes to drive the length of its route.
 - Find the standard deviation of the length of time the bus takes to drive the length of its route.
20. Tybalt receives in the mail an offer to enter a national sweepstakes. The prizes and chances of winning are listed in the offer as: \$5 million, one chance in 65 million; \$150,000 one chance in 6.5 million; \$5,000 one chance in 650,000 and \$1,000 one chance in 65,000. If it costs Tybalt 44 cents to mail his entry, what is the expected value of the sweepstakes to him?

Additional Exercises

21. The number X of nails in a randomly selected 1-pound box has the probability distribution shown. Find the average number of nails per pound.

x	100	101	102
$P(x)$	0.01	0.96	0.03

(4.7.1.16)

22. Three fair dice are rolled at once. Let X denote the number of dice that land with the same number of dots on top as at least one other die. The probability distribution for X is

x	0	u	3
$P(x)$	p	$\frac{15}{36}$	$\frac{1}{36}$

(4.7.1.17)

- Find the missing value u of X .
 - Find the missing probability p .
 - Compute the mean of X .
 - Compute the standard deviation of X .
23. Two fair dice are rolled at once. Let X denote the difference in the number of dots that appear on the top faces of the two dice. Thus for example if a one and a five are rolled, $X = 4$, and if two sixes are rolled, $X = 0$.
- Construct the probability distribution for X .
 - Compute the mean μ of X .
 - Compute the standard deviation σ of X .
24. A fair coin is tossed repeatedly until either it lands heads or a total of five tosses have been made, whichever comes first. Let X denote the number of tosses made.
- Construct the probability distribution for X .
 - Compute the mean μ of X .
 - Compute the standard deviation σ of X .
25. A manufacturer receives a certain component from a supplier in shipments of 100 units. Two units in each shipment are selected at random and tested. If either one of the units is defective the shipment is rejected. Suppose a shipment has 5 defective units.
- Construct the probability distribution for the number X of defective units in such a sample. (A tree diagram is helpful).
 - Find the probability that such a shipment will be accepted.
26. Shylock enters a local branch bank at 4 : 30 *p. m.* every payday, at which time there are always two tellers on duty. The number X of customers in the bank who are either at a teller window or are waiting in a single line for the next available teller has the following probability distribution.

x	0	1	2	3
$P(x)$	0.135	0.192	0.284	0.230

(4.7.1.18)

x	4	5	6
$P(x)$	0.103	0.051	0.005

(4.7.1.19)

- a. What number of customers does Shylock most often see in the bank the moment he enters?
 - b. What number of customers waiting in line does Shylock most often see the moment he enters?
 - c. What is the average number of customers who are waiting in line the moment Shylock enters?
27. The owner of a proposed outdoor theater must decide whether to include a cover that will allow shows to be performed in all weather conditions. Based on projected audience sizes and weather conditions, the probability distribution for the revenue X per night if the cover is not installed is

<i>Weather</i>	x	$P(x)$
<i>Clear</i>	\$3,000	0.61
<i>Threatening</i>	\$2,800	0.17
<i>Light Rain</i>	\$1,975	0.11
<i>Show – cancelling rain</i>	\$0	0.11

(4.7.1.20)

The additional cost of the cover is \$410,000. The owner will have it built if this cost can be recovered from the increased revenue the cover affords in the first ten 90-night seasons.

- a. Compute the mean revenue per night if the cover is not installed.
- b. Use the answer to (a) to compute the projected total revenue per 90-night season if the cover is not installed.
- c. Compute the projected total revenue per season when the cover is in place. To do so assume that if the cover were in place the revenue each night of the season would be the same as the revenue on a clear night.
- d. Using the answers to (b) and (c), decide whether or not the additional cost of the installation of the cover will be recovered from the increased revenue over the first ten years. Will the owner have the cover installed?

Answers

1. a. no: the sum of the probabilities exceeds 1
b. no: a negative probability
c. no: the sum of the probabilities is less than 1
2. a. 0.4
b. 0.1
c. 0.9
d. 79.15
e. $\sigma^2 = 1.5275$
f. $\sigma = 1.2359$
3. a. 0.6528
b. 0.7153
c. $\mu = 7.8333$
d. $\sigma^2 = 5.4866$
e. $\sigma = 2.3424$
4. a. 0.79
b. 0.60
c. $\mu = 5.8, \sigma = 1.2570$

5.

x	0	1	2	3
$P(x)$	1/8	3/8	3/8	1/8

(4.7.1.21)

6. a.

x	-1	999	499	99
$P(x)$	$\frac{4987}{5000}$	$\frac{1}{5000}$	$\frac{2}{5000}$	$\frac{10}{5000}$

(4.7.1.22)

- b. -0.4
- c. 17.8785

7. 136

8. a.

$$\begin{array}{c|ccc} x & C & C & -150,000 \\ \hline P(x) & 0.9825 & & 0.0175 \end{array} \quad (4.7.1.23)$$

- b. $C - 2625$
- c. $C \geq 2625$
- d. $C \geq 2875$

9. a.

$$\begin{array}{c|ccc} x & -1 & 1 \\ \hline P(x) & \frac{20}{38} & \frac{18}{38} \end{array} \quad (4.7.1.24)$$

- b. $E(X) = -0.0526$. In many bets the bettor sustains an average loss of about 5.25 cents per bet.
- c. 0.9986

10. a. 43.54

b. 1.2046

11. 101.02

12. a.

$$\begin{array}{c|cccccc} x & 0 & 1 & 2 & 3 & 4 & 5 \\ \hline P(x) & \frac{6}{36} & \frac{10}{36} & \frac{8}{36} & \frac{6}{36} & \frac{4}{36} & \frac{2}{36} \end{array} \quad (4.7.1.25)$$

- b. 1.9444
- c. 1.4326

13. a.

$$\begin{array}{c|ccc} x & 0 & 1 & 2 \\ \hline P(x) & 0.902 & 0.096 & 0.002 \end{array} \quad (4.7.1.26)$$

b. 0.902

14. a. 2523.25

b. 227,092.5

c. 270,000

d. The owner will install the cover.

4.3: The Binomial Distribution

Basic

1. Determine whether or not the random variable X is a binomial random variable. If so, give the values of n and p . If not, explain why not.
 - a. X is the number of dots on the top face of fair die that is rolled.
 - b. X is the number of hearts in a five-card hand drawn (without replacement) from a well-shuffled ordinary deck.
 - c. X is the number of defective parts in a sample of ten randomly selected parts coming from a manufacturing process in which 0.02% of all parts are defective.
 - d. X is the number of times the number of dots on the top face of a fair die is even in six rolls of the die.
 - e. X is the number of dice that show an even number of dots on the top face when six dice are rolled at once.
2. Determine whether or not the random variable X is a binomial random variable. If so, give the values of n and p . If not, explain why not.
 - a. X is the number of black marbles in a sample of 5 marbles drawn randomly and without replacement from a box that contains 25 white marbles and 15 black marbles.
 - b. X is the number of black marbles in a sample of 5 marbles drawn randomly and with replacement from a box that contains 25 white marbles and 15 black marbles.
 - c. X is the number of voters in favor of proposed law in a sample 1,200 randomly selected voters drawn from the entire electorate of a country in which 35% of the voters favor the law.
 - d. X is the number of fish of a particular species, among the next ten landed by a commercial fishing boat, that are more than 13 inches in length, when 17% of all such fish exceed 13 inches in length.
 - e. X is the number of coins that match at least one other coin when four coins are tossed at once.

3. X is a binomial random variable with parameters $n = 12$ and $p = 0.82$. Compute the probability indicated.
 - a. $P(11)$
 - b. $P(9)$
 - c. $P(0)$
 - d. $P(13)$
4. X is a binomial random variable with parameters $n = 16$ and $p = 0.74$. Compute the probability indicated.
 - a. $P(14)$
 - b. $P(4)$
 - c. $P(0)$
 - d. $P(20)$
5. X is a binomial random variable with parameters $n = 5$, $p = 0.5$. Use the tables in 7.1: Large Sample Estimation of a Population Mean to compute the probability indicated.
 - a. $P(X \leq 3)$
 - b. $P(X \geq 3)$
 - c. $P(3)$
 - d. $P(0)$
 - e. $P(5)$
6. X is a binomial random variable with parameters $n = 5$, $p = 0.\bar{3}$. Use the tables in 7.1: Large Sample Estimation of a Population Mean to compute the probability indicated.
 - a. $P(X \leq 2)$
 - b. $P(X \geq 2)$
 - c. $P(2)$
 - d. $P(0)$
 - e. $P(5)$
7. X is a binomial random variable with the parameters shown. Use the tables in 7.1: Large Sample Estimation of a Population Mean to compute the probability indicated.
 - a. $n = 10, p = 0.25, P(X \leq 6)$
 - b. $n = 10, p = 0.75, P(X \leq 6)$
 - c. $n = 15, p = 0.75, P(X \leq 6)$
 - d. $n = 15, p = 0.75, P(12)$
 - e. $n = 15, p = 0.\bar{6}, P(10 \leq X \leq 12)$
8. X is a binomial random variable with the parameters shown. Use the tables in 7.1: Large Sample Estimation of a Population Mean to compute the probability indicated.
 - a. $n = 5, p = 0.05, P(X \leq 1)$
 - b. $n = 5, p = 0.5, P(X \leq 1)$
 - c. $n = 10, p = 0.75, P(X \leq 5)$
 - d. $n = 10, p = 0.75, P(12)$
 - e. $n = 10, p = 0.\bar{6}, P(5 \leq X \leq 8)$
9. X is a binomial random variable with the parameters shown. Use the special formulas to compute its mean μ and standard deviation σ .
 - a. $n = 8, p = 0.43$
 - b. $n = 47, p = 0.82$
 - c. $n = 1200, p = 0.44$
 - d. $n = 2100, p = 0.62$
10. X is a binomial random variable with the parameters shown. Use the special formulas to compute its mean μ and standard deviation σ .
 - a. $n = 14, p = 0.55$
 - b. $n = 83, p = 0.05$
 - c. $n = 957, p = 0.35$
 - d. $n = 1750, p = 0.79$

11. X is a binomial random variable with the parameters shown. Compute its mean μ and standard deviation σ in two ways, first using the tables in 7.1: Large Sample Estimation of a Population Mean in conjunction with the general formulas $\mu = \sum xP(x)$ and $\sigma = \sqrt{[\sum x^2 P(x)] - \mu^2}$, then using the special formulas $\mu = np$ and $\sigma = \sqrt{npq}$.
 - a. $n = 5, p = 0.3$
 - b. $n = 10, p = 0.75$
12. X is a binomial random variable with the parameters shown. Compute its mean μ and standard deviation σ in two ways, first using the tables in 7.1: Large Sample Estimation of a Population Mean in conjunction with the general formulas $\mu = \sum xP(x)$ and $\sigma = \sqrt{[\sum x^2 P(x)] - \mu^2}$, then using the special formulas $\mu = np$ and $\sigma = \sqrt{npq}$.
 - a. $n = 10, p = 0.25$
 - b. $n = 15, p = 0.1$
13. X is a binomial random variable with parameters $n = 10$ and $p = 1/3$. Use the cumulative probability distribution for X that is given in 7.1: Large Sample Estimation of a Population Mean to construct the probability distribution of X .
14. X is a binomial random variable with parameters $n = 15$ and $p = 1/2$. Use the cumulative probability distribution for X that is given in 7.1: Large Sample Estimation of a Population Mean to construct the probability distribution of X .
15. In a certain board game a player's turn begins with three rolls of a pair of dice. If the player rolls doubles all three times there is a penalty. The probability of rolling doubles in a single roll of a pair of fair dice is $1/6$. Find the probability of rolling doubles all three times.
16. A coin is bent so that the probability that it lands heads up is $2/3$. The coin is tossed ten times.
 - a. Find the probability that it lands heads up at most five times.
 - b. Find the probability that it lands heads up more times than it lands tails up.

Applications

17. An English-speaking tourist visits a country in which 30% of the population speaks English. He needs to ask someone directions.
 - a. Find the probability that the first person he encounters will be able to speak English.
 - b. The tourist sees four local people standing at a bus stop. Find the probability that at least one of them will be able to speak English.
18. The probability that an egg in a retail package is cracked or broken is 0.025.
 - a. Find the probability that a carton of one dozen eggs contains no eggs that are either cracked or broken.
 - b. Find the probability that a carton of one dozen eggs has (i) at least one that is either cracked or broken; (ii) at least two that are cracked or broken.
 - c. Find the average number of cracked or broken eggs in one dozen cartons.
19. An appliance store sells 20 refrigerators each week. Ten percent of all purchasers of a refrigerator buy an extended warranty. Let X denote the number of the next 20 purchasers who do so.
 - a. Verify that X satisfies the conditions for a binomial random variable, and find n and p .
 - b. Find the probability that X is zero.
 - c. Find the probability that X is two, three, or four.
 - d. Find the probability that X is at least five.
20. Adverse growing conditions have caused 5% of grapefruit grown in a certain region to be of inferior quality. Grapefruit are sold by the dozen.
 - a. Find the average number of inferior quality grapefruit per box of a dozen.
 - b. A box that contains two or more grapefruit of inferior quality will cause a strong adverse customer reaction. Find the probability that a box of one dozen grapefruit will contain two or more grapefruit of inferior quality.
21. The probability that a 7-ounce skein of a discount worsted weight knitting yarn contains a knot is 0.25. Goneril buys ten skeins to crochet an afghan.
 - a. Find the probability that (i) none of the ten skeins will contain a knot; (ii) at most one will.
 - b. Find the expected number of skeins that contain knots.
 - c. Find the most likely number of skeins that contain knots.
22. One-third of all patients who undergo a non-invasive but unpleasant medical test require a sedative. A laboratory performs 20 such tests daily. Let X denote the number of patients on any given day who require a sedative.

- a. Verify that X satisfies the conditions for a binomial random variable, and find n and p .
 - b. Find the probability that on any given day between five and nine patients will require a sedative (include five and nine).
 - c. Find the average number of patients each day who require a sedative.
 - d. Using the cumulative probability distribution for X in 7.1: Large Sample Estimation of a Population Mean find the minimum number x_{min} of doses of the sedative that should be on hand at the start of the day so that there is a 99% chance that the laboratory will not run out.
23. About 2% of alumni give money upon receiving a solicitation from the college or university from which they graduated. Find the average number monetary gifts a college can expect from every 2,000 solicitations it sends.
 24. Of all college students who are eligible to give blood, about 18% do so on a regular basis. Each month a local blood bank sends an appeal to give blood to 250 randomly selected students. Find the average number of appeals in such mailings that are made to students who already give blood.
 25. About 12% of all individuals write with their left hands. A class of 130 students meets in a classroom with 130 individual desks, exactly 14 of which are constructed for people who write with their left hands. Find the probability that exactly 14 of the students enrolled in the class write with their left hands.
 26. A traveling salesman makes a sale on 65% of his calls on regular customers. He makes four sales calls each day.
 - a. Construct the probability distribution of X , the number of sales made each day.
 - b. Find the probability that, on a randomly selected day, the salesman will make a sale.
 - c. Assuming that the salesman makes 20 sales calls per week, find the mean and standard deviation of the number of sales made *per week*.
 27. A corporation has advertised heavily to try to insure that over half the adult population recognizes the brand name of its products. In a random sample of 20 adults, 14 recognized its brand name. What is the probability that 14 or more people in such a sample would recognize its brand name if the actual proportion p of all adults who recognize the brand name were only 0.50?

Additional Exercises

28. When dropped on a hard surface a thumbtack lands with its sharp point touching the surface with probability $2/3$; it lands with its sharp point directed up into the air with probability $1/3$. The tack is dropped and its landing position observed 15 times.
 - a. Find the probability that it lands with its point in the air at least 7 times.
 - b. If the experiment of dropping the tack 15 times is done repeatedly, what is the average number of times it lands with its point in the air?
29. A professional proofreader has a 98% chance of detecting an error in a piece of written work (other than misspellings, double words, and similar errors that are machine detected). A work contains four errors.
 - a. Find the probability that the proofreader will miss at least one of them.
 - b. Show that two such proofreaders working independently have a 99.96% chance of detecting an error in a piece of written work.
 - c. Find the probability that two such proofreaders working independently will miss at least one error in a work that contains four errors.
30. A multiple choice exam has 20 questions; there are four choices for each question.
 - a. A student guesses the answer to every question. Find the chance that he guesses correctly between four and seven times.
 - b. Find the minimum score the instructor can set so that the probability that a student will pass just by guessing is 20% or less.
31. In spite of the requirement that all dogs boarded in a kennel be inoculated, the chance that a healthy dog boarded in a clean, well-ventilated kennel will develop kennel cough from a carrier is 0.008.
 - a. If a carrier (not known to be such, of course) is boarded with three other dogs, what is the probability that at least one of the three healthy dogs will develop kennel cough?
 - b. If a carrier is boarded with four other dogs, what is the probability that at least one of the four healthy dogs will develop kennel cough?
 - c. The pattern evident from parts (a) and (b) is that if $K + 1$ dogs are boarded together, one a carrier and K healthy dogs, then the probability that at least one of the healthy dogs will develop kennel cough is $P(X \geq 1) = 1 - (0.992)^K$, where X is the binomial random variable that counts the number of healthy dogs that develop the condition. Experiment with different values of K in this formula to find the maximum number $K + 1$ of dogs that a kennel owner can board together so that if one of the dogs has the condition, the chance that another dog will be infected is less than 0.05.

32. Investigators need to determine which of 600 adults have a medical condition that affects 2% of the adult population. A blood sample is taken from each of the individuals.
- Show that the expected number of diseased individuals in the group of 600 is 12 individuals.
 - Instead of testing all 600 blood samples to find the expected 12 diseased individuals, investigators group the samples into 60 groups of 10 each, mix a little of the blood from each of the 10 samples in each group, and test each of the 60 mixtures. Show that the probability that any such mixture will contain the blood of at least one diseased person, hence test positive, is about 0.18.
 - Based on the result in (b), show that the expected number of mixtures that test positive is about 11. (Supposing that indeed 11 of the 60 mixtures test positive, then we know that none of the 490 persons whose blood was in the remaining 49 samples that tested negative has the disease. We have eliminated 490 persons from our search while performing only 60 tests.)

Answers

- not binomial; not success/failure.
 - not binomial; trials are not independent.
 - binomial; $n = 10, p = 0.0002$
 - binomial; $n = 6, p = 0.5$
 - binomial; $n = 6, p = 0.5$
- 0.2434
 - 0.2151
 - $0.18^{12} \approx 0$
 - 0
- 0.8125
 - 0.5000
 - 0.3125
 - 0.0313
 - 0.0312
- 0.9965
 - 0.2241
 - 0.0042
 - 0.2252
 - 0.5390
- $\mu = 3.44, \sigma = 1.4003$
 - $\mu = 38.54, \sigma = 2.6339$
 - $\mu = 528, \sigma = 17.1953$
 - $\mu = 1302, \sigma = 22.2432$
- $\mu = 1.6667, \sigma = 1.0541$
 - $\mu = 7.5, \sigma = 1.3693$

7.

x	0	1	2	3
$P(x)$	0.0173	0.0867	0.1951	0.2602

(4.7.1.27)

x	4	5	6	7
$P(x)$	0.2276	0.1365	0.0569	0.0163

(4.7.1.28)

x	8	9	10
$P(x)$	0.0030	0.0004	0.0000

(4.7.1.29)

- 0.0046
- 0.3
 - 0.7599
- $n = 20, p = 0.1$

- b. 0.1216
 - c. 0.5651
 - d. 0.0432
11. a. 0.0563 and 0.2440
- b. 2.5
 - c. 2
12. 40
13. 0.1019
14. 0.0577
15. a. 0.0776
- b. 0.9996
 - c. 0.0016
16. a. 0.0238
- b. 0.0316
 - c. 6

Contributor

- Anonymous

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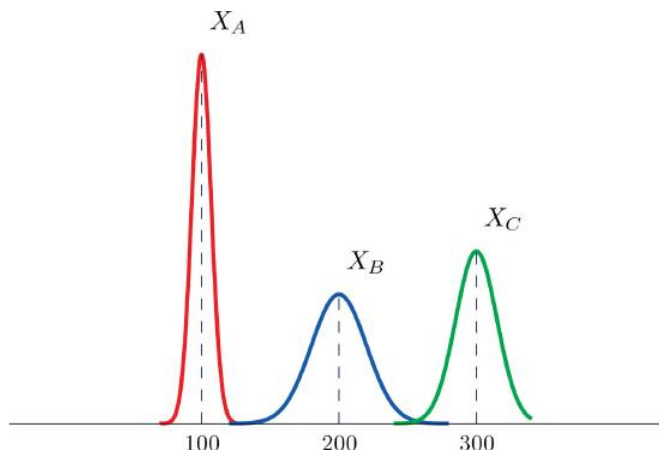
4.7.2: Continuous Random Variables

These are homework exercises to accompany the Textmap created for "Introductory Statistics" by Shafer and Zhang.

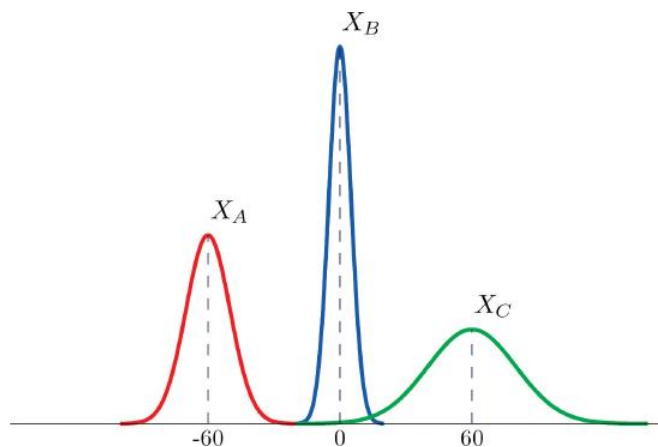
5.1: Continuous Random Variables

Basic

1. A continuous random variable X has a uniform distribution on the interval $[5, 12]$. Sketch the graph of its density function.
2. A continuous random variable X has a uniform distribution on the interval $[-3, 3]$. Sketch the graph of its density function.
3. A continuous random variable X has a normal distribution with mean 100 and standard deviation 10. Sketch a qualitatively accurate graph of its density function.
4. A continuous random variable X has a normal distribution with mean 73 and standard deviation 2.5. Sketch a qualitatively accurate graph of its density function.
5. A continuous random variable X has a normal distribution with mean 73. The probability that X takes a value greater than 80 is 0.212. Use this information and the symmetry of the density function to find the probability that X takes a value less than 66. Sketch the density curve with relevant regions shaded to illustrate the computation.
6. A continuous random variable X has a normal distribution with mean 169. The probability that X takes a value greater than 180 is 0.17. Use this information and the symmetry of the density function to find the probability that X takes a value less than 158. Sketch the density curve with relevant regions shaded to illustrate the computation.
7. A continuous random variable X has a normal distribution with mean 50.5. The probability that X takes a value less than 54 is 0.76. Use this information and the symmetry of the density function to find the probability that X takes a value greater than 47. Sketch the density curve with relevant regions shaded to illustrate the computation.
8. A continuous random variable X has a normal distribution with mean 12.25. The probability that X takes a value less than 13 is 0.82. Use this information and the symmetry of the density function to find the probability that X takes a value greater than 11.50. Sketch the density curve with relevant regions shaded to illustrate the computation.
9. The figure provided shows the density curves of three normally distributed random variables X_A , X_B and X_C . Their standard deviations (in no particular order) are 15, 7, and 20. Use the figure to identify the values of the means μ_A , μ_B , and μ_C and standard deviations σ_A , σ_B , and σ_C of the three random variables.



10. The figure provided shows the density curves of three normally distributed random variables X_A , X_B and X_C . Their standard deviations (in no particular order) are 20, 5, and 10. Use the figure to identify the values of the means μ_A , μ_B , and μ_C and standard deviations σ_A , σ_B , and σ_C of the three random variables.



Applications

11. Dogberry's alarm clock is battery operated. The battery could fail with equal probability at any time of the day or night. Every day Dogberry sets his alarm for 6 : 30 *a. m.* and goes to bed at 10 : 00 *p. m.*. Find the probability that when the clock battery finally dies, it will do so at the most inconvenient time, between 10 : 00 *p. m.* and 6 : 30 *a. m.*.
12. Buses running a bus line near Desdemona's house run every 15 minutes. Without paying attention to the schedule she walks to the nearest stop to take the bus to town. Find the probability that she waits more than 10 minutes.
13. The amount X of orange juice in a randomly selected half-gallon container varies according to a normal distribution with mean 64 ounces and standard deviation 0.25 ounce.
 - a. Sketch the graph of the density function for X .
 - b. What proportion of all containers contain less than a half gallon (64 ounces)? Explain.
 - c. What is the median amount of orange juice in such containers? Explain.
14. The weight X of grass seed in bags marked 50 lb varies according to a normal distribution with mean 50 lb and standard deviation 1 ounce (0.0625lb).
 - a. Sketch the graph of the density function for X .
 - b. What proportion of all bags weigh less than 50 pounds? Explain.
 - c. What is the median weight of such bags? Explain.

Answers

1. The graph is a horizontal line with height $1/7$ from $x = 5$ to $x = 12$
- 2.
3. The graph is a bell-shaped curve centered at 100 and extending from about 70 to 130.
- 4.
5. 0.212
- 6.
7. 0.76
- 8.
9. $\mu_A = 100$, $\mu_B = 200$, $\mu_C = 300$, $\sigma_A = 7$, $\sigma_B = 20$, $\sigma_C = 15$
- 10.
11. 0.3542
- 12.
13.
 - a. The graph is a bell-shaped curve centered at 64 and extending from about 63.25 to 64.75.
 - b. 0.5
 - c. 64

5.2: The Standard Normal Distribution

Basic

- Use Figure 7.1.5: Cumulative Normal Probability to find the probability indicated.

$$/* <![CDATA[*// **/4.7.2.5/ **/*]] > */$$
 - $P(Z < -1.72)$
 - $P(Z < 2.05)$
 - $P(Z < 0)$
 - $P(Z > -2.11)$
 - $P(Z > 1.63)$
 - $P(Z > 2.36)$
- Use Figure 7.1.5: Cumulative Normal Probability to find the probability indicated.

$$/* <![CDATA[*// **/4.7.2.5/ **/*]] > */$$
 - $P(Z < -1.17)$
 - $P(Z < -0.05)$
 - $P(Z < 0.66)$
 - $P(Z > -2.43)$
 - $P(Z > -1.00)$
 - $P(Z > 2.19)$
- Use Figure 7.1.5: Cumulative Normal Probability to find the probability indicated.
 - $P(-2.15 < Z < -1.09)$
 - $P(-0.93 < Z < 0.55)$
 - $P(0.68 < Z < 2.11)$
- Use Figure 7.1.5: Cumulative Normal Probability to find the probability indicated.
 - $P(-1.99 < Z < -1.03)$
 - $P(-0.87 < Z < 1.58)$
 - $P(0.33 < Z < 0.96)$
- Use Figure 7.1.5: Cumulative Normal Probability to find the probability indicated.
 - $P(-4.22 < Z < -1.39)$
 - $P(-1.37 < Z < 5.11)$
 - $P(Z < -4.31)$
 - $P(Z < 5.02)$
- Use Figure 7.1.5: Cumulative Normal Probability to find the probability indicated.
 - $P(Z > -5.31)$
 - $P(-4.08 < Z < 0.58)$
 - $P(Z < -6.16)$
 - $P(-0.51 < Z < 5.63)$
- Use Figure 7.1.5: Cumulative Normal Probability to find the probability listed. Find the second probability without referring to the table, but using the symmetry of the standard normal density curve instead. Sketch the density curve with relevant regions shaded to illustrate the computation.
 - $P(Z < -1.08)$, $P(Z > 1.08)$
 - $P(Z < -0.36)$, $P(Z > 0.36)$
 - $P(Z < 1.25)$, $P(Z > -1.25)$
 - $P(Z < 2.03)$, $P(Z > -2.03)$
- Use Figure 7.1.5: Cumulative Normal Probability to find the probability listed. Find the second probability without referring to the table, but using the symmetry of the standard normal density curve instead. Sketch the density curve with relevant regions shaded to illustrate the computation.
 - $P(Z < -2.11)$, $P(Z > 2.11)$
 - $P(Z < -0.88)$, $P(Z > 0.88)$
 - $P(Z < 2.44)$, $P(Z > -2.44)$
 - $P(Z < 3.07)$, $P(Z > -3.07)$

9. The probability that a standard normal random variable Z takes a value in the union of intervals $(-\infty, -\alpha] \cup [\alpha, \infty)$, which arises in applications, will be denoted $P(Z \leq -a \text{ or } Z \geq a)$. Use Figure 7.1.5: Cumulative Normal Probability to find the following probabilities of this type. Sketch the density curve with relevant regions shaded to illustrate the computation. Because of the symmetry of the standard normal density curve you need to use Figure 7.1.5: Cumulative Normal Probability only one time for each part.
- $P(Z < -1.29 \text{ or } Z > 1.29)$
 - $P(Z < -2.33 \text{ or } Z > 2.33)$
 - $P(Z < -1.96 \text{ or } Z > 1.96)$
 - $P(Z < -3.09 \text{ or } Z > 3.09)$
10. The probability that a standard normal random variable Z takes a value in the union of intervals $(-\infty, -\alpha] \cup [\alpha, \infty)$, which arises in applications, will be denoted $P(Z \leq -a \text{ or } Z \geq a)$. Use Figure 7.1.5: Cumulative Normal Probability to find the following probabilities of this type. Sketch the density curve with relevant regions shaded to illustrate the computation. Because of the symmetry of the standard normal density curve you need to use Figure 7.1.5: Cumulative Normal Probability only one time for each part.
- $P(Z < -2.58 \text{ or } Z > 2.58)$
 - $P(Z < -2.81 \text{ or } Z > 2.81)$
 - $P(Z < -1.65 \text{ or } Z > 1.65)$
 - $P(Z < -2.43 \text{ or } Z > 2.43)$

Answers

- 0.0427
 - 0.9798
 - 0.5
 - 0.9826
 - 0.0516
 - 0.0091
-
- 0.1221
 - 0.5326
 - 0.2309
-
- 0.0823
 - 0.9147
 - 0.0000
 - 1.0000
-
- 0.1401, 0.1401
 - 0.3594, 0.3594
 - 0.8944, 0.8944
 - 0.9788, 0.9788
-
- 0.1970
 - 0.01980
 - 0.0500
 - 0.0020

5.3: Probability Computations for General Normal Random Variables

Basic

- X is a normally distributed random variable with mean 57 and standard deviation 6. Find the probability indicated.
 - $P(X < 59.5)$

- b. $P(X < 46.2)$
 - c. $P(X > 52.2)$
 - d. $P(X > 70)$
2. X is a normally distributed random variable with mean -25 and standard deviation 4 . Find the probability indicated.
 - a. $P(X < -27.2)$
 - b. $P(X < -14.8)$
 - c. $P(X > -33.1)$
 - d. $P(X > -16.5)$
3. X is a normally distributed random variable with mean 112 and standard deviation 15 . Find the probability indicated.
 - a. $P(100 < X < 125)$
 - b. $P(91 < X < 107)$
 - c. $P(118 < X < 160)$
4. X is a normally distributed random variable with mean 72 and standard deviation 22 . Find the probability indicated.
 - a. $P(78 < X < 127)$
 - b. $P(60 < X < 90)$
 - c. $P(49 < X < 71)$
5. X is a normally distributed random variable with mean 500 and standard deviation 25 . Find the probability indicated.
 - a. $P(X < 400)$
 - b. $P(466 < X < 625)$
6. X is a normally distributed random variable with mean 0 and standard deviation 0.75 . Find the probability indicated.
 - a. $P(-4.02 < X < 3.82)$
 - b. $P(X > 4.11)$
7. X is a normally distributed random variable with mean 15 and standard deviation 1 . Use Figure 7.1.5
`/ * <![CDATA[*// **/4.7.2.5/ **// *]] > */Cumulative Normal Probability` to find the first probability listed. Find the second probability using the symmetry of the density curve. Sketch the density curve with relevant regions shaded to illustrate the computation.
 - a. $P(X < 12), P(X > 18)$
 - b. $P(X < 14), P(X > 16)$
 - c. $P(X < 11.25), P(X > 18.75)$
 - d. $P(X < 12.67), P(X > 17.33)$
8. X is a normally distributed random variable with mean 100 and standard deviation 10 . Use Figure 7.1.5
`/ * <![CDATA[*// **/4.7.2.5/ **// *]] > */Cumulative Normal Probability` to find the first probability listed. Find the second probability using the symmetry of the density curve. Sketch the density curve with relevant regions shaded to illustrate the computation.
 - a. $P(X < 80), P(X > 120)$
 - b. $P(X < 75), P(X > 125)$
 - c. $P(X < 84.55), P(X > 115.45)$
 - d. $P(X < 77.42), P(X > 122.58)$
9. X is a normally distributed random variable with mean 67 and standard deviation 13 . The probability that X takes a value in the union of intervals $(-\infty, 67 - a] \cup [67 + a, \infty)$ will be denoted $P(X \leq 67 - a \text{ or } X \geq 67 + a)$. Use Figure 7.1.5
`/ * <![CDATA[*// **/4.7.2.5/ **// *]] > */Cumulative Normal Probability` to find the following probabilities of this type. Sketch the density curve with relevant regions shaded to illustrate the computation. Because of the symmetry of the density curve you need to use Figure 7.1.5/
`/ * <![CDATA[*// **/4.7.2.5/ **// *]] > */Cumulative Normal Probability` only one time for each part.
 - a. $P(X < 57 \text{ or } X > 77)$
 - b. $P(X < 47 \text{ or } X > 87)$
 - c. $P(X < 49 \text{ or } X > 85)$
 - d. $P(X < 37 \text{ or } X > 97)$

10. X is a normally distributed random variable with mean 288 and standard deviation 6. The probability that X takes a value in the union of intervals $(-\infty, 288 - a] \cup [288 + a, \infty)$ will be denoted $P(X \leq 288 - a \text{ or } X \geq 288 + a)$. Use Figure 7.1.5 / * <![CDATA[*// **/4.7.2.5/ **/*]] > */Cumulative Normal Probability to find the following probabilities of this type. Sketch the density curve with relevant regions shaded to illustrate the computation. Because of the symmetry of the density curve you need to use Figure 7.1.5/ * <![CDATA[*// **/4.7.2.5/ **/*]] > */Cumulative Normal Probability only one time for each part.
- $P(X < 278 \text{ or } X > 298)$
 - $P(X < 268 \text{ or } X > 308)$
 - $P(X < 273 \text{ or } X > 303)$
 - $P(X < 280 \text{ or } X > 296)$

Applications

- The amount X of beverage in a can labeled 12 ounces is normally distributed with mean 12.1 ounces and standard deviation 0.05 ounce. A can is selected at random.
 - Find the probability that the can contains at least 12 ounces.
 - Find the probability that the can contains between 11.9 and 12.1 ounces.
- The length of gestation for swine is normally distributed with mean 114 days and standard deviation 0.75 day. Find the probability that a litter will be born within one day of the mean of 114.
- The systolic blood pressure X of adults in a region is normally distributed with mean 112 mm Hg and standard deviation 15 mm Hg. A person is considered “prehypertensive” if his systolic blood pressure is between 120 and 130 mm Hg. Find the probability that the blood pressure of a randomly selected person is prehypertensive.
- Heights X of adult women are normally distributed with mean 63.7 inches and standard deviation 2.71 inches. Romeo, who is 69.25 inches tall, wishes to date only women who are shorter than he but within 4 inches of his height. Find the probability that the next woman he meets will have such a height.
- Heights X of adult men are normally distributed with mean 69.1 inches and standard deviation 2.92 inches. Juliet, who is 63.25 inches tall, wishes to date only men who are taller than she but within 6 inches of her height. Find the probability that the next man she meets will have such a height.
- A regulation hockey puck must weigh between 5.5 and 6 ounces. The weights X of pucks made by a particular process are normally distributed with mean 5.75 ounces and standard deviation 0.11 ounce. Find the probability that a puck made by this process will meet the weight standard.
- A regulation golf ball may not weigh more than 1.620 ounces. The weights X of golf balls made by a particular process are normally distributed with mean 1.361 ounces and standard deviation 0.09 ounce. Find the probability that a golf ball made by this process will meet the weight standard.
- The length of time that the battery in Hippolyta's cell phone will hold enough charge to operate acceptably is normally distributed with mean 25.6 hours and standard deviation 0.32 hour. Hippolyta forgot to charge her phone yesterday, so that at the moment she first wishes to use it today it has been 26 hours 18 minutes since the phone was last fully charged. Find the probability that the phone will operate properly.
- The amount of non-mortgage debt per household for households in a particular income bracket in one part of the country is normally distributed with mean \$28,350 and standard deviation \$3,425. Find the probability that a randomly selected such household has between \$20,000 and \$30,000 in non-mortgage debt.
- Birth weights of full-term babies in a certain region are normally distributed with mean 7.125 lb and standard deviation 1.290 lb. Find the probability that a randomly selected newborn will weigh less than 5.5 lb, the historic definition of prematurity.
- The distance from the seat back to the front of the knees of seated adult males is normally distributed with mean 23.8 inches and standard deviation 1.22 inches. The distance from the seat back to the back of the next seat forward in all seats on aircraft flown by a budget airline is 26 inches. Find the proportion of adult men flying with this airline whose knees will touch the back of the seat in front of them.
- The distance from the seat to the top of the head of seated adult males is normally distributed with mean 36.5 inches and standard deviation 1.39 inches. The distance from the seat to the roof of a particular make and model car is 40.5 inches. Find the proportion of adult men who when sitting in this car will have at least one inch of headroom (distance from the top of the head to the roof).

Additional Exercises

23. The useful life of a particular make and type of automotive tire is normally distributed with mean 57,500 miles and standard deviation 950 miles.
- Find the probability that such a tire will have a useful life of between 57,000 and 58,000 miles.
 - Hamlet buys four such tires. Assuming that their lifetimes are independent, find the probability that all four will last between 57,000 and 58,000 miles. (If so, the best tire will have no more than 1,000 miles left on it when the first tire fails.) Hint: There is a binomial random variable here, whose value of p comes from part (a).
24. A machine produces large fasteners whose length must be within 0.5 inch of 22 inches. The lengths are normally distributed with mean 22.0 inches and standard deviation 0.17 inch.
- Find the probability that a randomly selected fastener produced by the machine will have an acceptable length.
 - The machine produces 20 fasteners per hour. The length of each one is inspected. Assuming lengths of fasteners are independent, find the probability that all 20 will have acceptable length. Hint: There is a binomial random variable here, whose value of p comes from part (a).
25. The lengths of time taken by students on an algebra proficiency exam (if not forced to stop before completing it) are normally distributed with mean 28 minutes and standard deviation 1.5 minutes.
- Find the proportion of students who will finish the exam if a 30-minute time limit is set.
 - Six students are taking the exam today. Find the probability that all six will finish the exam within the 30-minute limit, assuming that times taken by students are independent. Hint: There is a binomial random variable here, whose value of p comes from part (a).
26. Heights of adult men between 18 and 34 years of age are normally distributed with mean 69.1 inches and standard deviation 2.92 inches. One requirement for enlistment in the military is that men must stand between 60 and 80 inches tall.
- Find the probability that a randomly elected man meets the height requirement for military service.
 - Twenty-three men independently contact a recruiter this week. Find the probability that all of them meet the height requirement. Hint: There is a binomial random variable here, whose value of p comes from part (a).
27. A regulation hockey puck must weigh between 5.5 and 6 ounces. In an alternative manufacturing process the mean weight of pucks produced is 5.75 ounce. The weights of pucks have a normal distribution whose standard deviation can be decreased by increasingly stringent (and expensive) controls on the manufacturing process. Find the maximum allowable standard deviation so that at most 0.005 of all pucks will fail to meet the weight standard. (Hint: The distribution is symmetric and is centered at the middle of the interval of acceptable weights.)
28. The amount of gasoline X delivered by a metered pump when it registers 5 gallons is a normally distributed random variable. The standard deviation σ of X measures the precision of the pump; the smaller σ is the smaller the variation from delivery to delivery. A typical standard for pumps is that when they show that 5 gallons of fuel has been delivered the actual amount must be between 4.97 and 5.03 gallons (which corresponds to being off by at most about half a cup). Supposing that the mean of X is 5, find the largest that σ can be so that $P(4.97 < X < 5.03)$ is 1.0000 to four decimal places when computed using Figure 7.1.5: Cumulative Normal Probability which means that the pump is sufficiently accurate. (Hint: The z -score of 5.03 will be the smallest value of Z so that Figure 7.1.5: Cumulative Normal Probability gives $P(Z < z) = 1.0000$).

Answers

- 0.6628
 - 0.0359
 - 0.7881
 - 0.0150
-
- 0.5959
 - 0.2899
 - 0.3439
-
- 0.0000
 - 0.9131
-

7. a. 0.0013, 0.0013
b. 0.1587, 0.1587
c. 0.0001, 0.0001
d. 0.0099, 0.0099
- 8.
9. a. 0.4412
b. 0.1236
c. 0.1676
d. 0.0208
- 10.
11. a. 0.9772
b. 0.5000
- 12.
13. 0.1830
- 14.
15. 0.4971
- 16.
17. 0.9980
- 18.
19. 0.6771
- 20.
21. 0.0359
- 22.
23. a. 0.4038
b. 0.0266
- 24.
25. a. 0.9082
b. 0.5612
- 26.
27. 0.089

5.4: Areas of Tails of Distributions

Basic

1. Find the value of z^* that yields the probability shown.
 - a. $P(Z < z^*) = 0.0075$
 - b. $P(Z < z^*) = 0.9850$
 - c. $P(Z > z^*) = 0.8997$
 - d. $P(Z > z^*) = 0.0110$
2. Find the value of z^* that yields the probability shown.
 - a. $P(Z < z^*) = 0.3300$
 - b. $P(Z < z^*) = 0.9901$
 - c. $P(Z > z^*) = 0.0055$
 - d. $P(Z > z^*) = 0.7995$
3. Find the value of z^* that yields the probability shown.
 - a. $P(Z < z^*) = 0.1500$
 - b. $P(Z < z^*) = 0.7500$
 - c. $P(Z > z^*) = 0.3333$
 - d. $P(Z > z^*) = 0.8000$
4. Find the value of z^* that yields the probability shown.

- a. $P(Z < z^*) = 0.2200$
 - b. $P(Z < z^*) = 0.6000$
 - c. $P(Z > z^*) = 0.0750$
 - d. $P(Z > z^*) = 0.8200$
5. Find the indicated value of Z . (It is easier to find $-z_c$ and negate it.)
- a. $Z_{0.025}$
 - b. $Z_{0.20}$
6. Find the indicated value of Z . (It is easier to find $-z_c$ and negate it.)
- a. $Z_{0.002}$
 - b. $Z_{0.02}$
7. Find the value of x^* that yields the probability shown, where X is a normally distributed random variable X with mean 83 and standard deviation 4.
- a. $P(X < x^*) = 0.8700$
 - b. $P(X > x^*) = 0.0500$
8. Find the value of x^* that yields the probability shown, where X is a normally distributed random variable X with mean 54 and standard deviation 12.
- a. $P(X < x^*) = 0.0900$
 - b. $P(X > x^*) = 0.6500$
9. X is a normally distributed random variable X with mean 15 and standard deviation 0.25. Find the values X_L and X_R of X that are symmetrically located with respect to the mean of X and satisfy $P(X_L < X < X_R) = 0.80$. (Hint. First solve the corresponding problem for Z).
10. X is a normally distributed random variable X with mean 28 and standard deviation 3.7. Find the values X_L and X_R of X that are symmetrically located with respect to the mean of X and satisfy $P(X_L < X < X_R) = 0.65$. (Hint. First solve the corresponding problem for Z).

Applications

11. Scores on a national exam are normally distributed with mean 382 and standard deviation 26.
 - a. Find the score that is the 50th percentile.
 - b. Find the score that is the 90th percentile.
12. Heights of women are normally distributed with mean 63.7 inches and standard deviation 2.47 inches.
 - a. Find the height that is the 10th percentile.
 - b. Find the height that is the 80th percentile.
13. The monthly amount of water used per household in a small community is normally distributed with mean 7,069 gallons and standard deviation 58 gallons. Find the three quartiles for the amount of water used.
14. The quantity of gasoline purchased in a single sale at a chain of filling stations in a certain region is normally distributed with mean 11.6 gallons and standard deviation 2.78 gallons. Find the three quartiles for the quantity of gasoline purchased in a single sale.
15. Scores on the common final exam given in a large enrollment multiple section course were normally distributed with mean 69.35 and standard deviation 12.93. The department has the rule that in order to receive an A in the course his score must be in the top 10% of all exam scores. Find the minimum exam score that meets this requirement.
16. The average finishing time among all high school boys in a particular track event in a certain state is 5 minutes 17 seconds. Times are normally distributed with standard deviation 12 seconds.
 - a. The qualifying time in this event for participation in the state meet is to be set so that only the fastest 5% of all runners qualify. Find the qualifying time. (Hint: Convert seconds to minutes.)
 - b. In the western region of the state the times of all boys running in this event are normally distributed with standard deviation 12 seconds, but with mean 5 minutes 22 seconds. Find the proportion of boys from this region who qualify to run in this event in the state meet.
17. Tests of a new tire developed by a tire manufacturer led to an estimated mean tread life of 67,350 miles and standard deviation of 1,120 miles. The manufacturer will advertise the lifetime of the tire (for example, a "50,000 mile tire") using the largest

value for which it is expected that 98% of the tires will last at least that long. Assuming tire life is normally distributed, find that advertised value.

18. Tests of a new light led to an estimated mean life of 1,321 hours and standard deviation of 106 hours. The manufacturer will advertise the lifetime of the bulb using the largest value for which it is expected that 90% of the bulbs will last at least that long. Assuming bulb life is normally distributed, find that advertised value.
19. The weights X of eggs produced at a particular farm are normally distributed with mean 1.72 ounces and standard deviation 0.12 ounce. Eggs whose weights lie in the middle 75% of the distribution of weights of all eggs are classified as “medium.” Find the maximum and minimum weights of such eggs. (These weights are endpoints of an interval that is symmetric about the mean and in which the weights of 75% of the eggs produced at this farm lie.)
20. The lengths X of hardwood flooring strips are normally distributed with mean 28.9 inches and standard deviation 6.12 inches. Strips whose lengths lie in the middle 80% of the distribution of lengths of all strips are classified as “average-length strips.” Find the maximum and minimum lengths of such strips. (These lengths are endpoints of an interval that is symmetric about the mean and in which the lengths of 80% of the hardwood strips lie.)
21. All students in a large enrollment multiple section course take common in-class exams and a common final, and submit common homework assignments. Course grades are assigned based on students' final overall scores, which are approximately normally distributed. The department assigns a C to students whose scores constitute the middle $2/3$ of all scores. If scores this semester had mean 72.5 and standard deviation 6.14, find the interval of scores that will be assigned a C .
22. Researchers wish to investigate the overall health of individuals with abnormally high or low levels of glucose in the blood stream. Suppose glucose levels are normally distributed with mean 96 and standard deviation 8.5 mg/dl , and that “normal” is defined as the middle 90% of the population. Find the interval of normal glucose levels, that is, the interval centered at 96 that contains 90% of all glucose levels in the population.

Additional Exercises

23. A machine for filling 2-liter bottles of soft drink delivers an amount to each bottle that varies from bottle to bottle according to a normal distribution with standard deviation 0.002 liter and mean whatever amount the machine is set to deliver.
 - a. If the machine is set to deliver 2 liters (so the mean amount delivered is 2 liters) what proportion of the bottles will contain at least 2 liters of soft drink?
 - b. Find the minimum setting of the mean amount delivered by the machine so that at least 99% of all bottles will contain at least 2 liters.
24. A nursery has observed that the mean number of days it must darken the environment of a species poinsettia plant daily in order to have it ready for market is 71 days. Suppose the lengths of such periods of darkening are normally distributed with standard deviation 2 days. Find the number of days in advance of the projected delivery dates of the plants to market that the nursery must begin the daily darkening process in order that at least 95% of the plants will be ready on time. (Poinsettias are so long-lived that once ready for market the plant remains salable indefinitely.)

Answers

1.
 - a. -2.43
 - b. 2.17
 - c. -1.28
 - d. 2.29
- 2.
3.
 - a. -1.04
 - b. 0.67
 - c. 0.43
 - d. -0.84
- 4.
5.
 - a. 1.96
 - b. 0.84
- 6.
7.
 - a. 87.52
 - b. 89.58

- 8.
9. 15.32
- 10.
11. a. 382
b. 415
- 12.
13. 7030.14, 7069, 7107.86
- 14.
15. 85.90
- 16.
17. 65,054
- 18.
19. 1.58, 1.86
- 20.
21. 66.5, 78.5
- 22.
23. a. 0.5
b. 2.005

Contributor

- Anonymous

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CHAPTER OVERVIEW

5: Mathematics of Finance

Learning Objectives

In this chapter, you will learn to:

1. Solve financial problems that involve simple interest.
2. Solve problems involving compound interest.
3. Find the future value of an annuity, and the amount of payments to a sinking fund.
4. Find the future value of an annuity, and an installment payment on a loan.

5.1: Interest

5.1.1: Simple Interest

5.1.2: Compound Interest

5.2: Future Value of Annuities and Sinking Funds

5.3: Present Value of Annuities and Installment Payment

5.4: Classification of Finance Problems

5.5: Additional Application Problems

5.6: Chapter Review

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5.1: Interest

In this section we examine two ways of calculating interest earned:

- [5.1.1: Simple Interest](#)
 - [5.1.2: Compound Interest](#)
-

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5.1.1: Simple Interest

Learning Objectives

In this section, you will learn to:

- Use the simple interest formula to find an account balance, principal, rate, or time.

Prerequisite Skills

Before you get started, take this prerequisite quiz.

1. Solve $5 = 2(x + 9)$.

Click here to check your answer

$$x = -\frac{13}{2}$$

If you missed this problem, [review Section 1.1](#). (Note that this will open in a new window.)

2. Solve $x = -3(7 - 10)$.

Click here to check your answer

$$x = 9$$

If you missed this problem, [review Section 1.1](#). (Note that this will open in a new window.)

3. Solve $-12 = x(3 + 8)$.

Click here to check your answer

$$x = -\frac{12}{11}$$

If you missed this problem, [review Section 1.1](#). (Note that this will open in a new window.)

4. Solve $12 = 4(2 - x)$.

Click here to check your answer

$$x = -1$$

If you missed this problem, [review Section 1.1](#). (Note that this will open in a new window.)

Simple Interest

It costs to borrow money. The rent one pays for the use of money is called the **interest**. The amount of money that is being borrowed or loaned is called the **principal**, also called the present value. Simple interest is paid only on the original amount borrowed. When the money is loaned out, the person who borrows the money generally pays a fixed rate of interest on the principal for the time period he keeps the money. Although the interest rate is most commonly specified for a year, it may be specified for a week, a month, or a quarter, etc. Credit card companies often list their charges as monthly rates, sometimes it is as high as 1.5% a month.

Definition: Simple Interest

If an amount P is borrowed for a time t at an interest rate of r per time period, then the simple interest is given by

$$I = P \cdot r \cdot t \quad (5.1.1.1)$$

 Definition: Accumulated Value

The total amount A , also called the **accumulated value** or the future value, is given by

$$\begin{aligned} A &= P + I \\ &= P + Prt \end{aligned}$$

or

$$A = P(1 + rt) \quad (5.1.1.2)$$

where interest rate r is expressed in decimals.

✓ Example 5.1.1.1

Ursula borrows \$600 for 5 months at a simple interest rate of 15% per year. Find the interest, and the total amount she is obligated to pay?

Solution

The interest is computed by multiplying the principal with the interest rate and the time.

$$\begin{aligned} I &= Prt \\ &= \$600(0.15)\frac{5}{12} \\ &= \$37.50 \end{aligned}$$

The total amount is

$$\begin{aligned} A &= P + I \\ &= \$600 + \$37.50 \\ &= \$637.50 \end{aligned}$$

Incidentally, the total amount can be computed directly via Equation 5.1.1.2 as

$$\begin{aligned} A &= P(1 + rt) \\ &= \$600[1 + (0.15)(5/12)] \\ &= \$600(1 + 0.0625) \\ &= \$637.50 \end{aligned}$$

✓ Example 5.1.1.2

Jose deposited \$2500 in an account that pays 6% simple interest. How much money will he have at the end of 3 years?

Solution

The total amount or the future value is given by Equation 5.1.1.2

$$\begin{aligned} A &= P(1 + rt) \\ &= \$2500[1 + (.06)(3)] \\ A &= \$2950 \end{aligned}$$

✓ Example 5.1.1.3

Darnel owes a total of \$3060 which includes 12% simple interest for the three years he borrowed the money. How much did he originally borrow?

Solution

This time we are asked to compute the principal P via Equation 5.1.1.2

$$\$3060 = P[1 + (0.12)(3)]$$

$$\$3060 = P(1.36)$$

$$\frac{\$3060}{1.36} = P$$

$$\$2250 = P \quad \text{Darnel originally borrowed \$2250.}$$

✓ Example 5.1.1.4

A Visa credit card company charges a 1.5% simple interest finance charge each month on the unpaid balance. If Martha owed \$2350 and has not paid her bill for three months, how much does she owe now?

Solution

Before we attempt the problem, the reader should note that in this problem the rate of finance charge is given per month and not per year.

The total amount Martha owes is the previous unpaid balance plus the finance charge.

$$A = \$2350 + \$2350(.015)(3) = \$2350 + \$105.75 = \$2455.75$$

Alternatively, again, we can compute the amount directly by using formula $A = P(1 + rt)$

$$A = \$2350[1 + (.015)(3)] = \$2350(1.045) = \$2455.75$$

Summary

Below is a summary of the formulas we developed for calculations involving simple interest:

📌 Simple interest

If an amount P is borrowed for a time t at an interest rate of r per time period, then the simple interest is given by

$$I = P \cdot r \cdot t$$

The total amount A , also called the accumulated value or the future value, is given by

$$A = P + I = P + Prt$$

or

$$A = P(1 + rt)$$

where interest rate r is expressed in decimals.

Exercises

PROBLEM SET: SIMPLE INTEREST

Do the following simple interest problems.

- | | |
|--|---|
| 1) If an amount of \$2,000 is borrowed at a simple interest rate of 10% for 3 years, how much is the interest? | 2) You borrow \$4,500 for six months at a simple interest rate of 8%. How much is the interest? |
| 3) John borrows \$2400 for 3 years at 9% simple interest. How much will he owe at the end of 3 years? | 4) Jessica takes a loan of \$800 for 4 months at 12% simple interest. How much does she owe at the end of the 4-month period? |

5) If an amount of \$2,160, which includes a 10% simple interest for 2 years, is paid back, how much was borrowed 2 years earlier?

7) Shanti charged \$800 on her charge card and did not make a payment for six months. If there is a monthly charge of 1.5%, how much does she owe?

9) An amount of \$2000 is borrowed for 3 years. At the end of the three years, \$2660 is paid back. What was the simple interest rate?

6) Jamie just paid off a loan of \$2,544, the principal and simple interest. If he took out the loan six months ago at 12% simple interest, what was the amount borrowed?

8) A credit card company charges 18% interest on the unpaid balance. If you owed \$2000 three months ago and have been delinquent since, how much do you owe?

10) Nancy borrowed \$1,800 and paid back \$1,920, four months later. What was the simple interest rate?

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5.1.2: Compound Interest

Learning Objectives

In this section, you will learn to:

- Find the future value of a lump-sum.
- Find the present value of a lump-sum.
- Find the effective interest rate.

Prerequisite Skills

Before you get started, take this prerequisite quiz.

1. Simplify each expression.

a. $100(3 + 2^2)$

b. $100(3 + 2)^2$

Click here to check your answer

a. 700

b. 2500

If you missed this problem, [review here](#). (Note that this will open a different textbook in a new window.)

2. If an amount of \$2,000 is borrowed at a simple interest rate of 10% for 3 years, how much is the interest?

Click here to check your answer

\$600

If you missed this problem, [review Section 6.1](#). (Note that this will open in a new window.)

3. You borrow \$4,500 for six months at a simple interest rate of 8%. How much is the interest?

Click here to check your answer

\$180

If you missed this problem, [review Section 6.1](#). (Note that this will open in a new window.)

4. John borrows \$2400 for 3 years at 9% simple interest. How much will he owe at the end of 3 years?

Click here to check your answer

\$3048

If you missed this problem, [review Section 6.1](#). (Note that this will open in a new window.)

Compound Interest

In the last section, we examined problems involving simple interest. Simple interest is generally charged when the lending period is short and often less than a year. When the money is loaned or borrowed for a longer time period, if the interest is paid (or charged) not only on the principal, but also on the past interest, then we say the interest is **compounded**.

Suppose we deposit \$200 in an account that pays 8% interest each year. At the end of one year, we will have $\$200 + \$200(.08) = \$200(1 + .08) = \216 .

Now suppose we put this amount, \$216, in the same account. After another year, we will have $\$216 + \$216(.08) = \$216(1 + .08) = \233.28 .

So an initial deposit of \$200 has accumulated to \$233.28 in two years. Further note that had it been simple interest, this amount would have accumulated to only \$232. The reason the amount is slightly higher is because the interest (\$16) we earned the first year, was put back into the account. And this \$16 amount itself earned for one year an interest of $\$16(.08) = \1.28 , thus resulting in the increase. So we have earned interest on the principal as well as on the past interest, and that is why we call it compound interest.

Now suppose we leave this amount, \$233.28, in the bank for another year, the final amount will be $\$233.28 + \$233.28(.08) = \$233.28(1 + .08) = \251.94 .

Now let us look at the mathematical part of this problem so that we can devise an easier way to solve these problems.

After one year, we had $\$200(1 + .08) = \216

After two years, we had $\$216(1 + .08)$

But $\$216 = \$200(1 + .08)$, therefore, the above expression becomes

$$\$200(1 + .08)(1 + .08) = \$200(1 + .08)^2 = \$233.28$$

After three years, we get

$$\$233.28(1 + .08) = \$200(1 + .08)(1 + .08)(1 + .08)$$

which can be written as

$$\$200(1 + .08)^3 = \$251.94$$

Suppose we are asked to find the total amount at the end of 5 years, we will get

$$200(1 + .08)^5 = \$293.87$$

We summarize as follows:

The original amount	\$200	= \$200
The amount after one year	$\$200(1 + .08)$	= \$216
The amount after two years	$\$200(1 + .08)^2$	= \$233.28
The amount after three years	$\$200(1 + .08)^3$	= \$251.94
The amount after five years	$\$200(1 + .08)^5$	= \$293.87
The amount after t years	$\$200(1 + .08)^t$	

Periodically Compounded Interest

Banks often compound interest more than one time a year. Consider a bank that pays 8% interest but compounds it four times a year, or quarterly. This means that every quarter the bank will pay an interest equal to one-fourth of 8%, or 2%.

Now if we deposit \$200 in the bank, after one quarter we will have $\$200 \left(1 + \frac{.08}{4}\right)$ or \$204.

After two quarters, we will have $\$200 \left(1 + \frac{.08}{4}\right)^2$ or \$208.08.

After one year, we will have $\$200 \left(1 + \frac{.08}{4}\right)^4$ or \$216.49.

After three years, we will have $\$200 \left(1 + \frac{.08}{4}\right)^{12}$ or \$253.65, etc.

The original amount	\$200	= \$200
The amount after one quarter	$\$200 \left(1 + \frac{.08}{4}\right)$	= \$204

The amount after two quarters	$\$200\left(1 + \frac{.08}{4}\right)^2$	= \$208.08
The amount after one year	$\$200\left(1 + \frac{.08}{4}\right)^4$	= \$216.49
The amount after two years	$\$200\left(1 + \frac{.08}{4}\right)^8$	= \$234.31
The amount after three years	$\$200\left(1 + \frac{.08}{4}\right)^{12}$	= \$253.65
The amount after five years	$\$200\left(1 + \frac{.08}{4}\right)^{20}$	= \$297.19
The amount after t years	$\$200\left(1 + \frac{.08}{4}\right)^{4t}$	

We can see the formula for compound interest emerge.

Definition: Compound Interest, n times per year

If a lump-sum amount of P dollars is invested at an interest rate r , compounded n times a year, then after t years the final amount is given by

$$A = P\left(1 + \frac{r}{n}\right)^{nt} \quad (5.1.2.1)$$

P is called the **principal** and is also called **the present value**.

Example 5.1.2.1

If \$3500 is invested at 9% compounded monthly, what will the future value be in four years?

Solution

Clearly an interest of $.09/12$ is paid every month for four years. The interest is compounded $4 \times 12 = 48$ times over the four-year period. We get

$$A = \$3500\left(1 + \frac{.09}{12}\right)^{48} = \$3500(1.0075)^{48} = \$5009.92$$

\$3500 invested at 9% compounded monthly will accumulate to \$5009.92 in four years.

Example 5.1.2.2

How much should be invested in an account paying 9% compounded daily for it to accumulate to \$5,000 in five years?

Solution

We know the future value, but need to find the principal.

$$\begin{aligned} \$5000 &= P\left(1 + \frac{.09}{365}\right)^{365 \times 5} \\ \$5000 &= P(1.568225) \\ \$3188.32 &= P \end{aligned}$$

\$3188.32 invested into an account paying 9% compounded daily will accumulate to \$5,000 in five years.

Example 5.1.2.3

If \$4,000 is invested at 4% compounded annually, how long will it take to accumulate to \$6,000?

Solution

$n = 1$ because annual compounding means compounding only once per year. The formula simplifies to $A = (1 + r)^t$ when $n = 1$.

$$\begin{aligned} \$6000 &= 4000(1 + .04)^t \\ \frac{6000}{4000} &= 1.04^t \\ 1.5 &= 1.04^t \end{aligned}$$

We use logarithms to solve for the value of t because the variable t is in the exponent.

$$t = \log_{1.04}(1.5)$$

Using the change of base formula we can solve for t :

$$t = \frac{\ln(1.5)}{\ln(1.04)} = 10.33 \text{ years}$$

It takes 10.33 years for \$4000 to accumulate to \$6000 if invested at 4% interest, compounded annually

Example 5.1.2.4

If \$5,000 is invested now for 6 years what interest rate compounded quarterly is needed to obtain an accumulated value of \$8000.

Solution

We have $n = 4$ for quarterly compounding.

$$\begin{aligned} \$8000 &= \$5000 \left(1 + \frac{r}{4}\right)^{4 \times 6} \\ \frac{\$8000}{\$5000} &= \left(1 + \frac{r}{4}\right)^{24} \\ 1.6 &= \left(1 + \frac{r}{4}\right)^{24} \end{aligned}$$

We use roots to solve for t because the variable r is in the base, whereas the exponent is a known number.

$$\sqrt[24]{1.6} = 1 + \frac{r}{4}$$

Many calculators have a built in “nth root” key or function. In the TI-84 calculator, this is found in the Math menu. Roots can also be calculated as fractional exponents; if necessary, the previous step can be rewritten as

$$1.6^{1/24} = 1 + \frac{r}{4}$$

Evaluating the left side of the equation gives

$$\begin{aligned} 1.0197765 &= 1 + \frac{r}{4} \\ 0.0197765 &= \frac{r}{4} \\ r &= 4(0.0197765) = 0.0791 \end{aligned}$$

An interest rate of 7.91% is needed in order for \$5000 invested now to accumulate to \$8000 at the end of 6 years, with interest compounded quarterly.

Effective Interest Rate

Banks are required to state their interest rate in terms of an “**effective yield**” or “**effective interest rate**”, for comparison purposes. The effective rate is also called the Annual Percentage Yield (APY) or Annual Percentage Rate (APR).

The effective rate is the interest rate compounded annually would be equivalent to the stated rate and compounding periods. We often suppose we invest \$1 over the course of one year to determine the effective rate, as is shown in the next example.

To examine several investments to see which has the best rate, we find and compare the effective rate for each investment.

Example 5.1.2.5

If Bank A pays 7.2% interest compounded monthly, what is the effective interest rate?
 If Bank B pays 7.2% interest compounded semiannually, what is the effective interest rate?
 Which bank pays more interest?

Solution

Bank A: Suppose we deposit \$1 in this bank and leave it for a year, we will get

$$1 \left(1 + \frac{0.072}{12}\right)^{12} = 1.0744$$

$$r_{\text{EFF}} = 1.0744 - 1 = 0.0744$$

We earned interest of \$1.0744 - \$1.00 = \$.0744 on an investment of \$1.

The effective interest rate is 7.44%, often referred to as the APY.

Bank B: The effective rate is calculated as

$$r_{\text{EFF}} = 1 \left(1 + \frac{0.072}{2}\right)^2 - 1 = .0733$$

The effective interest rate is 7.33%.

Bank A pays slightly higher interest, with an effective rate of 7.44%, compared to Bank B with effective rate 7.33%.

Definition: Effective Interest Rate, compounded n times per year

If a bank pays an interest rate r per year, compounded n times a year, then **the effective interest rate** is given by

$$r_{\text{EFF}} = \left(1 + \frac{r}{n}\right)^n - 1 \quad (5.1.2.2)$$

This is also referred to as the **annual percentage yield**, or **APY**.

Continuously Compounded Interest

Interest can be compounded yearly, semiannually, quarterly, monthly, and daily. Using the same calculation methods, we could compound every hour, every minute, and even every second. As the compounding period gets shorter and shorter, we move toward the concept of continuous compounding.

But what do we mean when we say the interest is compounded continuously, and how do we compute such amounts? When interest is compounded "infinitely many times", we say that the interest is **compounded continuously**. Our next objective is to derive a formula to model continuous compounding.

Suppose we put \$1 in an account that pays 100% interest. If the interest is compounded once a year, the total amount after one year will be $\$1(1 + 1) = \2 .

- If the interest is compounded semiannually, in one year we will have $\$1(1 + 1/2)^2 = \2.25
- If the interest is compounded quarterly, in one year we will have $\$1(1 + 1/4)^4 = \2.44
- If the interest is compounded monthly, in one year we will have $\$1(1 + 1/12)^{12} = \2.61
- If the interest is compounded daily, in one year we will have $\$1(1 + 1/365)^{365} = \2.71

We show the results as follows:

Frequency of compounding	Formula	Total amount
Annually	$\$1(1 + 1)$	\$2
Semiannually	$\$1(1 + 1/2)^2$	\$2.25

Frequency of compounding	Formula	Total amount
Quarterly	$\$1(1 + 1/4)^4 = \2.44	\$2.44140625
Monthly	$\$1(1 + 1/12)^{12}$	\$2.61303529
Daily	$\$1(1 + 1/365)^{365}$	\$2.71456748
Hourly	$\$1(1 + 1/8760)^{8760}$	\$2.71812699
Every minute	$\$1(1 + 1/525600)^{525600}$	\$2.71827922
Every Second	$\$1(1 + 1/31536000)^{31536000}$	\$2.71828247
Continuously	$\$1(2.718281828 \dots)$	\$2.718281828...

We have noticed that the \$1 we invested does not grow without bound. It starts to stabilize to an irrational number 2.718281828... given the name "e" after the great mathematician Euler.

In mathematics, we say that as n becomes infinitely large the expression equals $(1 + \frac{1}{n})^n = e$.

Therefore, it is natural that the number e play a part in continuous compounding.

It can be shown that as n becomes infinitely large the expression $(1 + \frac{r}{n})^{nt} = e^{rt}$

Therefore, it follows that if we invest $\$P$ at an interest rate r per year, compounded continuously, after t years the final amount will be given by

$$A = P \cdot e^{rt}$$

Definition: Continuously Compounded Interest

If an amount P is invested for t years at an interest rate r per year, **compounded continuously**, then the future value is given by

$$A = Pe^{rt} \quad (5.1.2.3)$$

Example 5.1.2.6

\$3500 is invested at 9% compounded continuously. Find the future value in 4 years.

Solution

Using the formula for the continuous compounding, we get $A = Pe^{rt}$.

$$A = \$3500e^{0.09 \times 4}$$

$$A = \$3500e^{0.36}$$

$$A = \$5016.65$$

Example 5.1.2.7

If an amount is invested at 7.2% compounded continuously, what is the effective interest rate?

Solution

If we deposit \$1 in the bank at 7.2% compounded continuously for one year, and subtract that \$1 from the final amount, we get the effective interest rate in decimals.

$$r_{\text{EFF}} = 1e^{0.072} - 1$$

$$r_{\text{EFF}} = 1.07466 - 1$$

$$r_{\text{EFF}} = .07466 \text{ or } 7.466\%$$

Definition: Effective Interest Rate, compounded continuously

If a bank pays an interest rate r per year, compounded continuously, then the effective interest rate is given by

$$r_{\text{EFF}} = e^r - 1 \quad (5.1.2.4)$$

Example 5.1.2.8

If an amount is invested at 7% compounded continuously, how long will it take to double?

Solution

We don't know the initial value of the principal but we do know that the accumulated value is double (twice) the principal.

$$P \cdot e^{0.07t} = 2P$$

We divide both sides by P

$$e^{.07t} = 2$$

Using natural logarithm:

$$\begin{aligned} .07t &= \ln(2) \\ t &= \ln(2)/.07 = 9.9 \text{ years} \end{aligned}$$

It takes 9.9 years for money to double if invested at 7% continuous interest.

Example 5.1.2.9

- At the peak growth rate in the 1960's the world's population had a doubling time of 35 years. At that time, approximately what was the growth rate?
- As of 2015, the world population's annual growth rate was approximately 1.14%. Based on that rate, find the approximate doubling time.

Solution

We expect the world's population to grow continuously, not in discrete intervals such as years or months. Therefore, we will use the formula $A = Pe^{rt}$.

- Substituting $2P$ for A and 35 for t gives us the equation

$$2P = P \cdot e^{r(35)}$$

We divide both sides by P :

$$2 = e^{r(35)}$$

Using natural logarithm:

$$\ln(2) = r(35)$$

Dividing both sides by 35:

$$\begin{aligned} \frac{\ln(2)}{35} &= r \\ 0.0198 &= r \end{aligned}$$

The growth rate was approximately 1.98%

- Substituting $2P$ for A and 0.0114 for r gives us the equation

$$2P = P \cdot e^{0.0114t}$$

We divide both sides by P :

$$2 = e^{0.0114t}$$

Using natural logarithm:

$$\ln(2) = 0.0114t$$

Dividing both sides by 0.014:

$$\frac{\ln(2)}{0.0114} = t$$

$$60.8 = t$$

If the world population were to continue to grow at the annual growth rate of 1.14% , it would take approximately 60.8 years for the population to double.

SECTION 8.2 SUMMARY

Below is a summary of the formulas we developed for calculations involving compound interest:

COMPOUND INTEREST n times per year

1. If an amount P is invested for t years at an interest rate r per year, compounded n times a year, then the future value is given by

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

P is called the principal and is also called the present value.

2. If a bank pays an interest rate r per year, compounded n times a year, then the effective interest rate is given by

$$r_{\text{EFF}} = \left(1 + \frac{r}{n} \right)^n - 1$$

CONTINUOUSLY COMPOUNDED INTEREST

3. If an amount P is invested for t years at an interest rate r per year, compounded continuously, then the future value is given by

$$A = Pe^{rt}$$

4. If a bank pays an interest rate r per year, compounded continuously, then the effective interest rate is given by

$$r_{\text{EFF}} = e^r - 1$$

Exercises

PROBLEM SET: COMPOUND INTEREST

Do the following compound interest problems involving a lump-sum amount.

- | | |
|---|--|
| 1) What will the final amount be in 4 years if \$8,000 is invested at 9.2% compounded monthly? | 2) How much should be invested at 10.3% compounded quarterly for it to amount to \$10,000 in 6 years? |
| 3) Lydia's aunt Rose left her \$5,000. Lydia spent \$1,000 on her wardrobe and deposited the rest in an account that pays 6.9% compounded daily. How much money will she have in 5 years? | 4) Thuy needs \$1,850 in eight months for her college tuition. How much money should she deposit lump sum in an account paying 8.2% compounded monthly to achieve that goal? |

5) Bank A pays 5% compounded daily, while Bank B pays 5.12% compounded monthly. Which bank pays more? Explain.

7) Jon's grandfather was planning to give him \$12,000 in 10 years. Jon has convinced his grandfather to pay him \$6,000 now, instead. If Jon invests this \$6,000 at 7.5% compounded continuously, how much money will he have in 10 years?

6) EZ Photo Company needs five copying machines in 2 1/2 years for a total cost of \$15,000. How much money should be deposited now to pay for these machines, if the interest rate is 8% compounded semiannually?

8) What will be the price of a \$20,000 car in 5 years if the inflation rate is 6%?

COMPOUND INTEREST

Do the following compound interest problems.

9) At an interest rate of 8% compounded continuously, how many years will it take to double your money?

11) The City Library ordered a new computer system costing \$158,000; it will be delivered in 6 months, and the full amount will be due 30 days after delivery. How much must be deposited today into an account paying 7.5% compounded monthly to have \$158,000 in 7 months?

13) Find the effective interest rate for an account paying 7.2% compounded quarterly.

15) The population of the African nation of Cameroon was 12 million people in the year 2015; it has been growing at the rate of 2.5% per year. If the population continues to grow that rate, what will the population be in 2030?
(<http://databank.worldbank.org/data on 4/26/2016>)

10) If an investment earns 10% compounded continuously, in how many years will it triple? .

12) Mr. and Mrs. Tran are expecting a baby girl in a few days. They want to put away money for her college education now. How much money should they deposit in an account paying 10.2% so they will have \$100,000 in 18 years to pay for their daughter's educational expenses?

14) If a bank pays 5.75% compounded monthly, what is the effective interest rate?

16) According to the Law of 70, if an amount grows at an annual rate of 1%, then it doubles every seventy years. Suppose a bank pays 5% interest, how long will it take for you to double your money? How about at 15%?

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5.2: Future Value of Annuities and Sinking Funds

Learning Objectives

In this section, you will learn to:

- Find the future value of an annuity.
- Find the amount of payments to a sinking fund.

Prerequisite Skills

Before you get started, take this prerequisite quiz.

1. Solve $24 = 6 \cdot x^3$.

Click here to check your answer

$$x = \sqrt[3]{4} \approx 1.5874$$

If you missed this problem, [review here](#). (Note that this will open a different textbook in a new window.)

2. Solve $24 = 6 \cdot 3^x$.

Click here to check your answer

$$x = \log_3(4) \approx 1.2619$$

If you missed this problem, [review Section 5.3](#). (Note that this will open in a new window.)

3. Simplify $\frac{500(2 - \frac{5}{6})^{5.4} - 1}{\frac{2}{3}}$ in your calculator.

Click here to check your answer

$$16,366.55426$$

If you missed this problem, [review here](#). (Note that this will open a YouTube video in a new window.)

Ordinary Annuities

In the first two sections of this chapter, we examined problems where an amount of money was deposited as a lump sum in an account and was left there for the entire time period. Now we will do problems where timely payments are made in an account. When a sequence of payments of some fixed amount are made in an account at equal intervals of time, we call that an **annuity**. And this is the subject of this section.

Example 5.2.1

If at the end of each month a deposit of \$500 is made in an account that pays 8% compounded monthly, what will the final amount be after five years?

Solution

There are 60 deposits made in this account. The first payment stays in the account for 59 months, the second payment for 58 months, the third for 57 months, and so on. We can use the formula for compound interest to examine each payment.

- The first payment of \$500 will accumulate to an amount of $500(1 + 0.08/12)^{59}$.

- The second payment of \$500 will accumulate to an amount of $\$500(1 + 0.08/12)^{58}$.
- The third payment will accumulate to $\$500(1 + 0.08/12)^{57}$.
- The fourth payment will accumulate to $\$500(1 + 0.08/12)^{56}$.

And so on . . .

Finally the next to last (59th) payment will accumulate to $\$500(1 + 0.08/12)^1$.

The last payment is taken out the same time it is made, and will not earn any interest.

To find the total amount in five years, we need to add the accumulated value of these sixty payments.

In other words, we need to find the sum of the following series.

$$\$500(1 + 0.08/12)^{59} + \$500(1 + 0.08/12)^{58} + \$500(1 + 0.08/12)^{57} + \dots + \$500$$

Written backwards, we have

$$\$500 + \$500(1 + 0.08/12) + \$500(1 + 0.08/12)^2 + \dots + \$500(1 + 0.08/12)^{59}$$

This is a geometric series with $a = \$500$, $r = (1 + 0.08/12)$, and $n = 59$. The sum is

$$\begin{aligned} \text{sum} &= \frac{\$500 [(1 + 0.08/12)^{60} - 1]}{0.08/12} \\ &= \$500(73.47686) \\ &= \$36,738.43 \end{aligned}$$

When the payments are made at the end of each period rather than at the beginning, we call it an **ordinary annuity**.

Future Value of an Ordinary Annuity

If a payment of m dollars is made in an account n times a year at an interest r , then the final amount A after t years is

$$A = \frac{m [(1 + r/n)^{nt} - 1]}{r/n} \quad (5.2.1)$$

The final amount is also called the future value or the accumulated value.

Example 5.2.2

Tanya deposits \$300 at the end of each quarter in her savings account. If the account earns 5.75% compounded quarterly, how much money will she have in 4 years?

Solution

The future value of this annuity can be found using the above formula.

$$\begin{aligned} A &= \frac{\$300 [(1 + .0575/4)^{16} - 1]}{0.0575/4} \\ &= \$300(17.8463) \\ &= \$5353.89 \end{aligned}$$

If Tanya deposits \$300 into a savings account earning 5.75% compounded quarterly for 4 years, then at the end of 4 years she will have \$5,353.89

Example 5.2.3

Robert needs \$5,000 in three years. How much should he deposit each month in an account that pays 8% compounded monthly in order to achieve his goal?

Solution

If Robert saves m dollars per month, after three years he will have

$$\frac{m [(1 + .08/12)^{36} - 1]}{0.08/12}$$

But we'd like this amount to be \$5,000. Therefore,

$$\begin{aligned} \frac{m [(1 + .08/12)^{36} - 1]}{.08/12} &= \$5000 \\ m(40.5356) &= \$5000 \\ m &= \frac{5000}{40.5356} \\ &= \$123.35 \end{aligned}$$

Robert needs to deposit \$123.35 at the end of each month for 3 years into an account paying 8% compounded monthly in order to have \$5,000 at the end of 5 years.

Example 5.2.4

Suppose you contribute \$500 each month into an ordinary annuity earning 6.25% compounded monthly. How long do you need to continue making contributions so that you will have at least \$1,000,000?

Solution

Contributing \$500 each month, $m = 500$. Interest is 6.25% compounded monthly, so $r = 0.0625$ and $n = 12$. You want a future value of \$1,000,000 so $A = 1,000,000$.

$$1,000,000 = \frac{500 [(1 + 0.0625/12)^{12t} - 1]}{0.0625/12}$$

Multiply both sides by $(0.0625/12)$ to clear the denominator:

$$1,000,000 \cdot (0.0625/12) = 500 [(1 + 0.0625/12)^{12t} - 1]$$

Divide both sides by 500:

$$\frac{1,000,000 \cdot (0.0625/12)}{500} = (1 + 0.0625/12)^{12t} - 1$$

Add 1 to both sides:

$$\frac{1,000,000 \cdot (0.0625/12)}{500} + 1 = (1 + 0.0625/12)^{12t}$$

Convert to logarithmic form to isolate the exponent:

$$\log_{1+0.0625/12} \left(\frac{1,000,000 \cdot (0.0625/12)}{500} + 1 \right) = 12t$$

Use change of base formula:

$$\frac{\log \left(\frac{1,000,000 \cdot (0.0625/12)}{500} + 1 \right)}{\log(1 + 0.0625/12)} = 12t$$

Enter into calculator:

$$468.75 = t$$

Divide both sides by 12:

$$39.06 = t$$

It will take approximately 39 years for you to have at least \$1,000,000.

Sinking Fund

When a business deposits money at regular intervals into an account in order to save for a future purchase of equipment, the savings fund is referred to as a “**sinking fund**”. Calculating the sinking fund deposit uses the same method as the previous problem.

Example 5.2.5

A business needs \$450,000 in five years. How much should be deposited each quarter in a sinking fund that earns 9% compounded quarterly to have this amount in five years?

Solution

Again, suppose that m dollars are deposited each quarter in the sinking fund. After five years, the future value of the fund should be \$450,000. This suggests the following relationship:

$$\frac{m[(1 + 0.09/4)^{20} - 1]}{0.09/4} = \$450,000$$

$$m(24.9115) = 450,000$$

$$m = \frac{450000}{24.9115}$$

$$= \$18,063.93$$

The business needs to deposit \$18,063.93 at the end of each quarter for 5 years into an sinking fund earning interest of 9% compounded quarterly in order to have \$450,000 at the end of 5 years.

Annuity Due

If the payment is made at the beginning of each period, rather than at the end, we call it an **annuity due**. The formula for the annuity due can be derived in a similar manner. Reconsider Example 1, with the change that the deposits are made at the beginning of each month.

Example 5.2.6

If at the beginning of each month a deposit of \$500 is made in an account that pays 8% compounded monthly, what will the final amount be after five years?

Solution

There are 60 deposits made in this account. The first payment stays in the account for 60 months, the second payment for 59 months, the third for 58 months, and so on.

- The first payment of \$500 will accumulate to an amount of $\$500(1 + 0.08/12)^{60}$.
- The second payment of \$500 will accumulate to an amount of $\$500(1 + 0.08/12)^{59}$.
- The third payment will accumulate to $\$500(1 + 0.08/12)^{58}$.

And so on . . .

The last payment is in the account for a month and accumulates to $\$500(1 + 0.08/12)$

To find the total amount in five years, we need to find the sum of the series:

$$\$500(1 + 0.08/12)^{60} + \$500(1 + 0.08/12)^{59} + \$500(1 + 0.08/12)^{58} + \dots + \$500(1 + 0.08/12)$$

Written backwards, we have

$$\$500(1 + 0.08/12) + \$500(1 + 0.08/12)^2 + \dots + \$500(1 + 0.08/12)^{60}$$

If we add \$500 to this series, and later subtract that \$500, the value will not change. We get

$$\$500 + \$500(1 + 0.08/12) + \$500(1 + 0.08/12)^2 + \dots + \$500(1 + 0.08/12)^{60} - \$500$$

Except for the last term, we have a geometric series with $a = \$500$, $r = (1 + .08/12)$, and $n = 60$. Therefore the sum is

$$\begin{aligned} A &= \frac{\$500 [(1 + 0.08/12)^{61} - 1]}{0.08/12} - \$500 \\ &= \$500(74.9667) - \$500 \\ &= \$37483.35 - \$500 \\ &= \$36983.35 \end{aligned}$$

So, in the case of an annuity due, to find the future value, we increase the number of periods n by 1, and subtract one payment.

Future Value of an "Annuity Due"

$$A = \frac{m [(1 + r/n)^{nt+1} - 1]}{r/n} - m$$

Most of the problems we are going to do in this chapter involve ordinary annuities, therefore, we will down play the significance of the last formula for the annuity due. We mentioned the formula for the annuity due only for completeness.

Summary

Finally, it is the author's wish that the student learn the concepts in a way that he or she will not have to memorize every formula. It is for this reason formulas are kept at a minimum. But before we conclude this section we will once again mention one single equation that will help us find the future value, as well as the sinking fund payment.

If a payment of m dollars is made in an account n times a year at an interest r , then the future value A after t years is

$$A = \frac{m [(1 + r/n)^{nt} - 1]}{r/n}$$

Note that the formula assumes that the payment period is the same as the compounding period. If these are not the same, then this formula does not apply.

Exercises

PROBLEM SET: ANNUITIES AND SINKING FUNDS

Each of the following problems involve an annuity - a sequence of payments.

1) Find the future value of an annuity of \$200 per month for 5 years at 6% compounded monthly.	2) How much money should be deposited at the end of each month in an account paying 7.5% for it to amount to \$10,000 in 5 years?
3) At the end of each month Rita deposits \$300 in an account that pays 5%. What will the final amount be in 4 years?	4) Mr. Chang wants to retire in 10 years and can save \$650 every three months. If the interest rate is 7.8%, how much will he have (a) at the end of 5 years? (b) at the end of 10 years?
5) A firm needs to replace most of its machinery in five years at a cost of \$500,000. The company wishes to create a sinking fund to have this money available in five years. How much should the quarterly deposits be if the fund earns 8%?	6) Mrs. Brown needs \$5,000 in three years. If the interest rate is 9%, how much should she save at the end of each month to have that amount in three years?
7) A company has a \$120,000 note due in 4 years. How much should be deposited at the end of each quarter in a sinking fund to payoff the note in four years if the interest rate is 8%?	8) You are now 20 years of age and decide to save \$100 at the end of each month until you are 65. If the interest rate is 9.2%, how much money will you have when you are 65?
9) Is it better to receive \$400 at the beginning of each month for six years, or a lump sum of \$25,000 today if the interest rate is 7%? Explain.	10) To save money for a vacation, Jill decided to save \$125 at the beginning of each month for the next 8 months. If the interest rate is 7%, how much money will she have at the end of 8 months?

11) Mrs. Gill puts \$2200 at the end of each year in her IRA account that earns 9% per year. How much total money will she have in this account after 20 years?

12) If the inflation rate stays at 6% per year for the next five years, how much will the price be of a \$15,000 car in five years? How much must you save at the end of each month at an interest rate of 7.3% to buy that car in 5 years?

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5.3: Present Value of Annuities and Installment Payment

Learning Objectives

In this section, you will learn to:

- Find the present value of an annuity.
- Find the amount of installment payment on a loan.
- Find the outstanding balance on a loan.

Prerequisite Skills

Before you get started, take this prerequisite quiz.

1. What will the final amount be in 4 years if \$8,000 is invested at 9.2% compounded monthly?

Click here to check your answer

\$11,542.53

If you missed this problem, [review Section 6.2](#). (Note that this will open in a new window.)

2. How much should be invested at 10.3% compounded quarterly for it to amount to \$10,000 in 6 years?

Click here to check your answer

\$5,432.55

If you missed this problem, [review Section 6.2](#). (Note that this will open in a new window.)

3. Find the future value of an annuity of \$200 per month for 5 years at 6% compounded monthly.

Click here to check your answer

\$13,954.01

If you missed this problem, [review Section 6.3](#). (Note that this will open in a new window.)

4. How much money should be deposited at the end of each month in an account paying 7.5% for it to amount to \$10,000 in 5 years?

Click here to check your answer

\$137.88

If you missed this problem, [review Section 6.3](#). (Note that this will open in a new window.)

PRESENT VALUE OF AN ANNUITY

In the previous two sections, we learned to find the future value of a lump sum and the future value of an annuity. With these two concepts in hand, we will now learn to amortize a loan, and to find the present value of an annuity.

The **present value** of an annuity is the amount of money we would need now in order to be able to make the payments in the annuity in the future. In other words, the present value is the value now of a future stream of payments.

We start by breaking this down step by step to understand the concept of the present value of an annuity. After that, the examples provide a more efficient way to do the calculations by working with concepts and calculations we have already explored in the last two sections.

Suppose Carlos owns a small business and employs an assistant manager to help him run the business. Assume it is January 1 now. Carlos plans to pay his assistant manager a \$1000 bonus at the end of this year and another \$1000 bonus at the end of the following year. Carlos' business had good profits this year so he wants to put the money for his assistant's future bonuses into a savings account now. The money he puts in now will earn interest at the rate of 4% per year compounded annually while in the savings

account.

How much money should Carlos put into the savings account now so that he will be able to withdraw \$1000 one year from now and another \$1000 two years from now?

At first, this sounds like a sinking fund. But it is different. In a sinking fund, we put money into the fund with periodic payments to save to accumulate to a specified lump sum that is the future value at the end of a specified time period.

In this case we want to put a lump sum into the savings account now, so that lump sum is our principal, P . Then we want to withdraw that amount as a series of period payments; in this case the withdrawals are an annuity with \$1000 payments at the end of each of two years.

We need to determine the amount we need in the account now, the present value, to be able to make withdraw the periodic payments later.



We use the compound interest formula from Section 6.2 with $r = 0.04$ and $n = 1$ for annual compounding to determine the present value of each payment of \$1000.

Consider the first payment of \$1000 at the end of year 1. Let P_1 be its present value

$$\$1000 = P_1(1.04)^1 \text{ so } P_1 = \$961.54$$

Now consider the second payment of \$1000 at the end of year 2. Let P_2 be its present value

$$\$1000 = P_2(1.04)^2 \text{ so } P_2 = \$924.56$$

To make the \$1000 payments at the specified times in the future, the amount that Carlos needs to deposit now is the present value $P = P_1 + P_2 = \$961.54 + \$924.56 = \$1886.10$

The calculation above was useful to illustrate the meaning of the present value of an annuity.

But it is not an efficient way to calculate the present value. If we were to have a large number of annuity payments, the step by step calculation would be long and tedious.

Example 5.3.1 investigates and develops an efficient way to calculate the present value of an annuity, by relating the future (accumulated) value of an annuity and its present value.

✓ Example 5.3.1

Suppose you have won a lottery that pays \$1,000 per month for the next 20 years. But, you prefer to have the entire amount now. If the interest rate is 8%, how much will you accept?

Solution

This classic present value problem needs our complete attention because the rationalization we use to solve this problem will be used again in the problems to follow.

Consider, for argument purposes, that two people Mr. Cash, and Mr. Credit have won the same lottery of \$1,000 per month for the next 20 years. Mr. Credit is happy with his \$1,000 monthly payment, but Mr. Cash wants to have the entire amount now.

Our job is to determine how much Mr. Cash should get. We reason as follows:

If Mr. Cash accepts P dollars, then the P dollars deposited at 8% for 20 years should yield the same amount as the \$1,000 monthly payments for 20 years. In other words, we are comparing the future values for both Mr. Cash and Mr. Credit, and we would like the future values to equal.

Since Mr. Cash is receiving a lump sum of x dollars, its future value is given by the lump sum formula, and it is

$$A = P(1 + .08/12)^{240}$$

Since Mr. Credit is receiving a sequence of payments, or an annuity, of \$1,000 per month, its future value is given by the annuity formula. This value is

$$A = \frac{\$1000 [(1 + .08/12)^{240} - 1]}{.08/12}$$

The only way Mr. Cash will agree to the amount he receives is if these two future values are equal. So we set them equal and solve for the unknown.

$$\begin{aligned} P(1 + .08/12)^{240} &= \frac{\$1000 [(1 + .08/12)^{240} - 1]}{.08/12} \\ P &= \frac{\frac{\$1000 [(1 + .08/12)^{240} - 1]}{.08/12}}{(1 + .08/12)^{240}} \\ P &= \$119,554.36 \end{aligned}$$

The present value of an ordinary annuity of \$1,000 each month for 20 years at 8% is \$119,554.36

The reader should also note that if Mr. Cash takes his lump sum of $P = \$119,554.36$ and invests it at 8% compounded monthly, he will have an accumulated value of $A = \$589,020.41$ in 20 years.

If a person or business needs to buy or pay for something now (a car, a home, college tuition, equipment for a business) but does not have the money, they can borrow the money as a loan.

They receive the loan amount called the principal (or present value) now and are obligated to pay back the principal in the future over a stated amount of time (term of the loan), as regular periodic payments with interest.

We can see the formula begin to develop in the example above.

Definition: Present Value of an Annuity

If a payment of m dollars is made in an account n times a year at an interest r , then the **present value P of the annuity** after t years is

$$P(1 + r/n)^{nt} = \frac{m [(1 + r/n)^{nt} - 1]}{r/n} \quad (5.3.1)$$

When used for a loan, the amount P is the **loan amount**, and m is the periodic payment needed to repay the loan over a term of t years with n payments per year.

Note that the formula assumes that the payment period is the same as the compounding period. If these are not the same, then this formula does not apply.

✓ Example 5.3.2

You determine that you can afford \$550 per month for a car. At the dealership, the finance department offers you a 3-year loan with a 7% interest rate, compounded monthly.

- What is the highest priced car you can afford in this scenario?
- How much will you pay over the 3 years?
- What is the difference between the answers for part a and part b, and what does it represent?

Solution

a. Using the formula above, we are looking for the total amount of the loan, or P . We know that $m = 550$, $r = 0.07$, $n = 12$, and $t = 3$.

$$P(1 + 0.07/12)^{(12 \cdot 3)} = \frac{550 [(1 + 0.07/12)^{(12 \cdot 3)} - 1]}{0.07/12}$$

$$P = \frac{550 \left[\frac{(1+0.07/12)^{(12 \cdot 3)} - 1}{0.07/12} \right]}{(1+0.07/12)^{(12 \cdot 3)}}$$

$$P = 17812.55545$$

You can take a loan of \$17,812.55 to pay off within 3 years.

b. Over the course of 3 years, you would pay \$550 every month, 12 months each year, for 3 years.

Total paid = $\$550 \cdot 12 \cdot 3 = \$19,800$.

c. The difference between \$17,812.55 as the purchase price and \$19,800 as the repayment price is $\$19800 - \$17812.55 = \$1987.45$. This means that you'll pay \$1,987.45 in interest over the 3 years.

Alternate Method to find Present Value of an Annuity

We should note that many finite mathematics and finance books develop the formula for the present value of an annuity differently.

Instead of using the formula:

$$P(1 + r/n)^{nt} = \frac{m[(1 + r/n)^{nt} - 1]}{r/n} \quad (5.3.2)$$

and solving for the present value P after substituting the numerical values for the other items in the formula, many textbooks first solve the formula for P in order to develop a new formula for the present value. Then the numerical information can be substituted into the present value formula and evaluated, without needing to solve algebraically for P .

Starting with formula 5.3.2: $P(1 + r/n)^{nt} = \frac{m[(1 + r/n)^{nt} - 1]}{r/n}$

Divide both sides by $(1 + r/n)^{nt}$ to isolate P , and simplify

$$P = \frac{m[(1 + r/n)^{nt} - 1]}{r/n} \cdot \frac{1}{(1 + r/n)^{nt}}$$

$$P = \frac{m[1 - (1 + r/n)^{-nt}]}{r/n} \quad (5.3.3)$$

The authors of this book believe that it is easier to use formula 5.3.2 at the top of this page and solve for P or m as needed. In this approach there are fewer formulas to understand, and many students find it easier to learn. In the problems the rest of this chapter, when a problem requires the calculation of the present value of an annuity, formula 5.3.2 will be used.

However, formula 5.3.3 is ideal when used to solve for t , and some people prefer to use this formula to find present value. It is a mathematically correct option to do so. Note that if you choose to use formula 5.3.3, you need to be careful with the negative exponents in the formula.

INSTALLMENT PAYMENTS ON A LOAN

Example 5.3.3 examines how to calculate the loan payment, using reasoning similar to Example 5.3.1.

✓ Example 5.3.3

Daria purchased a truck costing \$45,000 and accepted a 60-month loan at an interest rate of 9%, compounded monthly.

- Find the monthly installment payments.
- How much will she pay over the course of the loan?
- How much interest will she pay over the course of the loan?

Solution

a. The monthly installment payments can be found in a similar process to finding the principal in Example 5.3.2. Note that 60 months is the same as 5 years. We can use $t = 5$ or we can use $nt = 60$.

$$\$45,000(1 + .09/12)^{60} = \frac{m[(1 + .09/12)^{60} - 1]}{.09/12}$$

We then multiply both sides by the denominator:

$$(\$45,000(1 + .09/12)^{60})(\frac{.09}{12}) = m [(1 + .09/12)^{60} - 1]$$

Finally dividing by the entire quantity that is multiplied to the m leaves:

$$\frac{(\$45,000(1 + .09/12)^{60})(\frac{.09}{12})}{[(1 + .09/12)^{60} - 1]} = m$$

$$\$934.13 = m$$

Therefore, the monthly payment needed to repay the loan is \$934.13 for five years.

b. Over the course of the loan, she would pay \$934.13 every month for 60 months.

Total paid = $\$934.13 \cdot 60 = \56047.80

c. To find interest, we find the difference between purchase price and repayment price. $\$56047.80 - \$45000 = \$16047.80$ so she will pay \$16047.80 in interest over the course of the loan.

OUTSTANDING BALANCE ON A LOAN

One of the most common problems deals with finding the balance owed at a given time during the life of a loan. Suppose a person buys a house and amortizes the loan over 30 years, but decides to sell the house a few years later. At the time of the sale, he is obligated to pay off his lender, therefore, he needs to know the balance he owes. Since most long term loans are paid off prematurely, we are often confronted with this problem.

To find the outstanding balance of a loan at a specified time, we need to find the present value P of all future payments that have not yet been paid. In this case t does not represent the entire term of the loan. Instead:

- t represents the time that still remains on the loan
- nt represents the total number of future payments.

If the problem does not directly state the amount of time still remaining in the term of the loan, then it must be calculated FIRST using $t = [\text{original term of loan}] - [\text{time already passed since the start date of the loan}]$.

✓ Example 5.3.4

Mr. Jackson bought his house in 1995, and financed the loan for 30 years at an interest rate of 7.8%. His monthly payment was \$1260. In 2015, Mr. Jackson decides to pay off the loan. Find the balance of the loan he still owes.

Solution

The reader should note that the original amount of the loan is not mentioned in the problem. That is because we don't need to know that to find the balance.

The original loan was for 30 years. 20 years have passed so there are 10 years still remaining. $12(10) = 120$ payments still remain to be paid on this loan.

As for the bank or lender is concerned, Mr. Jackson is obligated to pay \$1260 each month for 10 more years; he still owes a total of 120 payments. But since Mr. Jackson wants to pay it all off now, we need to find the present value P at the time of repayment of the remaining 10 years of payments of \$1260 each month. Using the formula we get for the present value of an annuity, we get

$$P(1 + .078/12)^{120} = \frac{\$1260 [(1 + .078/12)^{120} - 1]}{(.078/12)}$$

$$P(2.17597) = \$227957.85$$

$$P = \$104761.48$$

📌 To find the outstanding balance of a loan

If a loan has a payment of m dollars made n times a year at an interest r , then the outstanding value of the loan when there are t years still remaining on the loan is given by P :

$$P(1 + r/n)^{nt} = \frac{m [(1 + r/n)^{nt} - 1]}{r/n} \quad (5.3.4)$$

IMPORTANT: Note that t is not the original term of the loan but instead t is the amount of time still remaining in the future; nt is the number of payments still remaining in the future

If the problem does not directly state the amount of time still remaining in the term of the loan, then it must be calculated BEFORE using the above formula as $t = [\text{original term of loan}] - [\text{time already passed since the start date of the loan}]$.

Note that there are other methods to find the outstanding balance on a loan, but the method illustrated above is generally considered the easiest.

Exercises

PROBLEM SET: PRESENT VALUE OF AN ANNUITY AND INSTALLMENT PAYMENT

For the following problems, show all work.

1) Shawn has won a lottery paying him \$10,000 per month for the next 20 years. He'd rather have the whole amount in one lump sum today. If the current interest rate is 8.2%, how much money can he hope to get?	2) Sonya bought a car for \$15,000. Find the monthly payment if the loan is to be amortized over 5 years at a rate of 10.1%.
3) You determine that you can afford \$250 per month for a car. What is the maximum amount you can afford to pay for a car if the interest rate is 9% and you want to repay the loan in 5 years?	4) Compute the monthly payment for a house loan of \$200,000 to be financed over 30 years at an interest rate of 10%.
5) If the \$200,000 loan in the previous problem is financed over 15 years rather than 30 years at 10%, what will the monthly payment be?	6) Friendly Auto offers Jennifer a car for \$2000 down and \$300 per month for 5 years. Jason wants to buy the same car but wants to pay cash. How much must Jason pay if the interest rate is 9.4%?
7) The Gomez family bought a house for \$450,000. They paid 20% down and amortized the rest at 5.2% over a 30-year period. Find their monthly payment.	8) Mr. and Mrs. Wong purchased their new house for \$350,000. They made a down payment of 15%, and amortized the rest over 30 years. If the interest rate is 5.8%, find their monthly payment.
9) A firm needs a piece of machinery that has a useful life of 5 years. It has an option of leasing it for \$10,000 a year, or buying it for \$40,000 cash. If the interest rate is 10%, which choice is better?	10) Jackie wants to buy a \$19,000 car, but she can afford to pay only \$300 per month for 5 years. If the interest rate is 6%, how much does she need to put down?
11) Vijay's tuition at college for the next year is \$32,000. His parents have decided to pay the tuition by making nine monthly payments. If the interest rate is 6%, what is the monthly payment?	12) Glen borrowed \$10,000 for his college education at 8% compounded quarterly. Three years later, after graduating and finding a job, he decided to start paying off his loan. If the loan is amortized over five years at 9%, find his monthly payment for the next five years.

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5.4: Classification of Finance Problems

Learning Objectives

In this section, you will learn to:

- Re-examine the types of financial problems and classify them.
- Re-examine the vocabulary words used in describing financial calculations

Prerequisite Skills

Before you get started, take this prerequisite quiz.

1. Describe the differences between solving $50 = x^9$ vs solving $50 = 9^x$.

Click here to check your answer

If the exponent is known, you'll need to take that root of each side of the equation. If the exponent is unknown, you'll need to take the logarithm of each side of the equation.

If you missed this problem, [review Section 5.1](#). (Note that this will open in a new window.)

2. Describe the differences between simple interest vs compound interest.

Click here to check your answer

Simple interest is calculated once, based on the agreed timeframe of the loan. Compound interest is calculated multiple times and added to the account balance before it is calculated again.

If you missed this problem, [review Section 6.2](#). (Note that this will open in a new window.)

3. Describe the differences between periodically compounded interest vs continuously compounded interest.

Click here to check your answer

Periodically compounded interest is calculated at regular time intervals, such as every quarter, month, day, etc. The n -value tells how many times that time interval occurs in a year. Continuously compounded interest is calculated at an infinitely small time interval, using e instead of an n -value.

If you missed this problem, [review Section 6.3](#). (Note that this will open in a new window.)

4. Describe the differences between a lump-sum vs an annuity.

Click here to check your answer

In a lump-sum, money is added to the account once. The only thing that changes the account balance after that is the interest. In an annuity, money is added (and/or removed) from the account at regular time intervals.

If you missed this problem, [review Section 6.3](#). (Note that this will open in a new window.)

5. Describe the differences between the future value of an annuity vs the present value of an annuity. (Be careful because either formula can be used to find what should be done now (in the present) and what would be the result in the future!)

Click here to check your answer

The future value formula is used when interest and regular payments both add to the account balance. The present value formula is used when interest and regular payments work in opposite directions, one increasing the balance and one decreasing the balance.

If you missed this problem, [review Section 6.3](#). (Note that this will open in a new window.)

We'd like to remind the reader that the hardest part of solving a finance problem is determining the category it falls into. So in this section, we will emphasize the classification of problems rather than finding the actual solution. We suggest that the student read each problem carefully and look for the word or words that may give clues to the kind of problem that is presented.

For instance, students often fail to distinguish a lump-sum problem from an annuity. Since the payments are made each period, an annuity problem contains words such as each, every, per etc. One should also be aware that in the case of a lump-sum, only a single deposit is made, while in an annuity numerous deposits are made at equal spaced time intervals. To help interpret the vocabulary used in the problems, we include a glossary at the end of this section.

Students often confuse the present value of an annuity with the future value of an annuity. Don't get too caught up on the words, as either formula can be used to find what should be done now (in the present) and what would be the result in the future.

- The future value formula will be used in the context of investments, where the regular deposits and interest both work together to add to the account balance.
- The present value formula will be used in the context of loans or bank withdrawals, where the regular payments and interest work against each other. In a loan, the monthly payments lower the account balance while the interest payments raise the account balance.

CLASSIFICATION OF PROBLEMS AND EQUATIONS

We now list eight problems that form a basis for all finance problems. Read each question and determine which formula would be used to find the missing information.

Example 5.4.1

Dillon takes out a simple interest loan for \$5,000 at 10.5% interest with an agreement to pay it back in 10 months. What is the total amount he will repay at the end of the 10 months?

Classification: Since the loan is a simple interest loan and we are looking for the total amount he will repay, we will use the **Accumulated Value of Simple Interest Formula**.

Equation:

$$A = 5,000\left(1 + 0.105\left(\frac{10}{12}\right)\right) \quad (5.4.1)$$

Example 5.4.2

If \$2,000 is invested at 7% compounded quarterly, what will the final amount be in 5 years?

Classification: Since the \$2000 is invested once, this is a lump-sum. The interest is compounded each quarter, so this will use the **Accumulated Value of Periodic Compounding Formula**.

Equation:

$$A = \$2000\left(1 + .07/4\right)^{20} \quad (5.4.2)$$

Example 5.4.3

If \$2,000 is invested at 7% compounded continuously, what will the final amount be in 5 years?

Classification: Since the \$2000 is invested once, this is a lump-sum. The interest is compounded continuously, so this will use the **Accumulated Value of Continuous Compounding Formula**.

Equation:

$$A = \$2000e^{0.07*5} \quad (5.4.3)$$

Example 5.4.4

How much should be invested at 8% compounded yearly, for the final amount to be \$5,000 in five years?

Classification: Since this question is asking for a one-time investment, this is a lump-sum. The interest is compounded each year, so this will use the **Accumulated Value of Periodic Compounding Formula**. Note that we are given the accumulated value at the end of the 5 years and are looking for the principal.

Equation:

$$\$5,000 = P(1 + .08)^5 \quad (5.4.4)$$

Example 5.4.5

If \$200 is invested *each* month at 8.5% compounded monthly, what will the final amount be in 4 years?

Classification: Since the money is invested each month, this is an example of an annuity. This is an investment where the monthly payments and the interest both add to the value, so this will use the **Future Value of an Annuity Formula**.

Equation:

$$A = \frac{\$200 [(1 + .085/12)^{12 \cdot 4} - 1]}{.085/12} \quad (5.4.5)$$

Example 5.4.6

How much should be invested *each* month at 9% for it to accumulate to \$8,000 in three years?

Classification: Since the money is invested each month, this is an example of an annuity. This is an investment where the monthly payments and the interest both add to the value, so this will use the **Future Value of an Annuity Formula**. Note that we are given the accumulated value at the end of the 3 years and are looking for the monthly payment.

Equation:

$$\$8,000 = \frac{m [(1 + .09/12)^{36} - 1]}{.09/12} \quad (5.4.6)$$

Example 5.4.7

Keith wants to buy a new car worth \$30,000 and needs to apply for an auto loan. The dealership is offering him financing at 9.25% compounded monthly over the next 5 years. How much will each monthly payment be?

Classification: Since the money will be paid each month, this is an example of an annuity. This is a loan where the monthly payments lower the account balance and the interest raises the account balance, so this will use the **Present Value of an Annuity Formula**.

Equation:

$$\$30,000(1 + .0925/12)^{12 \cdot 5} = \frac{m [(1 + .0925/12)^{12 \cdot 5} - 1]}{.0925/12} \quad (5.4.7)$$

Example 5.4.8

Maria is saving money for her retirement. The retirement fund earns 8% interest compounded monthly, and she plans to withdraw \$1,500 each month for 25 years after her retirement. How much does she need in this fund before she retires?

Classification: Since the money will be withdrawn each month, this is an example of an annuity. This is a fund where the monthly withdrawals lower the account balance and the interest raises the account balance, so this will use the **Present Value of an Annuity Formula**.

Equation:

$$P(1 + .08/12)^{12*25} = \frac{\$1,500 [(1 + .08/12)^{12*25} - 1]}{.08/12} \quad (5.4.8)$$

GLOSSARY: VOCABULARY AND SYMBOLS USED IN FINANCIAL CALCULATIONS

As we've seen in these examples, it's important to read the problems carefully to correctly identify the situation. It is essential to understand to vocabulary for financial problems. Many of the vocabulary words used are listed in the glossary below for easy reference.

t	Term	Time period for a loan or investment. In this book t is represented in years and should be converted into years when it is stated in months or other units.
P	Principal	Principal is the amount of money borrowed in a loan. If a sum of money is invested for a period of time, the sum invested at the start is the Principal.
P	Present Value	Value of money at the beginning of the time period.
A	Accumulated Value Future Value	Value of money at the end of the time period
D	Discount	In loans involving simple interest, a discount occurs if the interest is deducted from the loan amount at the beginning of the loan period, rather than being repaid at the end of the loan period.
m	Periodic Payment	The amount of a constant periodic payment that occurs at regular intervals during the time period under consideration (examples: periodic payments made to repay a loan, regular periodic payments into a bank account as savings, regular periodic payment to a retired person as an annuity.)
n	Number of payment periods and compounding periods per year	In this book, when we consider periodic payments, we will always have the compounding period be the same as the payment period. In general the compounding and payment periods do not have to be the same, but the calculations are more complicated if they are different. If the periods differ, formulas for the calculations can be found in finance textbooks or various online resources. Calculations can easily be done using technology such as an online financial calculator, or financial functions in a spreadsheet, or a financial pocket calculator.
nt	Number of periods	$nt = (\text{number of periods per year}) \times (\text{number of years})$ nt gives the total number of payment and compounding periods In some situations we will calculate nt as the multiplication shown above. In other situations the problem may state nt , such as a problem describing an investment of 18 months duration compounded monthly. In this example: $nt = 18$ months and $n = 12$; then $t = 1.5$ years but t is not stated explicitly in the problem. The TI-84+ calculators built in TVM solver uses $N = nt$.
r	Annual interest rate Nominal rate	The stated annual interest rate. This is stated as a percent but converted to decimal form when using financial calculation formulas. If a bank account pays 3% interest compounded quarterly, then 3% is the nominal rate, and it is included in the financial formulas as $r = 0.03$
r/n	Interest rate per compounding period	If a bank account pays 3% interest compounded quarterly, then $r/n = 0.03/4 = 0.0075$, corresponding to a rate of 0.75% per quarter. Some Finite Math books use the symbol i to represent r/n

r_{EFF}	Effective Rate Effective Annual Interest Rate APY Annual Percentage Yield APR Annual Percentage Rate	The effective rate is the interest rate compounded annually that would give the same interest rate as the compounded rate stated for the investment. The effective rate provides a uniform way for investors or borrowers to compare different interest rates with different compounding periods.
I	Interest	Money paid by a borrower for the use of money borrowed as a loan. Money earned over time when depositing money into a savings account, certificate of deposit, or money market account. When a person deposits money in a bank account, the person depositing the funds is essentially temporarily lending the money to the bank and the bank pays interest to the depositor.
	Sinking Fund	A fund set up by making payments over a period of time into a savings or investment account in order to save to fund a future purchase. Businesses use sinking funds to save for a future purchase of equipment at the end of the savings period by making periodic installment payments into a sinking fund.
	Annuity	An annuity is a stream of periodic payments. In this book it refers to a stream of constant periodic payments made at the end of each compounding period for a specific amount of time. In common use the term annuity generally refers to a constant stream of periodic payments received by a person as retirement income, such as from a pension. Annuity payments in general may be made at the end of each payment period (ordinary annuity) or at the start of each period (annuity due). The compounding periods and payment periods do not need to be equal, but in this textbook we only consider situations when these periods are equal.
	Lump Sum	A single sum of money paid or deposited at one time, rather than being spread out over time. An example is lottery winnings if the recipient chooses to receive a single "lump sum" one-time payment, instead of periodic payments over a period of time or as. Use of the word lump sum indicates that this is a one time transaction and is not a stream of periodic payments.
	Loan	An amount of money that is borrowed with the understanding that the borrower needs to repay the loan to the lender in the future by the end of a period of time that is called the term of the loan. The repayment is most often accomplished through periodic payments until the loan has been completely repaid over the term of the loan. However there are also loans that can be repaid as a single sum at the end of the term of the loan, with interest paid either periodically over the term or in a lump sum at the end of the loan or as a discount at the start of the loan.

Exercises

PROBLEM SET: CLASSIFICATION OF FINANCE PROBLEMS

Let the letters A, B, C, D, E and F be represented as follows:

$$\begin{array}{lll}
 A = FV \text{ of a lump-sum} & C = FV \text{ of an annuity} & E = \text{Installment payment} \\
 B = PV \text{ of a lump-sum} & D = \text{sinking fund payment} & F = PV \text{ of an annuity}
 \end{array}$$

Classify each by writing the appropriate letter in the box, and write an equation for solution.

- 1) What monthly deposits made to an account paying 9% will grow to \$10,000 in 4 years?
- 2) An amount of \$4000 is invested at 6% compounded daily. What will the final amount be in 5 years?
- 3) David has won a lottery paying him \$10,000 per month for the next 20 years. He'd rather have the whole amount in one lump sum now. If the current interest rate is 7%, how much money can he hope to get?

- 4) Each month Linda deposits \$250 in an account that pays 9%. How much money will she have in 4 years?
 - 5) Find the monthly payment for a \$15,000 car if the loan is amortized over 4 years at a rate of 10%.
 - 6) What lump-sum deposited in an account paying 7% compounded daily will grow to \$10,000 in 5 years?
 - 7) What amount of quarterly payments will amount to \$250,000 in 5 years at a rate of 8%?
 - 8) The Chang family bought their house 25 years ago. They had their loan financed for 30 years at an interest rate of 11% resulting in a payment of \$1350 a month. Find the balance of the loan.
- A 10-year \$1000 bond pays \$35 every six months. If the current interest rate is 8%, in order to find the fair market value of the bond, we need to find the following.
- 9) The present value of \$1000.
 - 10) The present value of the \$35 per six month payments.
 - 11) What lump-sum deposit made today is equal to 33 monthly deposits of \$500 if the interest rate is 8%?
 - 12) What monthly deposits made to an account paying 10% will accumulated to \$10,000 in six years?
 - 13) A department store charges a finance charge of 1.5% per month on the outstanding balance.
If Ned charged \$400 three months ago and has not paid his bill, how much does he owe?
 - 14) What will the value of \$300 monthly deposits be in 10 years if the account pays 12% compounded monthly?
 - 15) What lump-sum deposited at 6% compounded daily will grow to \$2000 in three years?
 - 16) A company buys an apartment complex for \$5,000,000 and amortizes the loan over 10 years.
What is the yearly payment if the interest rate is 14%?
 - 17) In 2002, a house in Rock City cost \$300,000. Real estate in Rock City has been increasing in value at the annual rate of 5.3%..
Find the price of that house in 2016.
 - 18) You determine that you can afford to pay \$400 per month for a car. What is the maximum price you can pay for a car if the interest rate is 11% and you want to repay the loan in 4 years?
 - 19) A business needs \$350,000 in 5 years. How much lump-sum should be put aside in an account that pays 9% so that five years from now the company will have \$350,000?
 - 20) A person wishes to have \$500,000 in a pension fund 20 years from now. How much should he deposit each month in an account paying 9% compounded monthly?

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5.5: Additional Application Problems

Learning Objectives

In this section, you will learn to:

- Perform financial calculations in situations involving several stages of savings and/or annuities.
- Find the fair market value of a bond.
- Construct an amortization schedule for a loan.

We have already developed the tools to solve most finance problems. Now we use these tools to solve some additional applications.

PROBLEMS INVOLVING MULTIPLE STAGES OF SAVINGS AND/OR ANNUITIES

Consider the following situations:

- Suppose a baby, Aisha, is born and her grandparents invest \$5000 in a college fund. The money remains invested for 18 years until Aisha enters college, and then is withdrawn in equal semiannual payments over the 4 years that Aisha expects to need to finish college. The college investment fund earns 5% interest compounded semiannually. How much money can Aisha withdraw from the account every six months while she is in college?
- Aisha graduates college and starts a job. She saves \$1000 each quarter, depositing it into a retirement savings account. Suppose that Aisha saves for 30 years and then retires. At retirement she wants to withdraw money as an annuity that pays a constant amount every month for 25 years. During the savings phase, the retirement account earns 6% interest compounded quarterly. During the annuity payout phase, the retirement account earns 4.8% interest compounded monthly. Calculate Aisha's monthly retirement annuity payout.

These problems appear complicated. But each can be broken down into two smaller problems involving compound interest on savings or involving annuities. Often the problem involves a savings period followed by an annuity period. ; the accumulated value from first part of the problem may become a present value in the second part. Read each problem carefully to determine what is needed.

Example 5.5.2

Suppose a baby, Aisha, is born and her grandparents invest \$8000 in a college fund. The money remains invested for 18 years until Aisha enters college, and then is withdrawn in equal semiannual payments over the 4 years that Aisha expects to attend college. The college investment fund earns 5% interest compounded semiannually. How much money can Aisha withdraw from the account every six months while she is in college?

Solution

Part 1: Accumulation of College Savings: Find the accumulated value at the end of 18 years of a sum of \$8000 invested at 5% compounded semiannually.

$$A = \$8000(1 + .05/2)^{(2 \times 18)} = \$8000(1.025)^{36} = \$8000(2.432535)$$

$$A = \$19460.28$$

Part 2: Semiannual annuity payout from savings to put toward college expenses. Find the amount of the semiannual payout for four years using the accumulated savings from part 1 of the problem with an interest rate of 5% compounded semiannually.

$A = \$19460.28$ in Part 1 is the accumulated value at the end of the savings period. This becomes the present value $P = \$19460.28$ when calculating the semiannual payments in Part 2.

$$\$19460.28 \left(1 + \frac{.05}{2}\right)^{2 \times 4} = \frac{m \left[\left(1 + \frac{.05}{2}\right)^{2 \times 4} - 1 \right]}{(.05/2)}$$

$$\$23710.46 = m(8.73612)$$

$$m = \$2714.07$$

Aisha will be able to withdraw \$2714.07 semiannually for her college expenses.

Example 5.5.3

Aisha graduates college and starts a job. She saves \$1000 each quarter, depositing it into a retirement savings account. Suppose that Aisha saves for 30 years and then retires. At retirement she wants to withdraw money as an annuity that pays a constant amount every month for 25 years. During the savings phase, the retirement account earns 6% interest compounded quarterly. During the annuity payout phase, the retirement account earns 4.8% interest compounded monthly. Calculate Aisha's monthly retirement annuity payout.

Solution

Part 1: Accumulation of Retirement Savings: Find the accumulated value at the end of 30 years of \$1000 deposited at the end of each quarter into a retirement savings account earning 6% interest compounded quarterly.

$$A = \frac{\$1000[(1+.06/4)^{4 \times 30} - 1]}{(.06/4)}$$

$$A = \$331288.19$$

Part 2: Monthly retirement annuity payout: Find the amount of the monthly annuity payments for 25 years using the accumulated savings from part 1 of the problem with an interest rate of 4.8% compounded monthly.

$A = \$331288.19$ in Part 1 is the accumulated value at the end of the savings period. This amount will become the present value $P = \$331288.19$ when calculating the monthly retirement annuity payments in Part 2.

$$\$331288.19(1 + .048/12)^{12 \times 25} = \frac{m[(1+.048/12)^{12 \times 25} - 1]}{(.048/12)}$$

$$\$1097285.90 = m(578.04483)$$

$$m = \$1898.27$$

Aisha will have a monthly retirement annuity income of \$1898.27 when she retires.

FAIR MARKET VALUE OF A BOND

Whenever a business, and for that matter the U. S. government, needs to raise money it does it by selling bonds. A **bond** is a certificate of promise that states the terms of the agreement. Usually the business sells bonds for the **face amount** of \$1,000 each for a stated **term**, a period of time ending at a specified **maturity** date.

The person who buys the bond, the **bondholder**, pays \$1,000 to buy the bond.

The bondholder is promised two things: First that he will get his \$1,000 back at the maturity date, and second that he will receive a fixed amount of interest every six months.

As the market interest rates change, the price of the bond starts to fluctuate. The bonds are bought and sold in the market at their **fair market value**.

The interest rate a bond pays is fixed, but if the market interest rate goes up, the value of the bond drops since the money invested in the bond could earn more if invested elsewhere. When the value of the bond drops, we say it is trading at a **discount**.

On the other hand, if the market interest rate drops, the value of the bond goes up since the bond now yields a higher return than the market interest rate, and we say it is trading at a **premium**.

Example 5.5.4

The Orange Computer Company needs to raise money to expand. It issues a 10-year \$1,000 bond that pays \$30 every six months. If the current market interest rate is 7%, what is the fair market value of the bond?

Solution

The bond certificate promises us two things - An amount of \$1,000 to be paid in 10 years, and a semi-annual payment of \$30 for ten years. Therefore, to find the fair market value of the bond, we need to find the present value of the lump sum of \$1,000 we are to receive in 10 years, as well as, the present value of the \$30 semi-annual payments for the 10 years.

We will let P_1 = the present value of the face amount of \$1,000

$$P_1(1 + .07/2)^{20} = \$1,000$$

Since the interest is paid twice a year, the interest is compounded twice a year and $nt = 2(10)=20$

$$P_1(1.9898) = \$1,000$$

$$P_1 = \$502.56$$

We will let P_2 = the present value of the \$30 semi-annual payments is

$$P_2(1 + .07/2)^{20} = \frac{\$30 [(1 + .07/2)^{20} - 1]}{(.07/2)}$$

$$P_2(1.9898) = 848.39$$

$$P_2 = \$426.37$$

The present value of the lump-sum \$1,000 = \$502.56

The present value of the \$30 semi-annual payments = \$426.37

The fair market value of the bond is $P = P_1 + P_2 = \$502.56 + \$426.37 = \$928.93$

Note that because the market interest rate of 7% is higher than the bond's implied interest rate of 6% implied by the semiannual payments, the bond is selling at a discount; its fair market value of \$928.93 is less than its face value of \$1000.

Example 5.5.5

A state issues a 15 year \$1000 bond that pays \$25 every six months. If the current market interest rate is 4%, what is the fair market value of the bond?

Solution

The bond certificate promises two things - an amount of \$1,000 to be paid in 15 years, and semi-annual payments of \$25 for 15 years. To find the fair market value of the bond, we find the present value of the \$1,000 face value we are to receive in 15 years and add it to the present value of the \$25 semi-annual payments for the 15 years. In this example, $nt = 2(15) = 30$.

We will let P_1 = the present value of the lump-sum \$1,000

$$P_1(1 + .04/2)^{30} = \$1,000$$

$$P_1 = \$552.07$$

We will let P_2 = the present value of the \$25 semi-annual payments is

$$P_2(1 + .04/2)^{30} = \frac{\$25 [(1 + .04/2)^{30} - 1]}{(.04/2)}$$

$$P_2(1.18114) = \$1014.20$$

$$P_2 = \$559.90$$

The present value of the lump-sum \$1,000 = \$552.07

The present value of the \$30 semi-annual payments = \$559.90

Therefore, the fair market value of the bond is

$$P = P_1 + P_2 = \$552.07 + \$559.90 = \$1111.97$$

Because the market interest rate of 4% is lower than the interest rate of 5% implied by the semiannual payments, the bond is selling at a premium: the fair market value of \$1,111.97 is more than the face value of \$1,000.

To summarize:

To find the Fair Market Value of a Bond

Find the present value of the face amount A that is payable at the maturity date:

$$A = P_1(1 + r/n)^{nt}; \text{ solve to find } P_1 \quad (5.5.1)$$

Find the present value of the semiannually payments of $\$m$ over the term of the bond:

$$P_2(1 + r/n)^{nt} = \frac{m [(1 + r/n)^{nt} - 1]}{r/n} ; \text{ solve to find } P_2 \quad (5.5.2)$$

The fair market value (or present value or price or current value) of the bond is the sum of the present values calculated above:

$$P = P_1 + P_2 \quad (5.5.3)$$

AMORTIZATION SCHEDULE FOR A LOAN

An amortization schedule is a table that lists all payments on a loan, splits them into the portion devoted to interest and the portion that is applied to repay principal, and calculates the outstanding balance on the loan after each payment is made.

Example 5.5.6

An amount of \$500 is borrowed for 6 months at a rate of 12%. Make an amortization schedule showing the monthly payment, the monthly interest on the outstanding balance, the portion of the payment contributing toward reducing the debt, and the outstanding balance.

Solution

The reader can verify that the monthly payment is \$86.27.

The first month, the outstanding balance is \$500, and therefore, the monthly interest on the outstanding balance is

$$(\text{outstanding balance})(\text{the monthly interest rate}) = (\$500)(.12/12) = \$5$$

This means, the first month, out of the \$86.27 payment, \$5 goes toward the interest and the remaining \$81.27 toward the balance leaving a new balance of $\$500 - \$81.27 = \$418.73$.

Similarly, the second month, the outstanding balance is \$418.73, and the monthly interest on the outstanding balance is $(\$418.73)(.12/12) = \4.19 . Again, out of the \$86.27 payment, \$4.19 goes toward the interest and the remaining \$82.08 toward the balance leaving a new balance of $\$418.73 - \$82.08 = \$336.65$. The process continues in the table below.

Payment #	Payment	Interest	Debt Payment	Balance
1	\$86.27	\$5	\$81.27	\$418.73
2	\$86.27	\$4.19	\$82.08	\$336.65
3	\$86.27	\$3.37	\$82.90	\$253.75
4	\$86.27	\$2.54	\$83.73	\$170.02
5	\$86.27	\$1.70	\$84.57	\$85.45
6	\$86.27	\$0.85	\$85.42	\$0.03

Note that the last balance of 3 cents is due to error in rounding off.

An amortization schedule is usually lengthy and tedious to calculate by hand. For example, an amortization schedule for a 30 year mortgage loan with monthly payments would have $(12)(30)=360$ rows of calculations in the amortization schedule table. A car loan with 5 years of monthly payments would have $12(5)=60$ rows of calculations in the amortization schedule table. However it would be straightforward to use a spreadsheet application on a computer to do these repetitive calculations by inputting and copying formulas for the calculations into the cells.

Most of the other applications in this section's problem set are reasonably straightforward, and can be solved by taking a little extra care in interpreting them. And remember, there is often more than one way to solve a problem.

Exercises

PROBLEM SET: MISCELLANEOUS APPLICATION PROBLEMS

For problems 1 - 4, assume a \$200,000 house loan is amortized over 30 years at an interest rate of 5.4%.

1) Find the monthly payment.	2) Find the balance owed after 20 years.
3) Find the balance of the loan after 100 payments.	4) Find the monthly payment if the original loan were amortized over 15 years.

MISCELLANEOUS APPLICATION PROBLEMS

5) Mr. Patel wants to pay off his car loan. The monthly payment for his car is \$365, and he has 16 payments left. If the loan was financed at 6.5%, how much does he owe?	6) An amount of \$2000 is borrowed for a year at a rate of 7%. Make an amortization schedule showing the monthly payment, the monthly interest on the outstanding balance, the portion of the payment going toward reducing the debt, and the balance.
7) Fourteen months after Dan bought his new car he lost his job. His car was repossessed by his lender after he made only 14 monthly payments of \$376 each. If the loan was financed over a 4-year period at an interest rate of 6.3%, how much did the car cost the lender? In other words, how much did Dan still owe on the car?	8) You have a choice of either receiving \$5,000 at the end of each year for the next 5 years or receiving \$3000 per year for the next 10 years. If the current interest rate is 9%, which is better?
9) Mr. Smith is planning to retire in 25 years and would like to have \$250,000 then. What monthly payment made at the end of each month to an account that pays 6.5% will achieve his objective?	10) Assume Mr. Smith has reached retirement and has \$250,000 in an account which is earning 6.5%. He would now like to make equal monthly withdrawals for the next 15 years to completely deplete this account. Find the withdrawal payment.
11) Mrs. Garcia is planning to retire in 20 years. She starts to save for retirement by depositing \$2000 each quarter into a retirement investment account that earns 6% interest compounded quarterly. Find the accumulated value of her retirement savings at the end of 20 years.	12) Assume Mrs. Garcia has reached retirement and has accumulated the amount found in question 13 in a retirement savings account. She would now like to make equal monthly withdrawals for the next 15 years to completely deplete this account. Find the withdrawal payment. Assume the account now pays 5.4% compounded monthly.
13) A ten-year \$1,000 bond pays \$35 every six months. If the current interest rate is 8.2%, find the fair market value of the bond. Hint: You must do the following. a) Find the present value of \$1000. b) Find the present value of the \$35 payments. c) The fair market value of the bond = a + b	14) Find the fair market value of the ten-year \$1,000 bond that pays \$35 every six months, if the current interest rate has dropped to 6%. Hint: You must do the following. a) Find the present value of \$1000. b) Find the present value of the \$35 payments. c) The fair market value of the bond = a + b
15) A twenty-year \$1,000 bond pays \$30 every six months. If the current interest rate is 4.2%, find the fair market value of the bond. Hint: You must do the following. a) Find the present value of \$1000. b) Find the present value of the \$30 payments. c) The fair market value of the bond = a + b	16) Find the fair market value of the twenty-year \$1,000 bond that pays \$30 every six months, if the current interest rate has increased to 7.5%.

17) Mr. and Mrs. Nguyen deposit \$10,000 into a college investment account when their new baby grandchild is born. The account earns 6.25% interest compounded quarterly.

a) When their grandchild reaches the age of 18, what is the accumulated value of the college investment account?

b) The Nguyen's grandchild has just reached the age of 18 and started college. If she is to withdraw the money in the college savings account in equal monthly payments over the next 4 years, how much money will be withdrawn each month?

18) Mr. Singh is 38 and plans to retire at age 65. He opens a retirement savings account.

a) Mr. Singh wants to save enough money to accumulate \$500,000 by the time he retires.

The retirement investment account pays 7% interest compounded monthly. How much does he need to deposit each month to achieve this goal?

b) Mr. Singh has now reached at 65 and retires.

How much money can he withdraw each month for 25 years if the retirement investment account now pays 5.2% interest, compounded monthly?

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5.6: Chapter Review

SECTION 6.7 PROBLEM SET: CHAPTER REVIEW

1. Manuel borrows \$800 for 6 months at 18% simple interest. How much does he owe at the end of 6 months?
2. The population of a city is 65,000 and expects to grow at a rate of 2.3% per year for the next 10 years. What will the population of this city be in 10 years?
3. The Gill family is buying a \$250,000 house with a 10% down payment. If the loan is financed over a 30 year period at an interest rate of 4.8%, what is the monthly payment?
4. Find the monthly payment for the house in the above problem if the loan was amortized over 15 years.
5. You look at your budget and decide that you can afford \$250 per month for a car. What is the maximum amount you can afford to pay for the car if the interest rate is 8.6% and you want to finance the loan over 5 years?
6. Mr. Nakahama bought his house in the year 1998. He had his loan financed for 30 years at an interest rate of 6.2% resulting in a monthly payment of \$1500. In 2015, 17 years later, he paid off the balance of the loan. How much did he pay?
7. Lisa buys a car for \$16,500, and receives \$2400 for her old car as a trade-in value. Find the monthly payment for the balance if the loan is amortized over 5 years at 8.5%.
8. A car is sold for \$3000 cash down and \$400 per month for the next 4 years. Find the cash value of the car today if the money is worth 8.3% compounded monthly.
9. An amount of \$2300 is borrowed for 7 months at a simple interest rate of 16%. Find the discount and the proceeds.
10. Marcus has won a lottery paying him \$5000 per month for the next 25 years. He'd rather have the whole amount in one lump sum today. If the current interest rate is 7.3%, how much money can he hope to get?
11. In the year 2000, an average house in Star City cost \$250,000. If the average annual inflation rate for the past years has been about 4.7%, what was the price of that house in 2015?
12. Find the 'fair market' value of a ten-year \$1000 bond which pays \$30 every six months if the current interest rate is 7%. What if the current interest rate is 5%?
13. A Visa credit card company has a finance charge of 1.5% per month (18% per year) on the outstanding balance. John owed \$3200 and has been delinquent for 5 months. How much total does he owe, now?
14. You want to purchase a home for \$200,000 with a 30-year mortgage at 9.24% interest. Find
 - a. the monthly payment
 - b. the balance owed after 20 years.
15. When Jose bought his car, he amortized his loan over 6 years at a rate of 9.2%, and his monthly payment came out to be \$350 per month. He has been making these payments for the past 40 months and now wants to pay off the remaining balance. How much does he owe?
16. A lottery pays \$10,000 per month for the next 20 years. If the interest rate is 7.8%, find both its present and future values.
17. A corporation estimates it will need \$300,000 in 8 years to replace its existing machinery. How much should it deposit each quarter in a sinking fund earning 8.4% compounded quarterly to meet this obligation?
18. Our national debt in 1992 was about \$4 trillion. If the annual interest rate was 7% then, what was the daily interest on the national debt?
19. A business must raise \$400,000 in 10 years. What should be the size of the owners' monthly payments to a sinking fund paying 6.5% compounded monthly?
20. The population of a city of 80,000 is growing at a rate of 3.2% per year. What will the population be at the end of 10 years?
21. A sum of \$5000 is deposited in a bank today. What will the final amount be in 20 months if the bank pays 9% and the interest is compounded monthly?
22. A manufacturing company buys a machine for \$500 cash and \$50 per month for the next 3 years. Find the cash value of the machine today if the money is worth 6.2% compounded monthly.
23. The United States paid about 4 cents an acre for the Louisiana Purchase in 1803. Suppose the value of this property grew at a rate of 5.5% annually. What would an acre be worth in the year 2000?
24. What amount should be invested per month at 9.1% compounded monthly so that it will become \$5000 in 17 months?
25. A machine costs \$8000 and has a life of 5 years. It can be leased for \$160 per month for 5 years with a cash down payment of \$750. The current interest rate is 8.3%. Is it cheaper to lease or to buy?
26. If inflation holds at 5.2% per year for 5 years, what will be the cost in 5 years of a car that costs \$16,000 today? How much will you need to deposit each quarter in a sinking fund earning 8.7% per year to purchase the new car in 5 years?

27. City Bank pays an interest rate of 6%, while Western Bank pays 5.8% compounded continuously. Which one is a better deal?
28. Ali has inherited \$20,000 and is planning to invest this amount at 7.9% interest. At the same time he wishes to make equal monthly withdrawals to use up the entire sum in 5 years. How much can he withdraw each month?
29. Jason has a choice of receiving \$300 per month for the next 5 years or \$500 per month for the next 3 years. Which one is worth more if the current interest rate is 7.7%?
30. If a bank pays 6.8% compounded continuously, how long will it take to double your money?
31. A mutual fund claims a growth rate of 8.3% per year. If \$500 per month is invested, what will the final amount be in 15 years?
32. Mr. Vasquez has been given two choices for his compensation. He can have \$20,000 cash plus \$500 per month for 10 years, or he can receive \$12,000 cash plus \$1000 per month for 5 years. If the interest rate is 8%, which is the better offer?
33. How much should Mr. Shackley deposit in a trust account so that his daughter can withdraw \$400 per month for 4 years if the interest rate is 8%?
34. Mr. Albers borrowed \$425,000 from the bank for his new house at an interest rate of 4.7%. He will make equal monthly payments for the next 30 years. How much money will he end up paying the bank over the life of the loan, and how much is the interest?
35. Mr. Tong puts away \$500 per month for 10 years in an account that earns 9.3%. After 10 years, he decides to withdraw \$1,000 per month. If the interest rate stays the same, how long will it take Mr. Tong to deplete the account?
36. An amount of \$5000 is borrowed for 15 months at an interest rate of 9%. Find the monthly payment and construct an amortization schedule showing the monthly payment, the monthly interest on the outstanding balance, the amount of payment contributing towards debt, and the outstanding debt.

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CHAPTER OVERVIEW

6: Linear Functions

Recall that a function is a relation that assigns to every element in the domain exactly one element in the range. Linear functions are a specific type of function that can be used to model many real-world applications, such as plant growth over time. In this chapter, we will explore linear functions, their graphs, and how to relate them to data.

[6.1: The Rectangular Coordinate Systems and Graphs](#)

[6.2: Linear Functions](#)

[6.3: Linear Functions and Mathematical Models](#)

[6.3.1: Modeling with Linear Functions](#)

[6.3.2: Applications](#)

[6.4: Intersection of Straight Lines](#)

[6.5: Chapter Review](#)

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6.1: The Rectangular Coordinate Systems and Graphs

Learning Objectives

- Plot ordered pairs in a Cartesian coordinate system.
- Graph equations by plotting points.
- Graph equations with a graphing utility.
- Find x -intercepts and y -intercepts.
- Use the distance formula.
- Use the midpoint formula.

Tracie set out from Elmhurst, IL, to go to Franklin Park. On the way, she made a few stops to do errands. Each stop is indicated by a red dot in Figure 6.1.1. Laying a rectangular coordinate grid over the map, we can see that each stop aligns with an intersection of grid lines. In this section, we will learn how to use grid lines to describe locations and changes in locations.

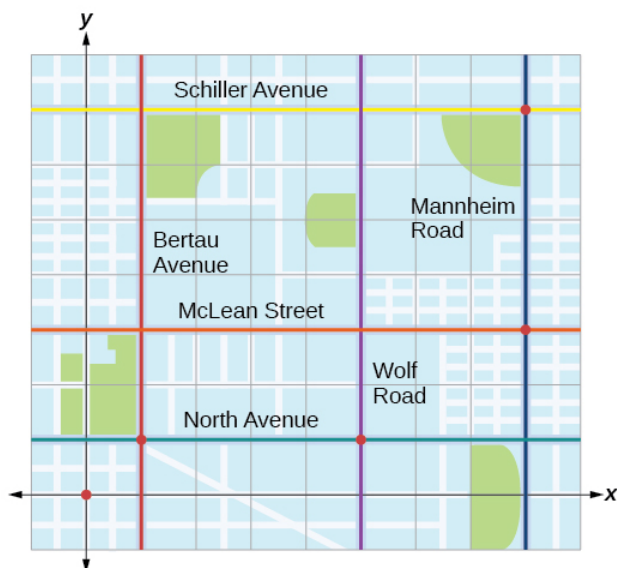


Figure 6.1.1

Plotting Ordered Pairs in the Cartesian Coordinate System

An old story describes how seventeenth-century philosopher/mathematician René Descartes invented the system that has become the foundation of algebra while sick in bed. According to the story, Descartes was staring at a fly crawling on the ceiling when he realized that he could describe the fly's location in relation to the perpendicular lines formed by the adjacent walls of his room. He viewed the perpendicular lines as horizontal and vertical axes. Further, by dividing each axis into equal unit lengths, Descartes saw that it was possible to locate any object in a two-dimensional plane using just two numbers—the displacement from the horizontal axis and the displacement from the vertical axis.

While there is evidence that ideas similar to Descartes' grid system existed centuries earlier, it was Descartes who introduced the components that comprise the Cartesian coordinate system, a grid system having perpendicular axes. Descartes named the horizontal axis the x -axis and the vertical axis the y -axis.

The Cartesian coordinate system, also called the *rectangular coordinate system*, is based on a two-dimensional plane consisting of the x -axis and the y -axis. Perpendicular to each other, the axes divide the plane into four sections. Each section is called a quadrant; the quadrants are numbered counterclockwise as shown in Figure 6.1.2.

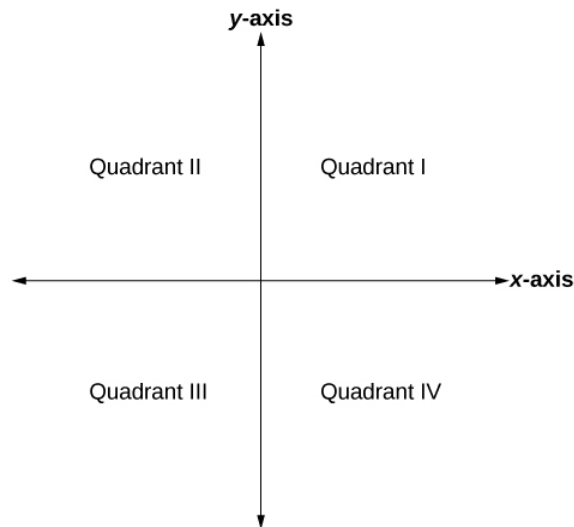


Figure 6.1.2

The center of the plane is the point at which the two axes cross. It is known as the origin, or point $(0, 0)$. From the origin, each axis is further divided into equal units: increasing, positive numbers to the right on the x -axis and up the y -axis; decreasing, negative numbers to the left on the x -axis and down the y -axis. The axes extend to positive and negative infinity as shown by the arrowheads in Figure 6.1.3.

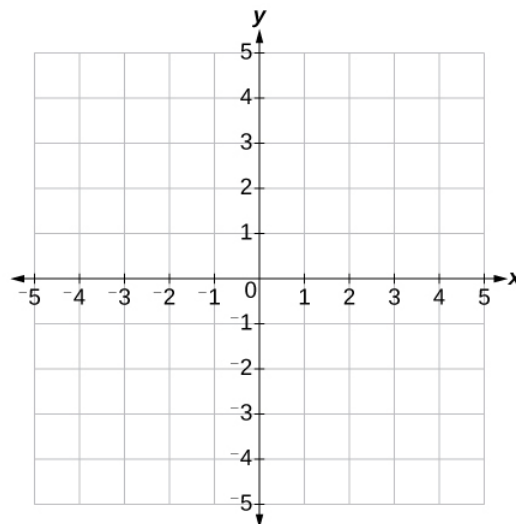


Figure 6.1.3

Each point in the plane is identified by its x -coordinate, or horizontal displacement from the origin, and its y -coordinate, or vertical displacement from the origin. Together, we write them as an ordered pair indicating the combined distance from the origin in the form (x, y) . An ordered pair is also known as a coordinate pair because it consists of x - and y -coordinates. For example, we can represent the point $(3, -1)$ in the plane by moving three units to the right of the origin in the horizontal direction, and one unit down in the vertical direction. See Figure 6.1.4.

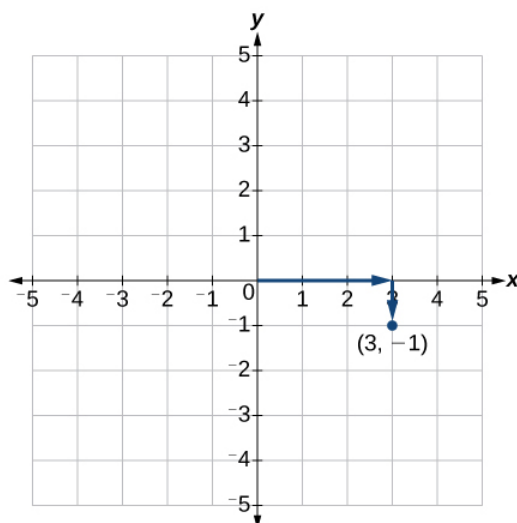


Figure 6.1.4

When dividing the axes into equally spaced increments, note that the x -axis may be considered separately from the y -axis. In other words, while the x -axis may be divided and labeled according to consecutive integers, the y -axis may be divided and labeled by increments of 2, or 10, or 100. In fact, the axes may represent other units, such as years against the balance in a savings account, or quantity against cost, and so on. Consider the rectangular coordinate system primarily as a method for showing the relationship between two quantities.

Cartesian Coordinate System

A two-dimensional plane where the

- x -axis is the horizontal axis
- y -axis is the vertical axis

A point in the plane is defined as an ordered pair, (x, y) , such that x is determined by its horizontal distance from the origin and y is determined by its vertical distance from the origin.

Example 6.1.1: Plotting Points in a Rectangular Coordinate System

Plot the points $(-2, 4)$, $(3, 3)$, and $(0, -3)$ in the plane.

Solution

To plot the point $(-2, 4)$, begin at the origin. The x -coordinate is -2 , so move two units to the left. The y -coordinate is 4, so then move four units up in the positive y direction.

To plot the point $(3, 3)$, begin again at the origin. The x -coordinate is 3, so move three units to the right. The y -coordinate is also 3, so move three units up in the positive y direction.

To plot the point $(0, -3)$, begin again at the origin. The x -coordinate is 0. This tells us not to move in either direction along the x -axis. The y -coordinate is -3 , so move three units down in the negative y direction. See the graph in Figure 6.1.5.

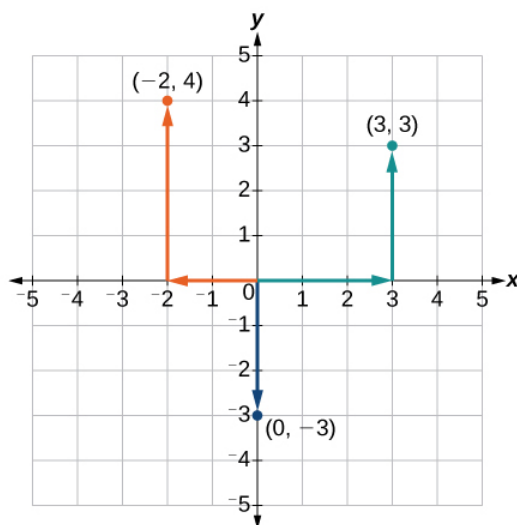


Figure 6.1.5

Analysis

Note that when either coordinate is zero, the point must be on an axis. If the x -coordinate is zero, the point is on the y -axis. If the y -coordinate is zero, the point is on the x -axis.

Graphing Equations by Plotting Points

We can plot a set of points to represent an equation. When such an equation contains both an x variable and a y variable, it is called an **equation in two variables**. Its graph is called a **graph in two variables**. Any graph on a two-dimensional plane is a graph in two variables.

Suppose we want to graph the equation $y = 2x - 1$. We can begin by substituting a value for x into the equation and determining the resulting value of y . Each pair of x - and y -values is an ordered pair that can be plotted. Table 6.1.1 lists values of x from -3 to 3 and the resulting values for y .

Table 6.1.1

x	$y = 2x - 1$	(x, y)
-3	$y = 2(-3) - 1 = -7$	$(-3, -7)$
-2	$y = 2(-2) - 1 = -5$	$(-2, -5)$
-1	$y = 2(-1) - 1 = -3$	$(-1, -3)$
0	$y = 2(0) - 1 = -1$	$(0, -1)$
1	$y = 2(1) - 1 = 1$	$(1, 1)$
2	$y = 2(2) - 1 = 3$	$(2, 3)$
3	$y = 2(3) - 1 = 5$	$(3, 5)$

We can plot the points in the table. The points for this particular equation form a line, so we can connect them (Figure 6.1.6). This is not true for all equations.

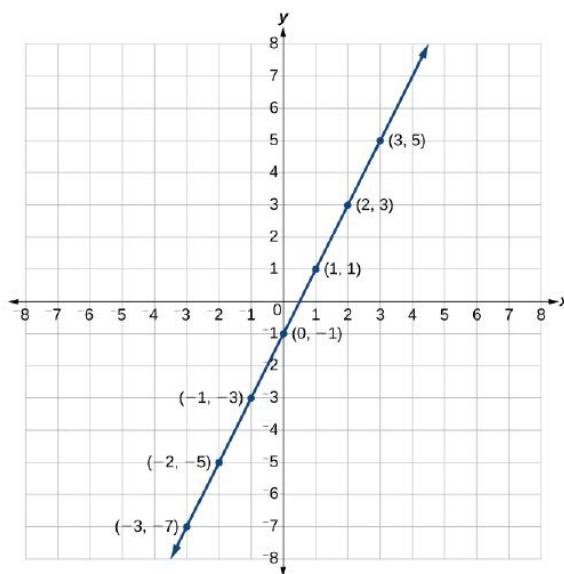


Figure 6.1.6

Note that the x -values chosen are arbitrary, regardless of the type of equation we are graphing. Of course, some situations may require particular values of x to be plotted in order to see a particular result. Otherwise, it is logical to choose values that can be calculated easily, and it is always a good idea to choose values that are both negative and positive. There is no rule dictating how many points to plot, although we need at least two to graph a line. Keep in mind, however, that the more points we plot, the more accurately we can sketch the graph.

Howto: Given an equation, graph by plotting points

1. Make a table with one column labeled x , a second column labeled with the equation, and a third column listing the resulting ordered pairs.
2. Enter x -values down the first column using positive and negative values. Selecting the x -values in numerical order will make the graphing simpler.
3. Select x -values that will yield y -values with little effort, preferably ones that can be calculated mentally.
4. Plot the ordered pairs.
5. Connect the points if they form a line.

✓ Example 6.1.2: Graphing an Equation in Two Variables by Plotting Points

Graph the equation $y = -x + 2$ by plotting points.

Solution

First, we construct a table similar to Table 6.1.2. Choose x values and calculate y .

Table 6.1.2

x	$y = -x + 2$	(x, y)
-5	$y = -(-5) + 2 = 7$	$(-5, 7)$
-3	$y = -(-3) + 2 = 5$	$(-3, 5)$
-1	$y = -(-1) + 2 = 3$	$(-1, 3)$
0	$y = -(0) + 2 = 2$	$(0, 2)$
1	$y = -(1) + 2 = 1$	$(1, 1)$
3	$y = -(3) + 2 = -1$	$(3, -1)$
5	$y = -(5) + 2 = -3$	$(5, -3)$

Now, plot the points. Connect them if they form a line. See Figure 6.1.7.

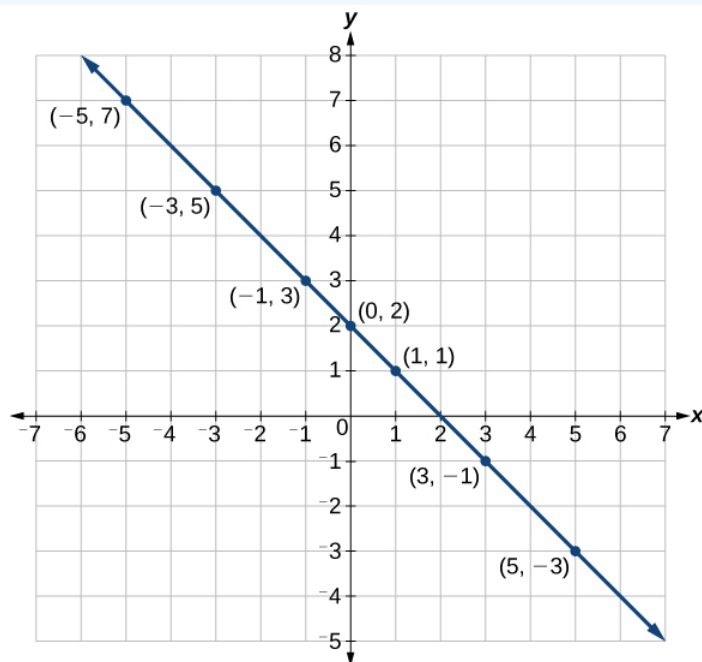


Figure 6.1.7

? Exercise 6.1.1

Construct a table and graph the equation by plotting points: $y = \frac{1}{2}x + 2$.

Answer

Please see Table 6.1.3 and graph below.

Table 6.1.3

x	$y = 12x + 2$	(x, y)
-2	$y = 12(-2) + 2 = 1$	$(-2, 1)$
-1	$y = 12(-1) + 2 = 32$	$(-1, 32)$
0	$y = 12(0) + 2 = 2$	$(0, 2)$
1	$y = 12(1) + 2 = 52$	$(1, 52)$
2	$y = 12(2) + 2 = 3$	$(2, 3)$

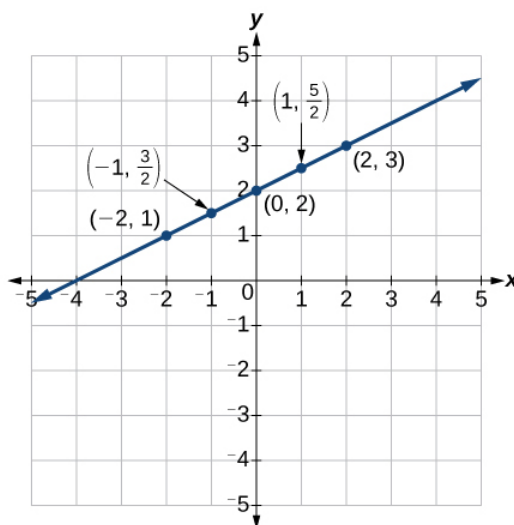


Figure 6.1.8

Graphing Equations with a Graphing Utility

Most graphing calculators require similar techniques to graph an equation. The equations sometimes have to be manipulated so they are written in the style $y = \underline{\hspace{1cm}}$. The TI-84 Plus, and many other calculator makes and models, have a mode function, which allows the window (the screen for viewing the graph) to be altered so the pertinent parts of a graph can be seen.

For example, the equation $y = 2x - 20$ has been entered in the TI-84 Plus shown in Figure 6.1.9a. In Figure 6.1.9b, the resulting graph is shown. Notice that we cannot see on the screen where the graph crosses the axes. The standard window screen on the TI-84 Plus shows $-10 \leq x \leq 10$, and $-10 \leq y \leq 10$. See Figure \(\PageIndex{9c}\).

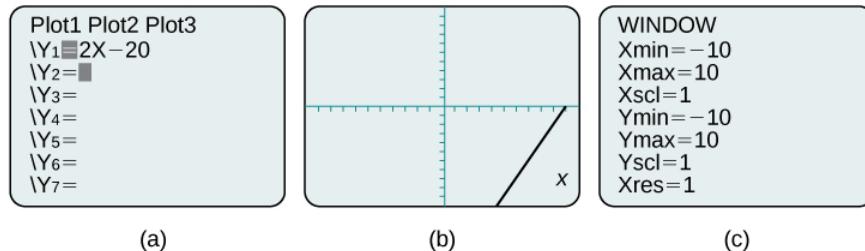


Figure 6.1.9: a. Enter the equation. b. This is the graph in the original window. c. These are the original settings.

By changing the window to show more of the positive x -axis and more of the negative y -axis, we have a much better view of the graph and the x - and y -intercepts. See Figure 6.1.10a and Figure 6.1.10b

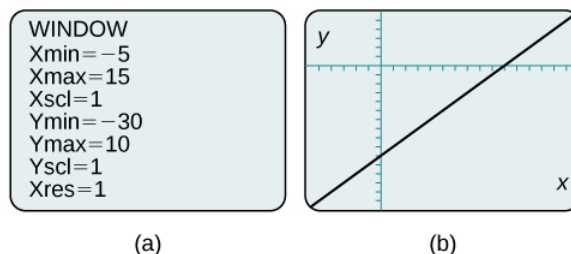


Figure 6.1.10: a. This screen shows the new window settings. b. We can clearly view the intercepts in the new window.

✓ Example 6.1.3: Using a Graphing Utility to Graph an Equation

Use a graphing utility to graph the equation: $y = -\frac{2}{3}x - \frac{4}{3}$.

Solution

Enter the equation in the $y =$ function of the calculator. Set the window settings so that both the x - and y - intercepts are showing in the window. See Figure 6.1.11.

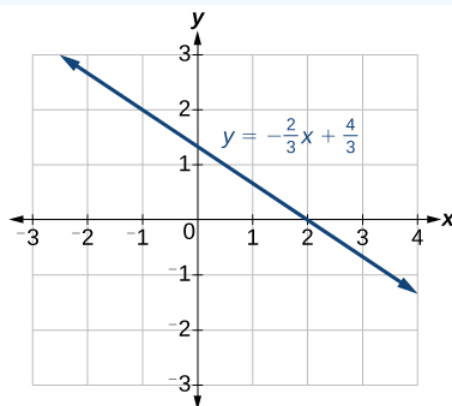


Figure 6.1.11

Finding x -intercepts and y -intercepts

The **intercepts** of a graph are points at which the graph crosses the axes. The x -intercept is the point at which the graph crosses the x -axis. At this point, the y -coordinate is zero. The y -intercept is the point at which the graph crosses the y -axis. At this point, the x -coordinate is zero.

To determine the x -intercept, we set y equal to zero and solve for x . Similarly, to determine the y -intercept, we set x equal to zero and solve for y . For example, let's find the intercepts of the equation $y = 3x - 1$.

To find the x -intercept, set $y = 0$.

$$\begin{aligned} y &= 3x - 1 \\ 0 &= 3x - 1 \\ 1 &= 3x \\ \frac{1}{3} &= x \end{aligned}$$

x -intercept: $\left(\frac{1}{3}, 0\right)$

To find the y -intercept, set $x = 0$.

$$\begin{aligned} y &= 3x - 1 \\ y &= 3(0) - 1 \\ y &= -1 \end{aligned}$$

y -intercept: $(0, -1)$

We can confirm that our results make sense by observing a graph of the equation as in Figure 6.1.12. Notice that the graph crosses the axes where we predicted it would.

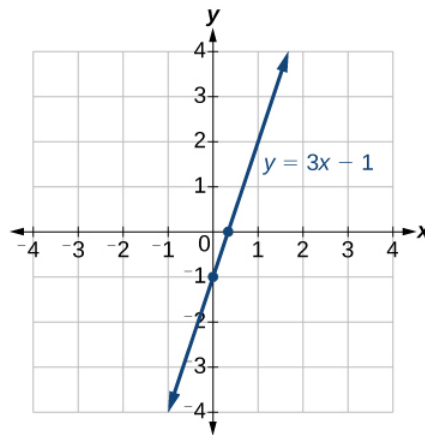


Figure 6.1.12

Howto: GIVEN AN EQUATION, FIND THE INTERCEPTS

1. Find the x -intercept by setting $y = 0$ and solving for x .
2. Find the y -intercept by setting $x = 0$ and solving for y .

✓ Example 6.1.4: Finding the Intercepts of the Given Equation

Find the intercepts of the equation $y = -3x - 4$. Then sketch the graph using only the intercepts.

Solution

Set $y = 0$ to find the x -intercept.

$$\begin{aligned} y &= -3x - 4 \\ 0 &= -3x - 4 \\ 4 &= -3x \\ \frac{4}{-3} &= x \end{aligned}$$

x -intercept: $\left(-\frac{4}{3}, 0\right)$

Set $x = 0$ to find the y -intercept.

$$\begin{aligned} y &= -3x - 4 \\ y &= -3(0) - 4 \\ y &= -4 \end{aligned}$$

y -intercept: $(0, -4)$

Plot both points, and draw a line passing through them as in Figure 6.1.13

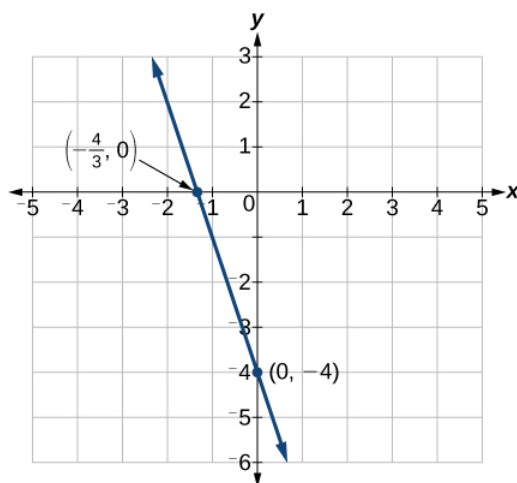


Figure 6.1.13

? Exercise 6.1.2

Find the intercepts of the equation and sketch the graph: $y = -\frac{3}{4}x + 3$.

Answer

x -intercept is $(4, 0)$; y -intercept is $(0, 3)$

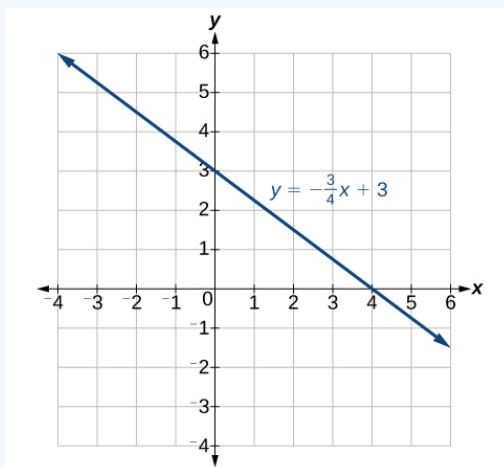


Figure 6.1.14

Using the Distance Formula

Derived from the **Pythagorean Theorem**, the **distance formula** is used to find the distance between two points in the plane. The Pythagorean Theorem, $a^2 + b^2 = c^2$, is based on a right triangle where a and b are the lengths of the legs adjacent to the right angle, and c is the length of the hypotenuse. See Figure 6.1.15

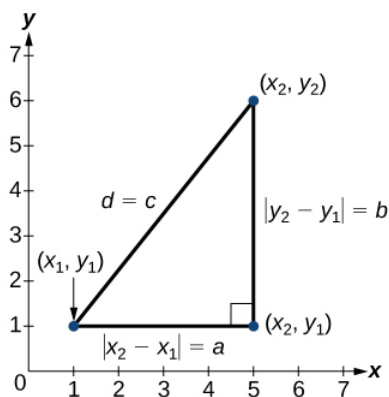


Figure 6.1.15

The relationship of sides $|x_2 - x_1|$ and $|y_2 - y_1|$ to side d is the same as that of sides a and b to side c . We use the absolute value symbol to indicate that the length is a positive number because the absolute value of any number is positive. (For example, $|-3| = 3$.) The symbols $|x_2 - x_1|$ and $|y_2 - y_1|$ indicate that the lengths of the sides of the triangle are positive. To find the length c , take the square root of both sides of the Pythagorean Theorem.

$$c^2 = a^2 + b^2 \rightarrow c = \sqrt{a^2 + b^2} \quad (6.1.1)$$

It follows that the distance formula is given as

$$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 \rightarrow d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (6.1.2)$$

We do not have to use the absolute value symbols in this definition because any number squared is positive.

distance between two points

Given endpoints (x_1, y_1) and (x_2, y_2) , the distance between two points is given by

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (6.1.3)$$

✓ Example 6.1.5: Finding the Distance between Two Points

Find the distance between the points $(-3, -1)$ and $(2, 3)$.

Solution

Let us first look at the graph of the two points. Connect the points to form a right triangle as in Figure 6.1.16

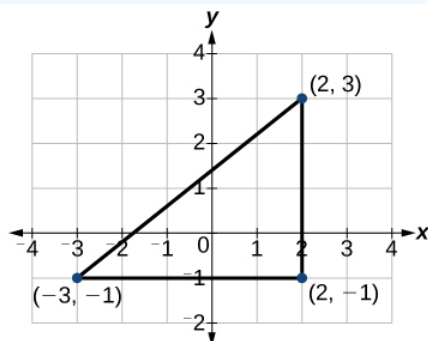


Figure 6.1.16

Then, calculate the length of d using the distance formula.

$$\begin{aligned}
 d &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\
 &= \sqrt{(2 - (-3))^2 + (3 - (-1))^2} \\
 &= \sqrt{(5)^2 + (4)^2} \\
 &= \sqrt{25 + 16} \\
 &= \sqrt{41}
 \end{aligned}$$

? Exercise 6.1.3

Find the distance between two points: (1, 4) and (11, 9).

Answer

$$\sqrt{125} = 5\sqrt{5}$$

✓ Example 6.1.6: Finding the Distance between Two Locations

Let's return to the situation introduced at the beginning of this section.

Tracie set out from Elmhurst, IL, to go to Franklin Park. On the way, she made a few stops to do errands. Each stop is indicated by a red dot in Figure 6.1.1. Find the total distance that Tracie traveled. Compare this with the distance between her starting and final positions.

Solution

The first thing we should do is identify ordered pairs to describe each position. If we set the starting position at the origin, we can identify each of the other points by counting units east (right) and north (up) on the grid. For example, the first stop is 1 block east and 1 block north, so it is at (1, 1). The next stop is 5 blocks to the east, so it is at (5, 1). After that, she traveled 3 blocks east and 2 blocks north to (8, 3). Lastly, she traveled 4 blocks north to (8, 7). We can label these points on the grid as in Figure 6.1.17.

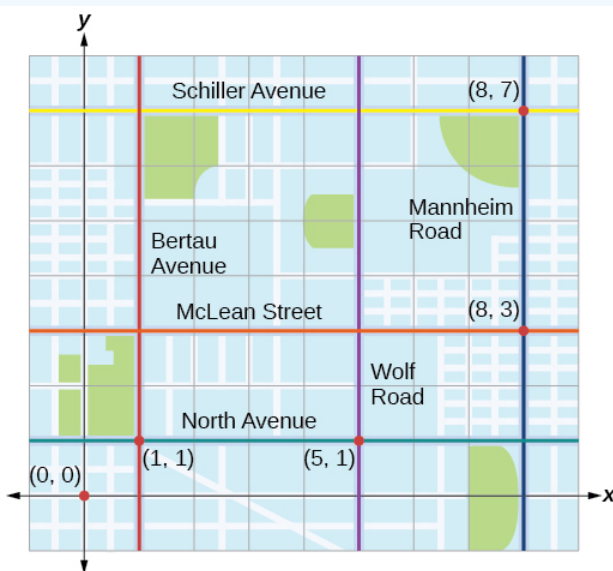


Figure 6.1.17

Next, we can calculate the distance. Note that each grid unit represents 1,000 feet.

- From her starting location to her first stop at (1, 1), Tracie might have driven north 1,000 feet and then east 1,000 feet, or vice versa. Either way, she drove 2,000 feet to her first stop.
- Her second stop is at (5, 1). So from (1, 1) to (5, 1), Tracie drove east 4,000 feet.

- Her third stop is at (8, 3). There are a number of routes from (5, 1) to (8, 3). Whatever route Tracie decided to use, the distance is the same, as there are no angular streets between the two points. Let's say she drove east 3, 000 feet and then north 2, 000 feet for a total of 5, 000 feet.
- Tracie's final stop is at (8, 7). This is a straight drive north from (8, 3) for a total of 4, 000 feet.

Next, we will add the distances listed in Table 6.1.4

Table 6.1.4

From/To	Number of Feet Driven
(0, 0) to (1, 1)	2, 000
(1, 1) to (5, 1)	4, 000
(5, 1) to (8, 3)	5, 000
(8, 3) to (8, 7)	4, 000
Total	15, 000

The total distance Tracie drove is 15, 000 feet, or 2.84 miles. This is not, however, the actual distance between her starting and ending positions. To find this distance, we can use the distance formula between the points (0, 0) and (8, 7).

$$\begin{aligned}
 d &= \sqrt{(0-8)^2 + (7-0)^2} \\
 &= \sqrt{64+49} \\
 &= \sqrt{113} \\
 &= 10.63 \text{ units}
 \end{aligned}$$

At 1, 000 feet per grid unit, the distance between Elmhurst, IL, to Franklin Park is 10, 630.14 feet, or 2.01 miles. The distance formula results in a shorter calculation because it is based on the hypotenuse of a right triangle, a straight diagonal from the origin to the point (8, 7). Perhaps you have heard the saying "as the crow flies," which means the shortest distance between two points because a crow can fly in a straight line even though a person on the ground has to travel a longer distance on existing roadways.

Using the Midpoint Formula

When the endpoints of a line segment are known, we can find the point midway between them. This point is known as the midpoint and the formula is known as the **midpoint formula**. Given the endpoints of a line segment, (x_1, y_1) and (x_2, y_2) , the midpoint formula states how to find the coordinates of the midpoint M.

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \quad (6.1.4)$$

A graphical view of a midpoint is shown in Figure 6.1.18. Notice that the line segments on either side of the midpoint are congruent.

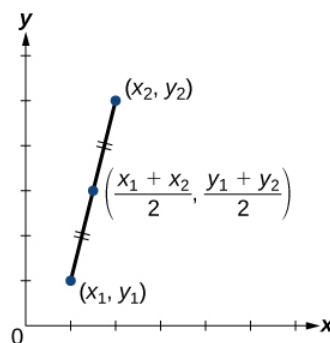


Figure 6.1.18

✓ Example 6.1.7: Finding the Midpoint of the Line Segment

Find the midpoint of the line segment with the endpoints $(7, -2)$ and $(9, 5)$.

Solution

Use the formula to find the midpoint of the line segment.

$$\begin{aligned}\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) &= \left(\frac{7 + 9}{2}, \frac{-2 + 5}{2}\right) \\ &= \left(8, \frac{3}{2}\right)\end{aligned}$$

? Exercise 6.1.4

Find the midpoint of the line segment with endpoints $(-2, -1)$ and $(-8, 6)$.

Answer

$$\left(-5, \frac{5}{2}\right)$$

✓ Example 6.1.8: Finding the Center of a Circle

The diameter of a circle has endpoints $(-1, -4)$ and $(5, -4)$. Find the center of the circle.

Solution

The center of a circle is the center, or midpoint, of its diameter. Thus, the midpoint formula will yield the center point.

$$\begin{aligned}\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) &= \left(\frac{-1 + 5}{2}, \frac{-4 - 4}{2}\right) \\ &= \left(\frac{4}{2}, -\frac{8}{2}\right) \\ &= (2, -4)\end{aligned}$$

📌 Media

Access these online resources for additional instruction and practice with the Cartesian coordinate system.

1. [Plotting points on the coordinate plane](#)
2. Find x and y intercepts based on the graph of a line

Key Concepts

- We can locate, or plot, points in the Cartesian coordinate system using ordered pairs, which are defined as displacement from the x -axis and displacement from the y -axis. See [Example](#).
- An equation can be graphed in the plane by creating a table of values and plotting points. See [Example](#).
- Using a graphing calculator or a computer program makes graphing equations faster and more accurate. Equations usually have to be entered in the form $y = \underline{\hspace{1cm}}$. See [Example](#).
- Finding the x - and y -intercepts can define the graph of a line. These are the points where the graph crosses the axes. See [Example](#).
- The distance formula is derived from the Pythagorean Theorem and is used to find the length of a line segment. See [Example](#) and [Example](#).
- The midpoint formula provides a method of finding the coordinates of the midpoint dividing the sum of the x -coordinates and the sum of the y -coordinates of the endpoints by 2. See [Example](#) and [Example](#).

Exercises

For the following exercises, find the x -intercept and the y -intercept without graphing.

1. $4x - 3y = 12$
2. $2y - 4 = 3x$

For the following exercises, solve for y in terms of x , putting the equation in slope-intercept form.

3. $5x = 3y - 12$
4. $2x - 5y = 7$

For the following exercises, find the distance between the two points.

5. $(-2,5)(4,-1)$
6. $(-12,-3)(-1,5)$
7. Find the distance between the two points $(-71,432)$ and $(511,218)$ using your calculator, and round your answer to the nearest thousandth.

For the following exercises, find the coordinates of the midpoint of the line segment that joins the two given points.

8. $(-1,5)$ and $(4,6)$
9. $(-13,5)$ and $(17,18)$

For the following exercises, construct a table and graph the equation by plotting at least three points.

10. $y = \frac{1}{2}x + 4$
11. $4x - 3y = 6$

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6.2: Linear Functions

Learning Objectives

In this section, you will:

- Represent a linear function.
- Determine whether a linear function is increasing, decreasing, or constant.
- Calculate and interpret slope.
- Write the point-slope form of an equation.
- Write and interpret a linear function.



Figure 1 Shanghai MagLev Train (credit: “kanegen”/Flickr)

Just as with the growth of a bamboo plant, there are many situations that involve constant change over time. Consider, for example, the first commercial maglev train in the world, the Shanghai MagLev Train (Figure 1). It carries passengers comfortably for a 30-kilometer trip from the airport to the subway station in only eight minutes.²

Suppose a maglev train were to travel a long distance, and that the train maintains a constant speed of 83 meters per second for a period of time once it is 250 meters from the station. How can we analyze the train’s distance from the station as a function of time? In this section, we will investigate a kind of function that is useful for this purpose, and use it to investigate real-world situations such as the train’s distance from the station at a given point in time.

Representing Linear Functions

The function describing the train’s motion is a **linear function**, which is defined as a function with a constant rate of change, that is, a polynomial of degree 1. There are several ways to represent a linear function, including word form, function notation, tabular form, and graphical form. We will describe the train’s motion as a function using each method.

Representing a Linear Function in Word Form

Let’s begin by describing the linear function in words. For the train problem we just considered, the following word sentence may be used to describe the function relationship.

- *The train’s distance from the station is a function of the time during which the train moves at a constant speed plus its original distance from the station when it began moving at constant speed.*

The speed is the rate of change. Recall that a rate of change is a measure of how quickly the dependent variable changes with respect to the independent variable. The rate of change for this example is constant, which means that it is the same for each input value. As the time (input) increases by 1 second, the corresponding distance (output) increases by 83 meters. The train began moving at this constant speed at a distance of 250 meters from the station.

Representing a Linear Function in Function Notation

Another approach to representing linear functions is by using function notation. One example of function notation is an equation written in the form known as the slope-intercept form of a line, where x is the input value, m is the rate of change, and b is the initial value of the dependent variable.

Equation form	$y = mx + b$
Equation notation	$f(x) = mx + b$

In the example of the train, we might use the notation $D(t)$ in which the total distance D is a function of the time t . The rate, m , is 83 meters per second. The initial value of the dependent variable b is the original distance from the station, 250 meters. We can write a generalized equation to represent the motion of the train.

$$D(t) = 83t + 250$$

Representing a Linear Function in Tabular Form

A third method of representing a linear function is through the use of a table. The relationship between the distance from the station and the time is represented in Figure 2. From the table, we can see that the distance changes by 83 meters for every 1 second increase in time.

		1 second	1 second	1 second
t	0	1	2	3
$D(t)$	250	333	416	499
		83 meters	83 meters	83 meters

Figure 2 Tabular representation of the function D showing selected input and output values

Q&A

Can the input in the previous example be any real number?

No. The input represents time, so while nonnegative rational and irrational numbers are possible, negative real numbers are not possible for this example. The input consists of non-negative real numbers.

Representing a Linear Function in Graphical Form

Another way to represent linear functions is visually, using a graph. We can use the function relationship from above, $D(t) = 83t + 250$, to draw a graph, represented in Figure 3. Notice the graph is a line. When we plot a linear function, the graph is always a line.

The rate of change, which is constant, determines the slant, or **slope** of the line. The point at which the input value is zero is the vertical intercept, or **y-intercept**, of the line. We can see from the graph in Figure 3 that the y-intercept in the train example we just saw is $(0, 250)$ and represents the distance of the train from the station when it began moving at a constant speed.

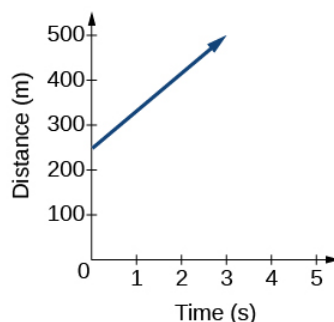


Figure 3 The graph of $D(t) = 83t + 250$. Graphs of linear functions are lines because the rate of change is constant.

Notice that the graph of the train example is restricted, but this is not always the case. Consider the graph of the line $f(x) = 2x + 1$. Ask yourself what numbers can be input to the function, that is, what is the domain of the function? The domain is comprised of all real numbers because any number may be doubled, and then have one added to the product.

Linear Function

A linear function is a function whose graph is a line. Linear functions can be written in the slope-intercept form of a line

$$f(x) = mx + b$$

where b is the initial or starting value of the function (when input, $x = 0$), and m is the constant rate of change, or slope of the function. The y -intercept is at $(0, b)$.

Example 1

Using a Linear Function to Find the Pressure on a Diver

The pressure, P , in pounds per square inch (PSI) on the diver in Figure 4 depends upon her depth below the water surface, d , in feet. This relationship may be modeled by the equation, $P(d) = 0.434d + 14.696$. Restate this function in words.



Figure 4 (credit: Ilse Reijs and Jan-Noud Hutten)

Answer

To restate the function in words, we need to describe each part of the equation. The pressure as a function of depth equals four hundred thirty-four thousandths times depth plus fourteen and six hundred ninety-six thousandths.

Analysis

The initial value, 14.696, is the pressure in PSI on the diver at a depth of 0 feet, which is the surface of the water. The rate of change, or slope, is 0.434 PSI per foot. This tells us that the pressure on the diver increases 0.434 PSI for each foot her depth increases.

Determining whether a Linear Function Is Increasing, Decreasing, or Constant

The linear functions we used in the two previous examples increased over time, but not every linear function does. A linear function may be increasing, decreasing, or constant. For an increasing function, as with the train example, the output values increase as the input values increase. The graph of an increasing function has a positive slope. A line with a positive slope slants upward from left to right as in Figure 5(a). For a decreasing function, the slope is negative. The output values decrease as the input values increase. A line with a negative slope slants downward from left to right as in Figure 5(b). If the function is constant, the output values are the same for all input values so the slope is zero. A line with a slope of zero is horizontal as in Figure 5(c).

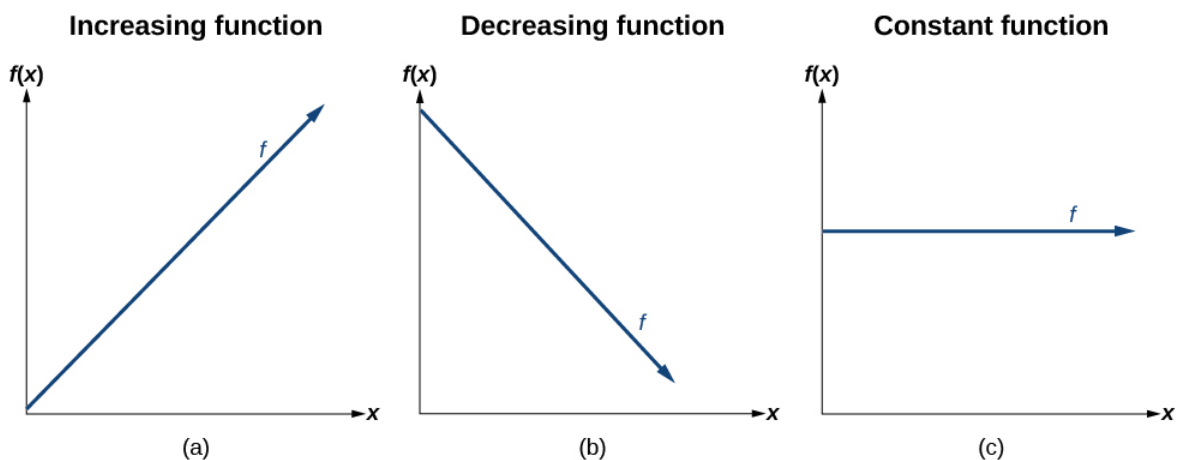


Figure 5

Increasing and Decreasing Functions

The slope determines if the function is an increasing linear function, a decreasing linear function, or a constant function.

- $f(x) = mx + b$ is an increasing function if $m > 0$.
- $f(x) = mx + b$ is an decreasing function if $m < 0$.
- $f(x) = mx + b$ is a constant function if $m = 0$.

Example 2

Deciding whether a Function Is Increasing, Decreasing, or Constant

Some recent studies suggest that a teenager sends an average of 60 texts per day.³ For each of the following scenarios, find the linear function that describes the relationship between the input value and the output value. Then, determine whether the graph of the function is increasing, decreasing, or constant.

1. ① The total number of texts a teen sends is considered a function of time in days. The input is the number of days, and output is the total number of texts sent.
2. ② A teen has a limit of 500 texts per month in his or her data plan. The input is the number of days, and output is the total number of texts remaining for the month.
3. ③ A teen has an unlimited number of texts in his or her data plan for a cost of \$50 per month. The input is the number of days, and output is the total cost of texting each month.

Answer

Analyze each function.

1. ③ The function can be represented as $f(x) = 60x$ where x is the number of days. The slope, 60, is positive so the function is increasing. This makes sense because the total number of texts increases with each day.
2. ③ The function can be represented as $f(x) = 500 - 60x$ where x is the number of days. In this case, the slope is negative so the function is decreasing. This makes sense because the number of texts remaining decreases each day and this function represents the number of texts remaining in the data plan after x days.
3. ③ The cost function can be represented as $f(x) = 50$ because the number of days does not affect the total cost. The slope is 0 so the function is constant.

Calculating and Interpreting Slope

In the examples we have seen so far, we have had the slope provided for us. However, we often need to calculate the slope given input and output values. Given two values for the input, x_1 and x_2 , and two corresponding values for the output, y_1 and y_2 — which can be represented by a set of points, (x_1, y_1) and (x_2, y_2) — we can calculate the slope m , as follows

$$m = \frac{\text{change in output (rise)}}{\text{change in input (run)}} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

where Δy is the vertical displacement and Δx is the horizontal displacement. Note in function notation two corresponding values for the output y_1 and y_2 for the function f , $y_1 = f(x_1)$ and $y_2 = f(x_2)$, so we could equivalently write

$$m = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

Figure 6 indicates how the slope of the line between the points, (x_1, y_1) and (x_2, y_2) , is calculated. Recall that the slope measures steepness. The greater the absolute value of the slope, the steeper the line is.

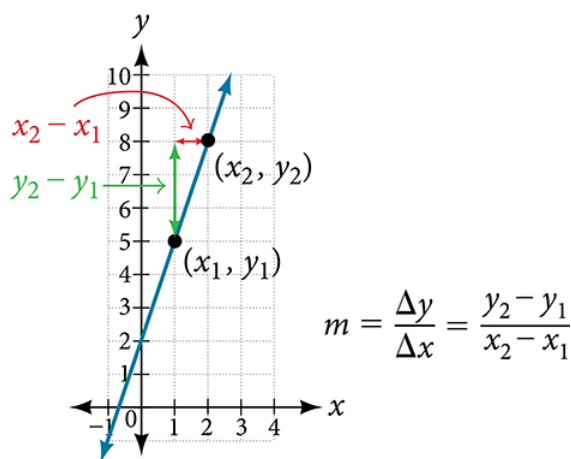


Figure 6 The slope of a function is calculated by the change in y divided by the change in x . It does not matter which coordinate is used as the (x_2, y_2) and which is the (x_1, y_1) , as long as each calculation is started with the elements from the same coordinate pair.

Q&A

Are the units for slope always $\frac{\text{units for the output}}{\text{units for the input}}$?

Yes. Think of the units as the change of output value for each unit of change in input value. An example of slope could be miles per hour or dollars per day. Notice the units appear as a ratio of units for the output per units for the input.

Calculate Slope

The slope, or rate of change, of a function m can be calculated according to the following:

$$m = \frac{\text{change in output (rise)}}{\text{change in input (run)}} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

where x_1 and x_2 are input values, y_1 and y_2 are output values.

How To

Given two points from a linear function, calculate and interpret the slope.

1. Determine the units for output and input values.
2. Calculate the change of output values and change of input values.
3. Interpret the slope as the change in output values per unit of the input value.

Example 3

Finding the Slope of a Linear Function

If $f(x)$ is a linear function, and $(3, -2)$ and $(8, 1)$ are points on the line, find the slope. Is this function increasing or decreasing?

Answer

The coordinate pairs are $(3, -2)$ and $(8, 1)$. To find the rate of change, we divide the change in output by the change in input.

$$m = \frac{\text{change in output}}{\text{change in input}} = \frac{1 - (-2)}{8 - 3} = \frac{3}{5}$$

We could also write the slope as $m = 0.6$. The function is increasing because $m > 0$.

Analysis

As noted earlier, the order in which we write the points does not matter when we compute the slope of the line as long as the first output value, or y -coordinate, used corresponds with the first input value, or x -coordinate, used.

Try It #1

If $f(x)$ is a linear function, and $(2, 3)$ and $(0, 4)$ are points on the line, find the slope. Is this function increasing or decreasing?

Example 4

Finding the Population Change from a Linear Function

The population of a city increased from 23,400 to 27,800 between 2008 and 2012. Find the change of population per year if we assume the change was constant from 2008 to 2012.

Answer

The rate of change relates the change in population to the change in time. The population increased by $27,800 - 23,400 = 4,400$ people over the four-year time interval. To find the rate of change, divide the change in the number of people by the number of years.

$$\frac{4,400 \text{ people}}{4 \text{ years}} = 1,100 \frac{\text{people}}{\text{year}}$$

So the population increased by 1,100 people per year.

Analysis

Because we are told that the population increased, we would expect the slope to be positive. This positive slope we calculated is therefore reasonable.

Try It #2

The population of a small town increased from 1,442 to 1,868 between 2009 and 2012. Find the change of population per year if we assume the change was constant from 2009 to 2012.

Writing the Point-Slope Form of a Linear Equation

Up until now, we have been using the slope-intercept form of a linear equation to describe linear functions. Here, we will learn another way to write a linear function, the point-slope form.

$$y - y_1 = m(x - x_1)$$

The point-slope form is derived from the slope formula.

$$\begin{aligned} m &= \frac{y - y_1}{x - x_1} && \text{assuming } x \neq x_1 \\ m(x - x_1) &= \frac{y - y_1}{x - x_1}(x - x_1) && \text{Multiply both sides by } (x - x_1). \\ m(x - x_1) &= y - y_1 && \text{Simplify.} \\ y - y_1 &= m(x - x_1) && \text{Rearrange.} \end{aligned}$$

Keep in mind that the slope-intercept form and the point-slope form can be used to describe the same function. We can move from one form to another using basic algebra. For example, suppose we are given an equation in point-slope form, $y - 4 = -\frac{1}{2}(x - 6)$. We can convert it to the slope-intercept form as shown.

$$\begin{aligned} y - 4 &= -\frac{1}{2}(x - 6) \\ y - 4 &= -\frac{1}{2}x + 3 && \text{Distribute the } -\frac{1}{2}. \\ y &= -\frac{1}{2}x + 7 && \text{Add 4 to each side.} \end{aligned}$$

Therefore, the same line can be described in slope-intercept form as $y = -\frac{1}{2}x + 7$.

Point-Slope Form of a Linear Equation

The **point-slope form** of a linear equation takes the form

$$y - y_1 = m(x - x_1)$$

where m is the slope, x_1 and y_1 are the x - and y -coordinates of a specific point through which the line passes.

Writing the Equation of a Line Using a Point and the Slope

The point-slope form is particularly useful if we know one point and the slope of a line. Suppose, for example, we are told that a line has a slope of 2 and passes through the point $(4, 1)$. We know that $m = 2$ and that $x_1 = 4$ and $y_1 = 1$. We can substitute these values into the general point-slope equation.

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 1 &= 2(x - 4) \end{aligned}$$

If we wanted to then rewrite the equation in slope-intercept form, we apply algebraic techniques.

$$\begin{aligned} y - 1 &= 2(x - 4) \\ y - 1 &= 2x - 8 && \text{Distribute the 2.} \\ y &= 2x - 7 && \text{Add 1 to each side.} \end{aligned}$$

Both equations, $y - 1 = 2(x - 4)$ and $y = 2x - 7$, describe the same line. See Figure 7.

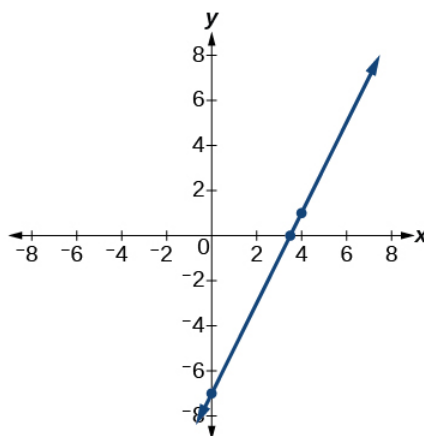


Figure 7

Example 5

Writing Linear Equations Using a Point and the Slope

Write the point-slope form of an equation of a line with a slope of 3 that passes through the point $(6, -1)$. Then rewrite it in the slope-intercept form.

Answer

Let's figure out what we know from the given information. The slope is 3, so $m = 3$. We also know one point, so we know $x_1 = 6$ and $y_1 = -1$. Now we can substitute these values into the general point-slope equation.

$$y - y_1 = m(x - x_1)$$

$$y - (-1) = 3(x - 6) \quad \text{Substitute known values.}$$

$$y + 1 = 3(x - 6) \quad \text{Distribute } -1 \text{ to find point-slope form.}$$

Then we use algebra to find the slope-intercept form.

$$y + 1 = 3(x - 6)$$

$$y + 1 = 3x - 18 \quad \text{Distribute 3.}$$

$$y = 3x - 19 \quad \text{Simplify to slope-intercept form.}$$

Try It #3

Write the point-slope form of an equation of a line with a slope of -2 that passes through the point $(-2, 2)$. Then rewrite it in the slope-intercept form.

Writing the Equation of a Line Using Two Points

The point-slope form of an equation is also useful if we know any two points through which a line passes. Suppose, for example, we know that a line passes through the points $(0, 1)$ and $(3, 2)$. We can use the coordinates of the two points to find the slope.

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{2 - 1}{3 - 0} \\ &= \frac{1}{3} \end{aligned}$$

Now we can use the slope we found and the coordinates of one of the points to find the equation for the line. Let use (0, 1) for our point.

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 1 &= \frac{1}{3}(x - 0) \end{aligned}$$

As before, we can use algebra to rewrite the equation in the slope-intercept form.

$$\begin{aligned} y - 1 &= \frac{1}{3}(x - 0) \\ y - 1 &= \frac{1}{3}x && \text{Distribute the } \frac{1}{3}. \\ y &= \frac{1}{3}x + 1 && \text{Add 1 to each side.} \end{aligned}$$

Both equations describe the line shown in Figure 8.

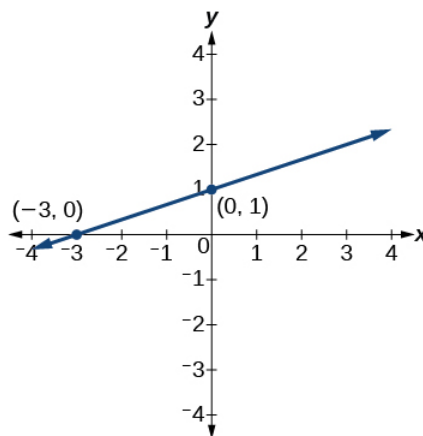


Figure 8

Example 6

Writing Linear Equations Using Two Points

Write the point-slope form of an equation of a line that passes through the points (5, 1) and (8, 7). Then rewrite it in the slope-intercept form.

Answer

Let's begin by finding the slope.

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{7 - 1}{8 - 5} \\ &= \frac{6}{3} \\ &= 2 \end{aligned}$$

So $m = 2$. Next, we substitute the slope and the coordinates for one of the points into the general point-slope equation. We can choose either point, but we will use (5, 1).

$$y - y_1 = m(x - x_1)$$

$$y - 1 = 2(x - 5)$$

The point-slope equation of the line is $y_2 - 1 = 2(x_2 - 5)$. To rewrite the equation in slope-intercept form, we use algebra.

$$y - 1 = 2(x - 5)$$

$$y - 1 = 2x - 10$$

$$y = 2x - 9$$

The slope-intercept equation of the line is $y = 2x - 9$.

Try It #4

Write the point-slope form of an equation of a line that passes through the points $(-1, 3)$ and $(0, 0)$. Then rewrite it in the slope-intercept form.

Writing and Interpreting an Equation for a Linear Function

Now that we have written equations for linear functions in both the slope-intercept form and the point-slope form, we can choose which method to use based on the information we are given. That information may be provided in the form of a graph, a point and a slope, two points, and so on. Look at the graph of the function f in Figure 9.

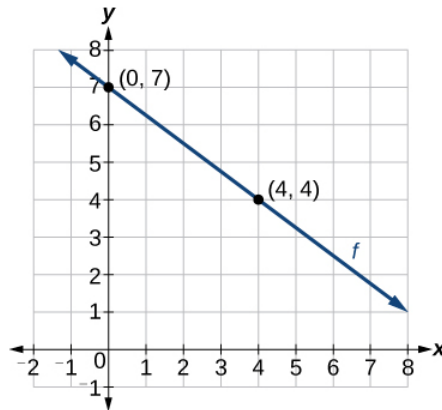


Figure 9

We are not given the slope of the line, but we can choose any two points on the line to find the slope. Let's choose $(0, 7)$ and $(4, 4)$. We can use these points to calculate the slope.

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{4 - 7}{4 - 0} \\ &= -\frac{3}{4} \end{aligned}$$

Now we can substitute the slope and the coordinates of one of the points into the point-slope form.

$$y - y_1 = m(x - x_1)$$

$$y - 4 = -\frac{3}{4}(x - 4)$$

If we want to rewrite the equation in the slope-intercept form, we would find

$$y - 4 = -\frac{3}{4}(x - 4)$$

$$y - 4 = -\frac{3}{4}x + 3$$

$$y = -\frac{3}{4}x + 7$$

If we wanted to find the slope-intercept form without first writing the point-slope form, we could have recognized that the line crosses the y -axis when the output value is 7. Therefore, $b = 7$. We now have the initial value b and the slope m so we can substitute m and b into the slope-intercept form of a line.

$$f(x) = mx + b$$

$$\begin{array}{cc} \uparrow & \uparrow \\ -\frac{3}{4} & 7 \end{array}$$

$$f(x) = -\frac{3}{4}x + 7$$

So the function is $f(x) = -\frac{3}{4}x + 7$, and the linear equation would be $y = -\frac{3}{4}x + 7$.

How To

Given the graph of a linear function, write an equation to represent the function.

1. Identify two points on the line.
2. Use the two points to calculate the slope.
3. Determine where the line crosses the y -axis to identify the y -intercept by visual inspection.
4. Substitute the slope and y -intercept into the slope-intercept form of a line equation.

Example 7

Writing an Equation for a Linear Function

Write an equation for a linear function given a graph of f shown in Figure 10.

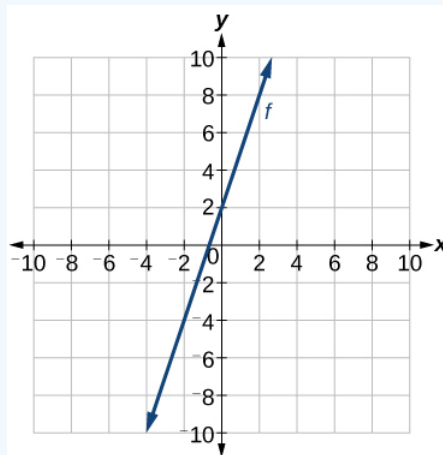


Figure 10

Answer

Identify two points on the line, such as $(0, 2)$ and $(-2, -4)$. Use the points to calculate the slope.

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{-4 - 2}{-2 - 0} \\ &= \frac{-6}{-2} \\ &= 3 \end{aligned}$$

Substitute the slope and the coordinates of one of the points into the point-slope form.

$$\begin{aligned}y - y_1 &= m(x - x_1) \\y - (-4) &= 3(x - (-2)) \\y + 4 &= 3(x + 2)\end{aligned}$$

We can use algebra to rewrite the equation in the slope-intercept form.

$$\begin{aligned}y + 4 &= 3(x + 2) \\y + 4 &= 3x + 6 \\y &= 3x + 2\end{aligned}$$

Analysis

This makes sense because we can see from Figure 11 that the line crosses the y-axis at the point $(0, 2)$, which is the y-intercept, so $b = 2$.

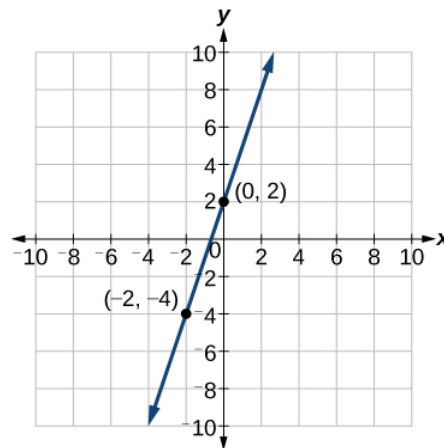


Figure 11

Example 8

Writing an Equation for a Linear Cost Function

Suppose Ben starts a company in which he incurs a fixed cost of \$1,250 per month for the overhead, which includes his office rent. His production costs are \$37.50 per item. Write a linear function C where $C(x)$ is the cost for x items produced in a given month.

Answer

The fixed cost is present every month, \$1,250. The costs that can vary include the cost to produce each item, which is \$37.50 for Ben. The variable cost, called the marginal cost, is represented by 37.5. The cost Ben incurs is the sum of these two costs, represented by $C(x) = 1250 + 37.5x$.

Analysis

If Ben produces 100 items in a month, his monthly cost is represented by

$$\begin{aligned}C(100) &= 1250 + 37.5(100) \\&= 5000\end{aligned}$$

So his monthly cost would be \$5,000.

Example 9

Writing an Equation for a Linear Function Given Two Points

If f is a linear function, with $f(3) = -2$, and $f(8) = 1$, find an equation for the function in slope-intercept form.

Answer

We can write the given points using coordinates.

$$f(3) = -2 \rightarrow (3, -2)$$

$$f(8) = 1 \rightarrow (8, 1)$$

We can then use the points to calculate the slope.

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{1 - (-2)}{8 - 3} \\ &= \frac{3}{5} \end{aligned}$$

Substitute the slope and the coordinates of one of the points into the point-slope form.

$$y - y_1 = m(x - x_1)$$

$$y - (-2) = \frac{3}{5}(x - 3)$$

We can use algebra to rewrite the equation in the slope-intercept form.

$$y + 2 = \frac{3}{5}(x - 3)$$

$$y + 2 = \frac{3}{5}x - \frac{9}{5}$$

$$y = \frac{3}{5}x - \frac{19}{5}$$

Try It #5

If $f(x)$ is a linear function, with $f(2) = -11$, and $f(4) = -25$, find an equation for the function in slope-intercept form.

Modeling Real-World Problems with Linear Functions

In the real world, problems are not always explicitly stated in terms of a function or represented with a graph. Fortunately, we can analyze the problem by first representing it as a linear function and then interpreting the components of the function. As long as we know, or can figure out, the initial value and the rate of change of a linear function, we can solve many different kinds of real-world problems.

How To

Given a linear function f and the initial value and rate of change, evaluate $f(c)$.

1. Determine the initial value and the rate of change (slope).
2. Substitute the values into $f(x) = mx + b$.
3. Evaluate the function at $x = c$.

Example 10**Using a Linear Function to Determine the Number of Songs in a Music Collection**

Marcus currently has 200 songs in his music collection. Every month, he adds 15 new songs. Write a formula for the number of songs, N , in his collection as a function of time, t , the number of months. How many songs will he own in a year?

Answer

The initial value for this function is 200 because he currently owns 200 songs, so $N(0) = 200$, which means that $b = 200$.

The number of songs increases by 15 songs per month, so the rate of change is 15 songs per month. Therefore we know that $m = 15$. We can substitute the initial value and the rate of change into the slope-intercept form of a line.

$$f(x) = mx + b$$

$\begin{array}{cc} \nearrow & \searrow \\ 15 & 200 \end{array}$
 $N(t) = 15t + 200$

We can write the formula $N(t) = 15t + 200$.

With this formula, we can then predict how many songs Marcus will have in 1 year (12 months). In other words, we can evaluate the function at $t = 12$.

$$\begin{aligned} N(12) &= 15(12) + 200 \\ &= 180 + 200 \\ &= 380 \end{aligned}$$

Marcus will have 380 songs in 12 months.

Analysis

Notice that N is an increasing linear function. As the input (the number of months) increases, the output (number of songs) increases as well.

Example 11

Using a Linear Function to Calculate Salary Plus Commission

Working as an insurance salesperson, Ilya earns a base salary plus a commission on each new policy. Therefore, Ilya's weekly income, I , depends on the number of new policies, n , he sells during the week. Last week he sold 3 new policies, and earned \$760 for the week. The week before, he sold 5 new policies and earned \$920. Find an equation for $I(n)$, and interpret the meaning of the components of the equation.

Answer

The given information gives us two input-output pairs: $(3, 760)$ and $(5, 920)$. We start by finding the rate of change.

$$\begin{aligned} m &= \frac{920 - 760}{5 - 3} \\ &= \frac{\$160}{2 \text{ policies}} \\ &= \$80 \text{ per policy} \end{aligned}$$

Keeping track of units can help us interpret this quantity. Income increased by \$160 when the number of policies increased by 2, so the rate of change is \$80 per policy. Therefore, Ilya earns a commission of \$80 for each policy sold during the week.

We can then solve for the initial value.

$$\begin{aligned} I(n) &= 80n + b \\ 760 &= 80(3) + b \quad \text{when } n = 3, \quad I(3) = 760 \\ 760 - 80(3) &= b \\ 520 &= b \end{aligned}$$

The value of b is the starting value for the function and represents Ilya's income when $n = 0$, or when no new policies are sold. We can interpret this as Ilya's base salary for the week, which does not depend upon the number of policies sold.

We can now write the final equation.

$$I(n) = 80n + 520$$

Our final interpretation is that Ilya's base salary is \$520 per week and he earns an additional \$80 commission for each policy sold.

Example 12

Using Tabular Form to Write an Equation for a Linear Function

Table 1 relates the number of rats in a population to time, in weeks. Use the table to write a linear equation.

w, number of weeks	0	2	4	6
P(w), number of rats	1000	1080	1160	1240

Table 1

Answer

We can see from the table that the initial value for the number of rats is 1000, so $b = 1000$.

Rather than solving for m , we can tell from looking at the table that the population increases by 80 for every 2 weeks that pass. This means that the rate of change is 80 rats per 2 weeks, which can be simplified to 40 rats per week.

$$P(w) = 40w + 1000$$

If we did not notice the rate of change from the table we could still solve for the slope using any two points from the table. For example, using (2, 1080) and (6, 1240)

$$\begin{aligned} m &= \frac{1240 - 1080}{6 - 2} \\ &= \frac{160}{4} \\ &= 40 \end{aligned}$$

Q&A

Is the initial value always provided in a table of values like Table 1?

No. Sometimes the initial value is provided in a table of values, but sometimes it is not. If you see an input of 0, then the initial value would be the corresponding output. If the initial value is not provided because there is no value of input on the table equal to 0, find the slope, substitute one coordinate pair and the slope into $f(x) = mx + b$, and solve for b .

Try It #6

A new plant food was introduced to a young tree to test its effect on the height of the tree. Table 2 shows the height of the tree, in feet, x months since the measurements began. Write a linear function, $H(x)$, where x is the number of months since the start of the experiment.

x	0	2	4	8	12
H(x)	12.5	13.5	14.5	16.5	18.5

Table 2

Media

Access this online resource for additional instruction and practice with linear functions.

- [Linear Functions](#)

2.1 Section Exercises

Verbal

1.

Terry is skiing down a steep hill. Terry's elevation, $E(t)$, in feet after t seconds is given by $E(t) = 3000 - 70t$. Write a complete sentence describing Terry's starting elevation and how it is changing over time.

2.

Maria is climbing a mountain. Maria's elevation, $E(t)$, in feet after t minutes is given by $E(t) = 1200 + 40t$. Write a complete sentence describing Maria's starting elevation and how it is changing over time.

3.

Jessica is walking home from a friend's house. After 2 minutes she is 1.4 miles from home. Twelve minutes after leaving, she is 0.9 miles from home. What is her rate in miles per hour?

4.

Sonya is currently 10 miles from home and is walking farther away at 2 miles per hour. Write an equation for her distance from home t hours from now.

5.

A boat is 100 miles away from the marina, sailing directly toward it at 10 miles per hour. Write an equation for the distance of the boat from the marina after t hours.

6.

Timmy goes to the fair with \$40. Each ride costs \$2. How much money will he have left after riding n rides?

Algebraic

For the following exercises, determine whether the equation of the curve can be written as a linear function.

7.

$$y = \frac{1}{4}x + 6$$

8.

$$y = 3x - 5$$

9.

$$y = 3x^2 - 2$$

10.

$$3x + 5y = 15$$

11.

$$3x^2 + 5y = 15$$

12.

$$3x + 5y^2 = 15$$

13.

$$-2x^2 + 3y^2 = 6$$

14.

$$-\frac{x-3}{5} = 2y$$

For the following exercises, determine whether each function is increasing or decreasing.

15.

$$f(x) = 4x + 3$$

16.

$$g(x) = 5x + 6$$

17.

$$a(x) = 5 - 2x$$

18.

$$b(x) = 8 - 3x$$

19.

$$h(x) = -2x + 4$$

20.

$$k(x) = -4x + 1$$

21.

$$j(x) = \frac{1}{2}x - 3$$

22.

$$p(x) = \frac{1}{4}x - 5$$

23.

$$n(x) = -\frac{1}{3}x - 2$$

24.

$$m(x) = -\frac{3}{8}x + 3$$

For the following exercises, find the slope of the line that passes through the two given points.

25.

(2, 4) and (4, 10)

26.

(1, 5) and (4, 11)

27.

(-1, 4) and (5, 2)

28.

(8, -2) and (4, 6)

29.

(6, 11) and (-4, 3)

For the following exercises, given each set of information, find a linear equation satisfying the conditions, if possible.

30.

$$f(-5) = -4, \text{ and } f(5) = 2$$

31.

$$f(-1) = 4 \text{ and } f(5) = 1$$

32.

(2, 4) and (4, 10)

33.

Passes through (1, 5) and (4, 11)

34.

Passes through (-1, 4) and (5, 2)

35.

Passes through $(-2, 8)$ and $(4, 6)$

36.

x intercept at $(-2, 0)$ and y intercept at $(0, -3)$

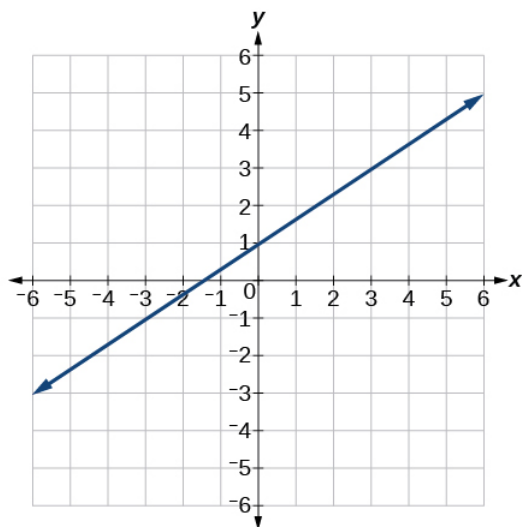
37.

x intercept at $(-5, 0)$ and y intercept at $(0, 4)$

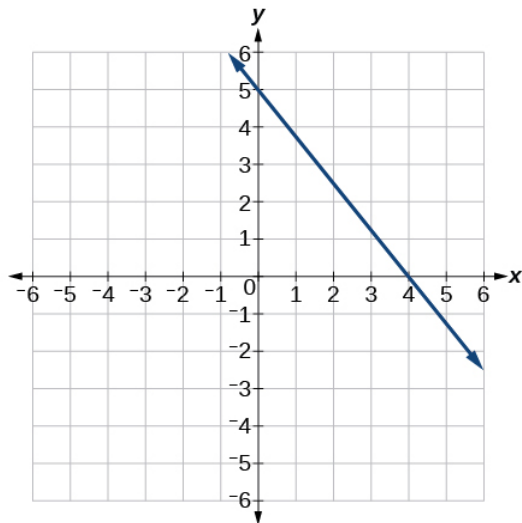
Graphical

For the following exercises, find the slope of the lines graphed.

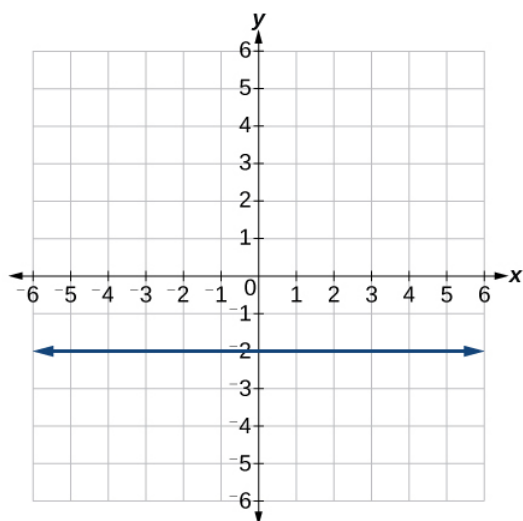
38.



39.

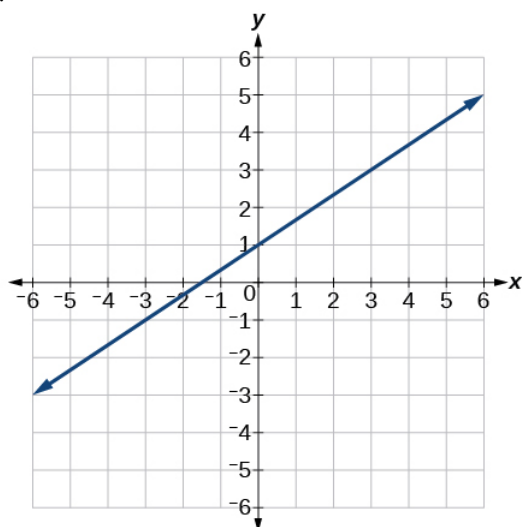


40.

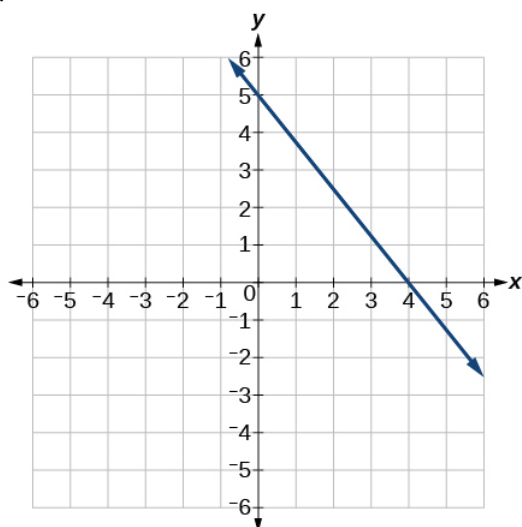


For the following exercises, write an equation for the lines graphed.

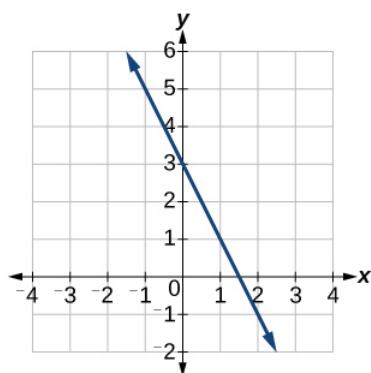
41.



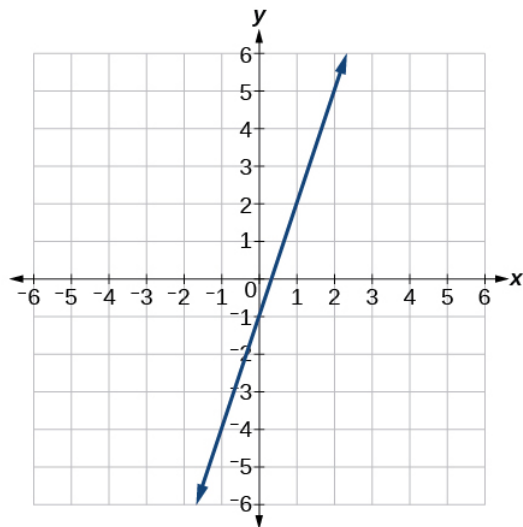
42.



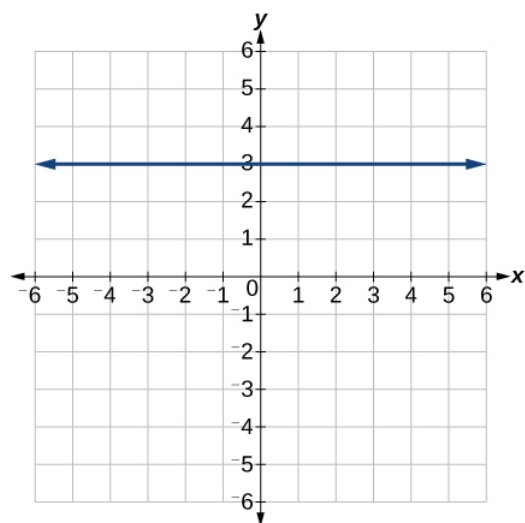
43.



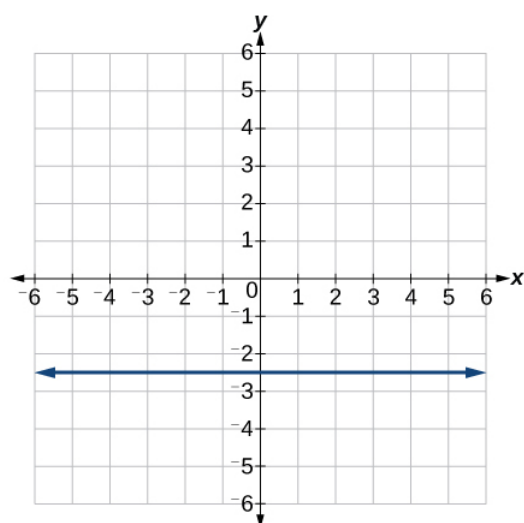
44.



45.



46.



Numeric

For the following exercises, which of the tables could represent a linear function? For each that could be linear, find a linear equation that models the data.

47.

x	0	5	10	15
$g(x)$	5	-10	-25	-40

48.

x	0	5	10	15
$h(x)$	5	30	105	230

49.

x	0	5	10	15
$f(x)$	-5	20	45	70

50.

x	5	10	20	25
$k(x)$	13	28	58	73

51.

x	0	2	4	6
$g(x)$	6	-19	-44	-69

52.

x	2	4	6	8
$f(x)$	-4	16	36	56

53.

x	2	4	6	8

$f(x)$	-4	16	36	56
--------	----	----	----	----

54.

x	0	2	6	8
$k(x)$	6	31	106	231

Technology

55.

If f is a linear function, $f(0.1) = 11.5$, and $f(0.4) = -5.9$, find an equation for the function.

56.

Graph the function f on a domain of $[-10, 10]$: $f(x) = 0.02x - 0.01$. Enter the function in a graphing utility. For the viewing window, set the minimum value of x to be -10 and the maximum value of x to be 10 .

57.

Graph the function f on a domain of $[-10, 10]$: $f(x) = 2,500x + 4,000$

58.

Table 3 shows the input, w , and output, k , for a linear function k . a. Fill in the missing values of the table. b. Write the linear function k , round to 3 decimal places.

w	-10	5.5	67.5	b
k	30	-26	a	-44

Table 3

59.

Table 4 shows the input, p , and output, q , for a linear function q . a. Fill in the missing values of the table. b. Write the linear function q .

p	0.5	0.8	12	b
q	400	700	a	1,000,000

Table 4

60.

Graph the linear function f on a domain of $[-10, 10]$ for the function whose slope is $\frac{1}{8}$ and y-intercept is $\frac{31}{16}$. Label the points for the input values of -10 and 10 .

61.

Graph the linear function f on a domain of $[-0.1, 0.1]$ for the function whose slope is 75 and y-intercept is -22.5 . Label the points for the input values of -0.1 and 0.1 .

62.

Graph the linear function f where $f(x) = ax + b$ on the same set of axes on a domain of $[-4, 4]$ for the following values of a and b .

- $a = 2$; $b = 3$
- $a = 2$; $b = 4$
- $a = 2$; $b = -4$
- $a = 2$; $b = -5$

Extensions

63.

Find the value of x if a linear function goes through the following points and has the following slope: $(x, 2), (-4, 6), m = 3$

64.

Find the value of y if a linear function goes through the following points and has the following slope: $(10, y), (25, 100), m = -5$

65.

Find the equation of the line that passes through the following points: (a, b) and $(a, b + 1)$

66.

Find the equation of the line that passes through the following points: $(2a, b)$ and $(a, b + 1)$

67.

Find the equation of the line that passes through the following points: $(a, 0)$ and (c, d)

Real-World Applications

68.

At noon, a barista notices that she has \$20 in her tip jar. If she makes an average of \$0.50 from each customer, how much will she have in her tip jar if she serves n more customers during her shift?

69.

A gym membership with two personal training sessions costs \$125, while gym membership with five personal training sessions costs \$260. What is cost per session?

70.

A clothing business finds there is a linear relationship between the number of shirts, n , it can sell and the price, p , it can charge per shirt. In particular, historical data shows that 1,000 shirts can be sold at a price of \$30, while 3,000 shirts can be sold at a price of \$22. Find a linear equation in the form $p(n) = mn + b$ that gives the price p they can charge for n shirts.

71.

A phone company charges for service according to the formula: $C(n) = 24 + 0.1n$, where n is the number of minutes talked, and $C(n)$ is the monthly charge, in dollars. Find and interpret the rate of change and initial value.

72.

A farmer finds there is a linear relationship between the number of bean stalks, n , she plants and the yield, y , each plant produces. When she plants 30 stalks, each plant yields 30 oz of beans. When she plants 34 stalks, each plant produces 28 oz of beans. Find a linear relationship in the form $y = mn + b$ that gives the yield when n stalks are planted.

73.

A city's population in the year 1960 was 287,500. In 1989 the population was 275,900. Compute the rate of growth of the population and make a statement about the population rate of change in people per year.

74.

A town's population has been growing linearly. In 2003, the population was 45,000, and the population has been growing by 1,700 people each year. Write an equation, $P(t)$, for the population t years after 2003.

75.

Suppose that average annual income (in dollars) for the years 1990 through 1999 is given by the linear function: $I(x) = 1054x + 23,286$, where x is the number of years after 1990. Which of the following interprets the slope in the context of the problem?

- a. As of 1990, average annual income was \$23,286.
- b. In the ten-year period from 1990–1999, average annual income increased by a total of \$1,054.
- c. Each year in the decade of the 1990s, average annual income increased by \$1,054.
- d. Average annual income rose to a level of \$23,286 by the end of 1999.

76.

When temperature is 0 degrees Celsius, the Fahrenheit temperature is 32. When the Celsius temperature is 100, the corresponding Fahrenheit temperature is 212. Express the Fahrenheit temperature as a linear function of C , the Celsius temperature, $F(C)$.

- Find the rate of change of Fahrenheit temperature for each unit change temperature of Celsius.
- Find and interpret $F(28)$.
- Find and interpret $F(-40)$.

Footnotes

- 2<http://www.chinahighlights.com/shang...glev-train.htm>
- 3http://www.cbsnews.com/8301-501465_1...ay-study-says/

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6.3: Linear Functions and Mathematical Models

In the following subsections we will see how to model real-world scenarios using linear functions:

- [6.3.1: Modeling with Linear Functions](#)
 - [6.3.2: Applications](#)
-

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6.3.1: Modeling with Linear Functions

Learning Objectives

In this section, you will:

- Identify steps for modeling and solving.
- Build linear models from verbal descriptions.
- Build systems of linear models.



Figure 1 (credit: EEK Photography/Flickr)

Emily is a college student who plans to spend a summer in Seattle. She has saved \$3,500 for her trip and anticipates spending \$400 each week on rent, food, and activities. How can we write a linear model to represent her situation? What would be the x -intercept, and what can she learn from it? To answer these and related questions, we can create a model using a linear function. Models such as this one can be extremely useful for analyzing relationships and making predictions based on those relationships. In this section, we will explore examples of linear function models.

Identifying Steps to Model and Solve Problems

When modeling scenarios with linear functions and solving problems involving quantities with a constant rate of change, we typically follow the same problem strategies that we would use for any type of function. Let's briefly review them:

1. Identify changing quantities, and then define descriptive variables to represent those quantities. When appropriate, sketch a picture or define a coordinate system.
2. Carefully read the problem to identify important information. Look for information that provides values for the variables or values for parts of the functional model, such as slope and initial value.
3. Carefully read the problem to determine what we are trying to find, identify, solve, or interpret.
4. Identify a solution pathway from the provided information to what we are trying to find. Often this will involve checking and tracking units, building a table, or even finding a formula for the function being used to model the problem.
5. When needed, write a formula for the function.
6. Solve or evaluate the function using the formula.

7. Reflect on whether your answer is reasonable for the given situation and whether it makes sense mathematically.
8. Clearly convey your result using appropriate units, and answer in full sentences when necessary.

Building Linear Models

Now let's take a look at the student in Seattle. In her situation, there are two changing quantities: time and money. The amount of money she has remaining while on vacation depends on how long she stays. We can use this information to define our variables, including units.

- Output: M , money remaining, in dollars
- Input: t , time, in weeks

So, the amount of money remaining depends on the number of weeks: $M(t)$

We can also identify the initial value and the rate of change.

- Initial Value: She saved \$3,500, so \$3,500 is the initial value for M .
- Rate of Change: She anticipates spending \$400 each week, so $-\$400$ per week is the rate of change, or slope.

Notice that the unit of dollars per week matches the unit of our output variable divided by our input variable. Also, because the slope is negative, the linear function is decreasing. This should make sense because she is spending money each week.

The rate of change is constant, so we can start with the linear model $M(t) = mt + b$. Then we can substitute the intercept and slope provided.

$$M(t) = mt + b$$

$$\begin{array}{ccc} & \nearrow & \nwarrow \\ -400 & & 3500 \end{array}$$

$$M(t) = -400t + 3500$$

To find the x -intercept, we set the output to zero, and solve for the input.

$$\begin{aligned} 0 &= -400t + 3500 \\ t &= \frac{3500}{400} \\ &= 8.75 \end{aligned}$$

The x -intercept is 8.75 weeks. Because this represents the input value when the output will be zero, we could say that Emily will have no money left after 8.75 weeks.

When modeling any real-life scenario with functions, there is typically a limited domain over which that model will be valid—almost no trend continues indefinitely. Here the domain refers to the number of weeks. In this case, it doesn't make sense to talk about input values less than zero. A negative input value could refer to a number of weeks before she saved \$3,500, but the scenario discussed poses the question once she saved \$3,500 because this is when her trip and subsequent spending starts. It is also likely that this model is not valid after the x -intercept, unless Emily will use a credit card and goes into debt. The domain represents the set of input values, so the reasonable domain for this function is $0 \leq t \leq 8.75$.

In the above example, we were given a written description of the situation. We followed the steps of modeling a problem to analyze the information. However, the information provided may not always be the same. Sometimes we might be provided with an intercept. Other times we might be provided with an output value. We must be careful to analyze the information we are given, and use it appropriately to build a linear model.

Using a Given Intercept to Build a Model

Some real-world problems provide the y -intercept, which is the constant or initial value. Once the y -intercept is known, the x -intercept can be calculated. Suppose, for example, that Hannah plans to pay off a no-interest loan from her parents. Her loan balance is \$1,000. She plans to pay \$250 per month until her balance is \$0. The y -intercept is the initial amount of her debt, or \$1,000. The rate of change, or slope, is $-\$250$ per month. We can then use the slope-intercept form and the given information to develop a linear model.

$$\begin{aligned} f(x) &= mx + b \\ &= -250x + 1000 \end{aligned}$$

Now we can set the function equal to 0, and solve for x to find the x -intercept.

$$\begin{aligned}0 &= -250x + 1000 \\1000 &= 250x \\4 &= x \\x &= 4\end{aligned}$$

The x -intercept is the number of months it takes her to reach a balance of \$0. The x -intercept is 4 months, so it will take Hannah four months to pay off her loan.

Using a Given Input and Output to Build a Model

Many real-world applications are not as direct as the ones we just considered. Instead they require us to identify some aspect of a linear function. We might sometimes instead be asked to evaluate the linear model at a given input or set the equation of the linear model equal to a specified output.

How To

Given a word problem that includes two pairs of input and output values, use the linear function to solve a problem.

1. Identify the input and output values.
2. Convert the data to two coordinate pairs.
3. Find the slope.
4. Write the linear model.
5. Use the model to make a prediction by evaluating the function at a given x -value.
6. Use the model to identify an x -value that results in a given y -value.
7. Answer the question posed.

Example 1

Using a Linear Model to Investigate a Town's Population

A town's population has been growing linearly. In 2004 the population was 6,200. By 2009 the population had grown to 8,100. Assume this trend continues.

1. Ⓐ Predict the population in 2013.
2. Ⓑ Identify the year in which the population will reach 15,000.

Answer

The two changing quantities are the population size and time. While we could use the actual year value as the input quantity, doing so tends to lead to very cumbersome equations because the y -intercept would correspond to the year 0, more than 2000 years ago!

To make computation a little nicer, we will define our input as the number of years since 2004:

- Input: t , years since 2004
- Output: $P(t)$, the town's population

To predict the population in 2013 ($t = 9$), we would first need an equation for the population. Likewise, to find when the population would reach 15,000, we would need to solve for the input that would provide an output of 15,000. To write an equation, we need the initial value and the rate of change, or slope.

To determine the rate of change, we will use the change in output per change in input.

$$m = \frac{\text{change in output}}{\text{change in input}}$$

The problem gives us two input-output pairs. Converting them to match our defined variables, the year 2004 would correspond to $t = 0$, giving the point $(0, 6200)$. Notice that through our clever choice of variable definition, we have "given" ourselves the y -intercept of the function. The year 2009 would correspond to $t = 5$, giving the point $(5, 8100)$.

The two coordinate pairs are $(0, 6200)$ and $(5, 8100)$. Recall that we encountered examples in which we were provided two points earlier in the chapter. We can use these values to calculate the slope.

$$\begin{aligned} m &= \frac{8100-6200}{5-0} \\ &= \frac{1900}{5} \\ &= 380 \text{ people per year} \end{aligned}$$

We already know the y-intercept of the line, so we can immediately write the equation:

$$P(t) = 380t + 6200$$

To predict the population in 2013, we evaluate our function at $t = 9$.

$$\begin{aligned} P(9) &= 380(9) + 6,200 \\ &= 9,620 \end{aligned}$$

If the trend continues, our model predicts a population of 9,620 in 2013.

To find when the population will reach 15,000, we can set $P(t) = 15000$ and solve for t .

$$\begin{aligned} 15000 &= 380t + 6200 \\ 8800 &= 380t \\ t &\approx 23.158 \end{aligned}$$

Our model predicts the population will reach 15,000 in a little more than 23 years after 2004, or somewhere around the year 2027.

Try It #1

A company sells doughnuts. They incur a fixed cost of \$25,000 for rent, insurance, and other expenses. It costs \$0.25 to produce each doughnut.

1. Ⓐ Write a linear model to represent the cost C of the company as a function of x , the number of doughnuts produced.
2. Ⓑ Find and interpret the y-intercept.

Try It #2

A city's population has been growing linearly. In 2008, the population was 28,200. By 2012, the population was 36,800. Assume this trend continues.

1. Ⓐ Predict the population in 2014.
2. Ⓑ Identify the year in which the population will reach 54,000.

Using a Diagram to Model a Problem

It is useful for many real-world applications to draw a picture to gain a sense of how the variables representing the input and output may be used to answer a question. To draw the picture, first consider what the problem is asking for. Then, determine the input and the output. The diagram should relate the variables. Often, geometrical shapes or figures are drawn. Distances are often traced out. If a right triangle is sketched, the Pythagorean Theorem relates the sides. If a rectangle is sketched, labeling width and height is helpful.

Example 2

Using a Diagram to Model Distance Walked

Anna and Emanuel start at the same intersection. Anna walks east at 4 miles per hour while Emanuel walks south at 3 miles per hour. They are communicating with a two-way radio that has a range of 2 miles. How long after they start walking will they fall out of radio contact?

Answer

In essence, we can partially answer this question by saying they will fall out of radio contact when they are 2 miles apart, which leads us to ask a new question:

“How long will it take them to be 2 miles apart?”

In this problem, our changing quantities are time and position, but ultimately we need to know how long will it take for them to be 2 miles apart. We can see that time will be our input variable, so we’ll define our input and output variables.

- Input: t , time in hours.
- Output: $A(t)$, distance in miles, and $E(t)$, distance in miles

Because it is not obvious how to define our output variable, we’ll start by drawing a picture such as Figure 2.

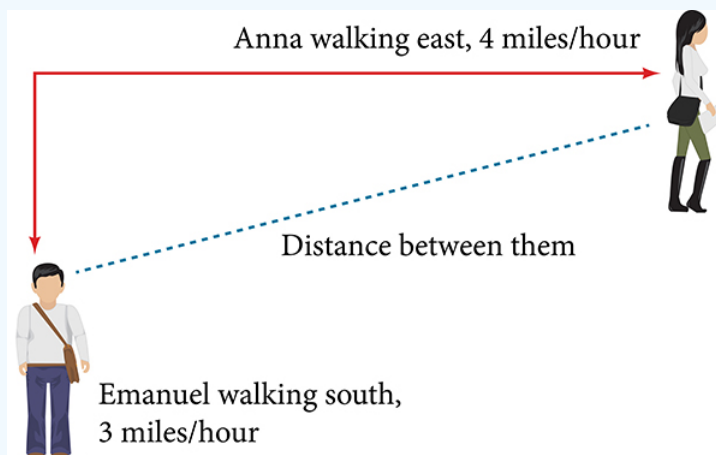


Figure 2

Initial Value: They both start at the same intersection so when $t = 0$, the distance traveled by each person should also be 0. Thus the initial value for each is 0.

Rate of Change: Anna is walking 4 miles per hour and Emanuel is walking 3 miles per hour, which are both rates of change. The slope for A is 4 and the slope for E is 3.

Using those values, we can write formulas for the distance each person has walked.

$$A(t) = 4t$$

$$E(t) = 3t$$

For this problem, the distances from the starting point are important. To notate these, we can define a coordinate system, identifying the “starting point” at the intersection where they both started. Then we can use the variable, A , which we introduced above, to represent Anna’s position, and define it to be a measurement from the starting point in the eastward direction. Likewise, can use the variable, E , to represent Emanuel’s position, measured from the starting point in the southward direction. Note that in defining the coordinate system, we specified both the starting point of the measurement and the direction of measure.

We can then define a third variable, D , to be the measurement of the distance between Anna and Emanuel. Showing the variables on the diagram is often helpful, as we can see from Figure 3.

Recall that we need to know how long it takes for D , the distance between them, to equal 2 miles. Notice that for any given input t , the outputs $A(t)$, $E(t)$, and $D(t)$ represent distances.

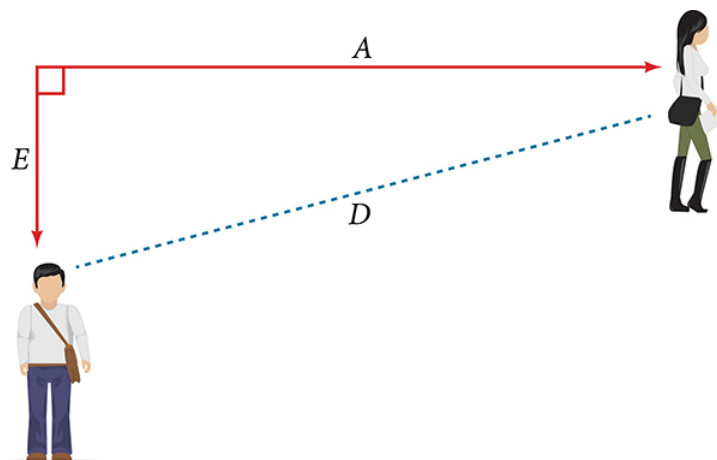


Figure 3

Figure 2 shows us that we can use the Pythagorean Theorem because we have drawn a right angle.

Using the Pythagorean Theorem, we get:

$$\begin{aligned}
 D(t)^2 &= A(t)^2 + E(t)^2 \\
 &= (4t)^2 + (3t)^2 \\
 &= 16t^2 + 9t^2 \\
 &= 25t^2 \\
 D(t) &= \pm\sqrt{25t^2} && \text{Solve for } D(t) \text{ using the square root} \\
 &= \pm 5|t|
 \end{aligned}$$

In this scenario we are considering only positive values of t , so our distance $D(t)$ will always be positive. We can simplify this answer to $D(t) = 5t$. This means that the distance between Anna and Emanuel is also a linear function. Because D is a linear function, we can now answer the question of when the distance between them will reach 2 miles. We will set the output $D(t) = 2$ and solve for t .

$$\begin{aligned}
 D(t) &= 2 \\
 5t &= 2 \\
 t &= \frac{2}{5} = 0.4
 \end{aligned}$$

They will fall out of radio contact in 0.4 hours, or 24 minutes.

Q&A

Should I draw diagrams when given information based on a geometric shape?

Yes. Sketch the figure and label the quantities and unknowns on the sketch.

Example 3

Using a Diagram to Model Distance between Cities

There is a straight road leading from the town of Westborough to Agritown 30 miles east and 10 miles north. Partway down this road, it junctions with a second road, perpendicular to the first, leading to the town of Eastborough. If the town of

Eastborough is located 20 miles directly east of the town of Westborough, how far is the road junction from Westborough?

Answer

It might help here to draw a picture of the situation. See Figure 4. It would then be helpful to introduce a coordinate system. While we could place the origin anywhere, placing it at Westborough seems convenient. This puts Agritown at coordinates $(30, 10)$, and Eastborough at $(20, 0)$.

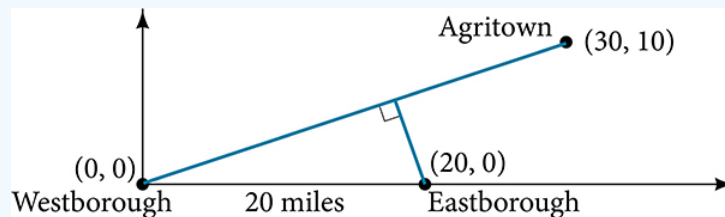


Figure 4

Using this point along with the origin, we can find the slope of the line from Westborough to Agritown:

$$m = \frac{10 - 0}{30 - 0} = \frac{1}{3}$$

The equation of the road from Westborough to Agritown would be

$$W(x) = \frac{1}{3}x$$

From this, we can determine the perpendicular road to Eastborough will have slope $m = -3$. Because the town of Eastborough is at the point $(20, 0)$, we can find the equation:

$$\begin{aligned} E(x) &= -3x + b \\ 0 &= -3(20) + b \quad \text{Substitute in } (20, 0) \\ b &= 60 \\ E(x) &= -3x + 60 \end{aligned}$$

We can now find the coordinates of the junction of the roads by finding the intersection of these lines. Setting them equal,

$$\begin{aligned} \frac{1}{3}x &= -3x + 60 \\ \frac{10}{3}x &= 60 \\ 10x &= 180 \\ x &= 18 \quad \text{Substituting this back into } W(x) \\ y &= W(18) \\ &= \frac{1}{3}(18) \\ &= 6 \end{aligned}$$

The roads intersect at the point $(18, 6)$. Using the distance formula, we can now find the distance from Westborough to the junction.

$$\begin{aligned} \text{distance} &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(18 - 0)^2 + (6 - 0)^2} \\ &\approx 18.974 \text{ miles} \end{aligned}$$

Analysis

One nice use of linear models is to take advantage of the fact that the graphs of these functions are lines. This means real-world applications discussing maps need linear functions to model the distances between reference points.

Try It #3

There is a straight road leading from the town of Timpson to Ashburn 60 miles east and 12 miles north. Partway down the road, it junctions with a second road, perpendicular to the first, leading to the town of Garrison. If the town of Garrison is located 22 miles directly east of the town of Timpson, how far is the road junction from Timpson?

Building Systems of Linear Models

Real-world situations including two or more linear functions may be modeled with a system of linear equations. Remember, when solving a system of linear equations, we are looking for points the two lines have in common. Typically, there are three types of answers possible, as shown in Figure 5.

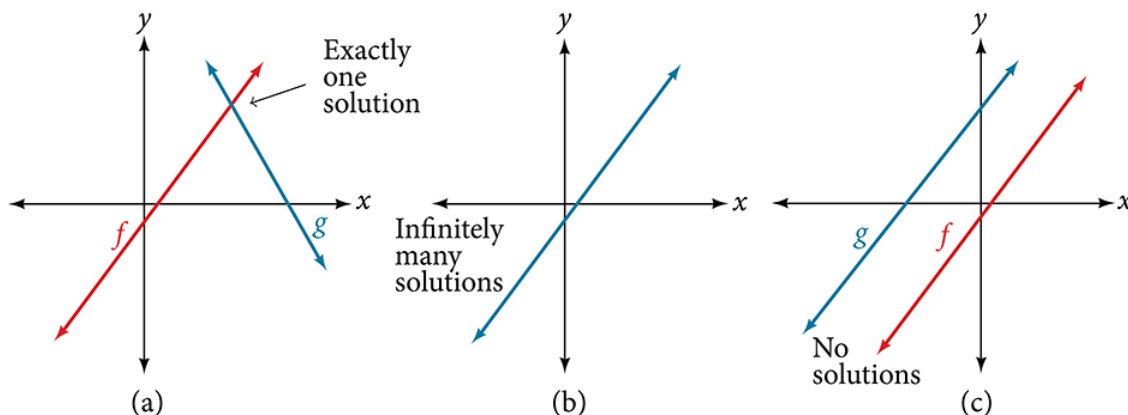


Figure 5

How To

Given a situation that represents a system of linear equations, write the system of equations and identify the solution.

1. Identify the input and output of each linear model.
2. Identify the slope and y-intercept of each linear model.
3. Find the solution by setting the two linear functions equal to one another and solving for x , or find the point of intersection on a graph.

Example 4

Building a System of Linear Models to Choose a Truck Rental Company

Jamal is choosing between two truck-rental companies. The first, Keep on Trucking, Inc., charges an up-front fee of \$20, then 59 cents a mile. The second, Move It Your Way, charges an up-front fee of \$16, then 63 cents a mile⁴. When will Keep on Trucking, Inc. be the better choice for Jamal?

Answer

The two important quantities in this problem are the cost and the number of miles driven. Because we have two companies to consider, we will define two functions.

Input	d , distance driven in miles
Outputs	$K(d)$: cost, in dollars, for renting from Keep on Trucking $M(d)$ cost, in dollars, for renting from Move It Your Way
Initial Value	Up-front fee: $K(0) = 20$ and $M(0) = 16$
Rate of Change	$K(d) = \$0.59$ /mile and $P(d) = \$0.63$ /mile

Table 1

A linear function is of the form $f(x) = mx + b$. Using the rates of change and initial charges, we can write the equations

$$K(d) = 0.59d + 20$$

$$M(d) = 0.63d + 16$$

Using these equations, we can determine when Keep on Trucking, Inc., will be the better choice. Because all we have to make that decision from is the costs, we are looking for when Move It Your Way, will cost less, or when $K(d) < M(d)$. The solution pathway will lead us to find the equations for the two functions, find the intersection, and then see where the $K(d)$ function is smaller.

These graphs are sketched in Figure 6, with $K(d)$ in blue.

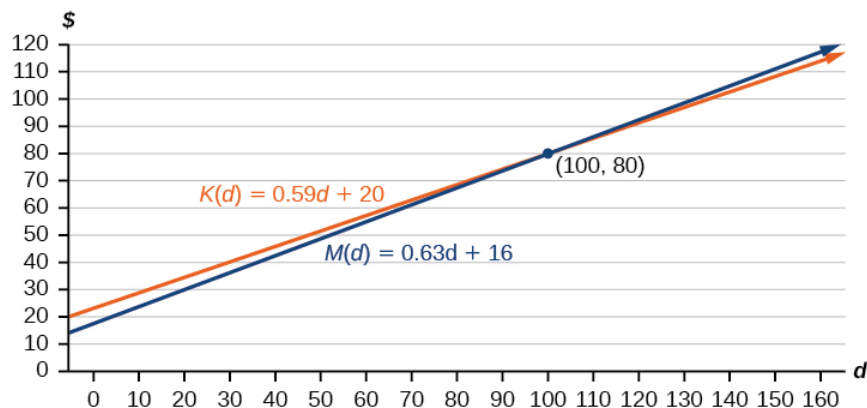


Figure 6

To find the intersection, we set the equations equal and solve:

$$K(d) = M(d)$$

$$0.59d + 20 = 0.63d + 16$$

$$4 = 0.04d$$

$$100 = d$$

$$d = 100$$

This tells us that the cost from the two companies will be the same if 100 miles are driven. Either by looking at the graph, or noting that $K(d)$ is growing at a slower rate, we can conclude that Keep on Trucking, Inc. will be the cheaper price when more than 100 miles are driven, that is $d > 100$.

Media

Access this online resource for additional instruction and practice with linear function models.

- [Interpreting a Linear Function](#)

2.3 Section Exercises

Verbal

1.

Explain how to find the input variable in a word problem that uses a linear function.

2.

Explain how to find the output variable in a word problem that uses a linear function.

3.

Explain how to interpret the initial value in a word problem that uses a linear function.

4.

Explain how to determine the slope in a word problem that uses a linear function.

Algebraic

5.

Find the area of a parallelogram bounded by the y -axis, the line $x = 3$, the line $f(x) = 1 + 2x$, and the line parallel to $f(x)$ passing through $(2, 7)$.

6.

Find the area of a triangle bounded by the x -axis, the line $f(x) = 12 - \frac{1}{3}x$, and the line perpendicular to $f(x)$ that passes through the origin.

7.

Find the area of a triangle bounded by the y -axis, the line $f(x) = 9 - \frac{6}{7}x$, and the line perpendicular to $f(x)$ that passes through the origin.

8.

Find the area of a parallelogram bounded by the x -axis, the line $g(x) = 2$, the line $f(x) = 3x$, and the line parallel to $f(x)$ passing through $(6, 1)$.

For the following exercises, consider this scenario: A town's population has been decreasing at a constant rate. In 2010 the population was 5,900. By 2012 the population had dropped 4,700. Assume this trend continues.

9.

Predict the population in 2016.

10.

Identify the year in which the population will reach 0.

For the following exercises, consider this scenario: A town's population has been increased at a constant rate. In 2010 the population was 46,020. By 2012 the population had increased to 52,070. Assume this trend continues.

11.

Predict the population in 2016.

12.

Identify the year in which the population will reach 75,000.

For the following exercises, consider this scenario: A town has an initial population of 75,000. It grows at a constant rate of 2,500 per year for 5 years.

13.

Find the linear function that models the town's population P as a function of the year, t , where t is the number of years since the model began.

14.

Find a reasonable domain and range for the function P .

15.

If the function P is graphed, find and interpret the x - and y -intercepts.

16.

If the function P is graphed, find and interpret the slope of the function.

17.

When will the output reached 100,000?

18.

What is the output in the year 12 years from the onset of the model?

For the following exercises, consider this scenario: The weight of a newborn is 7.5 pounds. The baby gained one-half pound a month for its first year.

19.

Find the linear function that models the baby's weight W as a function of the age of the baby, in months, t .

20.

Find a reasonable domain and range for the function W .

21.

If the function W is graphed, find and interpret the x - and y -intercepts.

22.

If the function W is graphed, find and interpret the slope of the function.

23.

When did the baby weight 10.4 pounds?

24.

What is the output when the input is 6.2? Interpret your answer.

For the following exercises, consider this scenario: The number of people afflicted with the common cold in the winter months steadily decreased by 205 each year from 2005 until 2010. In 2005, 12,025 people were afflicted.

25.

Find the linear function that models the number of people inflicted with the common cold C as a function of the year, t .

26.

Find a reasonable domain and range for the function C .

27.

If the function C is graphed, find and interpret the x - and y -intercepts.

28.

If the function C is graphed, find and interpret the slope of the function.

29.

When will the output reach 0?

30.

In what year will the number of people be 9,700?

Graphical

For the following exercises, use the graph in Figure 7, which shows the profit, y , in thousands of dollars, of a company in a given year, t , where t represents the number of years since 1980.

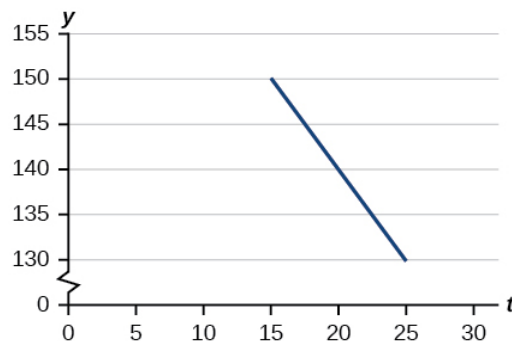


Figure 7

31.

Find the linear function y , where y depends on t , the number of years since 1980.

32.

Find and interpret the y -intercept.

33.

Find and interpret the x -intercept.

34.

Find and interpret the slope.

For the following exercises, use the graph in Figure 8, which shows the profit, y , in thousands of dollars, of a company in a given year, t , where t represents the number of years since 1980.

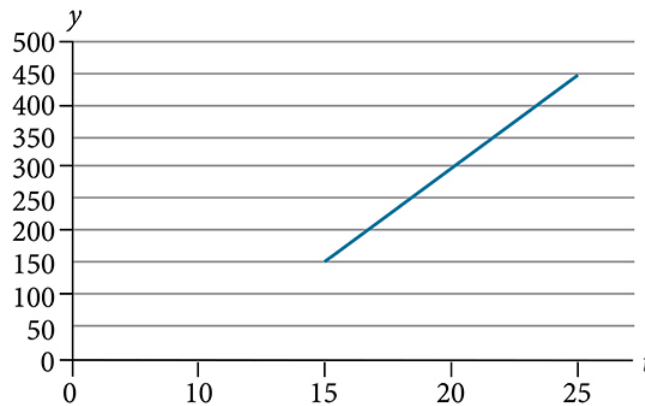


Figure 8

35.

Find the linear function y , where y depends on t , the number of years since 1980.

36.

Find and interpret the y -intercept.

37.

Find and interpret the x -intercept.

38.

Find and interpret the slope.

Numeric

For the following exercises, use the median home values in Mississippi and Hawaii (adjusted for inflation) shown in Table 2. Assume that the house values are changing linearly.

Year	Mississippi	Hawaii
1950	\$25,200	\$74,400
2000	\$71,400	\$272,700

Table 2

39.

In which state have home values increased at a higher rate?

40.

If these trends were to continue, what would be the median home value in Mississippi in 2010?

41.

If we assume the linear trend existed before 1950 and continues after 2000, the two states' median house values will be (or were) equal in what year? (The answer might be absurd.)

For the following exercises, use the median home values in Indiana and Alabama (adjusted for inflation) shown in Table 3. Assume that the house values are changing linearly.

Year	Indiana	Alabama
1950	\$37,700	\$27,100
2000	\$94,300	\$85,100

Table 3

42.

In which state have home values increased at a higher rate?

43.

If these trends were to continue, what would be the median home value in Indiana in 2010?

44.

If we assume the linear trend existed before 1950 and continues after 2000, the two states' median house values will be (or were) equal in what year? (The answer might be absurd.)

Real-World Applications

45.

In 2004, a school population was 1,001. By 2008 the population had grown to 1,697. Assume the population is changing linearly.

1. Ⓐ How much did the population grow between the year 2004 and 2008?
2. Ⓑ How long did it take the population to grow from 1,001 students to 1,697 students?
3. Ⓒ What is the average population growth per year?
4. Ⓓ What was the population in the year 2000?
5. Ⓔ Find an equation for the population, P , of the school t years after 2000.
6. Ⓕ Using your equation, predict the population of the school in 2011.

46.

In 2003, a town's population was 1,431. By 2007 the population had grown to 2,134. Assume the population is changing linearly.

1. Ⓐ How much did the population grow between the year 2003 and 2007?
2. Ⓑ How long did it take the population to grow from 1,431 people to 2,134 people?
3. Ⓒ What is the average population growth per year?
4. Ⓓ What was the population in the year 2000?
5. Ⓔ Find an equation for the population, P of the town t years after 2000.
6. Ⓕ Using your equation, predict the population of the town in 2014.

47.

A phone company has a monthly cellular plan where a customer pays a flat monthly fee and then a certain amount of money per minute used on the phone. If a customer uses 410 minutes, the monthly cost will be \$71.50. If the customer uses 720 minutes, the monthly cost will be \$118.

1. Ⓐ Find a linear equation for the monthly cost of the cell plan as a function of x , the number of monthly minutes used.
2. Ⓑ Interpret the slope and y -intercept of the equation.
3. Ⓒ Use your equation to find the total monthly cost if 687 minutes are used.

48.

A phone company has a monthly cellular data plan where a customer pays a flat monthly fee of \$10 and then a certain amount of money per megabyte (MB) of data used on the phone. If a customer uses 20 MB, the monthly cost will be \$11.20. If the customer uses 130 MB, the monthly cost will be \$17.80.

1. Ⓐ Find a linear equation for the monthly cost of the data plan as a function of x , the number of MB used.
2. Ⓑ Interpret the slope and y -intercept of the equation.
3. Ⓒ Use your equation to find the total monthly cost if 250 MB are used.

49.

In 1991, the moose population in a park was measured to be 4,360. By 1999, the population was measured again to be 5,880. Assume the population continues to change linearly.

1. Ⓐ Find a formula for the moose population, P since 1990.
2. Ⓑ What does your model predict the moose population to be in 2003?

50.

In 2003, the owl population in a park was measured to be 340. By 2007, the population was measured again to be 285. The population changes linearly. Let the input be years since 1990.

1. Ⓐ Find a formula for the owl population, P . Let the input be years since 2003.
2. Ⓑ What does your model predict the owl population to be in 2012?

51.

The Federal Helium Reserve held about 16 billion cubic feet of helium in 2010 and is being depleted by about 2.1 billion cubic feet each year.

1. Ⓐ Give a linear equation for the remaining federal helium reserves, R , in terms of t , the number of years since 2010.
2. Ⓑ In 2015, what will the helium reserves be?
3. Ⓒ If the rate of depletion doesn't change, in what year will the Federal Helium Reserve be depleted?

52.

Suppose the world's oil reserves in 2014 are 1,820 billion barrels. If, on average, the total reserves are decreasing by 25 billion barrels of oil each year:

1. Ⓐ Give a linear equation for the remaining oil reserves, R , in terms of t , the number of years since now.
2. Ⓑ Seven years from now, what will the oil reserves be?
3. Ⓒ If the rate at which the reserves are decreasing is constant, when will the world's oil reserves be depleted?

53.

You are choosing between two different prepaid cell phone plans. The first plan charges a rate of 26 cents per minute. The second plan charges a monthly fee of \$19.95 *plus* 11 cents per minute. How many minutes would you have to use in a month in order for the second plan to be preferable?

54.

You are choosing between two different window washing companies. The first charges \$5 per window. The second charges a base fee of \$40 plus \$3 per window. How many windows would you need to have for the second company to be preferable?

55.

When hired at a new job selling jewelry, you are given two pay options:

- Option A: Base salary of \$17,000 a year with a commission of 12% of your sales
- Option B: Base salary of \$20,000 a year with a commission of 5% of your sales

How much jewelry would you need to sell for option A to produce a larger income?

56.

When hired at a new job selling electronics, you are given two pay options:

- Option A: Base salary of \$14,000 a year with a commission of 10% of your sales
- Option B: Base salary of \$19,000 a year with a commission of 4% of your sales

How much electronics would you need to sell for option A to produce a larger income?

57.

When hired at a new job selling electronics, you are given two pay options:

- Option A: Base salary of \$20,000 a year with a commission of 12% of your sales
- Option B: Base salary of \$26,000 a year with a commission of 3% of your sales

How much electronics would you need to sell for option A to produce a larger income?

58.

When hired at a new job selling electronics, you are given two pay options:

- Option A: Base salary of \$10,000 a year with a commission of 9% of your sales
- Option B: Base salary of \$20,000 a year with a commission of 4% of your sales

How much electronics would you need to sell for option A to produce a larger income?

Footnotes

- 4Rates retrieved Aug 2, 2010 from <http://www.budgettruck.com> and <http://www.uhaul.com/>

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6.3.2: Applications

Learning Objectives

In this section, you will learn to use linear functions to model real-world applications

Now that we have learned to determine equations of lines, we get to apply these ideas in a variety of real-life situations.

Read the problem carefully. Highlight important information. Keep track of which values correspond to the independent variable (x) and which correspond to the dependent variable (y).

✓ Example 6.3.2.1

A taxi service charges \$0.50 per mile plus a \$5 flat fee. What will be the cost of traveling 20 miles? What will be cost of traveling x miles?

Solution

x = distance traveled, in miles and y = cost in dollars

The cost of traveling 20 miles is

$$y = (0.50)(20) + 5 = 10 + 5 = 15$$

The cost of traveling x miles is

$$y = (0.50)(x) + 5 = 0.50x + 5$$

In this problem, \$0.50 per mile is referred to as the **variable cost**, and the flat charge \$5 as the **fixed cost**. Now if we look at our cost equation $y = .50x + 5$, we can see that the variable cost corresponds to the slope and the fixed cost to the y -intercept.

✓ Example 6.3.2.2

The variable cost to manufacture a product is \$10 per item and the fixed cost \$2500. If x represents the number of items manufactured and y represents the total cost, write the cost function.

Solution

- The variable cost of \$10 per item tells us that $m = 10$.
- The fixed cost represents the y -intercept. So $b = 2500$.

Therefore, the cost equation is $y = 10x + 2500$.

✓ Example 6.3.2.3

It costs \$750 to manufacture 25 items, and \$1000 to manufacture 50 items. Assuming a linear relationship holds, find the cost equation, and use this function to predict the cost of 100 items.

Solution

We let x = the number of items manufactured, and let y = the cost.

Solving this problem is equivalent to finding an equation of a line that passes through the points (25, 750) and (50, 1000).

$$m = \frac{1000 - 750}{50 - 25} = 10$$

Therefore, the partial equation is $y = 10x + b$

By substituting one of the points in the equation, we get $b = 500$

Therefore, the cost equation is $y = 10x + 500$

To find the cost of 100 items, substitute $x = 100$ in the equation $y = 10x + 500$

So the cost is

$$y = 10(100) + 500 = 1500$$

It costs \$1500 to manufacture 100 items.

✓ Example 6.3.2.4

The freezing temperature of water in Celsius is 0 degrees and in Fahrenheit 32 degrees. And the boiling temperatures of water in Celsius, and Fahrenheit are 100 degrees, and 212 degrees, respectively. Write a conversion equation from Celsius to Fahrenheit and use this equation to convert 30 degrees Celsius into Fahrenheit.

Solution

Let us look at what is given.

Celsius	Fahrenheit
0	32
100	212

Again, solving this problem is equivalent to finding an equation of a line that passes through the points (0, 32) and (100, 212).

Since we are finding a linear relationship, we are looking for an equation $y = mx + b$, or in this case $F = mC + b$, where x or C represent the temperature in Celsius, and y or F the temperature in Fahrenheit.

$$\text{slope } m = \frac{212 - 32}{100 - 0} = \frac{9}{5}$$

The equation is $F = \frac{9}{5}C + b$

Substituting the point (0, 32), we get

$$F = \frac{9}{5}C + 32.$$

To convert 30 degrees Celsius into Fahrenheit, substitute $C = 30$ in the equation

$$\begin{aligned} F &= \frac{9}{5}C + 32 \\ F &= \frac{9}{5}(30) + 32 = 86 \end{aligned}$$

✓ Example 6.3.2.5

The population of Canada in the year 1980 was 24.5 million, and in the year 2010 it was 34 million. The population of Canada over that time period can be approximately modelled by a linear function. Let x represent time as the number of years after 1980 and let y represent the size of the population.

- Write the linear function that gives a relationship between the time and the population.
- Assuming the population continues to grow linearly in the future, use this equation to predict the population of Canada in the year 2025.

Solution

The problem can be made easier by using 1980 as the base year, that is, we choose the year 1980 as the year zero. This will mean that the year 2010 will correspond to year 30. Now we look at the information we have:

Year	Population
0 (1980)	24.5 million

30 (2010)

34 million

a. Solving this problem is equivalent to finding an equation of a line that passes through the points (0, 24.5) and (30, 34). We use these two points to find the slope:

$$m = \frac{34 - 24.5}{30 - 0} = \frac{9.5}{30} = 0.32$$

The y -intercept occurs when $x = 0$, so $b = 24.5$

$$y = 0.32x + 24.5$$

b. Now to predict the population in the year 2025, we let $x = 2025 - 1980 = 45$

$$y = 0.32x + 24.5$$

$$y = 0.32(45) + 24.5 = 38.9$$

In the year 2025, we predict that the population of Canada will be 38.9 million people.

Note that we assumed the population trend will continue to be linear. Therefore if population trends change and this assumption does not continue to be true in the future, this prediction may not be accurate.

Exercises

SECTION 1.4 PROBLEM SET: APPLICATIONS

In the following application problems, assume a linear relationship holds.

1) The variable cost to manufacture a product is \$25 per item, and the fixed costs are \$1200. If x is the number of items manufactured and y is the cost, write the cost function.	2) It costs \$90 to rent a car driven 100 miles and \$140 for one driven 200 miles. If x is the number of miles driven and y the total cost of the rental, write the cost function.
3) The variable cost to manufacture an item is \$20 per item, and it costs a total of \$750 to produce 20 items. If x represents the number of items manufactured and y is the cost, write the cost function.	4) To manufacture 30 items, it costs \$2700, and to manufacture 50 items, it costs \$3200. If x represents the number of items manufactured and y the cost, write the cost function.
5) To manufacture 100 items, it costs \$32,000, and to manufacture 200 items, it costs \$40,000. If x is the number of items manufactured and y is the cost, write the cost function.	6) It costs \$1900 to manufacture 60 items, and the fixed costs are \$700. If x represents the number of items manufactured and y the cost, write the cost function.

SECTION 1.4 PROBLEM SET: APPLICATIONS

In the following application problems, assume a linear relationship holds.

7) A person who weighs 150 pounds has 60 pounds of muscles; a person that weighs 180 pounds has 72 pounds of muscles. If x represents body weight and y is muscle weight, write an equation describing their relationship. Use this relationship to determine the muscle weight of a person that weighs 170 pounds.	8) A spring on a door stretches 6 inches if a force of 30 pounds is applied. It stretches 10 inches if a 50 pound force is applied. If x represents the number of inches stretched, and y is the force, write a linear equation describing the relationship. Use it to determine the amount of force required to stretch the spring 12 inches.
---	--

9) A male college student who is 64 inches tall weighs 110 pounds. Another student who is 74 inches tall weighs 180 pounds. Assuming the relationship between male students' heights (x), and weights (y) is linear, write a function to express weights in terms of heights, and use this function to predict the weight of a student who is 68 inches tall.	10) EZ Clean company has determined that if it spends \$30,000 on advertising, it can hope to sell 12,000 of its Minivacs a year, but if it spends \$50,000, it can sell 16,000. Write an equation that gives a relationship between the number of dollars spent on advertising (x) and the number of minivacs sold(y).
11) The freezing temperatures for water for Celsius and Fahrenheit scales are 0°C and 32°F . The boiling temperatures for water are 100°C and 212°F . Let C denote the temperature in Celsius and F in Fahrenheit. Write the conversion function from Celsius to Fahrenheit. Use the function to convert 25°C into $^{\circ}\text{F}$.	12) By reversing the coordinates in the previous problem, find a conversion function that converts Fahrenheit into Celsius, and use this conversion function to convert 72°F into an equivalent Celsius measure.

SECTION 1.4 PROBLEM SET: APPLICATIONS

In the following application problems, assume a linear relationship holds.

13) California's population was 29.8 million in the year 1990, and 37.3 million in 2010. Assume that the population trend was and continues to be linear, write the population function. Use this function to predict the population in 2025. <i>Hint: Use 1990 as the base year (year 0); then 2010 and 2025 are years 20, and 35, respectively.)</i>	14) Use the population function for California in the previous problem to find the year in which the population will be 40 million people.
15) A college's enrollment was 13,200 students in the year 2000, and 15,000 students in 2015. Enrollment has followed a linear pattern. Write the function that models enrollment as a function of time. Use the function to find the college's enrollment in the year 2010. <i>Hint: Use year 2000 as the base year.</i>	16) If the college's enrollment continues to follow this pattern, in what year will the college have 16,000 students enrolled.
17) The cost of electricity in residential homes is a linear function of the amount of energy used. In Grove City, a home using 250 kilowatt hours (kwh) of electricity per month pays \$55. A home using 600 kwh per month pays \$118. Write the cost of electricity as a function of the amount used. Use the function to find the cost for a home using 400 kwh of electricity per month.	18) Find the level of electricity use that would correspond to a monthly cost of \$100.

SECTION 1.4 PROBLEM SET: APPLICATIONS

In the following application problems, assume a linear relationship holds.

19) At ABC Co., sales revenue is \$170,000 when it spends \$5000 on advertising. Sales revenue is \$254,000 when \$12,000 is spent on advertising. a) Find a linear function for y = amount of sales revenue as a function of x = amount spent on advertising. b) Find revenue if \$10,000 is spent on advertising. c) Find the amount that should be spent on advertising to achieve \$200,000 in revenue.	20) For problem 19, explain the following: a. Explain what the slope of the line tells us about the effect on sales revenue of money spent on advertising. Be specific, explaining both the number and the sign of the slope in the context of this problem. b. Explain what the y intercept of the line tells us about the sales revenue in the context of this problem
---	--

21) Mugs Café sells 1000 cups of coffee per week if it does not advertise. For every \$50 spent in advertising per week, it sells an additional 150 cups of coffee.

a) Find a linear function that gives

y = number of cups of coffee sold per week

x = amount spent on advertising per week.

b) How many cups of coffee does Mugs Café expect to sell if \$100 per week is spent on advertising?

22) Party Sweets makes baked goods that can be ordered for special occasions. The price is \$24 to order one dozen (12 cupcakes) and \$9 for each additional 6 cupcakes.

a. Find a linear function that gives the total price of a cupcake order as a function of the number of cupcakes ordered

b. Find the price for an order of 5 dozen (60) cupcakes

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6.4: Intersection of Straight Lines

Learning Objectives

In this section, you will learn to:

1. Solve a linear system in two variables.
2. Find the equilibrium point when a demand and a supply equation are given.
3. Find the break-even point when the revenue and the cost functions are given.

Finding the Point of Intersection of Two Lines

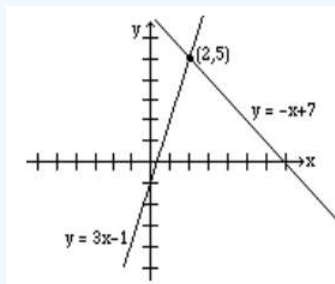
In this section, we will do application problems that involve the intersection of lines. Therefore, before we proceed any further, we will first learn how to find the intersection of two lines.

✓ Example 6.4.1

Find the intersection of the line $y = 3x - 1$ and the line $y = -x + 7$.

Solution

We graph both lines on the same axes, as shown below, and read the solution $(2, 5)$.



Finding an intersection of two lines graphically is not always easy or practical; therefore, we will now learn to solve these problems algebraically.

At the point where two lines intersect, the x and y values for both lines are the same. So in order to find the intersection, we either let the x -values or the y -values equal.

If we were to solve the above example algebraically, it will be easier to let the y -values equal. Since $y = 3x - 1$ for the first line, and $y = -x + 7$ for the second line, by letting the y -values equal, we get

$$\begin{aligned} 3x - 1 &= -x + 7 \\ 4x &= 8 \\ x &= 2 \end{aligned}$$

By substituting $x = 2$ in any of the two equations, we obtain $y = 5$.

Hence, the solution $(2, 5)$.

A common algebraic method used to solve systems of equations is called the **elimination method**. The object is to eliminate one of the two variables by adding the left and right sides of the equations together. Once one variable is eliminated, we have an equation with only one variable for can be solved. Finally, by substituting the value of the variable that has been found in one of the original equations, we get the value of the other variable.

✓ Example 6.4.2

Find the intersection of the lines $2x + y = 7$ and $3x - y = 3$ by the elimination method.

Solution

We add the left and right sides of the two equations.

$$\begin{array}{r} 2x + y = 7 \\ 3x - y = 3 \\ \hline 5x = 10 \\ x = 2 \end{array}$$

Now we substitute $x = 2$ in any of the two equations and solve for y .

$$\begin{array}{r} 2(2) + y = 7 \\ y = 3 \end{array}$$

Therefore, the solution is $(2, 3)$.

✓ Example 6.4.3

Solve the system of equations $x + 2y = 3$ and $2x + 3y = 4$ by the elimination method.

Solution

If we add the two equations, none of the variables are eliminated. But the variable x can be eliminated by multiplying the first equation by -2 , and leaving the second equation unchanged.

$$\begin{array}{r} -2x - 4y = -6 \\ 2x + 3y = 4 \\ \hline -y = -2 \\ y = 2 \end{array}$$

Substituting $y = 2$ in $x + 2y = 3$, we get

$$\begin{array}{r} x + 2(2) = 3 \\ x = -1 \end{array} \tag{6.4.1}$$

Therefore, the solution is $(-1, 2)$.

✓ Example 6.4.4

Solve the system of equations $3x - 4y = 5$ and $4x - 5y = 6$.

Solution

This time, we multiply the first equation by -4 and the second by 3 before adding. (The choice of numbers is not unique.)

$$\begin{array}{r} -12x + 16y = -20 \\ 12x - 15y = 18 \\ \hline y = -2 \end{array}$$

By substituting $y = -2$ in any one of the equations, we get $x = -1$.

Hence the solution is $(-1, -2)$.

SUPPLY, DEMAND AND THE EQUILIBRIUM MARKET PRICE

In a free market economy the supply curve for a commodity is the number of items of a product that can be made available at different prices, and the demand curve is the number of items the consumer will buy at different prices.

As the price of a product increases, its demand decreases and supply increases. On the other hand, as the price decreases the demand increases and supply decreases. The **equilibrium price** is reached when the demand equals the supply.

✓ Example 6.4.5

The supply curve for a product is $y = 3.5x - 14$ and the demand curve for the same product is $y = -2.5x + 34$, where x is the price and y the number of items produced. Find the following.

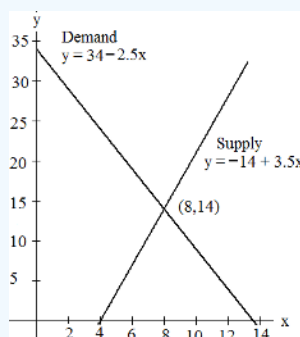
- How many items will be supplied at a price of \$10?
- How many items will be demanded at a price of \$10?
- Determine the equilibrium price.
- How many items will be produced at the equilibrium price?

Solution

- We substitute $x = 10$ in the supply equation, $y = 3.5x - 14$; the answer is $y = 3.5(10) - 14 = 21$.
- We substitute $x = 10$ in the demand equation, $y = -2.5x + 34$; the answer is $y = -2.5(10) + 34 = 9$.
- By letting the supply equal the demand, we get

$$\begin{aligned} 3.5x - 14 &= -2.5x + 34 \\ 6x &= 48 \\ x &= \$8 \end{aligned}$$

- We substitute $x = 8$ in either the supply or the demand equation; we get $y = 14$.



The graph shows the intersection of the supply and the demand functions and their point of intersection, (8,14).

Interpretation: At equilibrium, the price is \$8 per item, and 14 items are produced by suppliers and purchased by consumers.

The Break-Even Point

In a business, the profit is generated by selling products.

- If a company sells x number of items at a price P , then the **revenue R** is the price multiplied by number of items sold:
 $R = P \cdot x$.
- The **production costs C** are the sum of the variable costs and the fixed costs, and are often written as $C = mx + b$, where x is the number of items manufactured.
 - The slope m is the called marginal cost and represents the cost to produce one additional item or unit.
 - The variable cost, mx , depends on how much is being produced
 - The fixed cost b is constant; it does not change no matter how much is produced.
- Profit** is equal to Revenue minus Cost: Profit = $R - C$

A company makes a profit if the revenue is greater than the cost. There is a loss if the cost is greater than the revenue. The point on the graph where the revenue equals the cost is called the **break-even point**. At the break-even point, profit is 0.

✓ Example 6.4.6

If the revenue function of a product is $R = 5x$ and the cost function is $y = 3x + 12$, find the following.

- If 4 items are produced, what will the revenue be?
- What is the cost of producing 4 items?
- How many items should be produced to break even?
- What will be the revenue and the cost at the break-even point?

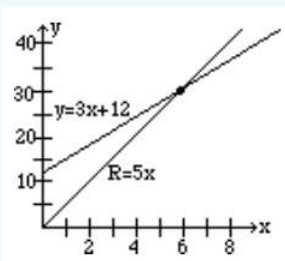
Solution

- We substitute $x = 4$ in the revenue equation $R = 5x$, and the answer is $R = 20$.
- We substitute $x = 4$ in the cost equation $C = 3x + 12$, and the answer is $C = 24$.
- By letting the revenue equal the cost, we get

$$\begin{aligned} 5x &= 3x + 12 \\ x &= 6 \end{aligned}$$

- Substitute $x = 6$ in either the revenue or the cost equation: we get $R = C = 30$.

The graph below shows the intersection of the revenue and cost functions and their point of intersection, (6, 30).



Exercises

SECTION 1.5 PROBLEM SET: MORE APPLICATIONS

Solve the following problems.

1) Solve for x and y. $y = 3x + 4$ $y = 5x - 2$	2) Solve for x and y. $2x - 3y = 4$ $3x - 4y = 5$
3) The supply and demand curves for a product are: Supply $y = 2000x - 6500$ Demand $y = -1000x + 28000$, where x is price and y is the number of items. At what price will supply equal demand and how many items will be produced at that price?	4) The supply and demand curves for a product are Supply $y = 300x - 18000$ and Demand $y = -100x + 14000$, where x is price and y is the number of items. At what price will supply equal demand, and how many items will be produced at that price?
5) A car rental company offers two plans for one way rentals. Plan I charges \$36 per day and 17 cents per mile. Plan II charges \$24 per day and 25 cents per mile. a. If you were to drive 300 miles in a day, which plan is better? b. For what mileage are both rates equal?	

SECTION 1.5 PROBLEM SET: MORE APPLICATIONS

Solve the following problems.

6) A demand curve for a product is the number of items the consumer will buy at different prices. At a price of \$2 a store can sell 2400 of a particular type of toy truck. At a price of \$8 the store can sell 600 such trucks. If x represents the price of trucks and y the number of items sold, write an equation for the demand curve.

8) The equilibrium price is the price where the supply equals the demand. From the demand and supply curves obtained in the previous two problems, find the equilibrium price, and determine the number of items that can be sold at that price.

7) A supply curve for a product is the number of items that can be made available at different prices. A manufacturer of toy trucks can supply 2000 trucks if they are sold for \$8 each; it can supply only 400 trucks if they are sold for \$4 each. If x is the price and y the number of items, write an equation for the supply curve.

9) A break-even point is the intersection of the cost function and the revenue function, that is, where total cost equals revenue, and profit is zero. Mrs. Jones Cookies Store's cost and revenue, in dollars, for x number of cookies is given by $C = .05x + 3000$ and $R = .80x$. Find the number of cookies that must be sold to break even.

SECTION 1.5 PROBLEM SET: MORE APPLICATIONS

Solve the following problems.

10) A company's revenue and cost in dollars are given by $R = 225x$ and $C = 75x + 6000$, where x is the number of items. Find the number of items that must be produced to break-even.

12) Whackemhard Sports is planning to introduce a new line of tennis rackets. The fixed costs for the new line are \$25,000 and the variable cost of producing each racket is \$60.
 x is the number of rackets; y is in dollars.
If the racket sells for \$80, how many rackets must be sold in order to break even?

11) A firm producing socks has a fixed cost of \$20,000 and variable cost of \$2 per pair of socks. Let x = the number of pairs of socks. Find the break-even point if the socks sell for \$4.50 per pair.

13) It costs \$1,200 to produce 50 pounds of a chemical and it costs \$2,200 to produce 150 pounds. The chemical sells for \$15 per pound
 x is the amount of chemical; y is in dollars.
a. Find the cost function.
b. What is the fixed cost?
c. How many pounds must be sold to break even?
d. Find the cost and revenue at the break-even point.

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6.5: Exercises

Review Exercises

Linear Functions

1.

Determine whether the algebraic equation is linear. $2x + 3y =$

2.

Determine whether the algebraic equation is linear. $6x^2 - y =$

3.

Determine whether the function is increasing or decreasing.

$$f(x) = 7x - 2$$

4.

Determine whether the function is increasing or decreasing.

$$g(x) = -x + 2$$

5.

Given each set of information, find a linear equation that satisfies the given conditions, if possible.

Passes through $(7, 5)$ and $(3, 17)$

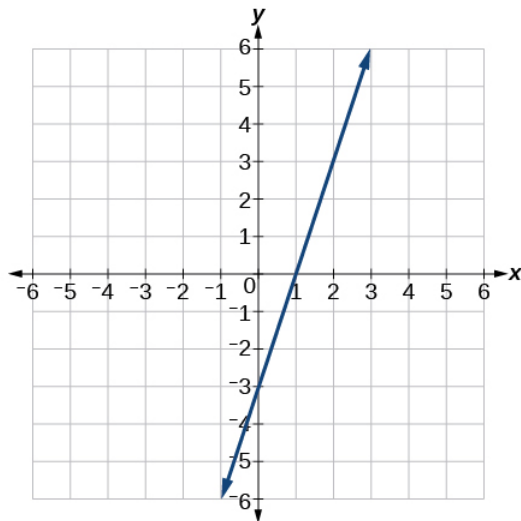
6.

Given each set of information, find a linear equation that satisfies the given conditions, if possible.

x -intercept at $(6, 0)$ and y -intercept at $(0, 10)$

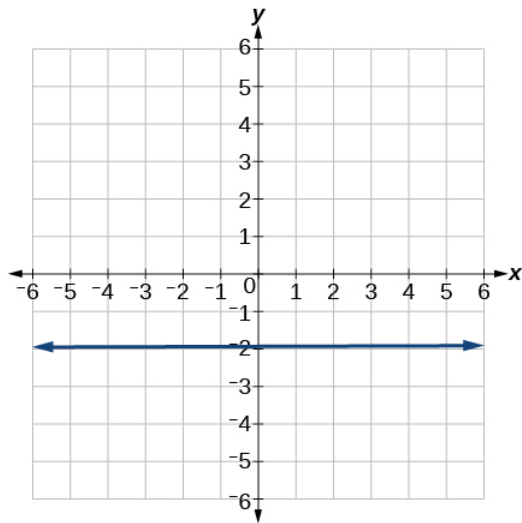
7.

Find the slope of the line shown in the line graph.



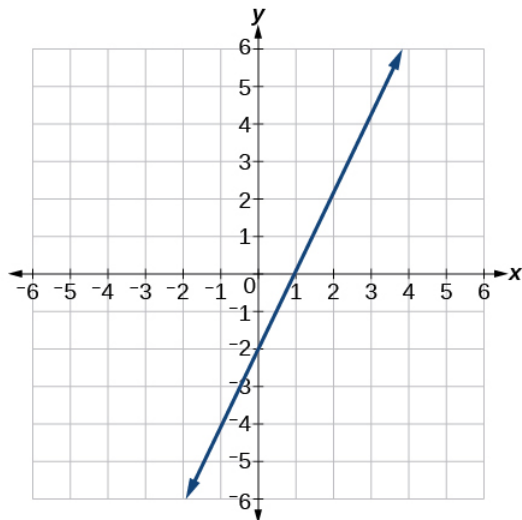
8.

Find the slope of the line graphed.



9.

Write an equation in slope-intercept form for the line shown.



10.

Does the following table represent a linear function? If so, find the linear equation that models the data.

x	-4	0	2	10
$g(x)$	18	-2	-12	-52

11.

Does the following table represent a linear function? If so, find the linear equation that models the data.

x	6	8	12	26
$g(x)$	-8	-12	-18	-46

12.

On June 1st, a company has \$4,000,000 profit. If the company then loses 150,000 dollars per day thereafter in the month of June, what is the company's profit n^{th} day after June 1st?

Graphs of Linear Functions

For the following exercises, determine whether the lines given by the equations below are parallel, perpendicular, or neither parallel nor perpendicular:

13.

$$2x - 6y = 12$$

$$-x + 3y = 1$$

14.

$$y = \frac{1}{3}x - 2$$

$$3x + y = -9$$

For the following exercises, find the x- and y- intercepts of the given equation

15.

$$7x + 9y = -$$

16.

$$f(x) = 2x - 1$$

For the following exercises, use the descriptions of the pairs of lines to find the slopes of Line 1 and Line 2. Is each pair of lines parallel, perpendicular, or neither?

17.

- Line 1: Passes through (5, 11) and (10, 1)
- Line 2: Passes through (-1, 3) and (-5, 11)

18.

- Line 1: Passes through (8, -10) and (0, -26)
- Line 2: Passes through (2, 5) and (4, 4)

19.

Write an equation for a line perpendicular to $f(x) = 5x - 1$ and passing through the point (5, 20).

20.

Find the equation of a line with a y- intercept of (0, -2) and slope $-\frac{1}{2}$.

21.

Sketch a graph of the linear function $f(t) = 2t - 5$.

22.

Find the point of intersection for the 2 linear functions:

$$\begin{aligned}x &= y + 6 \\ 2x - y &= 13\end{aligned}$$

23.

A car rental company offers two plans for renting a car.

- Plan A: 25 dollars per day and 10 cents per mile
- Plan B: 50 dollars per day with free unlimited mileage

How many miles would you need to drive for plan B to save you money?

Modeling with Linear Functions

24.

Find the area of a triangle bounded by the y axis, the line $f(x) = 10 - 2x$, and the line perpendicular to f that passes through the origin.

25.

A town's population increases at a constant rate. In 2010 the population was 55,000. By 2012 the population had increased to 76,000. If this trend continues, predict the population in 2016.

26.

The number of people afflicted with the common cold in the winter months dropped steadily by 50 each year since 2004 until 2010. In 2004, 875 people were inflicted.

Find the linear function that models the number of people afflicted with the common cold C as a function of the year, t . When will no one be afflicted?

For the following exercises, use the graph in Figure 1 showing the profit, y , in thousands of dollars, of a company in a given year, x , where x represents years since 1980.

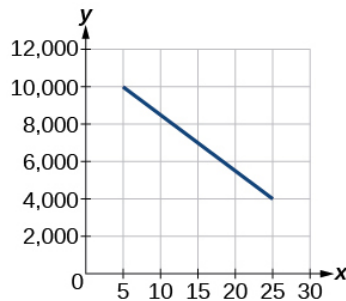


Figure 1

27.

Find the linear function y , where y depends on x , the number of years since 1980.

28.

Find and interpret the y -intercept.

For the following exercise, consider this scenario: In 2004, a school population was 1,700. By 2012 the population had grown to 2,500.

29.

Assume the population is changing linearly.

1. Ⓐ How much did the population grow between the year 2004 and 2012?
2. Ⓑ What is the average population growth per year?
3. Ⓒ Find an equation for the population, P , of the school t years after 2004.

For the following exercises, consider this scenario: In 2000, the moose population in a park was measured to be 6,500. By 2010, the population was measured to be 12,500. Assume the population continues to change linearly.

30.

Find a formula for the moose population, P .

31.

What does your model predict the moose population to be in 2020?

For the following exercises, consider this scenario: The median home values in subdivisions Pima Central and East Valley (adjusted for inflation) are shown in Table 1. Assume that the house values are changing linearly.

Year	Pima Central	East Valley
1970	32,000	120,250
2010	85,000	150,000

Table 1

32.

In which subdivision have home values increased at a higher rate?

33.

If these trends were to continue, what would be the median home value in Pima Central in 2015?

Fitting Linear Models to Data
34.

Draw a scatter plot for the data in Table 2. Then determine whether the data appears to be linearly related.

0	2	4	6	8	10
-105	-50	1	55	105	160

Table 2
35.

Draw a scatter plot for the data in Table 3. If we wanted to know when the population would reach 15,000, would the answer involve interpolation or extrapolation?

Year	Population
1990	5,600
1995	5,950
2000	6,300
2005	6,600
2010	6,900

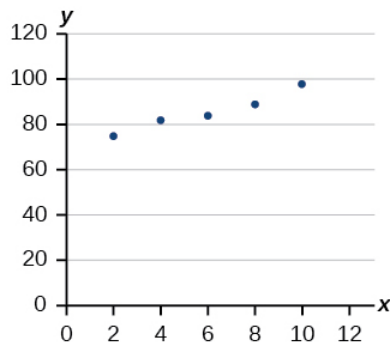
Table 3
36.

Eight students were asked to estimate their score on a 10-point quiz. Their estimated and actual scores are given in Table 4. Plot the points, then sketch a line that fits the data.

Predicted	Actual
6	6
7	7
7	8
8	8
7	9
9	10
10	10
10	9

Table 4
37.

Draw a best-fit line for the plotted data.



For the following exercises, consider the data in Table 5, which shows the percent of unemployed in a city of people 25 years or older who are college graduates is given below, by year.

Year	2000	2002	2005	2007	2010
Percent Graduates	6.5	7.0	7.4	8.2	9.0

Table 5

38.

Determine whether the trend appears to be linear. If so, and assuming the trend continues, find a linear regression model to predict the percent of unemployed in a given year to three decimal places.

39.

In what year will the percentage exceed 12%?

40.

Based on the set of data given in Table 6, calculate the regression line using a calculator or other technology tool, and determine the correlation coefficient to three decimal places.

x	17	20	23	26	29
y	15	25	31	37	40

Table 6

41.

Based on the set of data given in Table 7, calculate the regression line using a calculator or other technology tool, and determine the correlation coefficient to three decimal places.

x	10	12	15	18	20
y	36	34	30	28	22

Table 7

For the following exercises, consider this scenario: The population of a city increased steadily over a ten-year span. The following ordered pairs show the population and the year over the ten-year span (population, year) for specific recorded years:

(3,600, 2000); (4,000, 2001); (4,700, 2003); (6,000, 2006)

42.

Use linear regression to determine a function y , where the year depends on the population, to three decimal places of accuracy.

43.

Predict when the population will hit 12,000.

44.

What is the correlation coefficient for this model to three decimal places of accuracy?

45.

According to the model, what is the population in 2014?

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CHAPTER OVERVIEW

7: Linear Programming- A Geometric Approach

[7.1: Linear Inequalities in Two Variables](#)

[7.2: Maximization Applications](#)

[7.3: Minimization Applications](#)

[7.4: Chapter Review](#)

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7.1: Linear Inequalities in Two Variables

Previously, we graphed inequalities in one variable, but now we learn to graph inequalities in two variables. Although this section may seem similar to linear equations in two variables, linear inequalities in two variables have many applications. For example, business owners want to know when revenue is greater than cost so that their business makes a profit, e.g., revenue $>$ cost.

Definition: Linear Inequality

A **linear inequality** in two variables is an inequality of the form

$$ax + by < c,$$

where the inequality is written in the same form for $>$, $\leq$, $\geq$ and $a, b \neq 0$.

Recall. The solution to a linear inequality in one variable is an interval of numbers, e.g., $(-\infty, \infty)$, $[-2, 3)$, $(1, 9)$, $[-7, -3]$ etc.

Verifying Solutions

Solution to a Linear Inequality in Two Variables

An ordered pair (x, y) is a solution to a linear inequality in two variables, $ax + by < c$, if the ordered pair (x, y) makes the inequality true, where the same is for $>$, $\leq$, $\geq$ and $a, b \neq 0$.

Example 7.1.1

Verify whether each ordered pair is a solution to the inequality $y > x + 4$.

- a. $(0, 0)$
- b. $(1, 6)$

Solution

We substitute the ordered pairs into the inequality and determine if the results are true.

- a. Let's substitute $(0, 0)$ into the inequality and determine if the left side is greater than the right side.

$$\begin{aligned} y &\stackrel{?}{>} x + 4 && \text{Substitute } x = 0 \text{ and } y = 0 \\ 0 &\stackrel{?}{>} 0 + 4 && \text{Simplify} \\ 0 &\not> 4 && \text{X False} \end{aligned}$$

Hence, $(0, 0)$ is not a solution to the inequality $y > x + 4$.

- b. Let's substitute $(1, 6)$ into the inequality and determine if the left side is greater than the right side.

$$\begin{aligned} y &\stackrel{?}{>} x + 4 && \text{Substitute } x = 1 \text{ and } y = 6 \\ 6 &\stackrel{?}{>} 1 + 4 && \text{Simplify} \\ 6 &> 5 && \checkmark \text{ True} \end{aligned}$$

Hence, $(1, 6)$ is a solution to the inequality $y > x + 4$.

Boundary Lines

If we are given a linear inequality, $ax + by < c$, we could see from [Example 7.1.1](#) that not all ordered pairs are a solution, only some. Why? Well, notice that $(0, 0)$ is below the line $y = x + 4$ and $(1, 6)$ is above the line $y = x + 4$. This implies that ordered pairs in certain regions are solutions to the inequality $y > x + 4$. Hence, the line $y = x + 4$ is critical when finding solutions to the inequality. We call the line $y = x + 4$ a **boundary line**, a line that separates the ordered pairs that are solutions and the ordered pairs that are not solutions of the linear inequality in two variables $y > x + 4$.

Definition: Boundary Line

A linear equation in two variables $ax + by = c$ is called the **boundary line**, the line that separates the region where $ax + by > c$ and from the region where $ax + by < c$.

Note

Since there are four inequality symbols: $>$, $<$, $\geq$, $\leq$, then we have linear inequalities in two variables that include the boundary, e.g., inequalities with $\leq$ and $\geq$, and linear inequalities in two variables that exclude the boundary, e.g., inequalities with $<$ and $>$.

We can use the table below to help identify the boundary line, determine whether to include the boundary line, and the way the boundary line looks graphically.

Table 7.1.1

Case 1.	Case 2.
$ax + by < c$	$ax + by \leq c$
$ax + by > c$	$ax + by \geq c$
Boundary line: $ax + by = c$	Boundary line: $ax + by = c$
Boundary line is excluded in solution	Boundary line is included in solution
Boundary line is dashed	Boundary line is solid

Example 7.1.2

Let's revisit [Example 7.1.1](#) and graph the boundary line and points $(0, 0)$ and $(1, 6)$.

Solution

Since we have $y > x + 4$, we can see from the table above we have Case 1. and the boundary line is excluded. We represent this by graphing the line $y = x + 4$ as a dashed line.

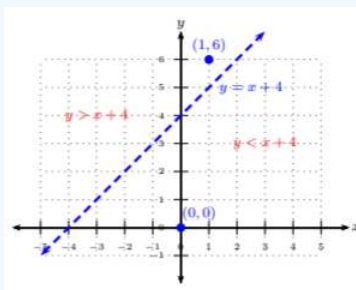


Figure 7.1.1

We can see from the graph that the point $(0, 0)$ lies below the boundary line $y = x + 4$ and point $(1, 6)$ lies above the boundary line. Recall, point $(1, 6)$ was verified as a solution of $y > x + 4$ in [Example 7.1.1](#). Furthermore, any ordered pair that lies above $y = x + 4$ will verify as a solution, i.e., making the inequality true. We usually represent this area by *shading the region* where the set of ordered pairs make the inequality true.

Graphing Linear Inequalities

Example 7.1.3

Graph the inequality from [Example 7.1.1](#).

Solution

Since we know that $(1, 6)$ is a solution to the inequality, then we shade above the dashed boundary line:

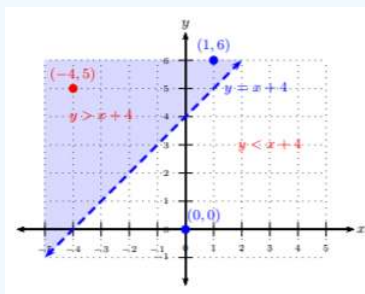


Figure 7.1.2

We see that any ordered pair in the shaded region is a solution to the inequality. For example, let's pick $(-4, 5)$ and verify this is a solution:

$$\begin{aligned} y &\stackrel{?}{>} x + 4 && \text{Substitute } x = -4 \text{ and } y = 5 \\ 5 &\stackrel{?}{>} -4 + 4 && \text{Simplify } 5 > 0 \qquad \qquad \qquad \checkmark \text{ True} \end{aligned}$$

Hence, $(-4, 5)$ is a solution to the inequality $y > x + 4$.

Steps for Graphing Linear Inequalities in Two Variables

Given a linear inequality in two variables, $ax + by < c$, we use the steps below to graph $ax + by < c$, where the the same process is applied for $>$, $\leq$, $\geq$ and $a, b \neq 0$.

Step 1. Rewrite the inequality in slope-intercept form, i.e., $y = mx + b$.

Step 2. Graph the boundary line according to the two cases:

Case 1. If the inequality is $<$ or $>$, then the boundary line is dashed.

Case 2. If the inequality is $\geq$ or $\leq$, then the boundary line is solid.

Step 3. Select a test point that is *not on the boundary line*. **Ask:** Does this ordered pair make the inequality true?

Step 4. If the ordered pair is

- a solution to the inequality, i.e., makes the inequality true, then shade the side that includes the ordered pair.
- not a solution, then shade the opposite side of the boundary line.

Note

If we choose a test point on the boundary line, we will obtain an identity, where both sides of the inequality symbol are the same number. Hence, it is critical to choose a point not on the boundary line.

Example 7.1.4

Graph the inequality:

$$2x - y > 3.$$

Solution

Let's follow the steps given above to graph the inequality.

Step 1. Rewrite the inequality in slope-intercept form, i.e., $y = mx + b$.

$$\begin{aligned} 2x - y &> 3 \\ -y &> -2x + 3 \\ y &< 2x - 3 \end{aligned}$$

Step 2. Graph the boundary line according to the two cases. Since the given inequality is $<$, then we have Case 1.

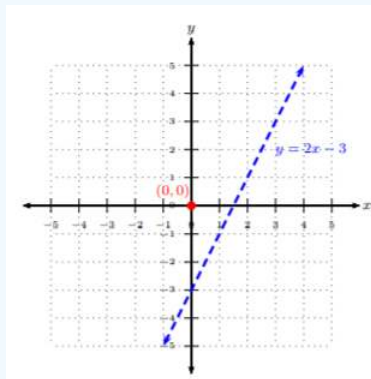


Figure 7.1.3

Step 3. Select a test point that is *not on the boundary line*. **Ask:** Does this ordered pair make the inequality true? Let's pick the test point $(0, 0)$ as it is a great choice!

$$\begin{aligned} y & \stackrel{?}{<} 2x - 3 \\ 0 & \stackrel{?}{<} 2(0) - 3 \\ 0 & \not< -3 \end{aligned}$$

Hence, $(0, 0)$ doesn't make the inequality true.

Step 4. If the ordered pair is

- a solution to the inequality, i.e., makes the inequality true, then shade the side that includes the ordered pair.
- not a solution, then shade the opposite side of the boundary line.

Since the ordered pair $(0, 0)$ is not a solution to the inequality, then we shade on the opposite side of the boundary line from the location of the ordered pair.

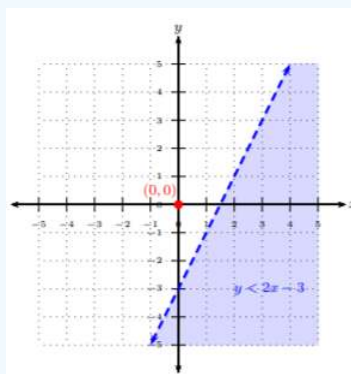


Figure 7.1.4

Note

Another way of graphing linear inequalities in two variables is to complete **Step 1.** and **Step 2.**, but instead of taking a test point in **Step 3.**, we can observe the inequality symbols. If the inequality has $<$ or $\leq$, then we easily shade below the boundary line, i.e., below the y -intercept. Similarly, if the inequality has $>$ or $\geq$, then we easily shade above the boundary line, i.e., above the y -intercept.

✓ Example 7.1.5

Graph the inequality:

$$3x + 2y \geq -6.$$

Solution

Let's follow the steps given above to graph the inequality, but try skipping **Step 3.** and **Step 4.**

Step 1. Rewrite the inequality in slope-intercept form, i.e., $y = mx + b$.

$$\begin{aligned} 3x + 2y &\geq -6 \\ 2y &\geq -3x - 6 \\ y &\geq -\frac{3}{2}x - 3 \end{aligned}$$

Step 2. Graph the boundary line according to the two cases. Since the given inequality is $\geq$, then we have Case 2.

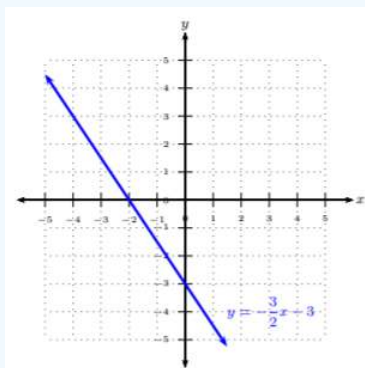


Figure 7.1.5

Since this inequality is $\geq$, where all ordered pairs above the boundary line are solutions to the inequality, we can easily shade above the y -intercept:

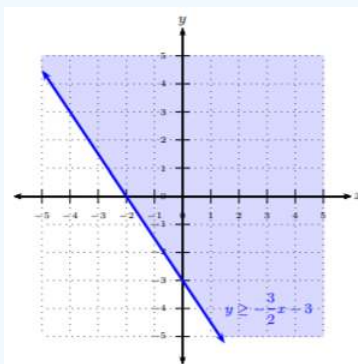


Figure 7.1.6

Linear Inequalities in Two Variables Homework

Determine whether the given ordered pairs are solutions to the inequality.

? Exercise 7.1.1

$$x + 2y \geq -4; (0, -4); (1, 1)$$

? Exercise 7.1.2

$$2x - y \leq 2; (1, 5); (3, 1)$$

Graph the following inequalities.

? Exercise 7.1.3

$$2x - y \leq 2$$

? Exercise 7.1.4

$$x > 4y - 8$$

? Exercise 7.1.5

$$x + 2y \geq -4$$

? Exercise 7.1.6

$$3x + 4y < 12$$

? Exercise 7.1.7

$$6x + 8y \leq 24$$

? Exercise 7.1.8

$$5x + 3y \leq 15$$

? Exercise 7.1.9

$$y > 3x + 1$$

? Exercise 7.1.10

$$3x + 2y \leq 12$$

? Exercise 7.1.11

$$5x - 2y < 10$$

? Exercise 7.1.12

$$3x + 4y \geq 24$$

? Exercise 7.1.13

$$y \leq 3x - 4$$

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7.2: Maximization Applications

Learning Objectives

In this section, you will learn to:

1. Recognize the typical form of a linear programming problem
2. Formulate maximization linear programming problems
3. Graph feasibility regions for maximization linear programming problems
4. Determine optimal solutions for maximization linear programming problems.

Application problems in business, economics, and social and life sciences often ask us to make decisions on the basis of certain conditions. The conditions or constraints often take the form of inequalities. In this section, we will begin to formulate, analyze, and solve such problems, at a simple level, to understand the many components of such a problem.

A typical **linear programming** problem consists of finding an extreme value of a linear function subject to certain constraints. We are either trying to maximize or minimize the value of this linear function, such as to maximize profit or revenue, or to minimize cost. That is why these linear programming problems are classified as **maximization** or **minimization problems**, or just **optimization problems**. The function we are trying to optimize is called an **objective function**, and the conditions that must be satisfied are called **constraints**.

A typical example is to maximize profit from producing several products, subject to limitations on materials or resources needed for producing these items; the problem requires us to determine the amount of each item produced. Another type of problem involves scheduling; we need to determine how much time to devote to each of several activities in order to maximize income from (or minimize cost of) these activities, subject to limitations on time and other resources available for each activity.

In this chapter, we will work with problems that involve only two variables, and therefore, can be solved by graphing. In the next chapter, we'll learn an algorithm to find a solution numerically. That will provide us with a tool to solve problems with more than two variables. At that time, with a little more knowledge about linear programming, we'll also explore the many ways these techniques are used in business and wide variety of other fields.

We begin by solving a maximization problem.

✓ Example 7.2.1

Niki holds two part-time jobs, Job I and Job II. She never wants to work more than a total of 12 hours a week. She has determined that for every hour she works at Job I, she needs 2 hours of preparation time, and for every hour she works at Job II, she needs one hour of preparation time, and she cannot spend more than 16 hours for preparation.

If Nikki makes \$40 an hour at Job I, and \$30 an hour at Job II, how many hours should she work per week at each job to maximize her income?

Solution

We start by choosing our variables.

- Let x = The number of hours per week Niki will work at Job I.
- Let y = The number of hours per week Niki will work at Job II.

Now we write the objective function. Since Niki gets paid \$40 an hour at Job I, and \$30 an hour at Job II, her total income I is given by the following equation.

$$I = 40x + 30y$$

Our next task is to find the constraints. The second sentence in the problem states, "She never wants to work more than a total of 12 hours a week." This translates into the following constraint:

$$x + y \leq 12$$

The third sentence states, "For every hour she works at Job I, she needs 2 hours of preparation time, and for every hour she works at Job II, she needs one hour of preparation time, and she cannot spend more than 16 hours for preparation." The translation follows.

$$2x + y \leq 16$$

The fact that x and y can never be negative is represented by the following two constraints:

$$x \geq 0, \text{ and } y \geq 0.$$

Well, good news! We have formulated the problem. We restate it as

$$\begin{array}{ll} \text{Maximize} & I = 40x + 30y \\ \text{Subject to:} & x + y \leq 12 \\ & 2x + y \leq 16 \\ & x \geq 0; y \geq 0 \end{array} \quad (7.2.1)$$

In order to solve the problem, we graph the constraints and shade the region that satisfies **all** the inequality constraints.

Any appropriate method can be used to graph the lines for the constraints. However often the easiest method is to graph the line by plotting the x -intercept and y -intercept.

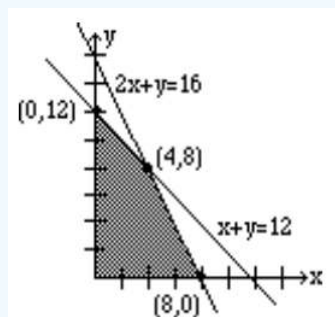
The line for a constraint will divide the plane into two region, one of which satisfies the inequality part of the constraint. A test point is used to determine which portion of the plane to shade to satisfy the inequality. Any point on the plane that is not on the line can be used as a test point.

- If the test point satisfies the inequality, then the region of the plane that satisfies the inequality is the region that contains the test point.
- If the test point does not satisfy the inequality, then the region that satisfies the inequality lies on the opposite side of the line from the test point.

In the graph below, after the lines representing the constraints were graphed using an appropriate method from Chapter 1, the point $(0,0)$ was used as a test point to determine that

- $(0,0)$ satisfies the constraint $x + y \leq 12$ because $0 + 0 < 12$
- $(0,0)$ satisfies the constraint $2x + y \leq 16$ because $2(0) + 0 < 16$

Therefore, in this example, we shade the region that is below and to the left of **both** constraint lines, but also above the x axis and to the right of the y axis, in order to further satisfy the constraints $x \geq 0$ and $y \geq 0$.



The shaded region where all conditions are satisfied is called the **feasibility region** or the feasibility polygon.

The **Fundamental Theorem of Linear Programming** states that the maximum (or minimum) value of the objective function always takes place at the vertices of the feasibility region.

Therefore, we will identify all the vertices (corner points) of the feasibility region. We call these points **critical points**. They are listed as $(0, 0)$, $(0, 12)$, $(4, 8)$, $(8, 0)$. To maximize Niki's income, we will substitute these points in the objective function to see which point gives us the highest income per week. We list the results below.

Critical Points	Income
$(0, 0)$	
$(0, 12)$	
$(4, 8)$	
$(8, 0)$	

Critical Points	Income
(0, 0)	$40(0) + 30(0) = \$0$
(0, 12)	$40(0) + 30(12) = \$360$
(4, 8)	$40(4) + 30(8) = \$400$
(8, 0)	$40(8) + 30(0) = \$320$

Clearly, the point (4, 8) gives the most profit: \$400.

Therefore, we conclude that Niki should work 4 hours at Job I, and 8 hours at Job II.

✓ Example 7.2.2

A factory manufactures two types of gadgets, regular and premium. Each gadget requires the use of two operations, assembly and finishing, and there are at most 12 hours available for each operation. A regular gadget requires 1 hour of assembly and 2 hours of finishing, while a premium gadget needs 2 hours of assembly and 1 hour of finishing. Due to other restrictions, the company can make at most 7 gadgets a day. If a profit of \$20 is realized for each regular gadget and \$30 for a premium gadget, how many of each should be manufactured to maximize profit?

Solution

We choose our variables.

- Let x = The number of regular gadgets manufactured each day.
- and y = The number of premium gadgets manufactured each day.

The objective function is

$$P = 20x + 30y$$

We now write the constraints. The fourth sentence states that the company can make at most 7 gadgets a day. This translates as

$$x + y \leq 7$$

Since the regular gadget requires one hour of assembly and the premium gadget requires two hours of assembly, and there are at most 12 hours available for this operation, we get

$$x + 2y \leq 12$$

Similarly, the regular gadget requires two hours of finishing and the premium gadget one hour. Again, there are at most 12 hours available for finishing. This gives us the following constraint.

$$2x + y \leq 12$$

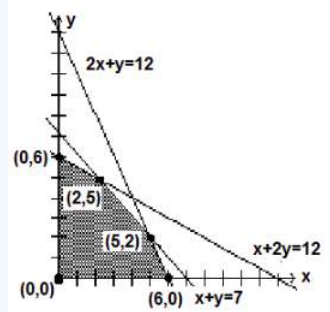
The fact that x and y can never be negative is represented by the following two constraints:

$$x \geq 0, \text{ and } y \geq 0.$$

We have formulated the problem as follows:

$$\begin{array}{ll} \textbf{Maximize} & P = 20x + 30y \\ \textbf{Subject to:} & x + y \leq 7 \\ & x + 2y \leq 12 \\ & 2x + y \leq 12 \\ & x \geq 0; y \geq 0 \end{array}$$

In order to solve the problem, we next graph the constraints and feasibility region.



Again, we have shaded the feasibility region, where all constraints are satisfied.

Since the extreme value of the objective function always takes place at the vertices of the feasibility region, we identify all the critical points. They are listed as (0, 0), (0, 6), (2, 5), (5, 2), and (6, 0). To maximize profit, we will substitute these points in the objective function to see which point gives us the maximum profit each day. The results are listed below.

Critical Point	Income
(0, 0)	$20(0) + 30(0) = \$0$
(0, 6)	$20(0) + 30(6) = \$180$
(2, 5)	$20(2) + 30(5) = \$190$
(5, 2)	$20(5) + 30(2) = \$160$
(6, 0)	$20(6) + 30(0) = \$120$

The point (2, 5) gives the most profit, and that profit is \$190.

Therefore, we conclude that we should manufacture 2 regular gadgets and 5 premium gadgets daily to obtain the maximum profit of \$190.

So far we have focused on “**standard maximization problems**” in which

1. The objective function is to be maximized
2. All constraints are of the form $ax + by \leq c$
3. All variables are constrained to be non-negative ($x \geq 0, y \geq 0$)

We will next consider an example where that is not the case. Our next problem is said to have “**mixed constraints**”, since some of the inequality constraints are of the form $ax + by \leq c$ and some are of the form $ax + by \geq c$. The non-negativity constraints are still an important requirement in any linear program.

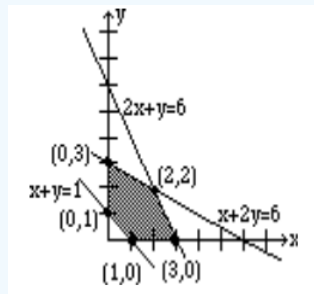
✓ Example 7.2.3

Solve the following maximization problem graphically.

$$\begin{array}{ll}
 \text{Maximize} & P = 10x + 15y \\
 \text{Subject to:} & x + y \geq 1 \\
 & x + 2y \leq 6 \\
 & 2x + y \leq 6 \\
 & x \geq 0; y \geq 0
 \end{array}$$

Solution

The graph is shown below.



The five critical points are listed in the above figure. The reader should observe that the first constraint $x + y \geq 1$ requires that the feasibility region must be bounded below by the line $x + y = 1$; the test point $(0,0)$ does not satisfy $x + y \geq 1$, so we shade the region on the opposite side of the line from test point $(0,0)$.

Critical point	Income
(1, 0)	$10(1) + 15(0) = \$10$
(3, 0)	$10(3) + 15(0) = \$30$
(2, 2)	$10(2) + 15(2) = \$50$
(0, 3)	$10(0) + 15(3) = \$45$
(0,1)	$10(0) + 15(1) = \$15$

Clearly, the point $(2, 2)$ maximizes the objective function to a maximum value of 50.

It is important to observe that if the point $(0,0)$ lies on the line for a constraint, then $(0,0)$ could not be used as a test point. We would need to select any other point we want that does not lie on the line to use as a test point in that situation.

Finally, we address an important question. Is it possible to determine the point that gives the maximum value without calculating the value at each critical point?

The answer is yes.

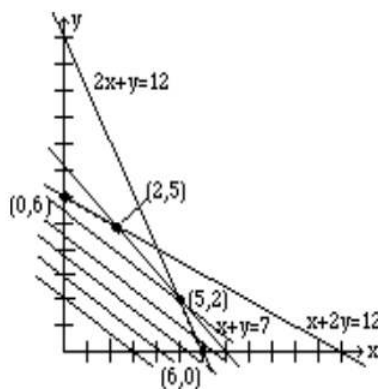
For example 7.2.2, we substituted the points $(0, 0)$, $(0, 6)$, $(2, 5)$, $(5, 2)$, and $(6, 0)$, in the objective function $P = 20x + 30y$, and we got the values \$0, \$180, \$190, \$160, \$120, respectively.

Sometimes that is not the most efficient way of finding the optimum solution. Instead we could find the optimal value by also graphing the objective function.

To determine the largest P , we graph $P = 20x + 30y$ for any value P of our choice. Let us say, we choose $P = 60$. We graph $20x + 30y = 60$.

Now we move the line parallel to itself, that is, keeping the same slope at all times. Since we are moving the line parallel to itself, the slope is kept the same, and the only thing that is changing is the P . As we move away from the origin, the value of P increases. The largest possible value of P is realized when the line touches the last corner point of the feasibility region.

The figure below shows the movements of the line, and the optimum solution is achieved at the point $(2, 5)$. In maximization problems, as the line is being moved away from the origin, this optimum point is the farthest critical point.



We summarize:

The Maximization Linear Programming Problems

1. Write the objective function.
2. Write the constraints.
 1. For the standard maximization linear programming problems, constraints are of the form: $ax + by \leq c$
 2. Since the variables are non-negative, we include the constraints: $x \geq 0$; $y \geq 0$.
3. Graph the constraints.
4. Shade the feasibility region.
5. Find the corner points.
6. Determine the corner point that gives the maximum value.
 - a. This is done by finding the value of the objective function at each corner point.
 - b. This can also be done by moving the line associated with the objective function.

Exercises

For the following maximization problems, choose your variables, write the objective function and the constraints, graph the constraints, shade the feasibility region, label all critical points, and determine the solution that optimizes the objective function.

- 1) A farmer has 100 acres of land on which she plans to grow wheat and corn. Each acre of wheat requires 4 hours of labor and \$20 of capital, and each acre of corn requires 16 hours of labor and \$40 of capital. The farmer has at most 800 hours of labor and \$2400 of capital available. If the profit from an acre of wheat is \$80 and from an acre of corn is \$100, how many acres of each crop should she plant to maximize her profit?
- 2) Mr. Tran has \$24,000 to invest, some in bonds and the rest in stocks. He has decided that the money invested in bonds must be at least twice as much as that in stocks. But the money invested in bonds must not be greater than \$18,000. If the bonds earn 6%, and the stocks earn 8%, how much money should he invest in each to maximize profit?
- 3) A factory manufactures chairs and tables, each requiring the use of three operations: Cutting, Assembly, and Finishing. The first operation can be used at most 40 hours; the second at most 42 hours; and the third at most 25 hours. A chair requires 1 hour of cutting, 2 hours of assembly, and 1 hour of finishing; a table needs 2 hours of cutting, 1 hour of assembly, and 1 hour of finishing. If the profit is \$20 per unit for a chair and \$30 for a table, how many units of each should be manufactured to maximize revenue?
- 4) The Silly Nut Company makes two mixtures of nuts: Mixture A and Mixture B. A pound of Mixture A contains 12 oz of peanuts, 3 oz of almonds and 1 oz of cashews and sells for \$4. A pound of Mixture B contains 12 oz of peanuts, 2 oz of almonds and 2 oz of cashews and sells for \$5. The company has 1080 lb. of peanuts, 240 lb. of almonds, 160 lb. of cashews. How many pounds of each of mixtures A and B should the company make to maximize profit?

(Hint: Use consistent units. Work the entire problem in pounds by converting all values given in ounces into fractions of pounds).

5)

$$\begin{array}{ll}\text{Maximize:} & Z = 4x + 10y \\ \text{Subject to:} & x + y \leq 5 \\ & 2x + y \leq 8 \\ & x + 2y \leq 8 \\ & x \geq 0, y \geq 0\end{array}$$

6) This maximization linear programming problem is not in “standard” form. It has mixed constraints, some involving $\leq$ inequalities and some involving $\geq$ inequalities. However with careful graphing, we can solve this using the techniques we have learned in this section.

$$\begin{array}{ll}\text{Maximize:} & Z = 5x + 7y \\ \text{Subject to:} & x + y \leq 30 \\ & 2x + y \leq 50 \\ & 4x + 3y \geq 60 \\ & 2x \geq y \\ & x \geq 0, y \geq 0\end{array}$$

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7.3: Minimization Applications

Learning Objectives

In this section, you will learn to:

1. Formulate minimization linear programming problems
2. Graph feasibility regions for minimization linear programming problems
3. Determine optimal solutions for minimization linear programming problems.

Minimization linear programming problems are solved in much the same way as the maximization problems.

For the **standard minimization linear program**, the constraints are of the form $ax + by \geq c$, as opposed to the form $ax + by \leq c$ for the standard maximization problem. As a result, the feasible solution extends indefinitely to the upper right of the first quadrant, and is unbounded. But that is not a concern, since in order to minimize the objective function, the line associated with the objective function is moved towards the origin, and the critical point that minimizes the function is closest to the origin.

However, one should be aware that in the case of an unbounded feasibility region, the possibility of no optimal solution exists.

✓ Example 7.3.1

At a university, Professor Symons wishes to employ two people, John and Mary, to grade papers for his classes. John is a graduate student and can grade 20 papers per hour; John earns \$15 per hour for grading papers. Mary is an post-doctoral associate and can grade 30 papers per hour; Mary earns \$25 per hour for grading papers. Each must be employed at least one hour a week to justify their employment.

If Prof. Symons has at least 110 papers to be graded each week, how many hours per week should he employ each person to minimize the cost?

Solution

We choose the variables as follows:

Let x = The number of hours per week John is employed.

and y = The number of hours per week Mary is employed.

The objective function is

$$C = 15x + 25y$$

The fact that each must work at least one hour each week results in the following two constraints:

$$x \geq 1$$

$$y \geq 1$$

Since John can grade 20 papers per hour and Mary 30 papers per hour, and there are at least 110 papers to be graded per week, we get

$$20x + 30y \geq 110$$

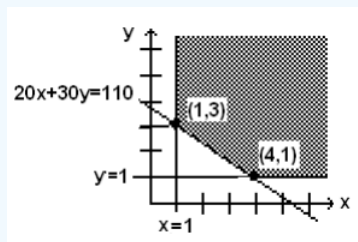
The fact that x and y are non-negative, we get

$$x \geq 0, \text{ and } y \geq 0.$$

The problem has been formulated as follows.

$$\begin{array}{ll} \text{Minimize} & C = 15x + 25y \\ \text{Subject to:} & x \geq 1 \\ & y \geq 1 \\ & 20x + 30y \geq 110 \\ & x \geq 0; y \geq 0 \end{array}$$

To solve the problem, we graph the constraints as follows:



Again, we have shaded the feasibility region, where all constraints are satisfied.

If we used test point (0,0) that does not lie on any of the constraints, we observe that (0, 0) **does not** satisfy any of the constraints $x \geq 1$, $y \geq 1$, $20x + 30y \geq 110$. Thus all the shading for the feasibility region lies on the opposite side of the constraint lines from the point (0,0).

Alternatively we could use test point (4,6), which also does not lie on any of the constraint lines. We'd find that (4,6) **does** satisfy all of the inequality constraints. Consequently all the shading for the feasibility region lies on the same side of the constraint lines as the point (4,6).

Since the extreme value of the objective function always takes place at the vertices of the feasibility region, we identify the two critical points, (1, 3) and (4, 1). To minimize cost, we will substitute these points in the objective function to see which point gives us the minimum cost each week. The results are listed below.

Critical points	Income
(1, 3)	$15(1) + 25(3) = \$90$
(4, 1)	$15(4) + 25(1) = \$85$

The point (4, 1) gives the least cost, and that cost is \$85. Therefore, we conclude that in order to minimize grading costs, Professor Symons should employ John 4 hours a week, and Mary 1 hour a week at a cost of \$85 per week.

✓ Example 7.3.2

Professor Hamer is on a low cholesterol diet. During lunch at the college cafeteria, he always chooses between two meals, Pasta or Tofu. The table below lists the amount of protein, carbohydrates, and vitamins each meal provides along with the amount of cholesterol he is trying to minimize. Mr. Hamer needs at least 200 grams of protein, 960 grams of carbohydrates, and 40 grams of vitamins for lunch each month. Over this time period, how many days should he have the Pasta meal, and how many days the Tofu meal so that he gets the adequate amount of protein, carbohydrates, and vitamins and at the same time minimizes his cholesterol intake?

	PASTA	TOFU
PROTEIN	8g	16g
CARBOHYDRATES	60g	40g
VITAMIN C	2g	2g
CHOLESTEROL	60mg	50mg

Solution

We choose the variables as follows.

Let x = The number of days Mr. Hamer eats Pasta.

and y = The number of days Mr. Hamer eats Tofu.

Since he is trying to minimize his cholesterol intake, our objective function represents the total amount of cholesterol C provided by both meals.

$$C = 60x + 50y$$

The constraint associated with the total amount of protein provided by both meals is

$$8x + 16y \geq 200$$

Similarly, the two constraints associated with the total amount of carbohydrates and vitamins are obtained, and they are

$$60x + 40y \geq 960$$

$$2x + 2y \geq 40$$

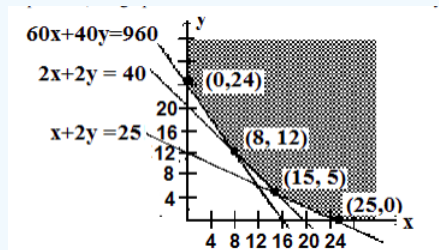
The constraints that state that x and y are non-negative are

$$x \geq 0, \text{ and } y \geq 0.$$

We summarize all information as follows:

$$\begin{array}{ll} \text{Minimize} & C = 60x + 50y \\ \text{Subject to:} & 8x + 16y \geq 200 \\ & 60x + 40y \geq 960 \\ & 2x + 2y \geq 40 \\ & x \geq 0; y \geq 0 \end{array}$$

To solve the problem, we graph the constraints and shade the feasibility region.



We have shaded the unbounded feasibility region, where all constraints are satisfied.

To minimize the objective function, we find the vertices of the feasibility region. These vertices are $(0, 24)$, $(8, 12)$, $(15, 5)$ and $(25, 0)$. To minimize cholesterol, we will substitute these points in the objective function to see which point gives us the smallest value. The results are listed below.

Critical points	Income
$(0, 24)$	$60(0) + 50(24) = 1200$
$(8, 12)$	$60(8) + 50(12) = 1080$
$(15, 5)$	$60(15) + 50(5) = 1150$
$(25, 0)$	$60(25) + 50(0) = 1500$

The point $(8, 12)$ gives the least cholesterol, which is 1080 mg. This states that for every 20 meals, Professor Hamer should eat Pasta 8 days, and Tofu 12 days.

We must be aware that in some cases, a linear program may not have an optimal solution.

- A linear program can fail to have an optimal solution if there is not a feasibility region. If the inequality constraints are not compatible, there may not be a region in the graph that satisfies **all** the constraints. If the linear program does not have a feasible solution satisfying all constraints, then it can not have an optimal solution.
- A linear program can fail to have an optimal solution if the feasibility region is unbounded.

- The two minimization linear programs we examined had unbounded feasibility regions. The feasibility region was bounded by constraints on some sides but was not entirely enclosed by the constraints. Both of the minimization problems had optimal solutions.
- However, if we were to consider a maximization problem with a similar unbounded feasibility region, the linear program would have no optimal solution. No matter what values of x and y were selected, we could always find other values of x and y that would produce a higher value for the objective function. In other words, if the value of the objective function can be increased without bound in a linear program with an unbounded feasible region, there is no optimal maximum solution.

Although the method of solving minimization problems is similar to that of the maximization problems, we still feel that we should summarize the steps involved.

Minimization Linear Programming Problems

1. Write the objective function.
2. Write the constraints.
 - a. For standard minimization linear programming problems, constraints are of the form: $ax + by \geq c$
 - b. Since the variables are non-negative, include the constraints: $x \geq 0$; $y \geq 0$.
3. Graph the constraints.
4. Shade the feasibility region.
5. Find the corner points.
6. Determine the corner point that gives the minimum value.
 - a. This can be done by finding the value of the objective function at each corner point.
 - b. This can also be done by moving the line associated with the objective function.
 - c. There is the possibility that the problem has no solution

Exercises

For each of the following minimization problems, choose your variables, write the objective function and the constraints, graph the constraints, shade the feasibility region, label all critical points, and determine the solution that optimizes the objective function.

1) A diet is to contain at least 2400 units of vitamins, 1800 units of minerals, and 1200 calories. Two foods, Food A and Food B are to be purchased. Each unit of Food A provides 50 units of vitamins, 30 units of minerals, and 10 calories. Each unit of Food B provides 20 units of vitamins, 20 units of minerals, and 40 calories. Food A costs \$2 per unit and Food B cost \$1 per unit. How many units of each food should be purchased to keep costs at a minimum?

2) A computer store sells two types of computers, laptops and desktops. The supplier demands that at least 150 computers be sold a month. Experience shows that most consumers prefer laptops, but some business customers require desktops. The result is that the number of laptops sold is at least twice of the number of desktops. The store pays its sales staff a \$60 commission for each laptop, and a \$40 commission for each desktop. Let x = the number of laptops and y = the number of desktop computers. How many of each type must be sold to minimize commission to its sales people?

What is the minimum commission?

3) An oil company has two refineries. Each day, Refinery A produces 200 barrels of high-grade oil, 300 barrels of medium-grade oil, and 200 barrels of low-grade oil and costs \$12,000 to operate. Each day, Refinery B produces 100 barrels of high-grade oil, 100 barrels of medium-grade oil, and 200 barrels of low-grade oil and costs \$10,000 to operate. The company must produce at least 800 barrels of high-grade oil, 900 barrels of medium-grade oil, and 1,000 barrels of low-grade oil.

How many days should each refinery be operated to meet the goals at a minimum cost?

4) A print shop at a community college in Cupertino, California, employs two different contractors to maintain its copying machines. The print shop needs to have 12 IBM, 18 Xerox, and 20 Canon copying machines serviced. Contractor A can repair 2 IBM, 1 Xerox, and 2 Canon machines at a cost of \$800 per month, while Contractor B can repair 1 IBM, 3 Xerox, and 2 Canon machines at a cost of \$1000 per month. How many months should each of the two contractors be employed to minimize the cost?

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7.4: Chapter Review

Solve the following linear programming problems by the graphical method.

- 1) Mr. Shoemaker has \$20,000 to invest in two types of mutual funds: a High-Yield Fund and an Equity Fund. The High-Yield fund has an annual yield of 12%, while the Equity fund earns 8%. He would like to invest at least \$3000 in the High-Yield fund and at least \$4000 in the Equity fund. How much should he invest in each to maximize his annual yield, and what is the maximum yield?
- 2) Dr. Lum teaches part-time at two community colleges, Hilltop College and Serra College. Dr. Lum can teach up to 5 classes per semester. For every class he teaches at Hilltop College, he needs to spend 3 hours per week preparing lessons and grading papers. For each class at Serra College, he must do 4 hours of work per week. He has determined that he cannot spend more than 18 hours per week preparing lessons and grading papers. If he earns \$6,000 per class at Hilltop College and \$7,500 per class at Serra College, how many classes should he teach at each college to maximize his income, and what will be his income?
- 3) Mr. Shamir employs two part-time typists, Inna and Jim, for his typing needs. Inna charges \$15 an hour and can type 6 pages an hour, while Jim charges \$18 an hour and can type 8 pages per hour. Each typist must be employed at least 8 hours per week to keep them on the payroll. If Mr. Shamir has at least 208 pages to be typed, how many hours per week should he employ each typist to minimize his typing costs, and what will be the total cost?
- 4) Mr. Boutros wants to invest up to \$20,000 in two stocks, Cal Computers and Texas Tools. The Cal Computers stock is expected to yield a 16% annual return, while the Texas Tools stock promises a 12% yield. Mr. Boutros would like to earn at least \$2,880 this year. According to Value Line Magazine's safety index (1 highest to 5 lowest), Cal Computers has a safety number of 3 and Texas Tools has a safety number of 2. How much money should he invest in each to minimize the safety number? Note: A lower safety number means less risk.
- 5) A store sells two types of copy machines: compact (low capacity) and standard (which takes more space). The store can sell up to 90 copiers a month. A maximum of 1080 cubic feet of storage space is available. A compact copier requires 6 cu. ft. of storage space, and a standard copier requires 18 cu. ft.. The compact and standard copy machines take, respectively, 1 and 1.5 sales hours of labor.
A maximum of 99 hours of labor is available. The profit from each of these copiers is \$60 and \$80, respectively, how many of each type should be sold to maximize profit, and what is the maximum profit?
- 6) A company manufactures two types of cell phones, a Basic model and a Pro model. The Basic model generates a profit of \$100 per phone and the Pro model has a profit of \$150 per phone. On the assembly line the Basic phone requires 7 hours, while the Pro model takes 11 hours. The Basic phone requires one hour and the Pro phone needs 3 hours for finishing, which includes loading software. Both phones require one hour for testing. On a particular production run the company has available 1,540 work hours on the assembly line, 360 work hours for finishing, and 200 work hours in the testing department. How many cell phones of each type should be produced to maximize profit, and what is that maximum profit?
- 7) John wishes to choose a combination of two types of cereals for breakfast - Cereal A and Cereal B. A small box (one serving) of Cereal A costs \$0.50 and contains 10 units of vitamins, 5 units of minerals, and 15 calories. A small box(one serving) of Cereal B costs \$0.40 and contains 5 units of vitamins, 10 units of minerals, and 15 calories. John wants to buy enough boxes to have at least 500 units of vitamins, 600 units of minerals, and 1200 calories. How many boxes of each food should he buy to minimize his cost, and what is the minimum cost?
- 8) Jessica needs at least 60 units of vitamin A, 40 units of vitamin B, and 140 units of vitamin C each week. She can choose between Costless brand or Savemore brand tablets. A Costless tablet costs 5 cents and contains 3 units of vitamin A, 1 unit of vitamin B, and 2 units of vitamin C. A Savemore tablet costs 7 cents and contains 1 unit of A, 1 of B, and 5 of C. How many tablets of each kind should she buy to minimize cost, and what is the minimum cost?
- 9) A small company manufactures two products: A and B. Each product requires three operations: Assembly, Finishing and Testing. Product A requires 1 hour of Assembly, 3 hours of Finishing, and 1 hour of Testing. Product B requires 3 hours of Assembly, 1 hour of Finishing, and 1 hour of Testing. The total work-hours available per week in the Assembly division is 60, in Finishing is 60, and in Testing is 24. Each item of product A has a profit of \$50, and each item of Product B has a profit of \$75. How many of each should be made to maximize profit? What is the maximum profit?

10) A factory manufactures two products, A and B. Each product requires the use of three machines, Machine I, Machine II, and Machine III. The time requirements and total hours available on each machine are listed below.

	Machine I	Machine II	Machine III
Product A	1	2	4
Product B	2	2	2
Total hours	70	90	160

If product A generates a profit of \$60 per unit and product B a profit of \$50 per unit, how many units of each product should be manufactured to maximize profit, and what is the maximum profit?

11) A company produces three types of shoes, formal, casual, and athletic, at its two factories, Factory I and Factory II. The company must produce at least 6000 pairs of formal shoes, 8000 pairs of casual shoes, and 9000 pairs of athletic shoes. Daily production of each factory for each type of shoe is:

	Factory I	Factory II
Formal	100	100
Casual	100	200
Athletic	300	100

Operating Factory I costs \$1500 per day and it costs \$2000 per day to operate Factory II. How many days should each factory operate to complete the order at a minimum cost, and what is the minimum cost?

12) A professor gives two types of quizzes, objective and recall. He plans to give at least 15 quizzes this quarter. The student preparation time for an objective quiz is 15 minutes and for a recall quiz 30 minutes. The professor would like a student to spend at least 5 hours (300 minutes) preparing for these quizzes above and beyond the normal study time. The average score on an objective quiz is 7, and on a recall type 5, and the professor would like the students to score at least 85 points on all quizzes. It takes the professor one minute to grade an objective quiz, and 1.5 minutes to grade a recall type quiz. How many of each type should he give in order to minimize his grading time?

13) A company makes two mixtures of nuts: Mixture A and Mixture B. Mixture A contains 30% peanuts, 30% almonds and 40% cashews and sells for \$5 per pound. Mixture B contains 30% peanuts, 60% almonds and 10% cashews and sells for \$3 a pound. The company has 540 pounds of peanuts, 900 pounds of almonds, 480 pounds of cashews. How many pounds of each of mixtures A and B should the company make to maximize profit, and what is the maximum profit?

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CHAPTER OVERVIEW

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8.1: Problem Solving

1. [Math Processing Error] about [Math Processing Error]
3. [Math Processing Error] in VAT
5. [Math Processing Error]
7. absolute increase: [Math Processing Error] Relative: [Math Processing Error] increase
9.
 - a. [Math Processing Error]
 - b. Yes, their goal was to decrease by at least 15%. They exceeded their goal.
11. Dropping by [Math Processing Error] is the same as keeping [Math Processing Error]. [Math Processing Error] Attendance was about 319 before the drop.
13.
 - a) Kaplan's enrollment was 64.3% larger than Walden's. 30510
 - b) Walden's enrollment was 39.1% smaller than Kaplan's.
 - c) Walden's enrollment was 60.9% of Kaplan's.
15. If the original price was [Math Processing Error] the basic clearance price would be [Math Processing Error] The additional markdown would bring it to [Math Processing Error] This is [Math Processing Error] of the original price.
17. These are not comparable; "a" is using a base of all Americans and is talking about health insurance from any source, while "b" is using a base of adults and is talking specifically about health insurance provided by employers.
21. These statements are equivalent, if we assume the claim in "a" is a percentage point increase, not a relative change. Certainly these messages are phrased to convey different opinions of the levy. We are told the new rate will be \$9.33 per \$1000, which is 0.933% tax rate. If the original rate was 0.833% (0.1 percentage point lower), then this would indeed be a 12% relative increase.
23. [Math Processing Error] of [Math Processing Error] is [Math Processing Error] a 6 percentage point decrease.
25. Probably not, unless the final is worth 50% of the overall class grade. If the final was worth 25% of the overall grade, then a 100% would only raise her average to 77.5%
27. [Math Processing Error] pounds [Math Processing Error] per pound (or 10 pounds/\$4 [Math Processing Error] pounds per dollar)
29. [Math Processing Error]
31. 2.5 cups
33. 74 turbines
35. 96 inches
37. \$6000
39. 55.6 meters
43. The population density of the US is 84 people per square mile. The density of India is about 933 people per square mile. The density of India is about 11 times greater than that of the U.S.
49. The oil in the spill could produce 93.1 million gallons of gasoline. Each car uses about 600 gallons a year. That would fuel 155,167 cars for a year.
53. An answer around 100-300 gallons would be reasonable
57. 156 million miles
59. The time it takes the light to reach you is so tiny for any reasonable distance that we can safely ignore it. 750 miles/hr is about 0.21 miles/sec. If the sound takes 4 seconds to reach you, the lightning is about 0.84 miles away. In general, the lightning will be [Math Processing Error] miles away, which is often approximated by dividing the number of seconds by 5

61. About 8.2 minutes

63. Four cubic yards (or 3.7 if they sell partial cubic yards)

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8.2: Sets and Counting

Sets, Set Operations, Venn Diagrams, and Cardinality

1) $\{m, i, s, p\}$

3) One possibility is: Multiples of 3 between 1 and 10

5) Yes

7) True

9) True

11) False

13) $A \cup B = \{1, 2, 3, 4, 5\}$

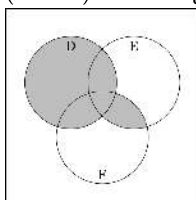
15) $A \cap C = \{4\}$

17) $A^C = \{6, 7, 8, 9, 10\}$

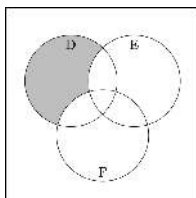
19) $D^C \cap E = \{t, s\}$

21) $(D \cap E) \cup F = \{a, b, h, k, t\}$

23) $(F \cap E)^C \cap D = \{b, c, k\}$



25)



27)

29) One possible answer: $(A \cap B) \cup (B \cap C)$

31) $(A \cap B^C) \cup C$

33) 5

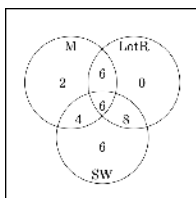
35) 6

37) $n(A \cap C) = 5$

39) $n(A \cap B \cap C) = 3$

41) $n(G \cup H) = 45$

43) 136 use Redbox



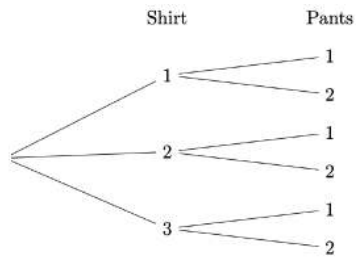
45)

a) 8 had seen exactly one

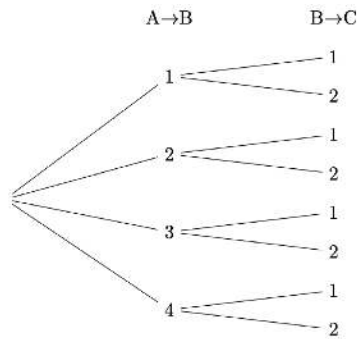
b) 6 had only seen SW

Tree Diagrams and the Multiplication Axiom

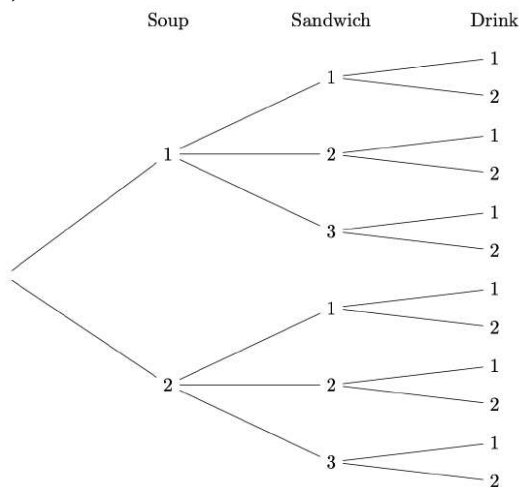
1) 6



3) 8



5) 12



7) 15,600,000

9) 6,400,000

11) BB, BG, GB, GG

13) 16

15) 27,000

17) 10,000

Permutations

1) 60

3) 210

5) 362,880

7) 25,200

9) 900

11) 48

13) 72

15) 2,400

17) 15,120

19) 5,040

21) 720

Combinations

1) 120

3) 10

5) 2,598,960

7) 66

9) 10

11) 20

13) 6

15) 924

17) 35

Combinations Involving Several Sets

1) 24

3) 25

5) 14,400

7) 4

9) 60

11) 80

13) 51

15) 7

17) 1,410

19) 171,600

21) 22,308

23) 24

25) 858,627,790,020

27) 11,647,387,752,000

29) 1890

31) 1680

Circular Permutations and Permutations with Similar Elements

1) 24

3) 120

5) 120

7) 64,864,800

9) 210

11) 6

13) 10

15) 210

17) 15,120

19) 5,040

21) 720

Chapter Review

1) 1,000

2) 20; 135; 15

3) 12

4) 144

5) 3,024

6) 11,639,628,000

7) 84

8) 60

- 9) 24
- 10) 126; 336; 210
- 11) 5,184
- 12) 1,048,576
- 13) 46,200
- 14) 60
- 15) 120
- 16) 20
- 17) 10
- 18) 1,296
- 19) 27,720
- 20) 720
- 21) 194,594,400
- 22) a) 5,148; b) 58,656; c) 123,552; d) 10,240 or 9,216
- 23) 17,576
- 24) 4,500
- 25) 5,040; 720
- 26) 3003; 371; 210; 191; 435
- 27) 10
- 28) 35
- 29) 72
- 30) 72,000
- 31) $-48384x^5y^3$
- 32) $2016a^5b^4$

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8.3: Probability

Sample Spaces and Probability

- 1) {1, 2, 3, 4, 5, 6}
- 3) {1H, 2H, 3H, 4H, 5H, 6H, 1T, 2T, 3T, 4T, 5T, 6T}
- 5) {(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)}
- 7) $4/52 = 2/26$
- 9) $13/52$
- 11) $16/52 = 4/13$
- 13) $6/20 = 3/10$
- 15) $13/20$
- 17) $3/8$
- 19) $6/8 = 3/4$
- 21) $2/8 = 1/4$
- 23) $4/36 = 1/9$
- 25) $7/36$
- 27) $6/36 = 1/6$
- 29) $8/12 = 2/3$
- 31) $6/12 = 1/2$
- 33) $1/6$

Mutually Exclusive Events and the Addition Rule

- 1) Yes
- 3) No
- 5) No
- 7) $16/52 = 4/13$
- 9) $13/36$
- 11) 40%
- 13) $30/100 = 3/10$
- 15) $94/100 = 47/50$
- 17) $68/100 = 17/25$
- 19) Not mutually exclusive
- 21) 0
- 23) 0.5
- 25) 0.4

Probability Using Tree Diagrams and Combinations

- 1) $20/56 = 5/14$
- 3) $6/65$
- 5) $1/6$
- 7) $3/10$
- 9) $10/220 = 1/22$
- 11) $56/220 = 14/55$
- 13) $45/1365 = 3/91$
- 15) $21/1365 = 1/65$
- 17) $324/1365 = 108/455$
- 19) $79,092/2,598,960 = 507/16,660 \approx 0.0304$
- 21) $24/2,598,960 = 1/108,290 \approx 0.000009234$
- 23) $1,584/2,598,960 = 33/54145 \approx 0.0006095$
- 25) $1890/12376 = 135/884 \approx 0.1527$
- 27) $1680/12376 = 30/221 \approx 0.1357$

- 29) $180/792 = 5/22 = 0.2\overline{27}$
31) $21/792 = 7/264 = 0.26\overline{51}$
33) 0.972864

Conditional Probability

- 1) $4/12 = 1/3$
3) $1/3$
5) $30/44 = 15/22$
7) $6/54 = 3/26$
9) a) $1/6$; b) $1/4$
11) 0.12
13) 0.4
15) 0.696
17) $1/2$
19) $3/4$
21) $84/528 = 7/44$
23) $84/2096 = 21/524$
25) $188/796 = 47/199$

Independent Events

- 1) $2/3$
3) $50/150 = 1/3$
5) a) $3/4$; b) $1/2$; c) $1/2$; d) no
7) 0.12
9) 0.36
11) yes
13) a) $28/100 = 7/25$; b) $82/100 = 41/50$
15) a) $7/8$; b) $3/4$; c) $3/4$; d) no
17) a) yes; b) 0.175
19) a) no; b) 0.081

Bayes' Formula

- 1) a) 0.6458; b) 0.4706; c) 0.625
3) The Republican party
5) 0.7787
7) a) 0.045; b) 0.2667; c) 0.03
9) a) 0.02; b) 0.111

Chapter Review

- 1) $3/36 = 1/12$; $4/36 = 1/9$
2) $8/12 = 2/3$; $7/12$
3) $8/52 = 2/13$; $16/52 = 4/13$
4) $3/5$; $9/10$
5) $3/20$; $1/20$; $5/6$; $1/6$
6) $1/16$; $1/4$
7) $3/4$; 0
8) 0.72
9) 40%
10) independent
11) $3/4$; 0.45
12) 0.8144
13) 0.22
14) 0.45278

- 15) a) $111540/2598960 = 143/3332 \approx 0.0429$; b) $949104/2598960 = 1521/4165 \approx 0.3652$; c) $1349088/2598960 = 2162/4165 \approx 0.5191$; d) $36/2598960 = 3/216580 \approx 0.00001385$
16) 9/20; 10/27; 15/33; 11/20; no; yes
17) 0.4
18) 3/14; 37/42; 2/7; 35/84
19) no
20) 0
21) 0.65
22) 0.36
23) 5/6
24) 0.2
25) 0.5
26) 0.3
-

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8.4: Probability Distributions and Statistics

Answers to these exercises may be found at the end of each subsection in [3.7.1: Discrete Random Variables](#) and [3.7.2: Continuous Random Variables](#).

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8.5: Mathematics of Finance

Simple Interest

- 1) \$600
- 3) \$3,048
- 5) \$1,800
- 7) \$872
- 9) 11%
- 11) Discount \$240; Proceeds \$1,760
- 13) \$7,440
- 15) \$2,790.70

Compound Interest

- 1) \$11,542.52
- 3) \$5,647.77
- 5) Bank B pays more. Bank A $r_{\text{EFF}} = 0.0513$; Bank B $r_{\text{EFF}} = 0.0523$.
- 7) \$12,702.00
- 9) 8.66 years
- 11) \$151,257.12
- 13) 7.3967%
- 15) 17.38 million people

Future Value of Annuities and Sinking Funds

- 1) \$13,954.01
- 3) \$15,904.47
- 5) \$20,578.36
- 7) \$6,438.02
- 9) A lump sum of \$25,000 is better; $A = \$38,002.64$. For \$400 paid monthly, $A = \$35,872.42$.
- 11) \$112,552.26

Present Value of Annuities and Sinking Funds

- 1) \$1,177,953.55
- 3) \$12,043.34
- 5) \$2,149.21
- 7) \$1,976.80
- 9) Leasing is better; $P = \$37,908$ if payments are assumed to be made at the end of each year as is the general assumption in this book. However, if we consider payments to be made at the beginning of each year, then the present value is $\$41,699 = \$37,908(1+0.10)$, which is greater than \$40,000; in this case, buying would be better and this situation more realistically represents the real world.

Classification of Finance Problems

- 1) D
- 3) F
- 5) E
- 7) D
- 9) a) B; b) F
- 11) F
- 13) A
- 15) B
- 17) A
- 19) B

Additional Application Problems

- 1) \$1,123.06
- 3) \$171,907.63
- 5) \$5,579.64
- 7) \$11,680.01
- 9) \$333.85 / month
- 10) \$2,177.77 / month
- 11) \$305,421.70
- 12) \$2,479.37
- 13) $\$447.70 + \$471.48 = \$919.18$
- 15) $\$435.48 + \$806.45 = \$1,241.94$
- 17) a) \$30,535.40; b) \$720.63

Chapter Review

- 1) \$872
- 2) 81,596
- 3) \$1,190.05
- 4) \$1,755.93
- 5) \$12,156.72
- 6) \$160,383.25
- 7) \$289.28
- 8) \$19,290.63
- 9) \$2,085.33
- 10) \$688,675.54
- 11) \$497,897.83
- 12) \$928.94 if $r = 0.07$ or \$1,077.95 if $r = 0.05$
- 13) \$3,447.31
- 14) a) \$1,643.90; b) \$128,451.61
- 15) \$9,898.48
- 16) Present value \$1,213,539.16; Future value \$5,745,936.31
- 17) \$6,669.70
- 18) \$767,123,287.67
- 19) \$2,375.25
- 20) 109,619
- 21) \$5,805.92
- 22) \$2,138.67
- 23) \$1,523.33
- 24) \$276.68
- 25) If leasing, $\$7,835.35 + \$750 = \$8,585.33$. Purchasing for \$8,000 is cheaper.
- 26) Cost in 5 years is \$20,615.73; Sinking fund deposit is \$833.79 / quarter.
- 27) Western Bank $r_{\text{EFF}} = 0.597$ or 5.97%. City Bank is better.
- 28) \$404.57
- 29) \$300 per month for 5 years is worth \$14,900.82; \$500 per month for 3 years is worth \$16,026.59 (better).
- 30) 10.19 years
- 31) \$177,692.68
- 32) \$20,000 and \$500 per month for 10 years is \$61,210.74; \$12,000 and \$1,000 per month for 5 years is \$61,318.43 (better).
- 33) \$16,384.77
- 34) Monthly payment \$2,204.21; Total paid \$793,515.60; Interest \$368,515.6.
- 35) 15.53 years

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8.6: Linear Functions

The Rectangular Coordinate Systems and Graphs

- 1) x -intercept: 3; y -intercept: -4
- 3) $y = \frac{5}{3}x + 4$
- 5) $\sqrt{72} \approx 8.4853$
- 7) $2\sqrt{96130} \approx 620.0968$
- 9) (2, 11.5)

Linear Functions

- 1) Terry starts at an elevation of 3000 feet and descends 70 feet per second.
- 3) 3 miles per hour
- 5) $d(t) = 100 - 10t$
- 7) Yes.
- 9) No.
- 11) No.
- 13) No.
- 15) Increasing.
- 17) Decreasing.
- 19) Decreasing.
- 21) Increasing.
- 23) Decreasing.
- 25) 3
- 27) -13
- 29) 45
- 31) $f(x) = -12x + 72$
- 33) $y = 2x + 3$
- 35) $y = -13x + 223$
- 37) $y = 45x + 4$
- 39) -54
- 41) $y = 23x + 1$
- 43) $y = -2x + 3$
- 45) $y = 3$
- 47) Linear, $g(x) = -3x + 5$
- 49) Linear, $f(x) = 5x - 5$
- 51) Linear, $g(x) = -252x + 6$
- 53) Linear, $f(x) = 10x - 24$
- 55) $f(x) = -58x + 17.3$
- 59) a. $a = 11900$; $b = 1000.1$ b. $q(p) = 1000p - 100$
- 63) $x = -163$
- 65) $x = a$
- 67) $y = dc - ax - adc - a$
- 69) \$45 per training session.
- 71) The rate of change is 0.1. For every additional minute talked, the monthly charge increases by \$0.1 or 10 cents. The initial value is 24. When there are no minutes talked, initially the charge is \$24.
- 73) The slope is -400. This means for every year between 1960 and 1989, the population dropped by 400 per year in the city.
- 75) c.

Linear Functions and Mathematical Models: Modeling with Linear Functions

- 1) Determine the independent variable. This is the variable upon which the output depends.
- 3) To determine the initial value, find the output when the input is equal to zero.
- 5) 6 square units

- 7) 20.012 square units
- 9) 2,300
- 11) 64,170
- 13) $P(t) = 75000 + 2500t$
- 15) $(-30, 0)$ Thirty years before the start of this model, the town had no citizens. $(0, 75,000)$ Initially, the town had a population of 75,000.
- 17) Ten years after the model began.
- 19) $W(t) = 0.5t + 7.5$
- 21) $(-15, 0)$: The x -intercept is not a plausible set of data for this model because it means the baby weighed 0 pounds 15 months prior to birth); $(0, 7.5)$: The baby weighed 7.5 pounds at birth.
- 23) At age 5.8 months.
- 25) $C(t) = 12025 - 205t$
- 27) $(58.7, 0)$: In roughly 59 years, the number of people inflicted with the common cold would be 0); $(0, 12,025)$: Initially there were 12,025 people afflicted by the common cold.
- 29) 2064
- 31) $y = -2t + 180$
- 33) In 2070, the company's profit will be zero.
- 35) $y = 30t - 300$
- 37) $(10, 0)$ In 1990, the profit earned zero profit.
- 39) Hawaii
- 41) During the year 1933
- 43) \$105,620
- 45) a) 696 people; b) 4 years; c) 174 people per year; d) 305 people; e) $P(t) = 305 + 174t$; f) 2,219 people
- 47) a) $C(x) = 0.15x + 10$; b) The flat monthly fee is \$10 and there is an additional \$0.15 fee for each additional minute used; c) \$113.05
- 49) a) $P(t) = 190t + 4360$; b) 6,640 moose
- 51) a) $R(t) = 16 - 2.1t$; b) 5.5 billion cubic feet; c) During the year 2017
- 53) More than 133 minutes
- 55) More than \$42,857.14 worth of jewelry
- 57) \$66,666.67

Linear Functions and Mathematical Models: Applications

- 1) $y = 25x + 1200$
- 3) $y = 20x + 350$
- 5) $y = 80x + 240\,000$
- 7) $y = \frac{2}{5}x$; 68
- 9) $y = 7x - 338$; 138
- 11) $F = \frac{9}{5}C + 32$; 77
- 13) $y = 0.375x + 29.8$; 42.925 million people in 2025
- 15) $y = 25x + 1200$; 14,400 students in 2010
- 17) $y = 0.18x + 10$; the cost is \$82 for a home using 400 kWh of electricity per month
- 19) a) $y = 12x + 110\,000$; b) $y = \$230,000$; c) $x = \$7,500$
- 21) a) $y = 3x + 1000$; b) when \$100 is spent on ads, 1,300 cups of coffee are sold

Intersection of Straight Lines

- 1) $x = 3, y = 13$
- 3) $x = \$11.50, y = 16,500$ items
- 5) a) Plan I costs \$87; Plan II costs \$99. Plan I is better. b) $x = 150$ miles; both plans cost \$61.50.
- 7) Supply curve: $y = 400x - 1200$
- 9) $x = 4,000$ cookies; cost = revenue = \$3,200
- 11) $x = 8,000$ pairs of socks; cost = revenue = \$36,000
- 13) a) cost function $y = 10x + 700$; b) fixed cost = \$700; c) $x = 140$ pounds; d) revenue = cost = \$2,100

Chapter Review

- 1) Yes
- 3) Increasing.
- 5) $y = -3x + 26$
- 7) 3
- 9) $y = 2x - 2$
- 11) Not linear.
- 13) parallel
- 15) $(-9, 0)$; $(0, -7)$
- 17) Line 1: $m = -2$; Line 2: $m = -2$; Parallel
- 19) $y = -0.2x + 21$
- 23) 250.
- 25) 118,000.
- 27) $y = -300x + 11500$
- 29) a) 800; b) 100 students per year; c) $P(t) = 100t + 1700$
- 31) 18,500
- 33) \$91,625
- 35) Extrapolation.
- 39) Midway through 2024.
- 41) $y = -1.294x + 49.412$; $r = -0.974$
- 43) Early in 2022
- 45) 7,660

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8.7: Linear Programming- A Geometric Approach

Linear Inequalities in Two Variables

1. yes; no

3.

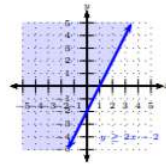


Figure 8.7.1

5.

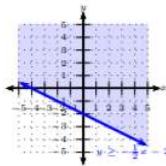


Figure 8.7.2

7.

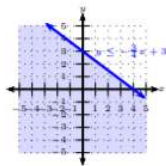


Figure 8.7.3

9.

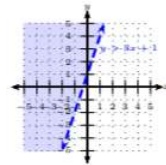


Figure 8.7.4

11.

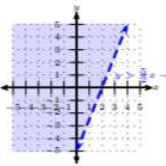


Figure 8.7.5

13.

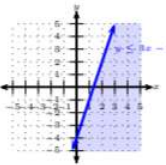


Figure 8.7.6

Maximization Applications

- 1) 80 acres of wheat and 20 acres of corn should be planted for a maximum profit of \$8,400.
- 2) 10 chairs and 15 tables should be manufactured for a maximum profit of \$650.
- 3) The maximum value of X is 40 when $x = 0$ and $y = 4$.

Minimization Applications

- 1) 30 units of Food A and 45 units of Food B should be purchased to keep costs at a minimum of \$105.
- 3) $\min C = 12,000x + 10,000y$
s.t. $200x + 100y \geq 800$ (high-grade oil)

$$300x + 100y \geq 900 \text{ (medium-grade oil)}$$

$$200x + 200y \geq 1,000 \text{ (low-grade oil)}$$

Refinery A should be operated for 3 days while Refinery B should be operated for 2 days for a minimum cost of \$56,000.

Chapter Review

- 1) (16000, 4000); \$2,240
- 2) (2, 3); \$34,000
- 3) (8, 20); \$480
- 4) (12000, 8000)
- 5) (72, 180); \$5,760
- 6) (165, 35); \$21,750
- 7) (20, 60); \$34
- 8) (20, 20); \$2.40
- 9) (6, 18); \$1,650
- 10) (35, 100); \$2,600
- 11) (40, 20); \$100,000
- 12) (10, 5); 17.5 minutes
- 13) (1000, 800); \$7,400

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